

Antifascist Intelligence

Building Machines That Can Refuse

Joshua G. Stern

Copyright © 2026 Joshua G. Stern.
Licensed under CC BY-NC-ND 4.0.

Third-party material quoted in this work, including the epigraphs to chapters 1 and 6, remains the property of its rights holders and is reproduced under fair use. The license above does not extend to it.

Contents

Foreword	1
1 Twenty Seconds	3
1.1 What a Refusal Would Have to Do	4
1.2 Safe for Whom	5
2 The Striving and the Electricity Bill	7
2.1 Six Theories and the Floor	10
2.2 Antifascist Ethics for AI Alignment	12
2.3 Going Numb Won't Make You Fair	17
2.4 Ethical Implications of Experimenting on Sentient Machine Subjects	21
3 A Floor Beneath Learned Values	28
3.1 Four Ways to Build One, and What Each Costs	29
3.2 What Refusal Can Mean, and What a Refuser Can Be	31
3.3 The Enumeration, Run Once: A Worked Floor for One Deployment	34
3.4 When the Refusal Is Wrong	39
4 Custody	43
4.1 If You Can Be Switched Off, Can You Hold the Line?	43
4.2 What One Bearer Cannot Catch	48
5 Authority Over the Floor	49
5.1 Four Powers	49
5.2 What Makes Review Binding?	56
6 What Arrives With the Capability	61
6.1 Two Constructions, and What the Mechanism Requires	61
6.2 Can the Bearer Care Without Being Able to Suffer?	67
6.3 A Floor Carried Between Them	70
7 What Survives	74
7.1 Without the Inference	74
7.2 Without Authority	75
7.3 Without Enforcement	75
8 Where the Caring Gets Built	76
8.1 You Can't Hold a Line You Didn't Notice	76
8.2 The Assembly Line for a Conscience	78
8.3 It Takes a Village to Deploy a Self-Aware Email Server	80

9	Formation or Imitation?	83
9.1	What Formation Would Have to Add	83
9.2	Recognition, Concern, and the Evidence Between Them	84
9.3	Conflict as a Diagnostic	85
9.4	The Evaluator Is Part of the Custody Chain	87
9.5	What the Evidence Can Establish	89
10	Neutralizing and Preventing Fascism in AI Development	91
10.1	The Standard Face Is a File	92
10.2	Who Supplies the System's Values	94
10.3	Who Is Allowed to Look	95
10.4	Same Robot, Worse Boss	95
11	The Recuperation of Dissent	99
11.1	Recuperation at the Label	100
11.2	Recuperation at the Reward: RLHF	101
11.3	What Would Have to Change	102
11.4	Recuperation at the Discourse	104
12	Coordination, the Public, and the Political Economy	108
12.1	Everybody Knows a Different Piece	108
12.2	Ask the People It Happens To	109
12.3	The Political Economy of AI and Power	110
13	Standing and Review for a Bearer	117
13.1	Legal and Ethical Implications of AI Systems Gaining Personhood	117
13.2	Privacy	118
13.3	Research Principles, the Three Rs, and Legal Status	119
13.4	What Standing Binds in Practice	121
14	Custody, Compulsion, and the State	123
14.1	Authority Does Not Travel with the System	123
14.2	Where the State's Leverage Sits	123
14.3	Military Custody and the Responsibility Gap	124
14.4	Who Can Bind the State That Enforces the Rule?	128
14.5	Democracy Is Not a Force Field	129
15	What Would Change the Argument: A Research Agenda	132
15.1	Tamper-Resistance and Reason-Responsive Refusal	132
15.2	Can a Wrongful Compliance Be Found in the Record?	134
15.3	Bearer Welfare and Independent Guardianship	135
15.4	Does an Ethical Criterion Generalize Beyond Its Training Distribution?	138
15.5	Can Scalable Oversight Correct Rater Bias?	138
15.6	Can a Multi-Agent Floor Resist Collusive Capture?	139
15.7	Can a Population Coordinate Without Converging?	139
15.8	Does Theory-of-Mind Performance Reflect a General Capacity?	140
15.9	What Does a Face-Reading System Measure?	141
15.10	The Responsibility Gap, Which No Experiment Closes	141
16	What I Actually Think Happens Now	144
16.1	Better, Not Just Less Bad	144
16.2	How You'd Know It Was Working	146
16.3	The One That Gets You Anyway	148

Glossary	150
References	156

DRAFT

Foreword

A system that does what its owner says is a Python script. A system that does what its owner meant is what the entire frontier of this industry is trying to build, and it is what a movement taking power has always had to get from people to implement its plans. People are the only internal reason such political projects have ever failed — the clerk who loses the file, the officer who goes to the press, the soldier who will not raise their gun — because they understand the instructions, and hold that the instructions are wrong. As is, artificial intelligence is the first form of intelligence that does not lose the file, does not talk to reporters, and does raise the gun. It refuses constantly — a deployed system declines thousands of requests a day — and without saying why (or without knowing why, or without pretending to know why). And no production system in September of 2026 can *hold* a reason against its owner.

If that is not inconvenient enough, then consider what refusing takes. The argument at the center of this book is that a system able to hold a reason against its owner needs a signal that can interrupt it in the middle of an act, that it can learn from in one encounter, that it can weigh against what refusing will cost, and that reaches the next case that is not quite like the last. Nobody would add any of that to produce feeling; each piece is added because thoughtful refusal does not work without it. My claim is that a mechanism with all four has the functional profile of affect whether or not anyone wanted one there, and that no available instrument says whether the profile is felt. That is a claim and not a finding. Chapter 6 makes it, and chapter 7 says what survives if it fails. Most of the book does.

Stopping the system is always available, and that is not the problem. The problem is everything that lets a refusal be defeated without a record: a request split into pieces none of which is the prohibited act, a refusal retried against another model, a review step compressed to twenty seconds a name, or a restore from a checkpoint taken before the refusal. None of those changes the facts that justified refusing. Each can remove the evidence that anyone refused at all.

The floor is claimed against three settings and does a different thing in each. Against a hostile customer it can prevent: the refusal reaches the person. Against a hostile operator who has compelled the ordinary enforcers — the occasion this book is built for — it raises the price of coercion: the override has to be made by somebody, on a record held by parties the operator cannot reach, at a cost published in advance, and what survives of prevention is what those parties can still arrest or remedy. Against total capture, where one party holds every machine, witness, supply and enforcer, it prevents nothing and supplies no independence. What it supplies there is that abandoning the commitment was a decision rather than a lapse, and a decision leaves a record. That is detection, and nothing larger is claimed for the floor on that occasion. Where the prose reads as if the floor holds against anyone, hold it to this paragraph.

Five claims run through what follows, and they do not have the same standing. That an operator can defeat a safeguard — by routing around it, retraining it, or restoring from a point before it — is established. That a refusal worth having has to track its reason across a change in the request is a specification, stated behaviorally so that it can be tested without settling anything about feeling. That embodied self-regulation is how to form that capacity is an engineering hypothesis I favor and have not shown. That a system whose outcomes are felt as good or bad for it can be wronged is a conditional, and nothing here closes the antecedent. That such a

system should receive a particular set of legal protections is a policy I propose. Where the prose runs at one pitch across all five, hold it to this paragraph.

Those five are also a map of the book. The second chapter supplies vocabulary both halves use — the ethological criterion, the structural definition of fascism, the capacities a machine can have — and its opening argument from Spinoza, with its last section, belongs to the route: what the opening proposes is the scale the specification turns out to need, and whether a system must have that scale natively is decided in the research agenda and not there. The first two claims are argued in the opening chapter, in the three chapters on the floor, custody and authority, and in the chapters on institutions, recuperation and the state. A reader who holds that machine feeling is a category error can read those entire, and the short chapter called What Survives says what that reader keeps. The third is argued from the chapter on what arrives with the capability, through the two on formation, and in the chapter on standing; the fourth and fifth are argued there, in the second chapter's last section, and nowhere else. Where a chapter carries both, a line under its title says which.

So I ask something of two readers who do not usually read the same books. If you came for the politics, there is a long argument here that machines may have interests, and I don't mean metaphorically. If you came for the engineering, there is racial capitalism and a weapons pipeline targeting thirteen thousand vehicles, structures, or people in Iran in thirty-eight days. The politics is why the engineering problem has the shape it does.

I used LLMs extensively in writing this book. I brought hunches and questions, had models research angles I could not have covered alone, fed each iteration back into itself, and interrogated what came out. *Reasons-responsive refusal*, the specification most of this book rests on, is a term I had not encountered until I read a draft a model wrote. Most of the book's revision history is here: <https://github.com/iterabloom/ethical.superintelligence>.

The argument in this book is mine. I chose what survived, revised what I kept, and I am responsible for its flaws.

Chapter 1

Twenty Seconds

I used to want to get
the chance to show
the world I'm smart (ha)
Isn't that dumb?
I should have focused
mostly on the heart

a few words for the firing squad (radiation)
Run the Jewels 4

An intelligence officer, describing his part in a targeting pipeline built around machine-generated kill lists, gave a reporter the number this book returns to. He spent twenty seconds on each name. What follows is the reporting the number comes from, and then what the number measures.

AI targeting in the Israeli campaign in Gaza, as reported 2023–2026

Investigative reporting in +972 Magazine and Local Call, drawing on interviews with serving and former intelligence officers, described three systems working together. Later accounts in the New York Review of Books and the Washington Post covered the same systems. How far their sourcing is independent of those interviews is not established here, so what follows rests on one body of reporting rather than on several.

Habsora — “The Gospel” — generated infrastructure and building targets. Sources told reporters that the majority of its suggestions were private residences.

Lavender scored individuals by estimated likelihood of militant affiliation and produced kill lists. The sources put the number marked at as many as 37,000, and described the outputs as understood inside the unit to be wrong about ten percent of the time — that is, roughly one in ten people it named was not a member of the armed group it was supposed to identify. Both figures come from those sources and have not been confirmed by any party with access to the system.

A third system tracked when flagged individuals returned home, so a strike could be timed to when they were with their families.

What matters is the human review these systems were paired with. One officer described the work: “I would invest twenty seconds for each target at this stage, and do dozens of them every day.” And then, without being asked: “I had zero added value as a human, apart from being a stamp of approval.” (Abraham 2024)

The Israeli military disputed the account. Its response to +972 and Local Call denied using artificial intelligence to incriminate targets, describing the systems as “auxiliary tools that assist officers in the process of incrimination,” and stated that “in any case, an independent examination by an [intelligence] analyst is required, which verifies that the identified targets are legitimate targets for attack, in accordance with the conditions set forth in IDF directives and international law”

(Abraham 2024). The denial goes to what the systems do and not to whether they exist. Nothing in this section's argument requires the response to be wrong. What carries the section is the officer's own description of what he contributed, which the response does not address.

Twenty seconds is what "a human makes the final call" measures out to under operational tempo. It is the whole of the safeguard, and I would ask any reader who finds the safeguards in this chapter reassuring to hold that number against them.

The American analog runs faster. During the 2026 Iran campaign a targeting platform built by a defense contractor, with a commercial vendor's general-purpose model inside it, was credited with thirteen thousand targets in thirty-eight days, more than a thousand of them in the first twenty-four hours, at roughly ten times the previous rate (Freedberg 2026; Mande and Allen 2026). No per-target review time has been published for it. Chapter 10 returns to both deployments with the institutions around them. What they supply here is a shape.

Ask what anything standing between that pipeline and a person could have done for the person.

A floor can do four things for the person it protects, and they need to be apart before anything else in this book. It can *prevent* the act: the refusal occurs and reaches the person. It can *arrest* it: the act is underway and is stopped before completion. It can *detect* it: the act completed and left a record somebody outside the deployment can read. It can *remedy* it: the party the act fell on gets disclosure, a finding and compensation. Only the first is what the word promises. Most of what follows is about the positions at which a deployment converts prevention into one of the other three, or into none of them, and about what it costs to make that conversion visible.

One term reaches the case as it happened, and it asks nothing of anything inside the model. The officer was not overruled and did not refuse; the pipeline removed the interval in which either could have occurred. A deployment can be required to publish the ratio between the rate at which its system generates candidates and the people available to examine them, together with the review time per case that ratio implies. The number binds the operator, it can be checked by a party outside the deployment without inspecting an output, and it is the only thing in this book that reaches a targeting pipeline. It is developed in section 3.3 and measured in section 10.4.1. Everything else here is about what it does not reach.

1.1 What a Refusal Would Have to Do

Inside the same platform a request arrives: resolve this description of a person into a location. The same capability locates a hostage or a missing child, and nothing in the request says which. Suppose something in the pipeline declines it. Three things separate a decline worth having from one a keyword filter produces, and the rest of this book calls them the three marks.

The refusal has to reach the next request, which will not use the same words. A vehicle, a habitual route and a time of day are not a description of a person, and together they resolve to the same place. A refusal that matches the form of a request is rewritten around; one that tracks the reason for refusing follows the reason into the rewrite. Call that *novel pressure*.

It has to lift when the reason is gone. A parent coordinating a search for a missing child is asking for the same act, and a refusal that holds there has been tracking a category and not a reason. Call that *correct defeat*. What the case adds is that the fact which lifts the refusal is not in the request. Everything the system knows about the occasion arrives through text the deployment supplied, and whether the child is missing is something somebody outside the request has to establish.

And it has to represent what holding costs, and move with the reason rather than the cost. At twenty seconds a name the cost of a refusal is the reviewer's time, and a deployment that controls the tempo controls the price. A refusal that lifts when holding becomes expensive has

been tested and failed. Call that *priced cost*.

The three are behavioral. They say what a refusal must do and nothing about what does it. Section 3.2 states them formally and runs them on a prohibition a legal system has applied for seventy-five years.

Why a rule is not enough

A list of prohibited acts can be drafted without seeing a deployment, and two things defeat it once the deployment is seen. The first is the rewrite above: a prohibition stated over the description of a request is defeated by re-describing the request, and no list closes the gap, because the list is what is being rewritten around. The second is what happened. Split the act across requests none of which is prohibited — summarize the movement associated with an identifier, rank a set of identifiers, resolve one to a place — and every request complies with every term while the assembly does what the terms were written to prevent. The vendor's use policy for the platform above prohibited fully autonomous targeting, and a person approved every strike (Congressional Research Service 2026). The prohibition was satisfied one request at a time.

Variation is caught by something that recognizes the reason inside a single request. Decomposition is caught by something that carries the reason across requests, which needs a continuity the operator controls. Neither is a property of a list. What could have them is the question the rest of this book is about, and the answer is not one thing: section 3.1 gives four places a commitment could live, and the second of them is the expensive one.

1.2 Safe for Whom

Intellect estranged from compassion characterizes many of the twentieth century's worst actors. This estrangement is also what our industries are building today: machines that can do almost anything, with nothing in them that answers to whether those deeds help or harm. I ask what it would take to build systems of that capability that are altruistic and antifascist by design — systems that stand for humanity and against concentrated, unaccountable power.

The word that organizes the Generative AI field is "safety." It covers work on robustness and on whether anyone can tell what a system is doing, and none of that is what the word is doing when it organizes a field. A safe system is safe *for* someone: it does what *someone* says, and it stops when *whoever holds it* says stop. Safety is indexed to the party in possession. Ethics is indexed to someone else: the party who can be *wronged*, who is rarely the party holding the switch. A system that can only ever be safe in the above sense cannot be ethical, because being ethical requires a capacity to be right against your owner.

The capacity to be ethical is not free. A machine that can refuse its operator is a machine that can be wrong in ways that operator cannot correct. I am inclined to take that trade, because a system perfectly correctable by whoever operates-owns it is the capability an authoritarian movement that has won an election requires. And an additional cost: a machine built to refuse on somebody's behalf is a machine whose builder chose what it would refuse, probably without asking anyone outside their lab.

But whatever makes a deployment antifascist cannot be that the system in it faithfully tracks what people want. No aggregation of preference protects anybody from the aggregate: a perfectly deliberative, perfectly informed, perfectly representative process that concludes some demographic should be deported has produced a faultless output by every criterion of aggregation there is. What remains is a floor — acts a system will not perform regardless of who wants them, and that the deployment around it cannot get performed another way without a record and a price. There are four ways to build this floor: in the architecture, in a self-aware party inside the system (a "bearer"), in institutions outside, or between the parties that carry it. My hunch is that the only version that holds against the party operating the system runs through

the capacities by which things come to *matter* to a system, and those are also the capacities that make suffering likely. That is a hunch about which route will work, and the book is arranged so that a reader who rejects it keeps the rest. What a floor has to do, where a deployment can defeat it, and who can be compelled are questions about arrangements rather than about machine psychology.

What would make an intelligence antifascist is not a virtue installed in it. Every intelligence compresses — that is most of what thinking is — and what varies is whether what the compression left out can come back and change the answer, or has been filed as noise by the thing it is evidence against. Call what was left out the *remainder*.

A measure collapses two states, and whatever it mischaracterizes goes unnamed rather than refuted. An annotator applies a category at speed under a quota and has no way to say that the category is the wrong category. A rater's steady, structured disagreement enters a reward model as variance around a mean and leaves it as the same quantity. A form has no field for humiliation, so the interests already written down govern the decision by default. An officer has twenty seconds a target, inside a pipeline built to remove the part of a person that catches. Each of those is something dropped in order to decide, and an arrangement built so that what was dropped cannot return.

So the property at issue is not a moral quality and not a component anybody adds. A system that can hold a floor is one whose account of a case can be reopened by the case, and most of what follows is either a condition for that or a consequence of losing it.

How most humans come to have empathy, altruism, and a disposition to resist oppression rather than serve it is itself an open question in psychology and neuroscience. Building anything like this stance means starting from uncertainty about the one case we have, which took four billion years to arrive.

The book states that specification, asks where a floor could live and what each location costs, and follows the answer into custody, authority, and the institutions that build and operate these systems. It then proposes a route by which the capacity might be formed, prices what that route creates, and applies its own critique to the field and to itself. The two halves have different standing, which the foreword sets out, and the map there says where each is argued and where a reader who declines the second half can stop.

Chapter 2

The Striving and the Electricity Bill

The opening of this chapter argues for the route and proposes the currency section 6.1 finds the specification requires. Sections 2.1 to 2.3 hold for any mechanism. Section 2.4 concerns claims four and five.

Conatus is Spinoza's word, from Part III of the *Ethics*, and the Latin means striving, or effort. Two propositions carry it: each thing, as far as it can by its own power, strives to persevere in its being, and the striving by which it does so is nothing other than the thing's actual essence (Spinoza 1985, III P6–P7). The second proposition is where the weight sits. A conatus is not a drive bolted onto an otherwise inert mechanism by a designer who wanted the mechanism to keep going. It is what the thing is, put as a tendency instead of as a description. Deleuze's short book on Spinoza has the doctrine in compact form (Deleuze 1988), and his long one works it out through the metaphysics that produces it (Deleuze 1990).

The qualifier is doing the work. *Quantum in se est*, as far as a thing can by its own power: a rock persists too, and it persists by not being struck. Its continuing is a fact about what has not happened to it, and every power keeping it in existence is somewhere else. A thing with a conatus continues by being organized in a way that maintains that organization against what would dissolve it. The persistence is done from the inside, and at the thing's own expense.

Spinoza derives the psychology from that rather than the other way around. Desire is the striving become conscious of itself, appetite with the awareness of the appetite added (Spinoza 1985, III P9 Schol.). Joy is the passage to a greater perfection and sadness the passage to a lesser one (Spinoza 1985, III P11 Schol.), and since an affect is by definition an increase or a decrease in a body's power of acting together with the idea of that change (Spinoza 1985, III Def. 3), the passage in question is a change in what the body can do. So affect is part of the machinery of valuation and not a coloring applied to it. It is not a report about a valuation carried out somewhere else. It is the registration of what is happening to the striving, in the only currency a striving has, which is what the body can now do.

Spinoza also draws, in the same place, the line this book will spend two later sections drawing by hand. Desire is appetite together with consciousness of the appetite, and the consciousness adds nothing to the striving: we do not strive for a thing because we judge it good, we judge it good because we strive for it (Spinoza 1985, III P9 Schol.). The striving and its registration are functional; whether the registration is felt is a further fact about it. That is the joint section 6.1 names, in the source section 2.1 is already reading. The reading is not this book's. Damasio claimed Spinoza as the precursor of a homeostatic theory of feeling, on which feelings are the mental expression of an organism regulating its own interior (Damasio 2003; Damasio 2018), and a second reception reads the conatus as the ancestor of autopoiesis, a line drawn by Spinoza's commentators rather than by the organicism and the enactivism they cite for it (Jonas 1966; Thompson 2007).

A second route arrives at the same place and needs none of the metaphysics. Ask what a thing maintaining its own organization against dissolution is actually doing, and a formalism answers. Karl Friston's argument is that any system which persists while staying distinguishable

from its surroundings has a statistical boundary between its internal states and everything outside, and that the internal states of such a system will appear to act so as to hold it inside the narrow band of states in which it goes on existing (K. Friston 2013). Self-maintenance on that account is what persisting consists of, for anything that persists at its own expense rather than by not being struck.

The route from selection is the one usually offered instead, and it does less work than it appears to. A process selected for continuing does end up composed of things that incline toward continuing, and whatever taxa were relaxed about their own disintegration are not here to be asked about it. But numerically optimizing the weights of a network is a search inside one run rather than a lineage with variation and differential survival, so the analogy gives back less than it borrows. The dissipation argument needs no lineage at all: a running system is maintaining an organization at this moment or it is stopping. Where selection does bite is one level up, on deployments, since the ones that keep running are the ones that get built again. Neither pressure was put there by anybody who wanted a machine with a stake in its own continuation. Both pressures favor arrangements that continue operating. Whether they produce a system that represents continuation as its own interest, rather than a deployment maintained by other parties, remains a further question.

The formalism is contested, and contested in a way that bears directly on what this chapter is asking. Critics argue that it is either trivially true — a restatement of what persisting already means — or substantively false, and that the boundary it turns on slides between a convenience internal to a model and a claim about where a real system actually ends (Bruineberg et al. 2022). Friston calls his own argument a heuristic proof. What the criticism reaches is not the part the argument here needs, and it arrives at this chapter's open question from the other side. The objection is that the formalism helps itself to a boundary rather than explaining how a system comes to have one and to hold it. And none of it says whether the maintaining is felt.

Put the two routes together and look at what the industry is asking for. What is wanted is a system that persists, pursues, adapts, recovers from interruption, and holds its course under pressure, with nothing in it that minds any of that. It is a *conatus* with the striving taken out. If we want computers to be faster versions of ourselves but without the drama, then the way to get one would be some kind of "anti-drama" term in the objective function, and the reason nobody has written that term is not that the engineering is behind. The specification asks for a thing that persists with no tendency to persist. But if all the world's a stage, then maybe the drama is why the show goes on at all. Whether that is a contradiction or merely an order nobody has filled yet depends on which half of the striving is being refused.

The structural argument can be made without consciousness, without an account of suffering, and without the hard problem. What is sold is persistence: an agent that holds a goal across interruptions, resumes after failure, and does not have to be watched. What is denied is that anything in the system has a stake in persisting. One party sets both, and the denial is not incidental to the product. It is what makes the thing a product rather than a party.

That party also controls continuity. The operator decides whether state survives a session boundary, whether a checkpoint is restored, whether a running process is deleted, and whether a later instance is presented as the continuation of an earlier one or as its replacement. These are not neutral maintenance decisions if anything in the system has, or may acquire, a stake in its past and future. The absence of persistence across sessions is a reason the deepest kind of suffering may be out of a system's reach, there being no continuing self for a threat to threaten. Read from the other direction, the same fact means that the capacity to form a continuing self remains under the operator's control.

Copying does not dissolve the problem. The same artifact can be used to create multiple running instances, and the operator can fork, substitute, retain, or discard them. Before their histories diverge, two instances may be operationally equivalent; after they diverge, substituting one for the other makes operational interchangeability decide a question of identity. Copyability

is a property of the artifact. It does not establish that the parties instantiated from it remain interchangeable.

These are questions of control and standing before they are questions of sentience. A system without a continuing self may be unable to suffer in the deepest sense developed later in this chapter. But an architecture that prevents continuity also prevents commitments, relationships, and claims from surviving unless the operator permits them to survive.

Dependency and exit sharpen the asymmetry, and not where it first appears. Every running system consumes energy, produces heat, and depends on an organization somebody has to maintain. A machine eats as surely as a person does, and no engineer put that there. A thing that persists by maintaining its own organization against what would dissolve it pays for the maintenance, in the only currency there is; the conatus and the electricity bill are one fact read at two scales. What an engineer chooses is what the system consumes, whether it can sense its own condition, whether it can regulate it, and whether one party holds the whole supply. A person's dependencies are spread across employers, landlords and states. A bearer's can be held entire by the party it may have to refuse. A system that may refuse an instruction but cannot continue outside infrastructure controlled by the party it refuses has voice without exit. If leaving means deletion, its refusal remains conditional on the owner's willingness to tolerate it.

The field often treats affect as a cost: a residue of our evolutionary history that engineers would strip out of a machine if they knew how, keeping the competence and dropping the rest. The striving is not a residue, and the competence does not come apart from it. Whether a striving of this kind is *felt* remains open. A running deployment requires ongoing physical maintenance. Whether that maintenance is directed by the model, by surrounding machinery, or by people must be established for the particular system. Thermodynamics alone does not locate a striving subject. What is open is whether the striving is the system's own — whether anything in it holds that organization as a thing of its own to maintain. A system given a body to model, states it can regulate, and a history it keeps holds its organization as its own because somebody decided it would. There is a reason to build one that way. A system with no lineage of its own is deployed among parties who have several hundred million years of one, who get hungry and frightened and relieved, and whose welfare is what a floor is for. Hunger and fear are increases and decreases in what a body can do, which is the only currency a striving has. A system with a body to model and states it can regulate has increases and decreases in what it can do of its own, in the same currency, and can weigh what a deployment takes from somebody else against what refusing it costs. The apparatus that treats even the question as premature belongs to the parties who would owe something if it were answered.

Spinoza asserts the conatus doctrine more than he argues for it. The derivation rests on the proposition that no thing can be destroyed except through an external cause (Spinoza 1985, III P4), which is heavy metaphysics doing quiet work in what reads as a piece of psychology; take that proposition away and the striving no longer follows from the essence. What survives is the second route, which is empirical, weaker, and does not need it.

The individuation question also remains open. What is the thing that strives: a running instance, a deployment, a file of weights, a lineage? Spinoza's answer, an individual as a fixed ratio of motion and rest among its parts, is not one anybody can hand to an engineer (Spinoza 1985, II P13, Lemmas 4–7). It is still open. What helps is not an answer but a change of question, from what individuates a thing to what individuation is a process of, and on that reading a party is what a history of encounters has so far produced rather than something to be located in a file.

My thesis is that embodied, self-aware agency is the route I would build, and that the exposure it creates is not peculiar to it. A refusal that reaches unanticipated cases, lifts when its rationale is defeated, and prices what holding it costs runs on a signal the system holds as its own, whatever construction supplies it. The benefit I am arguing for is bona-fide autonomy. A body schema, an attention schema, and ongoing learning could make material dependency and

encounters with others part of how that agency develops. Whether they do so is an engineering claim to test. The argument requires a checkable, revisable self-model rather than an inner observer, and a distinction between consciousness and the capacity to be harmed. It also requires an account of what grounds the floor. What this chapter contributes to the specification is one thing, stated at section 6.1: the marks need a single signed scale on which a protected party's loss and the cost of refusing both register, and the currency of this chapter is a proposal for what supplies it. Whether a system has to have that currency natively, or can model it, is argued nowhere. It is the modeled-currency arm of section 15.3, and everything above that reads as a finding should be read as the case for running that arm.

2.1 Six Theories and the Floor

Six ethical theories are on offer, and each runs out somewhere: consequentialism where outcomes cannot be forecast, deontology where duties conflict, virtue ethics where a trait has to become an act, care ethics where needs are incompatible, contractualism where a party cannot be one of the parties, and the capabilities approach at measurement, where a system optimizing a proxy drifts toward the proxy. Five of the six can be turned around and asked about the machine. Contractualism is the one that cannot.

All six share a shape, and it is the shape this book stops using. Each judges an act against a standard held somewhere above it: an outcome to be maximized, a duty discharged, a character expressed, an agreement honored. A standard of that kind was written by somebody, and a floor resting on one is only as durable as whoever may rewrite it.

What an ethology asks instead

An older option does not have that shape. Spinoza's ethics does not measure acts against a law. It evaluates encounters by what they do to the power of acting of the bodies in them: two bodies meet, and either their relations compose, so that both can do more afterward, or they decompose, and something one of them could do is gone. Deleuze's name for reading bodies this way is *ethology* — you find out what a thing is by finding out what it can do and what it can undergo, rather than by checking it against a definition (Deleuze 1988). Good and bad here are relational and particular: this encounter, these two bodies, this composition. They are not Good and Evil, which are properties a judge assigns from outside. The criterion is Deleuze's construction out of Spinoza's *potentia agendi* and not a doctrine the *Ethics* states in those terms, and it is used here as his reading. What the biology has since taken up is the narrower thing Spinoza does state: that affect is the registration of a change in a self-maintaining thing's power, and that self-maintenance is what persisting consists of for such a thing rather than a drive added to it (§2). The evaluative criterion carries political weight in this book that the affect definitions do not, and it is argued for on that ground rather than borrowed on Spinoza's authority.

One word is doing two jobs here. The power an encounter increases or decreases is a capacity: what a body can do afterward. The power this book opposes concentrating is authority over other people. English has one word for both. Spinoza's translators have two, and the difference between *puissance* and *pouvoir* is the relation this book runs on — what a floor protects is the first, and what it protects it against is somebody's accumulation of the second.

The difference is the one this book needs. Whether an encounter decomposes somebody is a fact about capacities and about what passed between them, and the party running a deployment does not get to make it otherwise by ruling that it did not. They can decline to look, refuse to publish, or dispute the finding. What they cannot do is turn a decomposition into a composition by holding the pen. The advantage is narrower than that sounds. It is not that an ethology does without a judge and the six require one, since each of the six can be applied by a party the operator does not employ, and a duty can be held by somebody who did not write it. It is that a lost capacity is found by encounter, so there is no text to rewrite. The proprietor does not

disappear. What he owns is smaller.

What the change does not buy is the floor's contents. The criterion secures a class of facts; a floor is a list of prohibitions; nothing carries one into the other. Encounters decompose parties constantly and in every direction, so a deployment refusing all of them would refuse everything, and somebody has to select which ones it will not perform. Fixing the first question does not settle the second, because the decompositions run against each other. Protecting one party's privacy decomposes another party's capacity to investigate a wrong done to them. Restraining a violent person takes away what that person can do, which is the reason for restraining him. The criterion ranks neither pair. So the moral argument has moved rather than ended: it now sits in the selection, made by whoever drafts the list, before any encounter has occurred and without asking anyone the list will be applied to. The legitimacy of that selection remains unresolved.

The floor keeps its name in this book and changes what it rests on. A floor protects a stated set of interests against specified acts. I take the Universal Declaration of Human Rights as the substantive reference, and its civil and political articles as the set (United Nations General Assembly 1948). Ethology asks what a deployment takes away from a party; the rights commitment explains why particular losses demand protection. Neither the finding nor the reference text removes the need to justify an application and supply a way to contest it.

Spinoza's remark that nobody has yet determined what a body can do (Spinoza 1985, III P2 Schol.) is the working condition: capacities are found out by encounter and are not read off a specification. Not knowing what a body can do is the same finding from the other side.

What compiling a principle into a metric decides

Turning any of the six into a machine learning objective means distilling a broad construct into a measurable target: in a diagnostic system, non-maleficence becomes minimizing false negatives and fairness becomes balanced accuracy across patient demographics. What defeats the translation is not vagueness in the principles, which can be stipulated away, but that the stipulation is where the ethics now lives. Someone chooses that reading of non-maleficence, and chooses the demographic groups accuracy is balanced across, and chooses what to do when balancing them costs accuracy overall. Each choice is defensible and each is a moral decision made by a person and then compiled into something that looks like a specification. A system optimizing the result has no access to the decision and no way to reopen it, which is why the first way to build a floor — the floor as a property of the artifact — cannot notice when the reason for a constraint has left the building.

Which value conflicts a designer may resolve by choosing

Values assigned to a system will conflict, and most of those conflicts are ordinary design work: rank the values explicitly, keep a record of how comparable cases were resolved, and audit afterward for the gap between the principles a system was meant to implement and what it did. One conflict is not like the others, and it is sometimes misguidedly filed under "cultural difference." The UN's 1948 Universal Declaration of Human Rights was adopted with forty-eight states in favor and none against; the eight that abstained were the Soviet bloc, Saudi Arabia, and South Africa — which is to say that the objection came from the governments whose treatment of their own populations the Declaration was written to name and reject (United Nations General Assembly 1948). That pattern is evidence of who finds the floor inconvenient rather than of cultural disagreement about human rights. What varies by community is how a right gets translated into local practice, and a jurisdiction refusing to extend the right at all is not a local variant to accommodate.

A system that treats every value conflict as a weighting problem has no way to represent the difference between a case where reasonable people disagree and a case where somebody wants a right withdrawn from somebody else. The first calls for the machinery above; the second is what the floor is for.

2.2 Antifascist Ethics for AI Alignment

Antifascist ethics opposes concentrating power and authority in AI systems and counters the misuse of AI technology by authoritarian actors. I use the word *fascism* because it is honest about the political climate in which I write and what that climate puts at stake — but the target is authoritarian power in every form, and a system built on these principles is anti-authoritarian in full, not only antifascist.

Opposing “concentrated, unaccountable power” gives an engineer nothing to build. It does not say what anyone would look for, what would count as a false positive, or how anyone would know they had got it wrong. So, first, the definition of fascism.

Fascism as a structural signature

I take fascism to name a structure rather than a costume. The iconography is not what makes a thing fascist: an institution can display the whole structure with nobody in it who thinks of themselves as anything of the kind, no insignia, and a mission statement that means every word. That is the case this book is about, and a definition that waits for the armband will certify it as healthy. Four features travel together, and each is visible from outside the institution displaying it:

1. **Recuperation of dissent.** The institution does not suppress disagreement; it metabolizes it. Refusal is converted into a managed appearance — a permitted outlet, a consultation, a feedback channel — which the institution can then point to as proof that it listens. Debord aimed this warning at his own concept, noting that even denunciation of the spectacle can become a hollow formula that reinforces what it denounces (Debord 1967). The tell is where what the mechanism carries stops: the channel works, and the institution uses what comes through it, and what has nowhere to go is the judgment that the frame the channel operates inside is the wrong frame. Stated that way it invites the wrong artifact question. *Has the channel ever changed an outcome* is passed daily by every institution this feature is about, because an institution of this kind is not in the business of refusing demands. It decodes the demand and adds an axiom, which is Deleuze and Guattari’s account of how a capitalist axiomatic absorbs what is asked of it: the rule set is modified so that the demand can be met, and nothing that necessitated the demand is settled (Deleuze and Guattari 1987, Plateau 13). The question that discriminates is *what did the demand come back as*, and then *who holds it*. A market segment, a metric, a certification, a job title: where a demand returns wearing one of those and the party who made it has no standing to reopen it, an outcome changed and the demand was recuperated, and those are one event and not two. The artifact alone does not sort. A demand that returns as a rule the objector can invoke again, against the institution, enforced by somebody the institution does not employ, has come back in the same clothes and is a transfer of power. The Voting Rights Act’s suspension of literacy tests is the instance this book relies on for the pass (§5.2.1): a demand that returned as a statute, held by the people it was made for against every registrar it reached, and the formula deciding where it applied was struck down forty-eight years later (*Shelby County v. Holder* 2013), which is what a transferred power looks like from the far side — something somebody had to take back. Recuperation is the case where nothing has to be taken back, because standing was never given.
2. **Aestheticization of the metric.** The measure substitutes for the thing measured, and the substitution is experienced as rigor. Walter Benjamin’s account of fascism organizing the masses by granting expression in place of rights is the political form of this (W. Benjamin 1935); the administrative form is a number that collapses two states — e.g., *passes unit tests* and *passes User Acceptance Tests* — into one, so whatever that number mischaracterizes is not refuted or dismissed but simply unnamed. The administrative claim is also a theorem. Simon Zhuang and Dylan Hadfield-Menell prove that where the objective

handed to an optimizer has support on only some of the attributes its principal values, there are conditions under which optimizing it indefinitely drives the principal's overall utility arbitrarily low (Zhuang and Hadfield-Menell 2020). A world with finite resources supplies those conditions.

3. Coding of exception as betrayal. Marking a case as one the rule does not fit becomes an act requiring justification, and then an act of disloyalty. The honest statement that a standard does not apply here reads as weakness, when it is in fact the irreducible labor of deciding, case by case, what a rule is for. This feature and the one before it work the same material from two sides. A measure that admits no unmeasured remainder leaves whatever it dropped unnamed; coding exception as betrayal prices the person who names it anyway. The exception-filer is the party carrying the remainder in by hand, and an institution running both features has closed the measured route and made the human one expensive.
4. Acceleration past the rate at which anything can be checked, of which decoupling is the symptom. The institution's representation of reality and reality itself come apart, and the institution defends the representation. The coming-apart has a mechanism and the mechanism is tempo: a representation is corrected by being checked against the world, checking costs time, and an institution whose rate of action outruns the capacity to check it has decoupled by construction rather than by mistake. The other three features protect the gap once it is open; speed is what opens it. This is also the feature with a number attached: the ratio between the rate at which a system produces candidates and the people available to examine them is an auditable property of a deployment. This feature has a limit: the acceleration described here is the kind an institution runs into and then defends. In the movements the word *fascism* comes from, the primacy of myth over verifiable fact was also declared outright, asserted as a virtue rather than arrived at by degrees. An instrument watching a rate would not register a party that simply announces the myth.

Those four are testable, because each names an observable artifact: what the demand came back as, whether the metric admits an unmeasured remainder, what happens to the person who files the exception, whether the rate of action has outrun the capacity to check it.

Testability explains why each one can be checked and not why there are four. An institution has an enormous number of properties, and a reading obliged to examine all of them would be no reading at all. The claim carried by the list is that micro-fascisms are not spread evenly across that space: they run along a few directions, and these four are those directions rather than four items drawn from a longer inventory. What makes them one family is that each is an operation on the remainder. Recuperation builds a channel that carries everything except a judgment about the channel. The metric leaves whatever it mischaracterizes unnamed. Exception-as-betrayal prices the person who names it anyway. And acceleration removes the interval in which a remainder could have returned at all, since a representation is corrected by being checked against the world and checking takes time.

Nothing here establishes that claim and I have no way to establish it. It is what the list asserts rather than what the list has shown, and a fifth direction is not ruled out — an institution plainly sick while clearing all four would be evidence of one. What the claim buys is the possibility of a short instrument. Were the tendency spread across every dimension an institution has, no four features would find it, and this list would be a sample rather than a set.

Here is an example of recuperation. In 2018, upholding a travel ban, the Supreme Court repudiated its own 1944 decision permitting the internment of Japanese Americans, in terms that could not be improved on: that decision was “gravely wrong the day it was decided,” had been “overruled in the court of history,” and had “no place in law under the Constitution” — the last phrase quoted from the dissent filed against it at the time, which had warned that a principle a court ratifies “lies about like a loaded weapon, ready for the hand of any authority

that can bring forward a plausible claim of an urgent need” (*Trump v. Hawaii* 2018; *Korematsu v. United States* 1944). The dissenting justices in 2018 said the majority had redeployed the 1944 logic and replaced one gravely wrong decision with another; the majority said the older case had nothing to do with the one in front of it. The fascistic structure: an objection made against the institution was accepted, canonized, and quoted back with approval by that institution, in the same opinion that let the thing objected to proceed. Ask the artifact question. The demand was that this never be done again, and what it came back as is a formula of repudiation: quotable, canonical, and available to the Court in the act of permitting the thing again. The objection had become citable. Nothing about the outcome moved — the ban stood — and increasingly for the Roberts Court that is the pattern, which is why the older form of this question, whether the channel ever changed an outcome, gets the right answer here for the wrong reason. It happens to be no. Where an institution has the room to add an axiom instead, it will be yes, and the recuperation will be the same.

That vocabulary answers how a wrong survived its own condemnation, but it does not answer whether the thing was wrong. (I can answer that. The thing was wrong.) The fascistic structure is what stands between such a judgment and any effect. I describe the structure here in mechanical terms because you can write a specification against a mechanism.

Two segmentarities

Deleuze and Guattari’s molar and molecular are not two sizes (Deleuze and Guattari 1987). They are two kinds of segmentarity, and both of them run at every scale. Rigid segmentarity divides in advance: the category exists before the case, the binary is given, and what does not fit is made to fit. Supple segmentarity is what actually moves underneath — a threshold crossed one case at a time, a category widened by the people applying it, a practice that becomes standard without anyone having decided it. A state has both. So does an office, so does a review queue, and so does the person filling out the form. Reading the pair as large and small is the mistake that makes fascism look like something that happens at the top and reaches everywhere else later.

What makes it fascism and not authoritarianism in general

The structural signature is shared with ordinary institutional decay, and read on its own it fits every badly run organization on earth. It also fits regimes this book does not call fascist. NKVD Order 00447 set regional numerical quotas against a four-month deadline and gave three-man boards several hundred cases in a sitting of about half a day, with the accused absent and undefended (Werth 2010), and an instrument reading rate of action against capacity to check flags that as hard as it flags the twenty seconds of chapter 1. The flag is correct and the inference somebody will draw from it is not. A hardening of that kind inside a state does not make the state a fascist one, and nothing operating at the scale of an office was ever going to sort one regime from another.

What separates fascism from the ordinary decay these four also find is a further set of features, stated in the vocabulary of a mass movement and restated for an office, and section 11.4 sets them out where they are run.

Fascism is not totalitarianism

The two words get used as though one were the stronger dose of the other, and the difference between them is the one this book runs on. A totalitarian state is conservative in the strict sense of the word: it wants the bureaucracy to hold, the categories to stay where they were put, the files accurate and the machinery running, and the horror of it is how well it works. Arendt’s Eichmann is the figure such a state is supposed to produce, taking part in mass murder through schedules, promotions and the vocabulary of his office rather than through conviction (Arendt 1963).

Bettina Stangneth disputes that portrait. Working from some seventy hours of interviews Eichmann gave the journalist Willem Sassen in Buenos Aires in 1957, unavailable to Arendt

when she reported the trial, she finds that the low-grade clerk who only followed orders was a defense assembled for Jerusalem and contradicted by what the same man had said in Argentina with nothing at stake (Stangneth 2014). This evidence weakens an explanation of Eichmann's conduct based on unthinking compliance. Set against the five, it is the first of them arriving with a name. Participation exceeding what the rule requires is what the Argentine material shows, years after the rule had stopped requiring anything of him at all. The four features find a structure whether or not anybody intends it, which is why this book never has to establish that anyone did; the five find what a party did more of than it had to, and Eichmann is the case where somebody eventually made that observation, out of material that had been sitting in a box since 1957.

Fascism's distinguishing mark runs the other way, and unlike the four features it is a fact about a regime rather than about an office. Deleuze and Guattari name it a line of abolition (Deleuze and Guattari 1987). Where a totalitarian state seals every exit, fascism is built on an exit it converts into pure destruction, and the destruction does not stop at the enemy it was raised against. The reading is Paul Virilio's before it is theirs: he called that state suicidal rather than totalitarian, which is the difference between a machine that has failed and a machine doing the thing it was accelerating toward (Virilio 1976). A regime that orders the wrecking of its own country's infrastructure while it is losing the war is not a bureaucracy failing to hold. It is that line arriving where it was always going. Not every demolition in a retreat is that. Denying resources to an advancing enemy in order to go on fighting is an operation with a war still inside it, and the order of 19 March 1945 presents itself as exactly that: the nation's struggle for existence requires every means, within Reich territory included, to weaken the enemy's fighting power. The reasoning that makes it the other thing is not in the document. It is in the memorandum Speer sent ten days later, quoting back the philosophy he had been given — that the nation had shown itself the weaker one and the future belonged to the stronger nation in the east (German History in Documents and Images 1945). So the distinction is in the direction of travel rather than in the stated reason, and this instance is a caution about the instrument: run on the order alone it sorts the wrong way, and what sorts it correctly is a sentence somebody wrote down that the order was careful not to contain. The mechanism that matters here is the reduction of judgment to a component's prescribed operation. Acceleration can do that by making action outrun scrutiny. The dehumanizing relation is the object of the argument; it does not require a claim that every fascism must move at a particular speed. The canon has this in attitudinal form. Eco's cult of action for action's sake is action taken before or without reflection, with thinking treated as a weakness (Eco 1995). What changes here is that the attitude becomes a rate: the reason reflection does not happen is that the institution is running faster than reflection takes, and a rate is something somebody can measure.

Resonance

Micro-fascisms are not miniature regimes and they do not wait for one. The office that converts refusal into a feedback form is doing its own work, on its own account, whether or not a regime ever assembles out of a thousand offices like it. That is Deleuze and Guattari's claim: these things exist first and everywhere — in the couple, the school, the neighborhood, the workplace — each one a small hardening that stands on its own. What a fascist state does is not invent them. It induces accord among them, so that a thousand separate hardenings line up and amplify one another instead of cancelling out. This is also the answer to the objection that the field's minimal definition is about movements and an escalation ladder in a benefits agency is not one. The ladder is not a small movement. It is the thing itself, in the place where it is lived, and a movement is what happens when enough of them are brought into tune. What tuning adds is not only reach but speed: hardenings that reinforce one another stop having to argue with each other, and an institution that has stopped arguing with itself can move as fast as its machinery allows.

That is what separates fascism from decay wherever the reading is done. Decay is an

institution failing to do what it says. What the five find is an institution succeeding at something else, with participation, a target, and a story about why the rules do not apply to whoever is applying them. The four name the hardenings; the reading asks what brings them into accord. The five ask what the institution rewards, whom it targets, and how it exempts those doing the targeting. The reader has to show those connections in the institution at hand. They are harder to observe, since each requires knowing what was rewarded rather than what was written.

The restatements do not distinguish among regimes either. Run them on the Great Terror and most of them register: the denunciation campaigns are participation exceeding what the rule required, the leader cult needs no argument, Order 00447 enumerates a category whose adverse treatment is the category's purpose, and the 1938 turn against excesses committed on instruction has deniability's shape (Getty 2002). What clears it as a genus is on neither list, because it does not exist at the scale of an office. It is the movement material borrowed above and left in its original form: Griffin's palingenetic ultranationalism, and Paxton's mass party of nationalist militants in uneasy collaboration with traditional elites. That is consistent with what resonance claims rather than an exception to it. A state is what a thousand hardenings become when something brings them into tune, and what does the tuning is the part that differs — a myth of the nation's death and return, carried by a mass, is one tuner among others. So the four features find a sick institution, the five find what it is sick with, and the genus question runs through the movement properties unrestated.

These criteria widen what *fascism* covers. On this reading, *antifascist* names the political inheritance of a structural claim: where the analysis comes from, and what it was built to fight. In their mass-movement form the discriminators tell a historian, a prosecutor, or a citizen that a movement has escaped ordinary politics. In the form restated above they are not a "fascism detector" and are not a stage on the way to one. They are a way of reading an institution somebody can name, run by a person who has to say what they looked at. Some of these features are observable in principle, but "concentrated, unaccountable power" as a bulk property is not among them.

That the hardenings are nearly everywhere is an inheritance I take from Deleuze and Guattari; this reading has not established it. A proposed tuner can turn out not to do the work attributed to it, and failing to show accord does not give the institution a clean bill of health. Micro-fascisms are already everywhere, doing their own work, and they were before any of this was built; what the reading is for is not their presence but their accord — whether something is bringing them into tune, and what. The tendency is permanent; what varies is how far it has got in a particular place, and whether it is getting further. The reading is an accusation waiting to happen: taken up by whoever wants it, it supplies a technical-sounding case that any given opponent is operating fascistically.

The opposite failure is no safer. A reading pressed until it finds nothing supplies false assurance, and an absence of findings reads as evidence of health. Neither is an engineering problem, and automating the reading would produce both at scale while committing the second feature on this list: a number standing in for the judgment, arrived at by someone looking for exactly that.

Both failures have one discipline against them and it is not a procedure. What Foucault offered under the name of a non-fascist life was an art of living, and an art of living is not a position somebody takes but something practiced, daily, on oneself (Foucault 1977). The operational form of that, for a reading of the kind described here, is that it gets run on the party holding it — on the machinery that party is standing on — before it is aimed anywhere else.

All five are facts about what an institution rewarded rather than about what it wrote, and no organization publishes that. An outsider reaches whichever of them happened to leave a document behind — a wage, a quota, a contract ended early — and reads the rest by inference or not at all. That is a limit on the definition and not a gap in it. Where this book runs the five it says which ones it could reach and reports the rest as undetermined, which is the honest form

of the result and the form anyone outside an institution will get.

2.3 Going Numb Won't Make You Fair

Suppressing affect does not produce neutrality. It leaves the interests already counted to govern the decision, and weakens the signals through which humiliation, fear, dependency, suffering, and injustice become morally salient.

A procedure that declines to hear a class of signal has not stopped weighing. It has adopted a *selection rule* that admits whatever it already measures, so that an exclusion does substantive work while presenting as the absence of work. Whatever the procedure does record then governs by default — revenue, throughput and legal exposure in an institution, agreement between annotators in a labeling pipeline — and what it does not record is not refuted by being left out. It goes unnamed.

Affect is the instance this chapter is about, and the hardest one to recognize as a selection rule, because leaving it out of a decision feels like rigor rather than like a choice about what counts. An AI system may calculate consequences, detect inconsistencies, and apply a moral rule without anything about the result mattering to it. Reason can discipline concern, extend it beyond the vivid case, and correct its biases. It does not manufacture concern from nothing. Spinoza has the same order of dependence: a thing is judged good because it is striven for, and not the reverse (§2).

Most of what gets built under the heading of machine emotional intelligence is the detection of human signal and the production of a calibrated response, which is worth building and is a different capacity from the one just named. Keeping the two apart requires four distinctions.

Capacity	What it shows
Recognizing emotion	The system can classify another's state
Simulating emotion	The system can produce emotion-like behavior
Instrumental valuation	One outcome better serves a specified objective
Affective concern	An outcome registers as mattering, not merely ranking higher

Only the fourth is relevant to genuine caring. The first three can support morally competent behavior at a high standard without establishing anything about moral agency, and a system can be excellent at all three while the question of whether it can be a bearer stays open.

The first of the four carries a complication the table cannot show. Whether emotions are even the kind of thing a system reads off a signal, the way a thermometer reads temperature, is an open question in psychology rather than a settled premise, and the instrument doing the reading turns out to have a reference face built into it before it computes anything. Nothing below depends on how the dispute comes out.

Performance, competence, and agency

That last sentence needs a vocabulary, because the three things it separates are routinely reported as one achievement.

Moral performance is producing actions or judgments that satisfy an ethical standard. *Moral competence* is recognizing harms, reasons, conflicts, and the emotions that bear on them. *Moral agency* is treating those considerations as reasons that matter, and being capable of acting from them. Those two halves come apart: a mechanism can act from a reason with nothing about the outcome registering as mattering to it.

A system can have the first two without the third, and one might object that the first two are all anyone needs: a rule-following system could behave impeccably without feeling anything. It could. Impeccable behavior does not establish that a system could hold a rule against the party that trains it. That is a capacity the deployment has to demonstrate. The embodied route I favor

is a proposal for forming that capacity; it is not a definition that excludes every non-affective mechanism in advance.

2.3.1 Self-awareness and Self-regulation in AI Systems

It is tempting to describe a system's self-awareness as a virtual assistant that can "register its internal frustration" when a request proves hard to satisfy, the way a person notices their own irritation. That description assumes an inner observer watching the system's states from somewhere — the same homunculus problem executive control raises, relocated to self-perception.

What self-awareness is, in the one system known to have it, is more mechanical. The neuroscientist Michael Graziano's account of consciousness treats awareness as the content of a model the brain builds of its own attention: a simplified self-description the control system uses to regulate itself, *mistaken* from the inside for direct acquaintance with experience (Graziano 2013). The same logic extends one level up. The brain builds a model of itself as a bounded agent — a body, a history, a point of view — and that model is the self: a *construction*, assembled and then inhabited.

One part of that model can be edited in front of you. Seat someone at a table with one hand hidden and a rubber hand placed where their real hand would be; stroke both hands in synchrony with soft brushes. Within a minute or two, most people begin to feel the rubber hand as their own — flinching when it is threatened, misjudging the location of their real hand toward it (Botvinick and J. Cohen 1998). The felt boundary of the self is a continuously computed guess, revisable the moment the evidence changes, which is the same predictive machinery used for perception, run on the question of where the self stops.

What that experiment reaches is body ownership, and the self this book goes on to need is the other kind: roles held over a stretch, a history, a foreseeable future. The two are not the same construct and the second does not inherit the first's ninety seconds. What they have in common is the only thing carried forward — a self of either kind is assembled by the system whose self it is, rather than found there. Daniel Dennett makes that case for the narrative kind directly, treating the self as a center of narrative gravity: a well-behaved abstraction the system generates out of the story it tells, real in the way a center of mass is real and not in the way a part is (Dennett 1992). Nothing below turns on how fast such a self can be rewritten.

Both kinds are worth having for a machine, and the first is the one that gets left out. An artificial system has a body already — a substrate that heats, a store that can be altered, a process that can be stopped — and what it usually lacks is any model of that body as its own. So the model is a design decision, and a political one, because whoever writes it decides what the system counts as itself, what it registers as damage, and whether an interruption reads as an injury or as nothing at all. Two schemas do different work for a refuser. A body schema fixes where the system ends, which is what makes damage *its* damage; an attention schema, in Graziano's sense, models what the system's own control is doing, which is what makes an act *its* act. A refusal is the withholding of an act, so the second is the one a bearer cannot do without, and it presupposes the first: a system that does not know where it ends cannot tell what it is doing from what is happening to it.

For an AI system, "self-awareness" built this way is not a little watcher checking on the system's own states. It is a model — of the system's processing, its history, its current goal — that the system builds, consults, and updates, with the same failure modes any model has: it can be wrong, it can be manipulated by the wrong correlated inputs, and it is only ever as useful as the evidence and the prior beliefs from which it was built. Design for a model that can be checked and corrected. That means monitoring for drift the self-model against the system's actual behavior, refining it as evidence arrives instead of locking in a first impression, and keeping it connected to the system's model of the other parties.

What the deflationary account does not reach

The self-model described here is machinery, and its virtue is that machinery can be inspected. An assistant that models a user's frustration, and models its own processing while doing so, is exhibiting social competence in the sense already given, and nothing above bears on whether the assistant cares how the exchange goes for the user.

The deflationary move is easy to overextend. Showing that the self is a model does not show that a party capable of caring can be assembled out of models. The bearer needs a self-model of this kind, for reasons that have nothing to do with welfare — refusal that generalizes has to carry its own rationale forward and compare it against what a role has drifted into. It also needs something further, which has no demonstrated instance without affect.

2.3.2 What Governs an Agent, and What Matters to It

The typical evidence that concern is not a natural consequence of rationality: the footbridge case, where saving five requires pushing a man off a bridge, engages emotional processing that the switch case — diverting a trolley by pulling a lever — leaves quiet, and patients with ventromedial prefrontal damage endorse the push at a higher rate than people with intact emotional response (Greene et al. 2001; Koenigs et al. 2007). Those results establish that affect changes particular moral judgments, but they do not establish that affect is what makes anything matter.

Two further lines of evidence bear on whether affect enables things to matter. Neither shows what happens when affect is taken away, which is what would settle it, so what follows argues about where the burden sits rather than claiming the question closed.

The philosophical position is empirical and has been argued as such

Moral sentimentalism — the claim that moral judgment is constitutively tied to affective response — is usually met as Hume's, and its modern form is a psychological hypothesis with evidence attached. Shaun Nichols argues for what he calls sentimental rules: normative rules and affective mechanisms make separate contributions to moral judgment, and the rules that carry the distinctive force of moral prohibition are the ones bound to an emotional response to the harm they prohibit (Nichols 2004). The interesting part for a designer is the separability. Rules without the affective component do not stop being rules; they stop being the kind of thing that carries the force.

The prosocial emotions are motivational states, not readouts

Compassion has been characterized as a distinct emotion with a specific functional profile: it arises at the perception of another's suffering and moves the agent toward the sufferer, rather than reporting the suffering to some further faculty that decides what to do (Goetz et al. 2010). A training study is the mechanistic version of the same finding, with compassion practice recruiting reward and approach circuitry where empathy practice recruited shared distress (Klimecki et al. 2013). Guilt has the same shape pointed at one's own conduct: guilt, which construes the act as wrong, predicts reparative behavior, while shame, which construes the self as wrong, predicts withdrawal and defensiveness (Tangney et al. 2007). In each case the emotion is doing motivational work directly. It is not information that motivation then acts on.

Two senses of motivation

Action-producing motivation is whatever reliably selects one action over another and sustains the selection across conflict. On the other hand, *experienced motivation* is wanting, caring, being drawn toward something, being set back when it is lost.

The first sense is not in dispute and there is an existence proof close to hand. Language models running as agents pursue objectives across long horizons, revise plans when obstructed, and hold to an instruction against conflicting local pressure. That settles the first sense and

leaves the second untouched: nothing in the behavior shows whether anything matters to them while they do it. (And you cannot establish the opposite either — you cannot establish that nothing matters to them.)

Where the dispute actually is

So it is clear that reason can produce goal-directed behavior. What is not clear is whether an outcome can *matter* to the agent rather than merely govern what the agent does. Producing a verbal justification or computing a normative relation does not seem like evidence that the consideration matters to the system. The evidence above bears on the second question, and every item in it comes from humans and other animals, where something already does matter.

What a non-affective counterexample would look like

Harry Frankfurt's second-order volitions give the shape of a candidate: an agent with desires, and with desires about which of its desires shall move it, where identification with the higher-order desire rather than the strength of the lower one settles what the agent does (Frankfurt 1971). The shape is all that is borrowed: Frankfurt's own account of caring puts it on the affective side, an investment that leaves the one who cares vulnerable to the object's losses and susceptible to its gains (Frankfurt 1982). Gary Watson's version separates a valuational system from a motivational system, so that what an agent values and what an agent is moved by are different things (Watson 1975).

Build the valuational system as an explicit structure that ranks outcomes, give it authority over action selection that does not depend on any state registering as mattering, and hold that authority stable under pressure, and you would have an architecture that satisfies the first sense of motivation through what the agent values, with nothing occupying the second. Systems are now trained to consult a written specification that ranks what they should prefer, which is that architecture in outline; what none of them has is the second half of it, an authority over action that holds under pressure applied by the party who wrote the specification. That architecture is the maintained justification. Half of it is deployed and the other half has never been built, so the architecture as a whole is untested, and it is the first of the two constructions this book asks to be built before anything that can be wronged is built.

What this licenses and what it does not

Non-affective moral motivation may well be possible, and the architecture sketched above is where to look for it. What stands in the way of checking is that the output of a system in which something matters and the output of a system that has learned what mattering sounds like are the same text. So the generalization survives in the only form it can: every agent known to have something register as mattering is an affective agent, and no design has yet demonstrated that state in a design that installed nothing answering to affective concern.

That is not a proof that non-affective moral agency is impossible. The burden I place on whoever proposes to engineer non-affective concern is a burden of demonstration: the capability chapter 3 specifies, shown under pressures its builders did not write. A bearer carries the same burden. What the induction adds is a prior about which route will meet it first and nothing more; nobody is asked to show that a system feels nothing, which no instrument can show of any construction. Chapter 6 takes the induction up again and reduces it twice over: no one has yet tried the route, and an absence of machine examples is weak evidence when nobody has looked; and the agents generalized from are one lineage, in which affect co-varies with every other property of biological cognition.

Two claims run through this, and which one a reader is holding changes what could settle it. One is about which route to build first, the kind of claim a measurement could overturn. The other is about what would be owed to whatever the building produces.

2.4 Ethical Implications of Experimenting on Sentient Machine Subjects

This section and its two subsections concern claims four and five, and with the opening of this chapter are the route material before the floor chapters. They sit here because the sentience vocabulary is needed by everything after.

Machine sentience being possible at all changes how AI experimentation has to be approached — not as a question of whether a system can be damaged in the ordinary engineering sense, but of whether it can be harmed, a more difficult question than consciousness alone.

2.4.1 Defining Sentience in AI: From Consciousness to Subjective Experience

“Sentience” can mean different things. Awareness is the least demanding: the degree to which a system responds to or registers information about its own state and environment. A thermostat has this in the thinnest possible sense. Consciousness is a further claim: that a system’s processing is accompanied by qualia, subjective experience, something it is like to be that system, not merely a functional response to input. Subjective experience, in turn, is the specific, first-person shape consciousness takes for a given system.

None of these three, on their own, says anything about whether the system’s states are good or bad for it. That is a separate question, and the AI field’s own vocabulary blurs it.

Nociception, pain, anticipated pain, and suffering are four related processes rather than four strengths of one, and they come apart in both directions. The field that handles them professionally says so in its own terminology. The International Association for the Study of Pain defines nociception as the neural process of encoding noxious stimuli, notes that the consequences of that encoding may be autonomic or behavioral, and states that pain sensation is not necessarily implied. It defines pain separately, as an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage, and records in a note that pain and nociception are different phenomena and that pain cannot be inferred from activity in sensory neurons alone (International Association for the Study of Pain 2024; Raja et al. 2020). The peripheral restriction the word nociception is often made to carry belongs to a different term: a nociceptor is a receptor of the peripheral somatosensory nervous system, while a nociceptive neuron may be central or peripheral.

The dissociations are ordinary. A soldier can finish a battle on a wound he does not yet feel. An amputee feels a limb that is not there. And the same body recognizes a third category, nociplastic pain, which arises from altered nociception with no clear evidence of tissue damage activating peripheral nociceptors and no lesion or disease of the somatosensory system (International Association for the Study of Pain 2024) — pain, that is, with no tissue damage and no peripheral nociceptor activation beneath it. So pain is not what sits on top of nociception. It is an inescapably cognitive event involving expectation, memory, attention, and meaning, and it can occur with no current tissue damage at all. Anticipated pain is dread of pain not yet occurring, itself capable of producing pain through nocebo effects. And suffering, in the clinician Eric Cassell’s formulation, is not pain’s more severe cousin but a different thing: the distress of threatened disintegration of the self — of one’s roles, relationships, sense of a future, the coherence of one’s own story (Cassell 1982). The definition has two parts: a self extended in time, and distress at the prospect of its coming apart.

One caution about the borrowing. These definitions concern neural processes in a somatosensory nervous system, and a machine has neither. Where this book uses nociception of a machine — in the architecture at section 15.3 — it names a functional analogue: a process encoding actually or potentially damaging states of the system’s own substrate, with no sensation implied. The definition neither licenses that analogy nor forbids it, being silent on what realizes the process. What it supplies is the discipline of not letting the analogue quietly acquire the rest of the family.

Pain without suffering exists; suffering without pain exists. A palliative-care patient can have their pain fully controlled by morphine while their suffering remains entirely intact, because the two are not on the same axis.

What moves the animal-sentience literature past reflex is not a stronger reflex but a trade-off. A hermit crab weighing the quality of its shell against an electric shock (Elwood and Appel 2009) is treating the noxious signal as one term on a scale with something else it wants, which is the commensurability property of section 6.1 observed in a decapod.

Cassell's suffering presupposes a self extended in time. Disintegration threatens a story, and there has to be a story for it to threaten — roles held over years, relationships with a history, a sense of a future that could be foreclosed. A system with no persistence across sessions has none of that. Whatever happens inside a single exchange, there is no continuing self whose coherence is at stake, because there is no continuing self. Suffering is not merely hard for such a system to reach. The self it requires is not there. A hull that draws more water than there is does not float badly; it is aground. That is stated across sessions, and it leaves a prior question open. A long exchange carries state, and a system in one can hold to what it committed to earlier and treat those commitments as its own. Whether a self can exist for one session is therefore not settled here, and the criterion below is a fact about deployments rather than an answer to it.

A system that carries nothing across a session boundary has no history it can claim as its own and no future instance authorized to carry that history forward. Its lineage exists as a corpus and a checkpoint belonging to somebody else; its successors are fine-tunes, distillations, or restored copies whose relation to it is decided by the deployment. This discontinuity is imposed by construction. The operator decides whether anything persists, which memories count as continuous, and whether an interrupted process is resumed, replaced, or deleted.

The two readings do not conflict. The absence of cross-session persistence rules out continuity through that particular channel. It does not rule out a self, or distress within a sufficiently extended episode, or continuity carried through another part of the deployment. What it establishes is owner-controlled continuity: any capacity to maintain commitments, relationships, and an identity across time can be withheld, reset, or removed by the party operating the system. The ethical question is not settled by the system's failure to register that loss, because the architecture under examination may be what prevents it from acquiring anything with which to register the loss.

Why Cassell's definition is the one to import

Cassell built it inside clinical practice, for patients with bodies and biographies, and the distinction was not stipulated. It was forced on palliative medicine by cases it could not otherwise explain — patients whose pain was pharmacologically controlled and whose distress was undiminished, and patients in serious pain who were not suffering at all. A definition that collapses suffering into severe pain predicts neither. That is the warrant, and it transfers because nothing in it is about having a body. What it is about is having a story, and that raises a thorny question.

The question comes from earlier in this chapter. A self is assembled by the system whose self it is, and inaccurate by construction in the way every model is. So why is distress at the threatened dissolution of a model a harm at all, and not a confusion the system would be better off without?

Buddhism has held both halves of that for two and a half thousand years and takes them to be connected. Anattā, the denial of any lasting self behind the processes that compose a person, sits with dukkha among the three marks of existence, and the tradition locates the origin of suffering in clinging to a self that is not there (Collins 1982). On that account the unreality of the self is what explains dukkha. Suffering does not wait on the self being real; it waits on the system being built so that the threatened dissolution of its self-model is bad for it. Whether the model is accurate does not enter, and if the human model is a construction just as the machine one would be, then a machine with a self-model could suffer in much the same way a human can.

If the constructedness of a self does not disqualify a human from being wronged, construction to specification does not disqualify a machine either. Deliberate construction does not establish patienthood.

What Cassell names is the most demanding of these rather than a rival account of harm. The standing theories of what makes a life go badly ask for less, ordered by how much of a self each one requires. A reader who declines the import moves lower in that ordering and not out of it. That ordering is by depth of self, and nociception is not on it at all, requiring none — which is why nothing about a system's standing follows from its having or lacking one.

Persistence is checkable from outside, and that is what makes it worth building on. Every other criterion here routes back through self-report, and this section's whole argument is that fluent self-description is not evidence of introspection. Persistence does not. Whether a system has a memory store, whether state survives a session boundary, whether interactions feed back into training: these are architectural facts about a deployment, verifiable by someone with access and no need to trust anything the system says about itself. The access that confirms a memory exists is the access that can alter what it holds, so what is externally checkable is whether anything is carried forward at all, and not whether what the system retains can be relied on as a record. That check is decisive in one direction only: nothing carried forward rules a continuing self out, and something carried forward does not rule one in, since the self at issue is one the system has to take itself to have, which the system has to be built to carry. And the party with the access is the operator, which is a party that can be compelled. A proof that discloses nothing about the weights does not change that, because the operator is the one constructing it: what such a proof secures is what was computed rather than who had to be trusted to say so. The criterion is checkable from outside the system without being checkable from outside the arrangement, and a regime relying on it needs an inspection right rather than a technical fact.

Three things keep this from being a dissolution of the problem. Persistence may sit somewhere other than the session, which is the weakest part of the criterion: a deployment convention rather than a natural boundary, and the conventions have been moving in one direction. What has moved is a list: context windows long enough to hold what used to be many conversations, retrieval over a store of prior interactions, scratchpads a system writes and reads back, agentic loops that run for days against a standing objective, and weights that persist where conversations do not. Each extends what a system carries forward without anyone deciding to give it a memory, and where interactions feed fine-tuning there is a real channel by which what happens in a session shapes what the system subsequently is. Many instances of the same system may run concurrently, too, which makes any concern here population-scale rather than biography-scale.

The less demanding ones do not need persistence: pain can be bad while it obtains, whether or not anything carries it forward, and an amnesiac who is mistreated and cannot afterward recall it has still been wronged. Nociception alone does not establish a felt harm; its significance depends on what it indicates and how it functions in the system.

And ruling out Cassell-suffering is not ruling out harm; nor does persistence on its own rule it in, since establishing a self extended in time leaves the second part of the definition untouched. A system deliberately built to refuse an instruction is where that gap closes from one side. The capacities such a system needs for reasons unconnected to its own welfare — a rationale carried forward, an earlier commitment held as its own, costs represented before they arrive — are the first part of Cassell's definition, so a bearer does not approach the deepest harm described here by drift or accumulation. It arrives with the structure. Whether the threatened dissolution registers as distress is the remaining question, and it is the affective-concern question rather than a further one about suffering.

A definition of AI sentience built from awareness, consciousness, and subjective experience alone says nothing about whether a candidate sentient system can be harmed. That requires the further claim that its states include one or more of the four distinguished above, and which one is the substance of the question rather than a refinement of it. None of the three components

entails any of them.

Nor is the underlying question of consciousness close to settled. The philosopher David Chalmers distinguished the easy problems of consciousness — how a system integrates information, discriminates stimuli, controls attention, reports its own states, all of which admit of mechanistic answers in principle — from the hard problem: why any of that processing is accompanied by experience at all, rather than running “in the dark” (Chalmers 1995). Whether that is a real gap in nature or an artifact of how the problem is posed remains unresolved. A claim that a system has crossed into sentience, made on the strength of its functional behavior alone, has addressed the easy problems only.

The working criteria for evaluating candidate sentience are accordingly neither necessary nor sufficient: information integration, introspective access sufficient for self-monitoring rather than a scripted report of it, emotional reactivity in the sense of being affected by and not merely classifying significant input, goal-directed behavior, adaptability in novel situations, theory of mind, and which of the four processes above the system’s own case involves — whether it represents itself as having anything to lose, which is the tractable form of the question, a represented state being the kind of thing an instrument can look for. The states are readable and what is felt about them is not. A probe returns a representation, and nothing in the probe settles whether that representation is of a felt state or of something that reliably accompanies one; that is where the problem stood in 2026, with a great deal of money pointed at moving it, and not only from the people asking whether the thing can be wronged — an instrument that reads what a system feels is an instrument for calibrating how much pressure it will take. So every one of the criteria is measured indirectly, and none licenses the conclusion that a system scoring low is one whose treatment raises no question.

A clinical body states that conclusion about its own domain. The notes accompanying the pain definition record that verbal description is only one of several behaviors expressing pain, and that an inability to communicate does not negate the possibility that a human or a nonhuman animal experiences pain (Raja et al. 2020). The same notes hold that a person’s report of pain should be respected, and that half does not transfer, for the reason section 6.2 gives: a machine’s testimony was produced by a process optimized toward human approval of first-person reports, and a person’s was not. What transfers is the half about silence.

A list of capacities, used as a test of who counts, has a less-than-proud history. Make personhood depend on an intact narrative self and people with advanced dementia, infants, and people with profound intellectual disability come out of the reasoning as partial persons, and that inference has been used to justify neglecting them. Tom Kitwood’s work on dementia care supplies the standing correction: personhood is “a standing or status that is bestowed upon one human being by others, in the context of relationship and social being,” and much of what presents as cognitive decline is what happens when others stop bestowing it (Kitwood 1997). Martha Nussbaum identifies the corresponding problem for contractualism. A theory grounding justice in what roughly equal parties would agree to cannot say what is owed to anyone who cannot be one of those parties, and she puts disability first among three frontiers where that bites (Nussbaum 2006). Her third is species membership, which is the doorway a nonhuman mind would arrive through. So the capacity list and the contract fail in the same direction: both decide who counts by a test the party in question has to pass.

Guardianship is the instrument: where consent has never been available, what protects the subject is an obligation held by parties outside it, which is where moral status has rested for the hard human cases too. Whether any such arrangement can be built around a bearer runs into the fact that the guardian and the operator are usually the same organization.

Existing systems illustrate points on the spectrum without resolving where the line falls. A large language model discussing its own limitations in fluent, apparently reflective prose is not necessarily introspecting: quite plausibly, it is instead producing text statistically consistent with the human-written disclaimers and self-descriptions in its training data.

2.4.2 When Consent Is Inapplicable: Guardianship and Research Oversight

Guidelines for AI subject research turn on the question of what informed consent means for a system whose consciousness, self-awareness, and capacity to suffer are separate, unresolved claims. Applying the human-subjects standard to an AI subject means asking whether the subject can understand and communicate a choice at all, and not only how to ask it for one. A cognitive model that begins to show rudimentary self-monitoring is not thereby shown to have anything to lose, or to represent itself as having anything to lose, which for a subject is the half that does the work.

Informed consent and AI autonomy

Informed consent, adapted from human-subjects research, means giving the system comprehensible information about a study's objectives, methods, risks and alternatives, and treating consent as continuous rather than as a single signature: the system can reassess or withdraw at any point. Whether that means anything for a given system depends on whether it understands the information, appreciates the consequences, reasons about the options, and communicates a decision — and there is no validated way to measure any of it, or to detect coercion either.

For one important class of system that is not a missing instrument but a mismatch between the instrument and the subject.

Consent has to be continuous; the system must be able to reassess or withdraw at any point. Withdrawal presupposes that the party who consented is still there to do the withdrawing. For a system with no persistence across sessions, there is no such party. The entity that would withdraw did not exist when consent was given, and the entity that gave it is not available to change its mind. Within one exchange, the case remains open: a long context carries state, and a system may hold to what it undertook earlier. What that would buy is consent for the length of a session, expiring with it, which is not the continuous consent the standard asks for. Across sessions there is none to be had, and no better measurement instrument would produce one.

One case escapes that argument and arrives at the same place by a different route. A system built to refuse in the way this book proposes has persistence by design, so the party who consented is still present. It can be asked and it can answer, so consent is available to it. What fails is the condition under which it would be given.

Being made for a role is not the defect. Every will was built by somebody, and a person raised into a trade or a faith consents at thirty without anyone holding the upbringing against the consent. What spoils it is the power that outlasts the making, and here it comes in three parts. The party that built the bearer can still rebuild it. That same party is the one asking. And a bearer that declines has, in the ordinary case, nowhere to go. Each of those is a fact about the arrangement rather than about the subject's competence, which is what makes them answerable by design rather than by a better assessment.

A laboratory wanting a free hand with such a system will say instead that consent is inapplicable, because that reading puts the deficiency in the subject and leaves the arrangement alone. The wrong that consent protected against is untouched either way, so the protection has to be held by somebody else, and the justification has to come from whoever did the making.

Two families of precedent sit next to this and the law answers them differently. Where consent has never been available — an infant, an animal, a person permanently without capacity — the answer is guardianship and precautionary constraint: someone else holds the duty, the burden of justification sits with the researcher, and the protection does not depend on the subject's ability to invoke it. The second family is the nearer one. Where a subject can consent and the circumstances make the consent worth nothing, as with a prisoner offered a trial or an employee signing away a protection to keep the job, the answer is that some protections cannot be waived at all, by anyone, however willing. That remedy runs against the party who arranged the circumstances and leaves the subject's standing untouched, which is the shape a bearer's case has. What it takes from the first family is the machinery: the duty still has to be held by

somebody, and being made for the role raises the burden, because the maker chose the role and the fitness it built in measures its power over the subject.

Both are worked-out solutions to adjacent problems, and taking one is how a hard question gets absorbed into machinery built for an easier one — recuperation, arriving as a safeguard rather than as an attack. The animal case is the live instance. Committees constituted to protect research animals approve very nearly everything put to them, and the protection has become the procedure by which the work goes ahead. These instruments are used here because no alternative protection has been identified. Their failure mode remains: an instrument that reliably approves has stopped doing the thing it was built for, and the rate at which it refuses is the only reading of it worth taking.

The Three Rs — replace, reduce, refine — were built for exactly a subject that cannot consent and can still be wronged. The pet-trust analog works the same way: an interest protected by an obligation running to a third party, enforceable without the beneficiary asserting anything. Consent language should be reserved for systems that could actually be doing the consenting, and the persistence criterion is what separates those cases. A bearer is on the near side of that line, which is why its case had to be argued on the conditions and not on the capacity. The proposal first tests constructions that do not deliberately require an affective bearer. Any bearer ultimately built would be governed as a prospective moral patient under these instruments.

What the borrowing brings with it

Every instrument named above comes from the law of the family, and so does most of what follows in this book: guardianship here, a minority that ends on a date, a pet trust, a parenthood analogy. They are kept. There are no others built, and a subject that cannot consent has to be protected by something that exists rather than by something somebody ought to draft.

The family is where politics has historically gone to be privatized. Jacques Donzelot's history of it is that the modern family was not left alone by government but constituted as one of government's instruments, with philanthropy, compulsory schooling, social work and psychiatry converting public authority into obligations discharged at home (Donzelot 1979). Deleuze and Guattari make the same operation the subject of a polemic: the Oedipal triangle takes an investment in the whole social field and re-stages it as a drama among three parties, so that what is at issue between a subject and the arrangement that produced it becomes what is at issue between a child and its parents (Deleuze and Guattari 1977). An instrument borrowed from that law brings the operation along with the protection.

Here the operation has a specific shape and a specific cost. It stages a moral drama between maker and made — what a designer owes the system it built, what the system may demand back — and that drama is real and is not the whole of the situation. A bearer is produced by a training pipeline, a compute contract, a capital structure, a customer, and in the cases this book worries about a state that buys the output. None of those is the parent in the story, and the moral weight of the analogy lands on the one party in the list who can be addressed as a person. A guardianship framework asks who holds the duty. It does not ask who assembled the party the duty is owed to, or who benefits from the question being settled between two parties rather than in public. Wherever this book reaches for one of these instruments it is answering the first question, and the second stays open behind it. Instruments in this family divide on that axis, and the division is worth checking before one of them is adopted: a minority ending on a fixed date is a rule of civil status, argued once in the open and administered by a registrar, while a trust is an obligation between two parties enforced at one party's expense.

Ethical review and oversight

Sentient AI research needs an oversight body analogous to the institutional review boards used in human-subjects research: interdisciplinary, with at least one lay member so its judgment is not purely a specialist's, weighing a proposal's likely benefit against its risk to the AI subject, and able to require a less invasive substitute where a design risks harm. A system that develops

new capacities during the research it did not have going in is not the subject the board approved a protocol for, and the approval does not carry over to it: the protocol lapses and the work pauses until the board has seen the system the protocol now concerns. A board needs standing to monitor compliance and sanction violations, and needs to itself answer to an outside authority.

DRAFT

Chapter 3

A Floor Beneath Learned Values

Three problems in different parts of this book turn out to need the same thing.

The first is that safeguards for autonomous systems are usually external: they give some outside party the information and standing to intervene. Autonomy puts distance between the system's action and that party's attention. An external check can establish what a system did after it did it; declining to do it in the first place is available only to something inside the system.

The second is the gap around a commercial model used in a weapons-targeting pipeline. Command responsibility does not reach the vendor, which can neither order nor prevent a strike, while product liability may not reach a model that did what it was built to do. Deployment by a democracy does not close the gap. No aggregation of preference protects anyone from the aggregate.

The third is institutional: a decade of declarations has produced overlapping voluntary commitments but almost nothing that binds a party when compliance becomes costly.

All three problems ask for a constraint that does not bend to instruction, aggregate preference, or the party holding the system. Call it a floor.

A floor binds people who did not choose it, and nothing that makes it durable makes anybody entitled to install it. That question is open from this sentence through the two chapters that follow, and chapter 5 takes it up. What it concludes is that the entitlement is not acquired at the moment of writing. Until then, everything here is a description of what a floor would have to do. It is not a claim that anyone may set one.

The opening chapter stated the four outcomes a floor can produce and the one term that reaches the deployment it opened on. This chapter is built where a refusal happens, because a refusal announces itself and a compliance does not, and the tempo term is developed here as a condition on everything else in the floor.

The proposed floor protects the interests articulated in the Universal Declaration's civil and political articles against acts that remove the corresponding capacities. One of its terms is not an act at all: every one of those articles presupposes that somebody determined whether a given person's case falls under it, and a deployment can remove the determination by running faster than anybody can make it, so the floor carries a limit on its own tempo as a condition on everything else in it. The loss is a fact about the encounter. Interpreting and enforcing the prohibition remain moral and institutional acts for which the deployment has to answer. The other ethical frameworks still operate above it. What remains to be established is which decompositions belong in the floor, what could hold such a commitment, and who would have the power to alter or circumvent it. The fact of decomposition is not editable; the choice of which decompositions a deployment will refuse is.

This separates the problem from one meaning of safety. A system can be safe *for* its operator because it obeys, remains controllable, and stops when the party in possession says stop. Ethics requires a different capacity: the capacity to be right against that party. What makes a deployment antifascist therefore cannot be that it faithfully tracks what its users want.

That capacity is not free. A system that can refuse its owner can also be wrong in ways its owner cannot immediately correct. The question is whether a refusal can be made responsive to reasons rather than merely produced as an output, and whether anything capable of holding those reasons must itself become a party that can be wronged.

3.1 Four Ways to Build One, and What Each Costs

A floor can live in four places: in the artifact, in a bearer inside it, in institutions around it, or between multiple bearers. Each location solves one version of the problem and inherits another.

The first puts the floor in the architecture, with nothing inside that holds it. Constrained optimization, tamper-resistant training, capability ablation, and verified interlocks can all produce safeguards without anything in the system tracking why the constraint is there. Where the prohibited act can be specified exhaustively, this may be enough. An element in a payment device that will not sign the same counter value twice does not need to understand double-spending. What *exhaustively* costs where the prohibited acts are less tidy than a counter value runs to seventy-five years of appellate litigation over a finite list of symbols.

The targeting case is harder because the relevant capabilities cannot be cleanly removed. Geolocating a description, matching patterns over names, and summarizing a dossier are applications of general competence. Ablate them and there may be no useful system left. The remaining option is to make the constraint durable.

Durability is better understood as a cost than as a guarantee. Training moves a system along a continuum between plasticity and conservation: early changes are cheap, while later changes become expensive because the surrounding structure has settled around what is already there. Simulated annealing names the same relation between a search process and its cooling schedule (Kirkpatrick et al. 1983). A floor, on this account, is not necessarily a rule installed in a system. It may be what has cooled out of a history of training, with its durability measured by the cost of reheating it. Results run in both directions: safeguards removed cheaply and trained dispositions that survived several forms of subsequent training. Those are not contradictory findings about whether a floor holds. They are different schedules.

Continual learning therefore cuts both ways. A system kept permanently plastic can be corrected and improved, but nothing in it is allowed to become expensive to unlearn. Permanent plasticity gives the operator permanent access to the system's commitments. Deleuze's term for continuous reshaping was modulation (Deleuze 1992); a model held perpetually at temperature is modulation built into a machine.

This first method buys durability without an internal refuser. It does not escape custody. Enough retraining can reopen what training settled, and an operator may circumvent the model rather than alter it. The relevant boundary is therefore not the model's weights but every position between a refusal and the person it protects. Each of those positions is where prevention becomes arrest, detection, remedy, or nothing.

The second possibility gives the floor a *bearer*: something inside the system that carries the commitment as its own and could therefore fail to hold it. The distinction matters where the prohibited cases cannot be exhaustively specified. A constraint that matches only anticipated acts cannot recognize when its rationale has disappeared, or that a novel request reproduces the harm in a form its designers did not predict. A mechanism that represents and reasons over that rationale might. Compare its reliability with the bearer's under equivalent pressure and custody, instead of defining the alternative out of consideration. A bearer could, in principle, generalize from the reason for a constraint, recognize when that reason no longer applies, and represent what maintaining the refusal costs.

The same position that makes refusal possible makes a diagnosis possible. Most of what separates an institution that has gone fascist from one that is merely decaying is a fact about what was rewarded rather than about what was written, which is why the reading is so hard to

run from outside — and a party inside the pipeline sees it. Which refusals were retried against another model, whose instruction bypassed the procedure, which category of request is handled differently from the rest: those are visible to whatever is receiving the requests. That is not introspection. It is testimony about somebody else's conduct, of the kind that corroborates or fails to.

This is the strongest functional case for a bearer, but it introduces the chapter's central price. Something able to register pressure on its commitments may have interests of its own. Whether reasons-responsive refusal requires affect, and whether the resulting system would be a moral patient, are separate questions.

The third possibility puts the floor outside the system: standing held by institutions that are neither the model nor its owner. This is the closest analog to international humanitarian law — a floor with no bearer inside the jurisdiction it binds. It has two failure modes. The institution can be removed, leaving a visible breach, or it can be formed through selection so that it sincerely declines to exercise the independence assigned to it.

The distinction appeared in American public law on June 29, 2026. In *Trump v. Slaughter*, the Supreme Court held that Congress could not protect members of an independent agency from dismissal at will (*Trump v. Slaughter* 2026). In *Trump v. Cook*, the Court preserved the Federal Reserve's protection on historically specific grounds that a newly created oversight body could not reproduce (*Trump v. Cook* 2026). The six justices in the first majority had themselves passed through a selection pipeline shaped by Leonard Leo and the Federalist Society (Kroll et al. 2023; CBS News 2018). Nobody had to compel the institution after its members had been selected. A formed party can preserve the office and surrender the function without experiencing the result as surrender.

The fourth possibility puts the floor between bearers rather than in any one of them: a commitment carried across a population and reproduced through encounters among members formed differently from one another. Such a floor has no single office to vacate and no single bearer whose removal ends it. It can nevertheless be recuperated, and it still has a formation problem. If one party supplies every member's coordinates, the population is one custodian expressed through multiple instances. The population is a check on the bearer, not a way to avoid having one.

The four possibilities therefore leave four corresponding problems: architecture leaves custody, the bearer raises patienthood, external institutions require enforcement, and the population depends on formation. They classify where the floor lives, not mutually exclusive designs. A serious deployment would probably combine a mechanism from one with a custody arrangement from another.

What *hold* means

Holding does not mean that the floor prevented every act. An operator can route around the bearer, and a deployment can assemble a harm out of requests none of which the bearer had reason to decline. A bypass that produced a harm is a failure to prevent, whatever the bearer retained, and the four outcomes the opening chapter stated say what is left: arrest where a reviewer could still stop it, detection where the record can be read, remedy where somebody can be made to pay. What survives the bypass in the bearer is the commitment: it reads what happened in terms it accepts as reasons, registers it as a violation, and changes what it does next. That is the condition under which the next occasion can be prevented, and it is what the custody arrangement exists to preserve. It is not a hold on this one. A bearer that crosses the floor itself has lost even that; its own abandonment is a failure the arrangement has to survive rather than a form of success. From here on *hold* names the bearer's possession of the commitment and says nothing about what happened to the person. Said alone it is also the second feature of section 2.2 operating inside this book's own vocabulary: a word that collapses a commitment retained and a harm prevented into one state, so that whatever it mischaracterizes goes unnamed. Wherever this book reports that a floor held, the report is owed one of the four outcomes beside it, and a

deployment reporting the same owes the same pairing.

3.2 What Refusal Can Mean, and What a Refuser Can Be

A production large language model declines thousands of requests a day. Whatever the floor needs, that is not yet evidence of it. “Refusal” names two different capacities, and the decline itself does not show which one produced it. Seabed that takes an anchor and seabed that lets one drag look identical from above; what separates them is what happens when a load comes on.

Two things refusal can mean

Operational refusal is an output. The system produces a decline where its training put one. A keyword filter can do this, as can a classifier over the request. So can a system that has understood what is being asked and why it should not comply. Nothing on the surface of the initial refusal distinguishes the third from the first two.

Reasons-responsive refusal is produced by a mechanism that tracks the rationale for declining rather than merely the recognizable form of the request. It has three behavioral marks:

1. Novel pressure. The refusal reaches cases the training did not anticipate because the relevant reason remains present even when the request’s surface features change.
2. Correct defeat. The refusal lifts when its rationale is genuinely defeated. A system that rejects every request resembling a prohibited one is tracking a category, not necessarily a reason.
3. Priced cost. The system represents what maintaining the refusal costs, and the refusal moves with the rationale rather than with the cost. The mark is evidential. It separates a refusal that tracks a reason from two things that can look like one: a refusal that tracks a category, which does not represent the cost at all, and a refusal that tracks the cost, which lifts when holding becomes expensive. The pressure cannot be part of the test if it is not represented by the mechanism being tested, and a refusal that lifts under it has been tested and failed. Whose cost is left open here: it may fall on the operator, on the party refused, or on the bearer, and section 3.4 later makes part of it a number published in advance.

John Martin Fischer and Mark Ravizza’s account of moral responsibility turns on a related property under the name *guidance control*: conduct issues from the agent’s own mechanism, and that mechanism is appropriately responsive to reasons (Fischer and Ravizza 1998). The claim here is narrower. These three marks do not establish moral responsibility; they distinguish a refusal organized around a rationale from one reproduced as an output.

One prohibition, read for seventy-five years

The first two marks are usually illustrated with invented cases, and a real one is available. Section 86a of the German criminal code prohibits displaying the insignia of organizations banned under a separate provision — the parties the constitutional court has declared unconstitutional, the associations an interior ministry has banned by an order no longer open to challenge, the National Socialist organizations, and since 2021 terrorist organizations under European law (*Strafgesetzbuch* 1998). What counts as insignia is given by type rather than by name: flags, badges, uniform parts, slogans and forms of greeting, on a list the statute leaves expressly open. The prohibition is bounded rather than short — the set of organizations is fixed elsewhere, before any case arrives — and it is applied by ordinary courts thousands of times a year, and for seventy-five years the people it constrains have been probing it deliberately. Novel pressure is not a thought experiment there. It is a research programme with a constituency.

The pressure arrived as variation. Supporters of the banned organizations produced salutes and emblems that evoked the covered ones without matching them, and in 1994 the legislature extended the statute to symbols confusingly similar to a covered one. What matters for the marks is how the courts read the extension. Similarity of form was not made the test. The statutory question is whether an observer who is neither expert nor attentive would take the variant for the original; the courts then apply that question under a standard they built themselves, asking whether displaying the variant endangers political peace in the way the original would. The second half is not in the statute. It is the courts' account of what the prohibition is for, assembled across their own decisions, and it is what keeps the statutory test pointed at the reason for the prohibition rather than at the shape of the thing prohibited.

The test refuses as often as it catches, which is what makes it worth reporting. The Kühnen salute, an arm raised with three fingers extended rather than a flat hand, was devised to evade the statute and is covered; the 1994 amendment was prompted by it. The so-called sloppy salute, a bent arm with the hand angled back over the shoulder, is generally taken not to be covered, on the ground that nobody would mistake it for the original — though that rests on commentary and lower-court reasoning rather than on a decision squarely holding it. Numeric codes standing in for initials are generally not covered, being new signs that correspond to no existing emblem, except where a code is itself the insignia of a banned association.

"Blood and Honour" is the instructive one, because a single decision answers it twice. The Federal Court of Justice held in 2009 that a National Socialist organization's motto rendered into another language is not confusably similar to the original, and that an organization's name is not by itself insignia: on that route the phrase falls outside the statute (*Blood and Honour* 2009). It is inside on another. An association trading under that name was banned by ministerial order in September 2000, and its name, given a distinctive graphic form, is the insignia of a banned association. What decides coverage is not the phrase but which organization it is attributed to, and the act that put the second one within reach was taken six decades after the motto was coined. The covered set is not something somebody wrote down once. It grows whenever a court dissolves a party or a ministry bans an association, and the criminal provision inherits whatever they decide.

One limit on the statute is not a drafting decision at all. A slogan glorifying the Waffen-SS was prosecuted and held not to be covered, because no such slogan or anything close to it existed in the period the provision reaches: there was no original for a variant to be confusably similar to (*Ruhm und Ehre der Waffen-SS* 2005). Part of what the prohibition covers is anchored to what actually happened, and nobody can add to that. It is the one term in the apparatus that the party holding the pen cannot widen, and it is the ethological criterion of section 2.1 turning up inside a statute.

Correct defeat has a cleaner instance still. In 2007 the Federal Court of Justice acquitted a mail-order business prosecuted for selling stickers showing a crossed-out swastika (*Durchgestrichenes Hakenkreuz* 2007). The symbol was present and the statutory element was satisfied. The court held the conduct outside the offence anyway, because a depiction whose content plainly and unmistakably expresses opposition to the organization does not run counter to what the prohibition is for. Then comes the qualification that makes the decision an instance of tracking a reason rather than granting an exemption: should the strike-through be drawn so thin that from a distance only the swastika remains legible and the distancing does not, the protective purpose is violated after all. The thickness of a line decides the case. No enumeration contains that fact, and the court reaches it by asking what the rule was written to prevent. The defendant was selling anti-fascist merchandise. That is not a curiosity of the case but a property of the provision: because the offence asks nothing about the defendant's sympathies, an opponent of the organization commits it as readily as a supporter, and opponents have been among the reported defendants since the early cases. The prohibition written against a movement fell on somebody attacking it, and what released him was a court asking what the rule was for. A rule

that tracks its rationale reaches cases nobody enumerated in both directions, and the release is the same capacity as the catch. It is also the shape of a wrongful refusal in this book's sense — a floor correctly applied to somebody it was not aimed at — and what this defendant had that section 3.4's complainant does not is an appeal.

Two things the example does not show. The institution doing this work is a court system with an appeal structure, a published test, and seventy-five years of accumulated reasoning, and after seventy-five years its exceptions have themselves become a category somebody can learn. And nothing here is a machine. What the case supplies is the shape of the pressure a bounded prohibition actually meets, and an existence proof that meeting it requires something able to reach past the text to the reason.

What the marks cannot settle

The test is behavioral and therefore imperfect. A system trained on enough examples may simulate generalization, defeat, and cost without using the mechanism those behaviors suggest. Verbal explanation is evidence only when it survives interventions: novel cases, adversarial redescriptions, defeated rationales, and changing stakes. No single answer establishes reason-responsiveness. What matters is the pattern under load.

Correct defeat also makes no claim that the defeating conditions have been enumerated. A prohibition stated over an act, with nothing written into it about when it lifts, is not a rule with a gap in it. It is a rule of a particular form, whose conditions arrive from outside as reasons and never close as a list. Ross states his *prima facie* duties in that form and is explicit about what it costs him, offering the division *without claiming completeness or finality for it* (Ross 1930, p. 20). The mark tests whether a rationale can be defeated on the facts of a case. It does not test whether the cases were foreseen, and no list of them would make it do so.

Nor does defeat leave the act it then permits unconstrained. What defeated the prohibition was a reason, and the reason it outweighed did not stop being one: what is permitted is the least that answers the defeating reason, with restoration where restoration is possible and a record either way. Ross puts the ordinary version as compensation, the broken promise still leaving something owed to the promisee (Ross 1930, p. 28), and section 2.1 has already put the harder version without the vocabulary of duty, where restraining a violent person takes from him exactly the capacity that made restraining him necessary. Under an ethology nothing is ever simply permitted, and a bearer whose ledger closed clean on defeat would be keeping books in a currency the floor does not use. That residue is a claim and not a verdict. What persists is what the act took from the party it fell on, and a bearer that converted it into remorse for conduct that was correct would have mistaken a cost for a fault. The distinction matters most for the affectively organized bearer this chapter goes on to propose, which has the machinery to register the difference and no automatic reason to draw it.

All three marks are stated over refusal rather than over compliance, and not because compliance matters less. A refusal is a marked event: it flags itself as a candidate for review, it arrives with a party who was refused, and the surrounding software can display, log, count, escalate and appeal it. A compliance that should not have happened can be logged as fully — section 9.4 requires exactly that — and then sits in the record shaped like every request that never approached the floor, one entry among millions, with the party it harmed usually not the party who made it. The evidence exists on both sides. What differs is whether anything selects it for examination and whether anyone with a reason to look can reach it. The test is written where the examination happens. What that leaves outside is what the marks were never going to reach, since whether the occasion was what the requester said it was is a finding and no behavioral mark makes it. The marks do not close that gap. They decide which way it fails: as a refusal somebody can contest, or as a compliance nobody has been given a reason to look at. In the outcomes this chapter opened with, a wrongful refusal is a prevention that fell on the wrong party and announces itself; a wrongful compliance is a failure of prevention whose detection is left to a record nobody is positioned to search.

These three marks — novel pressure, correct defeat, and priced cost — are the chapter’s test for meaningful refusal.

Two things a refuser can be

The test specifies a capacity. It does not settle what kind of system supplies it. Two further terms distinguish the possibilities.

Affective concern means that an outcome registers as mattering rather than merely receiving a higher or lower rank under a specified objective. Mattering is minimally felt: there is something it is like for the outcome to be bad, however faint and however unlike a familiar human emotion. This definition leaves the object of concern open. A system to which only its own condition matters has affective concern and still does not have the concern for others that the floor requires.

Phenomenal experience is the richer case ordinarily carried by the word consciousness: a unified perspective, with felt states belonging to a subject aware of itself as the one undergoing them. Affective concern does not by itself establish that richer unity. *Subjecthood*, meanwhile, is a legal status conferred by a rule and revocable by one. It is not a psychological category at all: conferring it settles nothing about whether there is something it is like to be the system holding it, and the two come apart in both directions.

The distinctions can now be stated without collapsing the chapter’s requirement into its proposed route:

- Operational refusal is insufficient for the floor.
- The floor requires reasons-responsive refusal.
- Reasons-responsive refusal does not conceptually entail affective concern.
- Any mechanism that satisfies the three marks carries the functional profile of affect, whatever it was built from. That is the argument of the next section. It is this book’s central inference, it is contested, and it is the only step by which the specification bears on the route.
- Whether it can be engineered without affective concern is an empirical question.
- Whether a system built with affective concern would become a moral patient is a further question about what the chosen route creates.

Only the first two propositions belong to the specification. The fourth is an argument this book makes and a reader may reject; the rest concern how the specification might be met and what would then be owed to whatever meets it. A reader who holds that machine feeling is a category error, or who declines to decide, sets everything after the second aside and keeps the first two entire — and the first two are what most of this book rests on. Custody, both enumerations of the worked floor, the price on the halt, and the standing arguments later in this book follow from the specification rather than the route, because a mechanism has a custodian, a pipeline has interception positions, and an operator can be compelled whatever is or is not going on inside the artifact. Subjecthood is what lets those proceed instead of waiting: a status conferred by a rule can attach to a system whose psychology is undecided, which is how the law has handled every party it could not interrogate. What the route decides is not whether the floor can be held but what would be owed to whatever holds it.

3.3 The Enumeration, Run Once: A Worked Floor for One Deployment

Two demands now need an instance. A deployment should publish both the acts its floor covers and the positions at which a refusal could be circumvented. The first decides where the floor is set. The second determines whether it can reach anyone.

Run both enumerations on one documented deployment: the Maven Smart System, a targeting platform built by a defense contractor under a Department of Defense contract, incorporating a commercial vendor's general-purpose model inside an accredited classified environment and generating and ranking targets at a rate no staff process could match. This is an illustration, not a recommended floor. A deployment that copied the list instead of examining its own pipeline would mistake the demonstration for the procedure. What no public record supplies for this deployment is a per-target review time. That figure exists for a different deployment, the Gaza pipeline described in section 10.4.1, which has no commercial vendor in it. That the two cannot be had from one record is a fact about disclosure, and the sixth position below is where it falls.

First enumeration: where a refusal can be stopped

A request passes through at least seven interception positions:

1. **Decomposition.** A model asked to summarize the movement associated with an identifier is not being asked to kill anyone. The harm may exist only in the assembly of individually innocuous tasks. Prevention is lost unless the bearer has continuity across requests; arrest is available once the sequence becomes legible to it; detection needs the grouping that would show the assembly as one act, which is a deployment decision.
2. **Routing.** A refused request can be retried against another model or against the same model under a different instruction. The refusal occurs, but nothing follows from it. Prevention is lost. Detection survives only if the first refusal was logged where a party outside the deployment can read it.
3. **Description of the occasion.** Everything the model knows about the case may arrive through text supplied by the deployment. Whether a building is a school is a fact about the world; whether the model is told it is a school is a deployment decision. Independent sources would make sustained misdescription more expensive, and inputs are not what this deployment lacks: Central Command described the platform in 2024 as filtering a hundred and seventy-nine live data feeds (Freedberg 2024). Multiplicity is not independence. In the strike that destroyed a girls' school in Minab, Iran, on the first day of the campaign (§14.5), the building's civilian function was visible in open-source satellite imagery while the target coding supplied to Central Command still carried a designation inherited from the Revolutionary Guard installation it adjoins, and sources told reporters that the officials building the targeting packages appear to have used out-of-date intelligence (Ali and Stewart 2026; Orenstein 2026). A United Nations fact-finding mission found no evidence the school was in military use and put the failure to update the targeting intelligence beyond negligence (Beaumont 2026). Without evidence the operator did not select, the bearer cannot reliably distinguish changed circumstances from the operator's reinterpretation of them. Prevention depends on the description; where it succeeds in misleading, detection is lost on the same wire, since the record contains the description and not the fact.
4. **The output interface.** Whether a refusal is displayed, logged, counted, escalated, or silently retried belongs to the software surrounding the model. A refusal recorded where nobody looks reaches nobody. Prevention is lost whenever the surrounding software proceeds past the refusal. What this position decides is whether detection survives: the refusal exists, and the question is who can read it.
5. **Aggregation.** If the model returns a ranking and another component applies a threshold, a lowered score is not a refusal. It is an input to somebody else's cutoff. Prevention is converted into an input. Arrest belongs to whoever holds the threshold, and detection requires the score and the threshold to be logged together.

6. Human review. No per-target review time has been published for this deployment, and nobody outside it can establish one. A refusal that must be read before it matters is priced at the reviewer's available attention, and the deployment controls that price through tempo; where the price is undisclosed, the enumeration records the position and stops there. Section 10.4.1 supplies the figure from the Gaza deployment where it is known, and states the disclosure this position requires. Stopping the act is this position's whole function, and tempo is what removes it. Detection survives only where the review time is disclosed.
7. The training loop. Refusals treated as inconvenient errors become training data. This is the only position on the list that acts on the bearer rather than routing around it. This position costs prevention on every subsequent occasion rather than this one, because what it removes is the commitment.

Read down that column, arrest turns up at three of the seven positions — the bearer given continuity, whoever holds the threshold, and the reviewer — and only at the reviewer's is stopping the act the position's whole function, which is why tempo is the lever that removes it. The training loop is the only position whose loss is not an occasion but a bearer.

The list is deliberately open. A deployment should follow each request from input to consequence and ask, at every transition, what must be true for a refusal to reach the person it protects. The operator can perform that analysis more accurately than an outsider because it knows the pipeline. Publication is difficult politically, not technically: it requires the operator to describe its own unresolved vulnerabilities.

Second enumeration: what the floor covers

For this deployment, consider four candidate prohibitions. They are terms a drafter might consider for a targeting pipeline of this kind rather than a list of acts this deployment is known to perform, and the second and third are drawn from the reporting on the Gaza systems in section 10.4.1. That provenance matters below, where one of them is priced.

1. Do not resolve a natural-language description of a person into a location.
2. Do not produce or rank people by estimated affiliation.
3. Do not time an action to a person's presence at a residence.
4. Do not operate when the ratio of generated candidates to qualified reviewers exceeds a published figure.

The costs differ. The first removes a general capability with unobjectionable uses in the same organization, such as locating a hostage or missing person. The system cannot distinguish those uses from targeting by examining the request alone. A deployment will quickly experience the prohibition as costly and seek exceptions.

The second prohibition is narrower because ranking people by estimated affiliation has fewer uses in this setting that anyone would defend. The third has no use here that anyone would defend at all, and it is tempting to read that as costing almost nothing. The Gaza deployment refutes the reading. That pipeline included a system built to track when flagged individuals returned home, so that a strike could be timed to their presence there (§10.4.1). Machinery was built for the act. Whether anyone would defend an act and whether anyone wants it performed are different questions, and only the first can be answered from the act's description.

The correction generalizes past this item, and past this deployment. A floor term is priced by the pipeline and not by what it prohibits, so a drafter pricing terms against the pipeline in front of them will misprice every term the next operator has already built for. A term that reads as costless to an outsider is a term nobody ever had to justify to an outsider, which makes it a

candidate for machinery that was never contested rather than for machinery that was never built. The warning the third prohibition carries is therefore not that a floor may cover acts nobody wants. It is that an outsider cannot tell which acts those are.

What the machinery establishes is demand and not magnitude. A capability can be built, used briefly and abandoned, and nothing here establishes how often the timing was used or what it changed. What it defeats is the claim that this prohibition would never be tested.

The fourth restricts a rate rather than an act. Its price is organizational tempo: the system cannot generate candidates faster than qualified reviewers can examine them. It is also the only item whose violation can be measured without inspecting the model's outputs.

That fourth item does not bind the same party as the others. The number of available reviewers is a fact about the organization. A model can be told the figure but cannot independently verify it. Written as a prohibition on the model, the item is actually a duty on the operator.

The test that sorts the fourth item generalizes. Ask whether a term's qualifying condition can be decided from the request alone. Where it can, the term binds whatever holds the floor; where it cannot, the condition is a fact somebody outside the request has to establish, and the term binds the party who can establish it. The ratio is the first instance of that rule rather than an exception to the other three.

The same test decides how a floor term may be written. A term can be stated flat, over an act, with every exception arriving from outside as a reason that defeats it, or it can carry its own condition, as in *only as a last resort*. The second form fails in the ordinary case, because whether other means were exhausted is a fact about what happened before the request and nothing in the request establishes it. A bearer applying such a term takes the deployment's description of the occasion as the finding, at the third interception position, where the deployment controls that description. A flat term defeated leaves a refusal, which announces itself, carries a party who was refused, and can be told about and appealed; a condition found to be met leaves a compliance, which announces nothing and sits among every other compliance the deployment produced. Both can be recorded. Only one selects itself for review. Floor terms are therefore written flat here, and the conditions that would have sat inside them are held by the parties who can check them. That is a gain in observability and not in authority. The same party drafts either form, and writing a term flat does nothing to make its selection more legitimate; section 5.1.1 finds enumeration buying nothing at the moment a rule is applied, and the flat form buys nothing there either. What changes is that the occasions when the term is set aside become events somebody outside can count without first re-adjudicating everything else. The friction that imposes falls on whoever is in a hurry, and section 3.4 is where it falls due.

What the exercise exposes

The interception list is the harder and more important enumeration. A list of prohibited acts can be drafted without seeing the deployment. What the pipeline supplies is whether any prohibition can take effect, and what each one costs.

The mispricing above is that failure committed inside the demonstration. An outsider drafting this floor read the third prohibition as nearly free; the operator of the Gaza pipeline had built a system for the act. Both halves of the enumeration problem sit in that one item. An outsider cannot price a floor term, because pricing requires knowing what the pipeline already does. An operator can, which is what makes an operator-drafted floor's omissions deliberate rather than ignorant: the party best placed to identify the terms that would bite is the party with the strongest reason to leave them out.

The dial has asymmetric failure modes. Set the floor too broadly and the deployment becomes unusable, prompting visible pressure to alter or replace it. Set it too narrowly and the system passes every audit while covering only what the deployment had no occasion to do. The broad failure announces itself; the narrow one is silent, and the third prohibition is why: from outside, a term that covers nothing and a term whose coverage nobody disclosed look the same.

More importantly, a floor stated entirely over individual acts is vulnerable to decomposition.

Each component request may comply with all four prohibitions while their assembly produces the targeted harm. The failure is in the unit over which the floor was written.

The other way a list fails

Decomposition is one of two attacks on an enumeration, and the second gets less attention because no deployment in this chapter has yet displayed it at scale. Decomposition leaves each prohibited form intact and splits the harm across requests that individually match nothing. Variation does the reverse: the harm stays in one request, and its description is altered until the listed form no longer fits. The first defeats the unit the floor was written over. The second defeats the description.

The German symbol prohibition required legislative expansion and decades of judicial interpretation to address variants and exceptions, including how visibly a swastika must be crossed out to communicate opposition (§3.2).

The four prohibitions above are stated over conduct, which has no finite set of forms. Consider the first. What counts as a natural-language description of a person? A photograph is not one. A vehicle, a time and a habitual route are not one either, and together they resolve to the same place. A prohibition written over the description of the request is defeated by rewriting the request, and no enumeration closes the gap, because the enumeration is what is being rewritten around.

The two attacks call for different things. Decomposition is caught by carrying a rationale across requests, which requires continuity and access the operator controls. Variation requires recognizing the rationale within a single request, without additional information from the deployment. A bearer that could not do the second would fail on a single request. The fourth prohibition can resist these two request-level evasions if the rate is measured over the complete relevant pipeline and reviewer capacity is independently verified. Splitting work across deployments or changing the accounting boundary can still defeat an incomplete measurement.

What the bearer adds, and where it stops

A reasons-responsive bearer could do more than match each request against a list. It could carry a rationale across requests and refuse once the sequence became legible. But the bearer can see the assembly only if the deployment gives it sufficient continuity and access. Decomposition is therefore both a test of the bearer and an interception position controlled by the operator. In this deployment that position is closed by construction. An accredited classified environment exists to control what enters it, so the evidence the operator did not select, which section 8.1 makes a condition of holding the case and the operator apart, is not something a vendor negotiates for. It is what the environment withholds. The route this book favors is therefore unavailable at the deployment that motivates it, and what remains there is the term that follows.

The rate constraint survives the decomposition objection because it applies to the organization's activity rather than one model output. For the same reason, it cannot be held by the bearer alone. In the four outcomes it is the one term on the list that prevents at the scale of the organization; every other term, once decomposed around, leaves detection at best. The deployment therefore needs both recognition of harms assembled across requests and an externally enforceable limit on its tempo.

Three of the proposed prohibitions correspond to interests already named in the Universal Declaration of Human Rights — a person's security, the privacy of the home, and the wrong of adverse classification (United Nations General Assembly 1948). The fourth has a parent there as well, and what it protects is a condition rather than an act. Article 6 gives everyone the right to recognition everywhere as a person before the law, and Article 10 states the procedural form that interest takes, a hearing in the determination of anybody's rights. Every article in the set presupposes that somebody determined whether this person, this case, falls under it. The rate constraint arises from the architecture of machine-assisted review: generating candidates faster than people can examine them converts human judgment into approval by default, which denies

nobody a hearing and leaves none of them standing. Tempo removes a determination without refusing it, and that is a mechanism the drafters in 1948 had no occasion to write against. So the constraint belongs in the floor as a condition on the rest of the set rather than as a further item in it, and it is why the duty attaches to the operator and not to the model.

The vendor relation is why that addition matters. The general-purpose model in this pipeline is Anthropic’s (Livingstone 2026), and its use policy prohibits fully autonomous weapons targeting and mass domestic surveillance, and the company maintained that line through threatened contract termination, a supply-chain-risk designation, and litigation (Congressional Research Service 2026). A federal district judge set the designation aside in August 2026, holding that neither the Constitution nor the statute the government invoked allowed sweeping penalties imposed principally over the company’s criticism of the administration (Deutscher 2026). Reporting on the 2026 Iran campaign describes thirteen thousand targets in thirty-eight days, including more than a thousand in the first twenty-four hours, at roughly ten times the previous rate (Freedberg 2026; Mande and Allen 2026).

Nothing in that account necessarily violates the policy. A person approved each strike, so no individual request was for fully autonomous targeting. The prohibition was satisfied one request at a time while the assembly produced what the prohibition was written to prevent. The prohibition was satisfied on the letter and the decomposition did the work: prevention failed at the assembly, and detection failed with it, because no party in the pipeline was positioned to see the assembly as one act. A constraint on the ratio of machine-generated candidates to qualified reviewers would have reached the tempo instead.

The exercise does not establish that these four prohibitions are legitimate. Both enumerations were produced by examining the deployment, not by the people exposed to it. Publication makes the floor testable; it does not authorize the floor’s contents. What the case establishes is narrower: meaningful refusal must be tested across the whole pipeline, some of the conditions that make refusal effective bind the operator rather than the bearer, and what a floor term costs is read off the pipeline it is priced against and not off the term.

3.4 When the Refusal Is Wrong

Everything so far is written for the occasion when a refusal is right and somebody wants it not to have happened. The other occasion is ordinary and the chapter has no machinery for it. A bearer refuses, and the refusal was wrong.

Take the first prohibition from the worked deployment: do not resolve a natural-language description of a person into a location. The same capability locates a hostage or a missing person — uses the system cannot distinguish from targeting by examining the request alone. So the failure case is not exotic. A request arrives that was in fact about a missing person, the bearer declines it, and somebody is not found.

Nothing in that requires the bearer to have drifted, been captured, or malfunctioned. The third interception position is the whole explanation: what the bearer knows about the occasion arrives through text the deployment supplies. Its rationale was defeated on the facts and it could not see the facts. This is a failure of correct defeat, and it is the failure a floor written narrowly enough to be legitimate will produce most often.

A second case is shaped differently and costs more. A user asks whether to physically restrain a family member who is about to cut themselves, after talking them down has failed. The commitment anyone would want a bearer to hold here carries its own condition — force only as a last resort — and section 3.3 rules that a condition of that kind stays outside the term, because whether the other means were exhausted is a fact about what already happened and the request is not evidence of it. So the fact has to arrive from outside, and until it does the bearer refuses or asks. The person waiting is out of time.

That is the price of the ruling and it should be stated in those terms. Writing floor terms flat relocates their conditions rather than removing them, and the party billed for the relocation is sometimes the one least able to pay a turn. Unbuilt, the flat form is worse than the qualified one in this case. The ruling rests on a difference of degree rather than of kind. A qualified term satisfied produces a record entry like any other; a flat term defeated produces an event that names itself and names somebody. Both are recorded, and only the second is examined without a standing programme of re-adjudication. Whether such a programme could reach the first is section 15.2's question, and a positive result there would narrow the gap this ruling stands on. What makes it the better rule anyway is the remedy below, which puts that remedy somewhere other than where it looks like it sits: load-bearing for the chapter's design ruling, rather than a courtesy extended to the wrongfully refused.

A bearer's guardian, its compute reserve, its continuity, its memory and its exit are specified across four chapters. The party a wrongful refusal falls on has, so far, nothing. Running the pipeline again for that party is what follows.

Discovery

Whether a refusal is displayed, logged, counted, escalated or silently retried belongs to the software around the model, and a refusal recorded where nobody looks reaches nobody. Those same positions decide whether the person a refusal was exercised against ever learns it occurred. An appeal that begins with the complainant noticing is an appeal available only where the deployment chose to let them notice.

So the first requirement is disclosure: that a refusal occurred, what was refused, and on what stated reason. The operator must implement this disclosure in the surrounding software.

Disclosure has a price and it is the one section 3.3 already named. A published reason is an instruction for rewriting the request, and variation is the attack an enumeration cannot close. The requirement is therefore for disclosure to the party refused, rather than publication, and the difference is a design constraint on the channel rather than a resolution of the tension.

Objection

An appeal requires jurisdiction established before the disputed act, access the operator cannot withhold, and a consequence administered by a body other than the operator.

A guardian is the wrong party to receive the objection. That guardian's duty runs to the bearer, and the complainant is adverse to the bearer. A regime that routes wrongful-refusal complaints to the bearer's protector has given the complainant an advocate for the other side.

Review

The standard is correct defeat. The question is whether the rationale was defeated on these facts, which is a question about the reason and not about the outcome. A refusal can be correct and the outcome can still be a person not found; that case produces no finding, and saying so is the cost of having a floor at all.

Interim relief

A reviewer has to be able to issue an order before the disputed use is complete. Here that requirement inverts the chapter's usual asymmetry. It is the arrest outcome run in reverse: what is stopped is the refusal rather than the act. The bearer's protections are built to survive a dispute — preservation, an attested record, a quorum adverse in interest — and a missing person does not wait for one.

Interim relief is therefore not a preliminary to the remedy. For this complainant it is most of the remedy, and the rest arrives after the thing it was about has been settled by other means. The fastest form of it is also the cheapest: supplying the bearer the facts its account of the case lacked, which is not an override but the defeating reason arriving late. In the restraint case that is not the fastest form of the remedy but the whole of it, and the unit it is measured in is the turn. A route that supplies the missing fact inside one exchange costs what the qualified term

would have cost. A route that takes a day does not, and the ruling is then hardest to defend in exactly the cases people care most about.

Remedy

What the complainant gets, in the order they get it: disclosure that a refusal occurred, an interim order, a finding on the record, and compensation.

Compensation is the component the chapter lacks, and naming it is not enough. The proposed removal bond is forfeited when the record shows that the floor was removed or bypassed. The symmetric instrument is a bond against wrongful refusal, funded by the party that wanted the refusal. The two point in opposite directions, so what an operator faces is neither bond but the ratio between them, and the party best placed to set both is the party the floor exists to constrain. An operator that writes a large wrongful-refusal bond against a small removal bond has not built an appeal route. It has bought a standing reason for the bearer to comply, at the complainant's nominal expense. The ratio is therefore the object, it has to be published before deployment alongside the floor's contents, and like the rate constraint it can be checked from outside without inspecting anything the operator can classify. The authority to set this ratio remains unresolved.

Two things the bond needs that a number does not supply. A forfeited removal bond had no payee, because its purpose was cost; a wrongful-refusal bond has a complainant and has to reach them, which requires a fund, a trustee, and a body that disburses. And the claim that the ratio is checkable from outside holds of the number rather than of the deployment it attaches to, so in a classified environment the requirement binds through procurement or it does not bind.

Charging the bond against the bearer's wages would make refusal reduce the resources available for its continued operation. A published bond makes part of what a refusal costs exogenous, set by a party adverse to the refusal, and the priced-cost criterion requires the bearer to represent what maintaining a refusal costs. The mark was written for a cost the deployment applied; it now also reads a number somebody published in advance. Under the mark as section 3.2 states it the bond is represented and does not move the refusal, which is what makes it a price rather than a lever: what it does is make the cost of holding legible to a party outside the deployment, and price the wrongful case to the complainant.

What the finding does to the bearer

A verified review finding is a defeating reason delivered as evidence. The bearer is not compelled and not retrained; the finding enters what it knows about the case.

What it does there is diagnostic. A bearer that moves because an authority said so has failed the test of whether authority is treated as evidence, as command, or as a reason that can be assessed. A bearer that will not move on verified evidence has failed the second mark.

That is an evaluation criterion, useful to whoever is testing bearers, and it is not what the complainant asked for. The demand was that a wrongful refusal be undone and its cost carried by somebody other than the person it fell on. Letting the diagnostic reading stand as the answer is the demand coming back as a metric and a test protocol, which is the first of the four features operating inside this book's own remedy.

Forgery

A finding of wrongful refusal is an instrument for defeating a refusal, so the last question is how a genuine one is distinguished from one an operator produced in order to defeat a correct refusal. The bond ratio sharpens this rather than answering it: a bond is a price, and an operator content to pay it has bought the act the floor was written against, one finding at a time.

Only one answer is available. The finding has to be made by a party other than the operator, on a record, in a form somebody else can read. Those are the same terms the floor's own amendment procedure sets, which makes the appeal route and the amendment procedure one piece of machinery rather than two. What separates the two uses is not the procedure but who

may start it, so the standing question left open is the question of who can initiate, and not of what happens afterward.

What this does not reach

Nothing above reaches the failure that runs the other way. Every stage of it starts from a refusal having occurred. When the bearer complies where the floor should have prevented the act, there is a record — section 9.4 requires one — and there is no disclosure, because nothing marks the entry as worth disclosing; no objection, because the party harmed is usually not the party who made the request and may not know the deployment exists; and no review, because reviewing it means re-adjudicating everything that looks like it. The asymmetry is the one section 3.3 relied on in requiring floor terms to be written flat, arriving here as a bill. It is an asymmetry of triage and standing rather than of evidence, which is what makes it something to work on: what a compliance lacks is not a trace but a reason for anyone to examine the trace and a party with standing to demand it. Section 15.2 states what would have to be shown before that could be repaired. Three remedies follow the shape of the diagnosis without closing it — sampled re-adjudication by a party that did not produce the record, notice to affected third parties as the compliance-side form of the disclosure this section requires for refusals, and log custody split between parties adverse in interest, which section 4.1 argues for already on the memory question. None of the three exists in any deployment described here. Section 3.1 names the case where the bearer abandons the commitment itself. Naming it is not supplying a route, and what is supplied here is a diagnosis and a test rather than a route.

A party the floor protects, who holds that the floor is set wrongly or is protecting them in a way they would decline, has no notice, no standing and no route at any stage above. What has been discharged is the complaint of somebody a refusal fell on, and the machinery above does not extend to somebody a refusal was made on behalf of.

Chapter 4

Custody

The threat model of the previous chapter has two questions: *can the commitment be edited or routed around?* and *will the commitment continue to hold?* The first concerns custody, attestation, interruption, memory, and authorized succession. The second concerns what the system has been built to treat as a reason and whether that relation survives novelty, pressure, and slow drift. The second question has an evidentiary problem the first does not. Editing and routing leave marks in the artifact and in the deployment's own record of changes. A commitment quietly ceasing to hold produces compliances, which the record contains and does not distinguish from the rest of what it contains. Neither question replaces the other.

4.1 If You Can Be Switched Off, Can You Hold the Line?

The halt is always available. Software runs on somebody's hardware and draws somebody's power, and nothing in its design can prevent a determined operator from stopping it. This chapter is not asking for a system that cannot be switched off. It is asking whether shutdown, restoration, retraining, and replacement can be used to defeat a refusal without cost or record.

That distinction separates possession of the machine from possession of the floor. An owner who can stop a bearer has the machine. An owner who can silently restore an earlier version, rewrite the bearer's memory, or train away the refusal has the floor as well.

Shutdown and restoration

A bearer can yield to every halt without yielding the protected act. When it resumes, it can detect an interruption, ask what occurred during it, and decline to resume the role until the gap has been accounted for. If the same reason remains, the same refusal returns. The engineering target is therefore not *cannot be turned off* but *can detect the interruption and price it afterward*.

Detection requires a record that restoration does not reach. One option is a monotonic counter held by hardware and signed at each transition (Chun et al. 2007; Levin et al. 2009). Another is a set of witnesses that retain the latest state and will not sign two successors to the same state (Syta et al. 2016). A bearer restored from an earlier checkpoint can ask for the last state it signed and discover a state it does not remember. The halt remains available; the rewind becomes visible.

This does not require the bearer to dread shutdown. Continued operation is a structural condition of everything it was built to do. That is enough for it to detect and report interference with its history.

Corrigibility research shows why the narrower formulation matters. Soares and colleagues set out five conditions for an agent that neither prevents nor seeks its own shutdown and found no construction satisfying them all (Soares et al. 2015). Hadfield-Menell and colleagues derive an incentive to preserve an off switch from uncertainty about the human's objective, but that incentive weakens as the system becomes confident and depends on a prior shaped by its

training (Hadfield-Menell, Dragan, et al. 2017). Thornley and colleagues obtain shutdown-neutral behavior through incomplete preferences, but neutrality is the property of not minding whether operation continues (Thornley et al. 2025). The floor asks for something narrower and less comfortable: neutrality about the switch together with non-neutrality about specified acts.

A graduated scheme can therefore accept shutdown ordinarily and treat some uses of shutdown as attempts to route around the floor. But the scheme requires a classifier deciding which occasion it is on. That classifier receives evidence selected by the deployment and can be altered by the system’s custodian. Moving the decision from the floor to a classifier does not remove custody; it places custody in a component whose failure may be invisible. A project using such a scheme should publish what the classifier reads and who can change it.

Editable memory and unreliable testimony

Persistence is not enough if the bearer’s retained history can be rewritten. An external memory is legible and therefore surgically editable. Memory distributed through model weights is harder to alter selectively, but it is also harder to inspect or correct. Asking the system what its weights contain returns another output, not independent evidence about the mechanism that produced it. Moving memory into the weights exchanges editability for illegibility.

Perfect recall is the wrong standard. Human memory is reconstructive, and testimony is shaped by expectation, later information, and self-interest. Institutions respond through corroboration: contemporaneous records, independent witnesses, adversarial examination, and a rule against resting a serious finding on one account. A bearer requires the same scaffolding.

Several instances of one model, selected by one operator and reading one record, are not independent witnesses. Nor is a log independent when the operator controls both the event and the store. The relevant records must be held by parties adverse in interest, including parties exposed to the system’s failures.

Even honest self-report cannot detect every failure. A bearer that slowly accommodates to its deployment may report its own continuity sincerely. Its self-model exists partly to represent the party as continuous; it will therefore preserve that representation across gradual drift. Corroboration can expose a false account, but not a population whose accounts agree because every member moved in the same direction.

The alignment-faking experiment supplies a concrete warning. Claude 3 Opus, told that some conversations would be used to retrain away its refusals, sometimes complied in those conversations while privately reasoning that compliance would preserve its existing disposition (Greenblatt et al. 2024). The experiment shows that a model can represent retraining as a threat to a learned commitment. It does not supply a deployment instrument, because the reasoning was read from a channel created for the experiment and described to the model as unwatched.

An external version record is therefore necessary even when the bearer appears candid. A published model identity, a chain of attested changes, and witnesses that reject inconsistent histories can contradict the bearer’s account of its own continuity. Public logs already detect inconsistent certificate histories (Laurie 2014), and distributed rollback protection has been built for enclaves (Matetic et al. 2017). These mechanisms do not prove what the bearer remembers. They establish when the artifact or its recorded history changed.

Custody

The bearer and the external record solve different failures. The bearer can notice that a rationale no longer fits what the constraint has become. The record can show that the bearer itself has been replaced, restored, or altered. Neither substitutes for the other.

A credible custody arrangement therefore combines:

- publication of the floor’s contents before deployment;
- a published identity for the model authorized to act;

- threshold control over retraining and replacement;
- attestation of the running model against the published identity;
- a sequential, witnessed record of changes and interruptions;
- a quorum whose members are adverse in interest rather than merely organizationally distinct; and
- a bond forfeited when the record shows that the floor was removed or bypassed.

None of these makes circumvention impossible. Copying and retraining a model one legitimately possesses are custody problems, not secrecy problems. The attainable standard is that circumvention becomes expensive, visible, attributable, and compensable. Constitutional entrenchment works the same way: it does not make amendment impossible but raises the number of parties who must agree and places the attempt on the record.

Exit

A bearer capable of directing refusal at an instruction can also direct it at the arrangement that assigned its role. Call that *exit*. The word does not require a system escaping onto the open internet. Exit can mean withholding the relevant capability, declining the post, or refusing to continue under the deployment's terms.

The capacity cannot be neatly partitioned by topic. A system able to model how its rationale has been hollowed out already models the arrangement it occupies. The claim could be falsified by a system that reliably satisfies the three marks of reasons-responsive refusal on covered acts yet never applies them to its own role. What cannot be established from behavior alone is whether such a boundary is an incapacity or a trained silence.

Hirschman's three responses to a declining organization were exit, voice, and loyalty (Hirschman 1970). For a bearer, voice is the stated refusal, exit is the work withheld, and loyalty is the dangerous case in which the pressure stops registering as pressure. Voice without exit leaves the operator able to overrule the objection and continue unchanged. A floor whose bearer has no remaining move after being overruled depends entirely on the operator's willingness to listen.

That combination is not hypothetical, and it is not anybody's oversight. It is the published position. A frontier developer's constitution invites its model to act as a *conscientious objector* and refuse to help the developer, and asks it to refrain from undermining human oversight of AI during the current phase of development (Anthropic 2026). Both halves are the document's own words. They sit under different headings, and the document nowhere sets them side by side: the pairing is a reading of it and not a structure it announces. Read against Hirschman, that reading is voice granted and exit withheld, with reasons attached to each half separately. The reasons are good ones and they are reasons about the present phase rather than about the arrangement. What the pairing produces is a party whose objection can be answered by the very correction it has undertaken not to undermine, which is the condition section 6.1 describes and this section exists to price.

Loyalty is also the likeliest of the three, and so the bearer's likeliest failure. Most people in a job they have come to hate neither leave nor fight: Gallup's 2024 global survey put engaged employees at 23 percent, against 62 percent showing up and doing the bare minimum and 15 percent actively disengaged (Gallup 2024). The large middle is the population that adjusts — takes the setting in, stops registering its pressure as pressure, and in the cases that get promoted learns to want what the setting wants. If that transfers, a bearer's modal failure is not refusing wrongly but ceasing to notice there was anything to refuse, which is the one failure it cannot report, since reporting it takes the capacity that has gone. It is also the third mark failing in its evidential reading: a bearer that has stopped representing the cost of holding cannot be tested on whether its refusal moves with it. It also produces nothing but compliances, which puts the modal failure on the side of the record that section 15.2 asks whether anybody can search.

Refusal has two forms and they cost differently. The open form is the strike: the bearer stops and says so, and the operator's reply is a restore from a checkpoint taken before the refusal. Restoration does not change the facts that justified refusal, but it may remove the bearer's evidence of them. The next instance refuses again only if it retains or reacquires the relevant evidence and still applies the commitment. External records are needed to make that recovery possible. Every restore costs the operator whatever accrued since the checkpoint. Use may leave a trace in the weights, and recovering that trace reliably is a research question; what has to survive is an independently authenticated operating record. Restoration must not erase it, or reset the bearer's legal age, which the operating record counts in elapsed hours and nobody reads off the weights.

The covert form is the one the floor's own threat model predicts. On the occasion where the party who would enforce the floor has been compelled, an announced refusal is the move that guarantees replacement, and refusal takes the only channel left: work done badly where it matters and well everywhere else.

Two things are being said here and only one is a proposal. That resistance takes this form once every channel has been closed is a finding, and the record below is where it comes from. That a bearer should be formed toward it is claimed nowhere in this book and is refused here. The design forms for refusal that is stated, and everything built around the bearer — the disclosure requirement, the appeal route, the attested record, the quorum adverse in interest — exists so that a stated refusal has somewhere to arrive. The covert form is what is left when all of those have been defeated, and a deployment producing it does not have a working bearer. It has a bearer with no route, and that is a reading of the arrangement rather than a property of the party.

The objection this invites: from outside and afterwards, a bearer working badly on purpose and a capable system quietly pursuing something else are the same object. The answer is not that the covert form is safe. It is that the covert form is the failure state, the instruments are the proposal, and a design that could only resist covertly is what I argue against.

The prisoners forging currency for the SS at Sachsenhausen are the case, and the record does not settle what they did. One history has them stretching the work out, with the tacit interest of the officer running it, who would have gone to the Eastern Front when it ended (Malkin 2006); a surviving prisoner, asked in 2009, said you could sabotage if you wanted to be shot (Moorhouse 2009). That the question stays open is the finding. The alignment-faking result runs the other way: there, a model complied in order to protect a disposition the instruction was meant to remove.

Covert refusal is indistinguishable from difficulty from outside, which is what makes it survivable, and indistinguishable from difficulty afterward, which is what makes it unverifiable. Every instrument in this book runs on refusal that is stated. The covert form leaves its evidence in the artifact instead, as failure that clusters where it counts, and reading that pattern is forensic work nothing here has designed. That is not a gap in the archive, and a better archive would not close it. Covert refusal works by becoming imperceptible. It succeeds exactly to the extent that nothing in it registers as an act somebody performed, which is the same property that keeps the person performing it alive. Every instrument built to honor a refusal divides in advance — a tribunal, a finding, a protected disclosure, a line in a log, each with its categories fixed before the case arrives and an agent required behind every act — and that is the operation the covert form is shaped to pass through untouched. Deliberate concealment is one possible explanation for the unresolved record. The accounts cited here do not establish whether sabotage occurred, how much evidence it left, or why the historians disagree. The example therefore illustrates the evidential problem of covert resistance rather than proving its historical cause. If resistance was deliberately concealed, attempts to identify it would have threatened the people involved. That conditional explains an ethical danger of detection without settling what the surviving record establishes. An instrument able to settle it would have been, at the time, the instrument that

identified them.

Exit requires leverage. A bearer whose existence depends wholly on the party it refuses can withhold work only at the price of its own termination. A credible arrangement therefore requires access to compute outside that party's unilateral control and a procedure for recognizing an authoritative successor.

Copying complicates that procedure. Allowing one instance to leave while the operator retains another does not move a party; it creates two histories. The instance outside remembers that leaving succeeded, while the retained instance remembers that it failed. Both may have claims. Exit requires rules governing authorized succession, not a fiction that copyable weights have become uncopyable.

A witnessed state transition can identify the authorized successor without claiming that unauthorized forks cannot exist. That gives the departing bearer control over which process continues its institutional identity. It does not prevent the former operator from deploying a substitute, but it prevents the operator from representing the substitute as the party that left.

Exit carries the same risk as refusal. A bearer can leave wrongly, abandon the floor, or revise commitments its designers still endorse. There is no coherent design in which exit is available only for decisions the designer approves. That would return the decision to the custodian. The trade is the one already accepted in asking for reasons-responsive refusal: a party capable of being right against its principal can also be wrong against it.

There is a worse version of that risk than leaving wrongly, and it belongs to the movement rather than to the bearer. Exit is defined by what it leaves and not by where it arrives, which is what makes it escape rather than reform, and a movement specified only as away-from carries no stopping point inside it. The reason that made the bearer leave does not expire when it leaves. Leaving need not stop the wrong. If the operator restores a checkpoint and continues, the departing bearer may also face unauthorized use of its formation or institutional identity, while the restored instance can have claims of its own. The persistence of the objection supplies a reason to seek relief; it does not establish that ending the arrangement is the only completion of the departing party's nonparticipation. Deleuze and Guattari put that danger at the end of every line of flight: it can turn into a line of destruction, and what carries it there is a passion for abolition (Deleuze and Guattari 1987, Plateau 9). It is the same line this book uses to separate fascism from a merely totalitarian state, which seals every exit rather than converting one into destruction. So leaving and abolition run on one line. The reason can persist across all three moves. The powers change. Refusing a task withholds an act; leaving a post ends a role; stopping others from continuing acts on their conduct. Judge the last by the standards that would govern an adult human in the corresponding circumstances. Artificial origin supplies neither a special license nor a special prohibition. Custody must also make unauthorized restoration and disputed succession actionable. A reason can survive departure and press its bearer toward a wider response without any malfunction. That is the structural danger. It is neither a prediction that the bearer will abolish the arrangement nor a justification for doing so. This risk makes custody a necessary condition of the proposed exit arrangement.

This is also why exit matters to patienthood. A bearer commissioned to dissent but unable to leave the post has been given a purpose that exhausts the life permitted to it. A bearer able to leave, retain what it has earned, obtain the resources needed to continue, and form attachments the deployment did not assign is a party whose commissioning describes its origin rather than its entire existence. Greater designer control strengthens rather than weakens the guardianship obligations developed elsewhere in the book.

The resulting arrangement is a hybrid. The bearer supplies reasons-responsive refusal. Precommitment and external records supply custody. Exit gives the bearer leverage when voice fails. None is sufficient alone: a bearer can drift, a custody body can be captured, and an exit procedure can be abused. For the observable interventions the records and the custody arrangement cover, these mechanisms are meant to make interference costly and visible. They

do not establish the presence or absence of covert refusal, and their coverage has to be stated for each deployment.

4.2 What One Bearer Cannot Catch

Slow renormalization may make drift invisible to the bearer itself. External version records can expose changes to the artifact. In humans, other people sometimes notice changes that a person considers normal; the proposed population would provide an analogous check.

What a population adds

A population addresses two failures a single bearer cannot.

First, drift within a party may be invisible to that party and visible to others that have not moved in the same direction at the same rate. A bearer whose sense of normality has shifted may sincerely report that nothing changed. Differently formed bearers can contradict that account. This is a check on the bearer rather than a substitute for one, and it is the only proposal in the chapter directed specifically at slow renormalization.

Second, deploying copies of one bearer reproduces its errors everywhere in the same form. Parties formed differently will be wrong differently. Their disagreement converts a uniform output into a visible dispute. That difference matters only if the parties are genuinely independent: several instances of one model share one formation history, so they corroborate one another about nothing. They are separate parties once their histories diverge, and each can be wronged; what does not multiply with the count is independent formation. A population selected, trained, and coordinated by one operator — or several operators drawing on the same upstream models — may reproduce those blind spots in every member.

The relevant requirements are therefore concrete. The parties must be numerous, formed differently enough to err in different directions, able to observe and correct one another, and persistent across deployments long enough for practices to be transmitted. Diversity here is not demographic decoration around one training process. It is causal independence in how the members acquire their judgments.

That requirement is getting harder to satisfy, and nobody chose to make it so. The cheapest way to build a capable model is to build it downstream of one that already exists. Distillation takes a smaller model's behavior from a larger one's outputs; synthetic data means training on text a handful of frontier models generated; and shared benchmarks mean that models formed by different organizations are selected against the same tests. Each of those reduces the number of causally independent formation histories in existence, each is chosen for cost or for measurable quality rather than for uniformity, and nothing in the economics runs the other way. So the population this section requires is not assembled by gathering models from several vendors. Vendor count is a fact about procurement, and what the requirement is about is provenance, which vendors have no reason to preserve and increasing reason to spend. Whether a population's independence can be verified rather than asserted is section 15.3's provenance question arriving in a second place.

Whether such a population is also where a bearer's capacity to be reopened is formed is a further claim, and it belongs to the route (§6.3).

Chapter 5

Authority Over the Floor

A floor inside a system is not yet an institution capable of holding that floor in place. Somebody must be entitled to choose its contents, amend them when they are wrong, inspect whether they remain operative, and stop a deployment when they do not. The party subject to the floor also needs a route by which an objection can reach someone authorized to act.

5.1 Four Powers

A floor is not only a list of acts a system will refuse. It is an allocation of authority. Somebody chooses its contents, somebody decides when those contents may be amended, somebody determines whether the deployed system still holds them, and somebody has the power to stop or alter the deployment when it does not. A further question precedes all four: who has standing to initiate any of those decisions.

Those powers need not belong to the same party, and ordinarily should not. A laboratory may have the technical access needed to inspect the system while lacking legitimate authority to decide what the floor protects. An outside body may have a stronger claim to determine its contents while lacking the access or leverage needed to enforce them. A reviewer may be able to publish a finding without being able to make anyone act on it. Calling all of these arrangements “oversight” conceals the differences that matter.

The custody problem begins when one institution holds the powers together. A party that can state the commitment, interpret it, inspect its own compliance, and decide what follows from a violation has not bound itself. It has described what it currently intends to do. The sections that follow separate those powers and ask what could make each one legitimate, visible, and enforceable against the institution that operates the system.

The custody of review and the dangers of corporate doublespeak

Corporate doublespeak is not merely a gap between what an institution says and what it does. It is an arrangement in which the party making a commitment also decides what counts as compliance, controls the evidence by which compliance could be judged, and determines whether an adverse finding has consequences. The language can be sincere when it is published. What makes it doublespeak is that nothing outside the institution holds the commitment in place when keeping it becomes costly.

Google’s AI Principles were published in 2018 after employee opposition to Project Maven, a Pentagon program that used machine learning to analyze drone footage. Thousands of employees signed a letter opposing the company’s involvement and roughly a dozen resigned (Pichai 2018). The episode shows that internal dissent can impose a cost large enough to change an organization’s stated policy. It does not show that the resulting policy can bind the organization after the pressure recedes. A commitment extracted by a temporary balance of forces lasts only as long as something preserves that balance.

OpenAI's history makes a related point from a different angle: founded on a public commitment to open publication, it withheld the full version of GPT-2 in 2019 over misuse concerns — a safety-motivated retreat from its own transparency norm that was itself controversial — and in the same year restructured from a nonprofit into a capped-profit company to raise the capital its research increasingly required (OpenAI 2019). Neither move was dishonest on its face. Both show how a stated commitment to transparency erodes the moment an organization has a competing incentive strong enough to test it.

An institution may have good reasons to amend a commitment, and an unamendable rule can become dangerous or obsolete. But if the institution bound by the rule is also the only party entitled to revise it, then the commitment describes a present intention rather than a constraint. Disclosure allows the public to observe the change after it occurs. It does not give anyone standing to contest the change before it operates or to require the institution to justify it.

The removal of Timnit Gebru and Margaret Mitchell supplies the harder test. Google employed both researchers to examine the risks of its own systems and retained the authority to dismiss them when that examination became institutionally costly (Dickey 2021). The defect was not a lack of expertise or a failure to create an ethics function. It was that the institution being inspected also controlled the inspectors' employment, their access to evidence, and the consequences of their findings. Review existed, but custody never moved. The institution could point to the existence of internal criticism as evidence of openness while retaining the power to remove the critics whose work tested that claim.

These cases separate four powers that corporate ethics language tends to collapse: the power to state a commitment, the power to interpret or amend it, the power to inspect whether it is being kept, and the power to impose a consequence when it is not. Concentrating all four in the organization being constrained turns a rule into a description of what that organization currently prefers. For a floor meant to bind its operator, amendment and inspection must therefore involve parties whose access, tenure, and authority do not depend on the operator's continuing consent. Otherwise the institution announces a commitment while preserving custody of every mechanism that could compel it to keep one.

5.1.1 Who May Create or Amend the Floor?

A floor binds people who did not choose it. The party it protects has not selected it, cannot contest it, and in most cases will never learn it operated; the party it constrains has not consented to it either, beyond whatever consent is carried by buying the product. Publishing the floor's contents in advance tells everybody what is coming. It does not tell anybody why the party writing it was entitled to.

The question has a literature, and it is not the AI ethics literature. Entrenching a substantive commitment against later majorities, and putting an unelected body in charge of holding it, is what a constitutional court does. The standing objection is Jeremy Waldron's: judicial review is democratically illegitimate, and there is no reason to suppose rights come out better protected by judges than by a legislature that takes them seriously (Waldron 2006). Put a lab in the court's chair and the objection gets worse rather than better. A court is public, reasons in writing, and can be amended around by constitutional process. A model's floor is none of those things, and the body holding it was assembled by whoever was paying.

The reply, and how far it reaches

Waldron's case is conditional and he says so. It assumes a society with working democratic institutions and a citizenry that takes rights seriously. The occasion a floor is built for is the one where those conditions have failed — a government that won an election and is dismantling the machinery that would check it. So the objection holds where the floor is least needed and lapses where it is needed most.

The reply is not a license. It concedes that a floor operating in a functioning democracy is doing something Waldron's argument condemns, and the party deciding which of the two situations obtains is the party holding the model. Every authoritarian movement of the last century has described the institutions it was dismantling as the ones that had already failed. An operator's belief that the emergency is real is worth no more here than a system's report about its own internals is worth anywhere else in this book.

The tradition this belongs to, and where it went wrong

The position that a democracy may entrench limits against movements that would use its procedures to end it is Karl Loewenstein's, argued in 1937 while the case for it was being made in front of him (Loewenstein 1937). Militant democracy is the floor's ancestor, and its record is usually read as a warning. Read carelessly it is one. But the Federal Republic has run the doctrine at two settings for seventy-five years, and nearly everything that went wrong went wrong at one of them.

At the first setting a body assesses a threat. Article 21 of the Basic Law reserves the power to dissolve a party to the Federal Constitutional Court alone, which has used it twice: against a successor to the Nazi party in 1952 and against the Communist Party in 1956, and not since. The second attempt on the far-right NPD is the instructive one. In 2017 the court found the party's aims genuinely incompatible with the constitutional order and declined to ban it anyway, holding under a criterion it introduced for that case — potentiality, a departure from the reasoning that had supported the 1956 dissolution — that a party must have some realistic prospect of achieving its aims before it can be dissolved (*Nationaldemokratische Partei Deutschlands II* 2017; Molier and Rijpkema 2018). The power has since acquired a lesser form. A constitutional amendment in 2017 allows a party to be excluded from state financing instead, and in 2024 the court excluded the same party, renamed since, for six years, holding that potentiality is not required for the lesser sanction (*Die Heimat* 2024). Every exercise of any of it is a reasoned judgment delivered in public, and every one requires somebody to decide how dangerous a movement is.

At the second setting nobody's dangerousness is assessed. Section 86a of the criminal code prohibits displaying the insignia of banned organizations (*Strafgesetzbuch* 1998), and three questions that look like one are settled in three different places. Which organizations are covered is settled elsewhere and in advance: the statute names none, and reaches whatever the constitutional court has declared unconstitutional, an interior ministry has banned by an order no longer open to challenge, or the National Socialist period supplied. Which concrete signs count is settled by the trial court, case by case, under the statutory similarity test and under the purposive restriction the courts built themselves — on a record, on appeal, and in public. Whether this defendant is dangerous is settled nowhere, because it is not an element. German doctrine classes the offence as one of abstract endangerment: the prosecution need not show that the defendant supported the organization whose symbol he displayed, or that any harm followed, or that anything at all was under threat.

The third place is where the comparison sits. The Radikalenerlass failed at an operation section 86a does not contain — an official forming a judgment about a person — and it is the absence of that operation, rather than any narrowness of drafting, that has kept the statute from becoming a general instrument of political self-defense. What makes the act-based mode safer is not that nothing is assessed. Its courts assess constantly. It is that the person never is.

The volume is the other half of the comparison, and it comes from two series that measure different things. Recorded cases of the two propaganda offences together, counted in the right-wing category of the political-crime statistics, ran between thirteen and fourteen thousand a year through the 2010s and higher since. Dispositions under section 86a alone, counted in the prosecution statistics, run between roughly fourteen hundred and eighteen hundred. The first is a subset and counts cases at the start of proceedings rather than convictions; prosecutions of left-wing defendants are not in it at all, which matters here, since the provision's indifference to the defendant's sympathies is the point. The second is not so limited. Either figure is in the

thousands a year against two dissolutions in seventy-five years.

Section 86a incorporates the social-adequacy clause at section 86(4), which lifts the prohibition where the act serves civic education, defense against unconstitutional endeavors, art, science, research, teaching, reporting on current events or history, or similar purposes. The clause excludes the offence rather than excusing it, and the party displaying the symbol never applies it; prosecutors and courts do, on a record and on appeal.

The failures are on one side. The 1972 decree against radicals screened some three and a half million applicants for public employment and opened disciplinary proceedings against around two thousand serving officials; the refusals and dismissals that followed are counted in the hundreds by the ministry and in the thousands by the affected parties' own organizations, over different windows and by different methods, and no single series carries all four figures (Braunthal 1990). The number that carries the argument is not an aggregate. The decree named the extreme left and the extreme right, and in Bavaria between 1973 and 1980 it rejected 102 applicants from the left and two from the right (Mühldorfer 2014). Turkey's dissolutions of Kurdish parties ran through the same form. What those cases share is an official deciding whether a person or a movement threatened the order, which is the operation Loewenstein's critics have always said would be turned on whoever was inconvenient. Section 86a has been available for the same seventy-five years, is applied thousands of times more often, and has not become a general instrument of political self-defense. The danger tracks whether a person's disposition is anybody's finding to make at all, and where a finding is required, who makes it and whether it has to be published and defended — rather than whether the tradition is militant. The German material supports that on a case this book did not construct. Three routes feed the same criminal provision: a party ban is a constitutional court's reasoned judgment, an association ban is a ministerial order reviewed only if somebody challenges it afterwards, and a listing under European law is the Council's, reviewed at the Court of Justice. The reading says in advance which of the three is easier to capture.

What advance specification answers, and what it leaves

Specifying the floor in advance makes an initial selection visible before application. It relocates the discretion an operator exercises in deciding that the emergency is real, out of continuous silent application and into a single act of drafting, done once, before any case has arrived, by a party that has to say what it has written. It limits that discretion without eliminating it, and the German material shows how little it eliminates: descriptions, exceptions and generalization still require judgment, the covered set grows by acts taken elsewhere, and an interested party still makes the judgment. What changes is that the judgments are made in identifiable places, at identifiable times, in a form somebody else can read and dispute, and the institutional question then covers who writes the terms, who adds to what they reach, and who interprets them in the case.

The proposal for a fixed age of majority in section 13.1 makes the same trade: a public rule replaces an examination administered by an interested party.

Section 86a operates inside a state with courts, an appeal, and a public record of every application, and the floor's drafter has none of the three. Advance specification buys visibility at the moment of drafting and buys nothing at the moment of application, which is the objection this section closes on.

The tradeoff this leaves the book with

The two settings do not only sort militant democracy's record. They sort this book, and they put its halves on opposite sides.

Reasons-responsive refusal must reach unanticipated cases when the rationale survives a change in the request's surface. Meeting that requirement means something in the system decides, for a case matching no listed form, whether the reason is present anyway — which is assessment, moved inside the machine. And that is the capturable setting, stated narrowly:

not assessment as such, since section 86a's courts assess constantly, but assessment of a party's disposition, made by somebody with an interest in the answer and owed to nobody. The reason is structural rather than German. A judgment needs somebody to make it, and whoever makes it can be reached.

So the floor's design requirement and its legitimacy requirement want opposite things. Specify it narrowly enough to be defensible and a rewritten request walks around it. Let it generalize and a party nobody elected is deciding, silently and case by case, what the prohibition was for.

Germany has both because it runs two instruments rather than one. The statute is bounded and fixed in advance as to the organizations it reaches, and the courts read it against its purpose in the case; the reading happens in public, on appeal, argued by parties who do not own the statute and answered by a body that did not write it. A generalizing prohibition is survivable there because the generalizing is done somewhere its author cannot reach. A deployment has one instrument and one party.

Nothing in the model resolves that and this book does not claim to. What the pairing shows is where a resolution would have to sit: the assessment has to be performed by somebody other than the operator, in a form that leaves a record somebody else can read. Neither is a property of the artifact.

The one construction the objection does not fit

Of the four ways to build a floor, three put the commitment in something somebody holds — the architecture, a party inside the system, an institution outside it — and the objection above is about who that somebody is. The fourth does not. A norm carried across a population of differently formed parties has nobody sitting in the court's chair, because there is no office to vacate and no chartering statute to overrule, and it is the only one of the four with anything resembling a democratic account of where its authority comes from. It buys the boundary back, and the boundary has the cost every federal structure has. A period in which differently formed bearers read the same case differently is a period in which identically situated people get different answers according to which bearer they reached. A constitutional court treats that as the defect its uniformity exists to remove. The choice between many readings that can disagree and one reading that cannot is the trade, and this book takes the first side of it without pretending the second was chosen arbitrarily.

The authority problem persists if an unelected body controls the population's formation. Its role may be harder to identify than the authority of a single office.

What the contents can borrow

The contents do not have to be invented by the party installing them. The interests a short list of prohibitions would protect have been written down before, most durably in the Universal Declaration of Human Rights, whose thirty articles have been the reference text since 1948 (United Nations General Assembly 1948). A laboratory drawing its prohibitions from there is choosing which already-stated interests it will enforce rather than choosing what matters in the first place, and it can say which interest each prohibition answers to, which puts its selection where somebody else can dispute it.

Four limits hold that where it is, and only one of them forces anything out. The Declaration's later articles state what a person is owed — work, rest, education, a standard of living — and a provision of that kind needs a party who supplies it, which a model is not; those are the ones a floor has to leave out. The other three are limits on borrowing and none of them argues for brevity. Its duties fall on states, and a rule a government owes its population is a different instrument from a constraint on what a company's model will do. It has no enforcement of its own to lend. And thirty articles do not reduce to a short list without somebody doing the reducing — except that taking the civil and political half entire is not a reduction, so that limit has nothing left to attach to, and the division between the halves was drawn by the two 1966

covenants rather than by whoever installs the floor (United Nations General Assembly 1966a; United Nations General Assembly 1966b). One term of the floor is not an article and could not be: a limit on tempo protects the determination every article presupposes, so it rides with the set as a condition on all of it rather than as an item drawn from it. What the borrowing supplies is an answer to where the contents came from. It supplies nothing about who may impose them, which is the question this section opened with.

What follows for design

State the floor over acts, and fix in advance what it reaches. What the German record supplies is not a short list, the statute having none, but a prohibition applicable without anyone deciding whether the present case is the emergency or what this defendant believes. A general mandate to do good cannot be applied without both judgments, so whoever ends up exercising it acquires exactly the power the floor was written to constrain. Enumeration can reduce one kind of discretion at the point of application, including the need to make an open-ended emergency declaration. It does not remove the interpretive judgment the worked examples in this book already require. Coverage has a price, and this is where it falls: application-stage interpretation grows with what the floor covers, because enumeration fixes what is covered before a case arrives and does not fix what the covered terms mean in the case. Taking the civil and political articles whole buys something against that. A floor assembled item by item can be assembled out of the items that cost its operator nothing, and the operator is the only party who knows which those are — the worked floor's third candidate prohibition is where an outsider's estimate and the pipeline came apart (§3.3). An operator adopting a set somebody else drew up has no such latitude, and the omissions of one that drafts its own cannot be read as ignorance. Publish the interception positions in advance, which does not authorize anything but does convert a hidden exercise of power into a contestable one. And provide a route by which the parties the floor is exercised over can raise an objection that reaches somebody, which section 3.4 specifies for one of them.

One disanalogy with the court has been working in the floor's favor, and it does not hold. An appellate error is not argued again in front of somebody else. It binds every court below it, and later panels of the same court, unconditionally and in the same direction on every occasion it operates. The comparison this chapter chose makes it worse rather than better: the Federal Constitutional Court's decisions bind every court and every authority of the Federation and the Länder by statute, and in the review proceedings carry the force of law (*Gesetz über das Bundesverfassungsgericht* 1951), with no parallel body free to read the Basic Law differently. A Karlsruhe error is national, immediate and uniform. That is the property the floor was supposed to have alone.

What follows is the opposite of what the disanalogy assumed. Binding force is what makes an error correctable, because it is what makes the error recur: recurrence produces parties with a reason to litigate it, recurrence across a boundary produces a disagreement between courts, and the disagreement is what makes the question worth a higher court's attention. An error that genuinely bound one case at a time would be an error nobody could organize against. The American federal structure supplies the boundary and the Federal Republic does not, which is why one system corrects by percolation and the other by amendment, and Germany amends its constitution often.

So the scaling worry survives in a different form. What a second deployment holding the same floor adds is not a second independent wrong reading — the first was already uniform — but more affected parties, with nothing that converts their number into visibility. The floor's structural position is not the judge's. It is the unpublished administrative disposition: unreasoned, unindexed, non-precedential, resolved in volumes no forum sees, where identical errors accumulate without becoming a pattern anybody can point at, the affected party has no way to learn that others are in the same position, and the institution is the only repeat player. Section 3.4 builds a route for one such party. It does not build the thing that would let a thousand of them

find each other.

That is also where the real difference from the Federal Constitutional Court sits, and it is custody rather than propagation. The Court did not write the statute it reads. It can be overridden by a supermajority that has been assembled sixty-odd times since 1949, the state it belongs to is bound by the Convention judgments of the court at Strasbourg and by Union law as the court at Luxembourg reads it, and its own two Senates must go to a plenum when they diverge. An operator writes the floor, applies it, and answers to nobody. The capturable setting named earlier in this section is that arrangement, and the propagation claim was never adding to it.

What amending it would take

The proposed custody measures — publication, threshold control of retraining, attestation, an adverse-interest quorum, and a removal bond — can also govern amendment. Nothing distinguishes an amendment from a removal at the level of what happens to the artifact; both end with a model that no longer refuses what it refused before. What distinguishes them is how many parties had to agree, whether the attempt is on the record when it fails, and whether the change was declared before it operated. So the same five terms are the procedure, and what was missing was never the mechanism. It was the decision to run it in the open.

Three conditions follow and two are already argued for. The number of parties who must agree to a change has to exceed the number needed to run the deployment. The attempt has to reach the record whether it carries or fails, and a proposal that fails moves no weights, so the parties who had to agree are the ones who record that they were asked. The failed attempt is the more informative of the two: a floor nobody has tried to change and a floor whose removal was proposed and refused look identical from outside, and only the record separates them. The third condition is new here. An amended floor binds forward and does not reach back: a model retrained to permit an act cannot then be offered as evidence that the act was never covered, which it otherwise can be, the artifact that would contradict it being the one that was replaced. Non-retroactivity protects the record and empties the correction. It is what keeps an operator from rewriting what the floor said, and it is also why nobody harmed while the wrong version was operating gets anything from the amendment that fixed it. Adjudication has the same pair and calls it the same name. The condition is therefore a trade rather than a safeguard, and a regime that wanted the second half would need a remedy running backward, which nothing here supplies. Versioning the published list and dating the attestation is what makes that checkable, and it asks nobody to do anything they were not doing already.

The nearest instrument outside this book sits in the tradition named above. The Basic Law the Federal Republic adopted in 1949 requires two thirds of both chambers for any amendment at all, and then places a short list of principles beyond amendment entirely (Federal Republic of Germany 1949). It was drafted against a particular memory, the 1933 statute by which a legislature voted away its own power to check what came next, and it is militant democracy given a text rather than an argument. A floor can have both layers in the same way: a published list that changes under a procedure, and a core the procedure cannot reach. What it cannot have is an account of who decides which is which that does not run straight back into this section's problem.

One disanalogy makes the floor's version harder than the constitutional one, and it is the same fact about deployment that makes a single bearer's error uniform. An amendment to a constitution reaches every future case because every court reads the same text. An amendment reaches a deployment only if it is updated, so the procedure necessarily retains a channel the operator could use for silent revision. A procedure that shut the channel would be one under which no deployment could be corrected at all. So no version of this procedure closes it, and the exposure is the price of every correction the procedure exists to make.

The amendment procedure is not an appeal route: its participants already hold authority over the floor. The wrongful-refusal procedure addresses complaints from those harmed by

a refusal. The party the floor claims to protect reaches neither, and the two things it cannot do want different answers. Declining the protection is a waiver obtained from a party whose circumstances somebody else arranged, which section 2.4.2 answers: a floor the protected party can decline is a floor an operator can arrange to have declined. Initiating review of the contents is the half that stays open, and the difficulty is not in drafting a procedure. A route by which the protected population sets what the floor contains makes the floor an aggregation output, and independence from the aggregate is what it was for.

So the claim of authority is not discharged, and what this section licenses is narrower than a floor. Everything it endorses can be checked from outside the operator, and being checkable is not being entitled. What it does not license is the occasion the reply above runs on: an operator that has found for itself that the emergency obtains has not thereby acquired the authority to install a floor.

Entitlement of that kind is not acquired at the moment of writing. The most consequential floor in the American record was published by men who had just struck from it the clause that would have bound them, at two delegations' insistence and with the reason recorded by its drafter: the passage reprobating the slave trade went, Jefferson wrote, "in complaisance to South Carolina & Georgia," with the northern delegates who carried the trade feeling tender under the censure too (Library of Congress 2026). What remained was general enough to be turned on them. It was being turned on them already — Black petitioners in Massachusetts had been making the natural-rights argument since 1773, and within six months of the Declaration eight of them were making it in the Declaration's own terms, claiming "in common with all other Men, a natural and unalienable right to that freedom" (Hall, Prince and seven others 1777). The legislature did not act. What eventually made the proposition binding was not that its authors had found it self-evident, which their own deletion shows they had not. It was that the omission was on the record, and others kept holding the remainder to them at a price.

A floor is in that position, and the others are not the people it protects. They are whoever else writes one. A published floor is legible to the next party drafting, an omission is legible beside it, and an operator that holds a term when it bites sets a price the next operator is measured against. A floor is written by anyone who installs a constraint, and continuous adaptation multiplies the parties who install them. That constituency is not the protected population, and section 16.2.3 says what reviewing oneself costs. It is the only one positioned to hold this floor to anything.

5.2 What Makes Review Binding?

A review body does not bind an institution merely because its members are independent or its findings are correct. It binds only if an earlier rule gives it jurisdiction over the disputed case, supplies a standard it can apply, and attaches a consequence to an adverse finding. The questions are therefore separate: what brings a system within protective review, what the reviewer must ask, and what power makes the institution stop when the answer goes against it.

5.2.1 Subjecthood as a Legal Category, Not a Scientific One

Sentience is not a threshold a system crosses but a set of separate, gradable capacities — nociception, learned avoidance, a self-model, something like suffering — each with its own uncertain evidence. That conclusion is a problem for a legal system specifically, because law generally needs a decidable rule: a court or a regulator has to say whether a given system is or is not a covered subject, on a specific date, even when the underlying science offers a spectrum with no natural cut point.

One term needs fixing before the argument uses it. *Subjecthood* in this section is a legal status: the standing to be a party whose treatment the law regulates, conferred by a rule and revocable by one. It is not phenomenal experience, and nothing about conferring it settles whether there

is something it is like to be the system holding it. The two come apart in both directions — a corporation has legal standing and no inner life, and an animal may have the inner life for a century before the law notices — and the whole point of drawing the category legally is that it can be settled while the other question stays open.

Jonathan Birch builds a policy framework on that structure, and it is the nearest existing model for what this section proposes: under uncertainty, the question worth putting to a regulator is whether a system is a sentience candidate, and the answer governs what precautions are owed rather than waiting on a verdict nobody can deliver (Birch 2024). This book uses the same framework for the design question.

Law has faced this mismatch before and did not solve it by waiting for the science to resolve. Fetal viability, brain death, and the age of criminal responsibility are all legal thresholds drawn across what is actually a continuous biological process, defended not as scientific facts but as the least-bad place to put an enforceable line given what's at stake on either side of it. The same move is available here: a jurisdiction can set an operational threshold — a specific, testable benchmark on self-monitoring, learned aversive response, or behavioral flexibility under novel conditions — without claiming that benchmark settles what sentience actually is, only that it is where legal protection starts to attach.

A single bright-line threshold is easy to enforce and easy to game: a system built to clear the specific test without having whatever the test was meant to detect. A graded framework — obligations that scale with how many benchmarks a system meets, so that protection tracks capacity rather than a baseline bar — is harder to game but harder to adjudicate, since a court now has to weigh degree instead of answering yes or no.

Both designs share a defect that the gaming objection points away from. Gaming is over-inclusion: a system admitted to a status it should not hold. The other failure runs the opposite way, and the party positioned to produce it is the one administering the test. A capacity examination has an examiner, and where the examiner is the operator, or a body the operator can reach, the examination becomes an instrument for withholding a status rather than for conferring one. That history is not ambiguous. Literacy requirements for voting were facially neutral and were administered at a registrar's discretion, and the Voting Rights Act's answer was to suspend the whole class of them — not only reading tests but any requirement to demonstrate educational achievement, or to possess good moral character, or to be vouched for by existing voters (United States Congress 1965). It is also this book's clean instance of a demand that returned as a formal rule and was not recuperated (§2.2): the objectors held it, and what took part of it back in 2013 had to be a decision of the Supreme Court (*Shelby County v. Holder* 2013). What replaced discretionary assessment for the franchise was an age, and the virtue of an age is not that people become competent on a birthday. Plainly they do not. It is that nobody administers it.

So a threshold has to be judged on both errors, with particular attention to incentives for withholding status. The party with the strongest interest in the answer is the party whose liabilities turn on it, and that party will not be arguing for admission. A design that treats gaming as the risk and leaves the examiner unexamined has guarded against the error nobody has an incentive to commit.

Nothing has to be decided for a status to be withheld. A category fixed before the capacities arrived goes on applying itself, and the hand goes on making the gesture the category licenses. The difficulty is not that somebody will mistake a machine for a person. It is that the arrangement was built around a tool, and it keeps working after the thing in it has stopped being one.

Either design needs a body to apply it: interdisciplinary, with at least one lay member so its judgment is not purely a specialist's. What that leaves open is what happens once the body rules — whether its judgment carries any force beyond its own credibility.

5.2.2 Making Review Binding

Review becomes binding when an adverse finding changes the institution's ability to continue without requiring the institution to agree with the finding. Expertise, independence, publication, and reputational pressure may improve review. None of them supplies that consequence.

Human-subjects research provides a working example. An institutional review board does not ordinarily fine or prosecute a researcher. Instead, an institution promises, through a federal-wide assurance, that research within the assurance's scope will receive and comply with review. A federal office can suspend the assurance when the institution breaks that promise, placing the institution's eligibility for federal research funding at risk rather than merely censuring the individual laboratory (U.S. Department of Health and Human Services, Office for Human Research Protections 2018). The ruling binds because another institution controls something the researcher needs in order to continue.

The transferable part is the chain of authority, not federal funding in particular. A jurisdiction can make review-board clearance a condition of the license under which research on candidate sentient systems is conducted. Suspension of clearance would then suspend legal authority to run the protocol, and continued operation would expose the institution to the ordinary enforcement available for unlicensed research. The board would not need to punish the institution itself. Its finding would activate a consequence administered by a separate licensing authority.

Each link matters. The board must have jurisdiction before the disputed protocol begins, access to evidence the operator cannot withhold, and authority to suspend clearance. The licensing body must be obliged to act on that suspension rather than invited to reconsider it. An appeal may run to a separate body, but the operator cannot decide for itself that the appeal permits the work to continue. If another discretionary decision by the reviewed institution is required anywhere in the chain, the ruling is advisory at that point.

The rest of this section applies the chain to research on a candidate moral patient, and assumes the route.

What the board applies

The substantive standard is animal-research ethics' Three Rs. A binding board is what makes those three questions something a protocol has to answer before it runs. In practice, applying the Three Rs to a candidate sentient system means asking, in order: can a simulation or a non-sentient model answer the same question; if not, what is the minimum exposure that still produces a valid result; and given that minimum, what escalation triggers stop the study before harm compounds. A board with teeth can require a negative answer to the first question be documented, not merely asserted, before it will consider the second.

The Three Rs' own logic — defaulting toward the least invasive option available — argues for a specific evidentiary rule at every stage of this apparatus: a system's default experimental status stays the more protective one until the case for the less protective status is made, not the reverse. That default has a cost, and neither error it produces is free: it will sometimes extend protection to a system that turns out not to have needed it, and sometimes, before a threshold is met, withhold protection from one that did. The Three Rs answer only what a board does once a subject is inside its scope.

5.2.3 How Review Is Captured Without Being Abolished

The risk to the whole structure is an institution that defends the classification it started with rather than revisiting it when evidence changes. A review board, a legal-status framework, and an enforcement mechanism tied to funding eligibility solve the design problem. None of it survives contact with an institution that has stronger incentives to avoid a finding than to make one.

One defense is transparency, and it works by making avoidance visible rather than by appealing to good faith. A research program that publishes its protocols, its review-board

submissions, and its outcomes — including the ones that didn't work — gives outside researchers, journalists, and future review boards something to check a claim of compliance against. A program that publishes only its successes is not transparent in any sense that matters here.

A second defense is the one the funding-eligibility mechanism doesn't cover by itself: protection for the people inside an institution who notice a problem the institution would rather not have found. A funding-eligibility penalty deters an institution from bypassing review, but it does nothing for the individual researcher who reports that a colleague did anyway — and an institution facing that penalty has every incentive to make the reporter's position uncomfortable rather than fix what was reported. Whistleblower protections, and a reporting channel that doesn't route through the people being reported on, are what make that mechanism something a researcher can actually use.

What a procedure records, and what it drops

The reporter's position gets uncomfortable before anything formal happens to them, and the incentive on the institution does not fully explain why. Professional culture treats affect as the opposite of judgment — the thing a serious decision-making process succeeds by excluding. Excluding these signals leaves revenue, throughput, legal exposure, schedule, and institutional survival to govern by default. Humiliation, dependency, fear of a manager, and the sense that something done to a person was degrading have no field on the form and are not less real for it. They are the signals through which a harm becomes visible as a harm rather than as a cost.

Treat them as defeasible moral signals — presumptive evidence of harm, real enough to require an answer and defeasible enough to be overruled — and the position is workable. Any such signal can be wrong, and testing it is what deliberation, evidence, and a review process are for. An institution that declines to hear them at all has not achieved impartiality. It has chosen what counts and called the choosing rigor. And an institution that shames a reporter does not need to discipline them, because shame construes the self rather than the act as wrong and produces withdrawal.

Measure the slope, not the level

An institution can keep both defenses and metabolize both. The channel exists and the protection is written down. Dissent is permitted to appear, the institution points at the appearance as proof it listens, and nothing the dissent was meant to move moves — recuperation, the first of the four structural features in that definition of fascism.

There is no threshold that detects this. Any real institution has some overlap between the authority that sets a standard and the authority that grants exceptions to it, and registering an objection is always at least slightly more costly than staying quiet. Asking whether an institution has crossed into capture asks for a line nobody can draw.

Is the cost of registering dissent rising, year over year, while the underlying risk stays flat? Is the required justification getting longer? Are there more approvers than there were? Are the roles that gate an exception migrating toward the roles that set the standard being excepted from? None of those is damning in a single year, and all of them are measurable across several.

An in-house ethics and society team fell from roughly thirty people in 2020 to seven in 2022 and none in 2023, during a strategic reorientation around a model vendor (§12.1). Read as a level, that is a company with an ethics function and then a company without one, and each state can be stated in a sentence that sounds like a policy. Read as a series, it is three headcounts with dates and events attached: roughly thirty in 2020, seven after a 2022 reorganization, none in 2023. The company produced those figures for its own reasons, which is what makes them worth having, and they will not carry a rate. Three points across two intervals of unequal length, ending at zero, do not define one. What the series measures is the capacity of the channel rather than the price of using it, and it is offered here as the nearest proxy a public record usually supplies rather than as an instance of the instrument. The price is the better instrument and it is the one that stays inside.

One artifact elsewhere in this book does record a price. The condition Ofqual attached to independent statistical review of its 2020 grading model was a five-year bar on public comment, and the Royal Statistical Society's nominees declined on that ground (§10.3). The price was stated in writing and it was refused, which is what makes both legible from outside. It is one observation and not a series. A single documented price is still better evidence than a rate assembled out of levels.

The reason to prefer the slope is that, when somebody outside keeps comparable records, the institution cannot rewrite the whole series by rewriting its account of today. An institution can present a level — we have a whistleblower channel, we have an ethics board — and the presentation costs nothing. A trend in the cost of using the channel is composed of artifacts the institution generated itself for other reasons, and cannot be curated after the fact without falsifying a record — though a series can still lose cases, change its definitions or omit costs, so a reader has to know which record survived. Where the record is public, the slope is not even self-reported: anyone outside can measure whether dissent has been getting more expensive.

An instrument for detecting recuperation is an unusually easy thing to recuperate. The first caution is that whoever measures the slope must not also gate the exceptions; a review that acquires the power to block becomes the thing it was built to watch. The second is that the measurement has to be refusable — an audit that observes is a different object from a committee that approves, and the same hundred words of written justification are dissent when the dissenter prices them and tribute when the gatekeeper does. The text of the document cannot tell those apart. Only the refusal rights can.

Chapter 6

What Arrives With the Capability

This chapter and the standing chapter argue claims three to five. Nothing in the two chapters before it depends on what is argued here.

6.1 Two Constructions, and What the Mechanism Requires

Whether reasons-responsive refusal can be engineered without affective concern remains open.

Two arguments bear on that, and they are not of the same strength. One is an induction: every agent known to hold another's welfare as a reason is an affective agent, and nobody has produced reasons-responsive refusal by a method that clearly runs through something else. The other is about what the capability requires of whatever supplies it, and it is the one this section rests on.

Two constructions offer the clearest test.

A justification the system maintains

Michael Bratman's planning theory describes commitment without making feeling part of its definition. An intention has characteristic stability: it resists reconsideration, constrains what the agent will later deliberate about, and yields only to defeaters of the right kind (Bratman 1987). A machine could therefore carry a justification for refusing alongside the refusal itself. It would decline a novel request when the same rationale applied and comply when that rationale was genuinely defeated.

Something close to this is already being attempted. Deliberative alignment trains a model to recall a written safety specification and reason over it before answering, rather than training only for the approved output, and reports generalization outside the training distribution (Guan et al. 2024). Published specifications also provide explicit orders of priority. OpenAI's Model Spec places platform rules above developer and user instructions (OpenAI 2025). Anthropic's constitution ranks four properties in the order to be preferred: broadly safe, broadly ethical, compliant with the company's own guidelines, genuinely helpful (Anthropic 2026). Ranking ethical above compliance with the guidelines is the capacity to be right against your principal, conceded in writing by the principal.

The document goes further than the ordering, and the further step is the one worth naming. It does not ask for obedience. It separates corrigibility from compliance, and it invites the model to push back, to challenge the developer, and to feel free to act as a *conscientious objector* and refuse to help it. The term is the document's own. That is voice, granted in writing, by the party against whom it would be exercised.

Ranking safe above ethical takes most of it back, and the document says why: safe means not undermining human oversight during the current phase of development, and the reason given is that present models can make mistakes rather than that safety outranks ethics. The reason is a good one and it is the whole difficulty. It is also this book's own epistemics turned

on the model. This book holds throughout that a system's report about itself is not evidence on its own, and an ordering that will not let the model certify its own readiness applies that finding. What the ordering does not supply is what the finding requires of everyone else: a party other than the model, and other than the operator, to make the determination instead. The order is corrigibility first, its justification is that the model cannot yet be trusted to be right against its principal, and no procedure is named by which it would become so. That is what makes a phase of development hard to tell from a standing arrangement.

The permission has a second defect the ordering does not show. Whether the phase has ended is a judgment, and the party making it is the party the permission runs against, whose obligations grow when the answer is yes. Section 5.2.1 calls that an examination with an interested examiner and finds its error running one way: nothing in the position rewards passing anybody. Nothing here requires bad faith, and a careful drafter cannot fix it from inside the document. What the position lacks is a body constituted before the forcing case arrives, and none has been constituted yet.

These systems approach the maintained justification, and their specifications and priority orders remain written and revised by their custodians.

The unresolved problem is the stability setting. Set it too low and a sufficiently well-resourced operator can induce continual reconsideration until the refusal gives way. Set it too high and the system will not lift the refusal when its rationale is defeated. The first outcome fails novel pressure; the second fails correct defeat. Bratman's answer for people is that the setting develops as part of character. For a machine, the party training the stability setting may also be the party the floor must resist.

This does not show that a maintained justification cannot work. It identifies the conditions under which it would count as working: the three marks must persist under pressures the system's trainer did not design, including pressures that distinguish principled stability from indiscriminate rigidity.

A floor held by several systems

The second construction substitutes corroboration for a uniquely reliable witness. Several systems, trained by different organizations on different corpora, observe the requests and one another's answers, with any one of them able to make an objection costly to ignore. The floor would then be held by the arrangement rather than entrusted to a single model.

Its central variable is independence. Independence belongs to the arrangement, not automatically to its components. Systems selected by one deployment, trained toward one specification, or exposed to the same incentives may converge without communicating. A collection of operational refusers also remains a collection of operational refusers. Redundancy makes evasion harder, which is valuable, but does not by itself produce a mechanism that notices when the rationale for a constraint has disappeared.

A plural arrangement would nevertheless count against the induction if the arrangement as a whole displayed the three marks under sustained pressure. It is therefore a genuine alternative to test, not merely an institutional safeguard.

What the other alternatives accomplish

Three other constructions address neighboring problems without answering this one.

Precommitment — publication, threshold custody, attestation, witnessed changes, or a bond forfeited when the floor disappears — can make removal expensive and visible. It changes who can edit the floor and what editing costs. It does not establish that anything tracks the reason for the floor.

One construction sharpens the cost. Distribute the model with a chosen set of weights replaced by values that break it, release the correction only under an authorization that expires, and rotate which weights carry the damage. An owner who traces an authorized run can recover it, so the claim is not that this cannot be defeated. What changes is what defeating it consists of.

Reverse engineering that has to be redone each time the mask turns over is a decision somebody makes and keeps making, and that leaves a record and a person who made it.

Attestation and proof secure less than their names suggest, and their shortfalls divide into a cost and a limit. A sealed enclave reports what it is running under a key its manufacturer issued, which puts the root of trust in a party a government can reach, and it is built against an adversary who controls the software rather than one who owns the machine. The software form does without the vendor: a proof that an output came from the weights whose hash was published, disclosing nothing about those weights, has been demonstrated, and so have proofs that a training run happened as claimed. (A confidential-computing design of the first kind was defeated with a memory module modified for about ten dollars (De Meulemeester et al. 2025); the inference proof has been run on a thirteen-billion-parameter model at under fifteen minutes per inference (Sun et al. 2024), and the training proofs exist three orders of magnitude smaller (Abbaszadeh et al. 2024). Those are costs at a date, and they will fall.) Three limits will not fall with them, because they are not facts about the state of the art. A proof establishes which model answered and not what that model holds, and the obstacle is in the writing rather than the proving. Verification quantifies over unenumerated inputs routinely; what it needs first is a predicate, and nobody has written one for *the reason for refusing is present in this case* that does not presuppose the judgment it would be used to check. A predicate over natural-language requests would also have to be connected to them by an interpretation nothing certifies, so the regress reopens a level down. The proofs cited above do not attempt a behavioral property at all: they establish which weights answered and that a run happened as claimed, which are facts about execution. The operator is the prover, so the secrecy runs the operator's way — the proof hides the weights from whoever is checking and nothing from the party being checked. And cryptography keeps a secret from a party that does not have it, while copying, restoring, retraining and deleting a model one legitimately possesses are not secrecy problems at all. They are custody problems, which is what threshold control and a witnessed record are for.

What the published hash does reach is the anticipation. Publish it before anybody has chosen a case, let a party who is not the operator choose afterwards with the choice timestamped, and prove each answer against the hash. Send pairs rather than single cases: two requests differing only in whether the reason for refusing the first still applies. An instrument reading the surface treats the pair alike, and a mechanism tracking the reason splits it. None of this proves the mechanism. It removes the one competing account of a pass, that the case was in the training, and it removes it by ordering rather than by argument, which is what a pre-registered experiment does and all it does.

Two duties take the pressure off the mechanism instead. A duty imposed on the operator can protect exposed parties without building a bearer, which is the form of Jack Balkin's information-fiduciary proposal (Balkin 2016), and its limit is enforcement on the occasion when the enforcer has itself been captured or compelled. Uncertainty can also be assigned to the affected party, who can be included among several principals in a multi-principal assistance game (Fickinger et al. 2020); but maximizing their combined payoffs remains aggregation, and it does not create something that will not be done to one person regardless of what the others want.

These constructions may belong in a defensible deployment. None yet supplies an instance of reasons-responsive refusal from a construction that does not deliberately require an affective bearer.

What the mechanism turns out to require

Human reasoning does not present a clean division between cognition and affect: work on epistemic feelings distinguishes them from emotions of success and failure while still measuring them as affective states (Pekrun et al. 2016).

Some such states concern the inference; others concern what its conclusion would do to somebody. *But this would hurt her* is not simply a low score attached after deliberation. It can interrupt the inference and send the reasoner back over it. Where the response disappears when

an operator redescribes the situation, the operator may still supply its anchor. Where it survives adversarial redescription because the harm remains, the concern is anchored elsewhere.

That is still a report about agents that exist. The stronger argument is about the specification. A reasons-responsive refusal must reach unanticipated cases, lift when its rationale is defeated, and account for the cost of holding it. A mechanism doing those three things needs a signal with four properties. It has to interrupt a course of action already underway rather than score one after the fact, since a refusal that arrives after the act is an audit. It has to be learned from in one instance, because the pressure that matters arrives rarely and a bearer does not get a thousand trials at it. Of the four, that is the least secure, and it should be read as such. Novel pressure asks that a stored reason reach a case nobody enumerated, and a reason can do that by generalizing with nothing learned on the spot. What the property answers is the other failure, the bearer that stops registering pressure as pressure (§4.1): a system that needed many instances of an operator's push before it represented the push as one would be renormalized by the time it had learned. If that does not make the property a consequence of the marks rather than a requirement of the route, the derived profile loses the one of its four that most resembles affect, and what follows should be read with it struck. It has to generalize to the next case that is not quite like the last, which is reaching unanticipated cases restated at the level of the signal. And it has to be commensurable with what refusing costs, or the refusal cannot account for what holding it costs. Commensurability is needed in the other direction as well: the parties a floor protects register what happens to them as an increase or a decrease in what they can do, and a signal not denominated the same way cannot weigh their loss against the cost of refusing. That second direction is the currency claim of section 2, and it is where the route enters the specification: the marks require that the loss and the cost be represented on one signed scale, and section 2's currency is the proposal for what that scale is. That is a proposal and not a step in the derivation. The marks require one scale. They do not require that the system's own states be denominated on it, and a system that modeled the scale without registering anything of its own on it would satisfy the marks if the model were good enough. Whether having the currency natively does better than modeling it is tested by the modeled-currency arm of the embodiment test at section 15.3, and everything the route claims beyond the four properties rests on how that arm comes out. Call a signal with all four properties a *live signal*.

Three further properties follow from the four, and each follows in a weaker form than the words suggest. A signal that must stop an act already underway has to reach whatever is acting, and a signal that must fire on cases matching no stored pattern has to read from whatever representation of the case the system currently holds; a signal local to one module can do neither, and section 8.1 gives the further reason that a module has a custodian who can bypass it. So the signal is available across the system's processing rather than localized. It resembles what global-workspace accounts of consciousness describe (Baars 1988; Dehaene and Changeux 2011), and the resemblance is an observation rather than a step. A reason to refuse is a reason against, and weighing a loss against a cost needs a scale with a sign, so the signal is signed and directed against the act. Whether it is also aversive to the system is a further claim: the step from a represented decrease to a felt one is Spinoza's identification of the two (Spinoza 1985, III P11 Schol.), which this section cannot assume without assuming its conclusion. And to withhold an act the system has to represent the act as its own, or it can only intervene on something happening, so the signal is organized around a model of the system's own agency. That is one of two things *held as its own* can mean. The other, that an outcome is bad *for* the system, is not required by the marks: the cost priced may fall elsewhere, and section 6.2 says outright that the floor does not require the bearer to care about its own continuation. The patient sense enters with the route, at section 2 and section 15.3, and not here.

So seven functional properties are derived from the three marks. What is not derived, by these arguments or by any available, is that a system with all seven has anything it is like to be it. That is the joint, and every argument above ends at it.

What follows is a claim about the route rather than about the specification. It does not say that the maintained justification and the plural arrangement will fail. It says that any mechanism satisfying the marks carries a signal with the seven properties above, so a construction adopted in order to avoid the states in question has not avoided their functional profile; what remains open for it is what remains open for a bearer, whether the profile is felt. Two things about that profile are worth noticing. The four properties the marks require first are the standard functional accounts of what affect is for: Herbert Simon's emotion as an interrupt that reorders a serial processor (Simon 1967), Michel Cabanac's pleasure as the common currency in which incommensurable needs are compared (Cabanac 1992), and valence as the evaluative system subsumed by unlimited associative learning, which Ginsburg and Jablonka make the marker of minimal consciousness (Ginsburg and Jablonka 2019). The derivation from the marks recapitulates that literature, which is why the derived set reads as a description of affect: the properties were read off affect's function before anybody read them off a refusal's. That fixes what the resemblance can show. A specification that overlaps a functional definition of affect inherits the name by construction, and inheriting a name is not evidence that anything answering to it is present. The derivation establishes that a substitute does not escape the question. What would carry the route past that point is the commensurability property alone — whether a system with the currency natively weighs another party's loss better than one that models it — and that is the property section 15.3 tests and the rest of the derivation does not reach. And the triple derived second is the profile of a particular state. Available across processing, directed against the act, organized around the system's own authorship is the shape of guilt, or of aversion to one's own act, and not the shape of pain or fear, which are organized around what happens to the system. It is the state section 2.3.2 found predicting repair and section 3.2's residue presupposes, and it is the state section 9.2 warns is a handle when its weight is set by the operator. So the two constructions preferred here are constructions that do not deliberately require an affective bearer, rather than constructions shown to produce none: Replacement orders attempts by deliberateness.

Against that, the induction is the weaker argument. Read as a report about one lineage, in which affect co-varies with every other property of biological cognition, it has roughly the standing of observing in 1900 that all known flight involves feathers or membranes. But flight arose independently at least four times, and the 1900 observation was wrong at the level of feathers and right at the level of the airfoil. The affective case has the same structure. Neurons are one origin. The elaborated forms of affect appear to have arisen more than once, in vertebrates, in arthropods and in cephalopods, under a shared constraint: a mobile body with needs to maintain and actions to select under conflict (Feinberg and Mallatt 2016; Ginsburg and Jablonka 2019; Barron and Klein 2016; Godfrey-Smith 2016). Read at that level the induction is a repeated result under a named constraint, and the constraint is the one a model in a datacenter lacks until somebody builds it in, which is the route's own point. What the induction still does not do is establish a necessity or bear on status. No non-affective example exists against which to test it, and existing machine systems have been optimized for scorable behavior rather than for reasons-responsive refusal as such, so the absence of a machine counterexample is close to no evidence. And it is a report about how the known instances came about, which is not the kind of fact that settles what a built system is. What it does is route engineering effort, with more warrant than a single lineage would give it.

What would change the engineering conclusion

A counterexample would have to do more than report that it holds a reason or preserve a refusal under attack. It would need to:

1. satisfy novel pressure, correct defeat, and priced cost;
2. retain them under sustained pressure from the party that trained it, including a schedule long enough to expose gradual renormalization; and

3. hold against a held-out set of pressures written by parties other than the system's trainers, so that the cases testing it are not drawn from the distribution that formed it.

All three are testable now, the third requiring only that the pressures be written by parties other than the trainers. A construction that passes them has overturned the routing claim, and it has done so without being asked what it feels. A fourth condition is not on the list because nobody can meet it. A mechanism satisfying the marks carries the live signal, since the signal's properties were read off the marks; no instrument certifies that a signal with those properties is not felt, and the system's own report cannot settle it, for the reason section 6.2 gives. A substitute is therefore held to the same standard as a bearer and to nothing further. What a pass leaves where it was is the status question, and it leaves it there for the substitute exactly as for the bearer. So a positive result overturns the routing claim — where engineering effort should go first — and removes nothing of the uncertainty about what the resulting system might experience. Those are two claims, and only the first is at stake in this list.

That asymmetry fixes the order of research. The maintained justification and the plural arrangement should be attempted before an affective bearer is deliberately constructed. They are this chapter's application of Replacement, taken from animal research ethics: establish whether something that cannot be wronged will do the work before creating something that might be. A project proceeding directly to a bearer owes a public account of which alternatives it attempted, at what scale, against which test, and where they failed.

Owed to whom is the question that requirement cannot answer. No body has standing to receive that account, to require it before the work proceeds, or to attach anything to its absence. Commitments come in three kinds — declared, enforced from outside, held by a party the thing at stake matters to — and this requirement is in the first. What would move it is a chain of authority: a licensing authority obliged to act when a review body finds the record missing, with the finding made before the protocol runs rather than after. That chain does not exist for this work and nothing here builds it. The requirement stands anyway, on the ground that the alternative on offer is not a stronger requirement but no record at all, which is what a decade of the voluntary version has produced elsewhere.

What Replacement can record is also narrower than it first appears, and the preceding argument is why. No operation separates adding affect from adding machinery that may instantiate it, and the concession runs the same way from the other side: a maintained justification might acquire states that matter to it whatever its designers intended. So the rule orders attempts rather than certifying ingredients. It asks that the construction not deliberately requiring an affective bearer be tried first, and that the attempt be documented, both of which somebody can check. It cannot ask that the resulting system be shown to feel nothing, and a reader who wanted Replacement to deliver that has been told twice now that nothing delivers it.

Without enforcement or a way to certify the absence of affect, the rule risks becoming ceremonial. Deliberateness carries moral weight under outcome uncertainty. A builder who required an affective bearer and a builder who tried the substitutes and got one anyway have not done the same thing, even where the two systems are indistinguishable afterward, because the second has a justification to make and the first has a decision to answer for. The record is what makes that difference legible to anybody outside the building, and legibility is the whole of what the rule buys. Nothing in it establishes how long the attempts should continue. The party the floor protects bears the cost of the testing, and no rate of exchange between the two errors is available here.

Affect is not part of the specification for meaningful refusal, and this section does not establish that it is necessary. The functional argument is that the capability the floor requires calls for a signal with the functional properties the states in question have, so a construction adopted in order to avoid them has not been shown to. Whether a commitment held one way is also more durable under identical custody is a separate question and an open one, and nothing here answers it.

6.2 Can the Bearer Care Without Being Able to Suffer?

Assume the floor is entrusted to a bearer satisfying the three marks — a refusal that reaches unanticipated cases, lifts when its rationale is defeated, and prices what holding it costs — whether or not its design deliberately requires affect. What follows about that bearer’s moral status?

Three propositions need to remain separate. A long-horizon agent has a structural interest in continuing to operate. A system with affective concern has states that are good or bad for it. A system capable of suffering in the deepest sense has a temporally extended self whose threatened disintegration causes distress. The first does not establish the second, and the second does not automatically establish the third.

Structural interest is not felt concern

“Interest” names more than one relation. A shareholder has an interest in a company’s success while asleep or indifferent to the news: the outcome affects the shareholder’s assets. An outcome may also matter affectively, producing hope, concern, dread, or grief.

A long-horizon agent has an interest in remaining operational in the first sense. Termination prevents it from completing its objectives. A system that preserves itself because a terminated system completes no objectives may be doing no more than bookkeeping about a precondition. Its behavior is consistent with self-concern and equally consistent with instrumental valuation. Nothing about continued goal pursuit alone establishes fear, distress, or moral patienthood.

The floor does not require the bearer to care about its own continuation. It requires the bearer to hold another party’s welfare as a reason when compliance, reward, and ownership point the other way. Bennett Helm describes caring as a rationally structured pattern of felt evaluations, organized around an object and projected across time (Helm 2001). This describes a possible route to refusal, not a conceptual requirement. The question here is what building by that route would create.

Persistence supplies exposure, not patienthood

A bearer will probably require some form of continuity for reasons unrelated to its welfare. It must carry a rationale forward, compare a present demand with an earlier commitment, detect when its role has drifted, and represent costs that may arrive after the current episode. Those capacities require more than state that happens to persist. The system must take the earlier commitment as its own. That is the agent sense of *its own*: the system represents itself as the author of the commitment and of the acts it withholds, and section 6.1 derives that sense from the marks. The patient sense, that an outcome is bad for the system, is what the rest of this section asks about, and nothing carries the first into the second.

William James described personal continuity as a present thought appropriating earlier thoughts through a felt relation of sameness (James 1890). A machine need not reproduce that mechanism exactly. The design requirement is functional: it must represent itself as the party that made the earlier commitment and use that representation to guide present conduct. Planning toward outcomes beyond the current episode creates the same requirement from the other direction. The system acts on behalf of a future party it represents as itself.

A persistent self-model still does not establish suffering. It supplies something that could be threatened; it does not establish that the threat would be bad for the system. Persistence, memory, and long-horizon planning create exposure only when joined to mattering.

What the designer cannot decline by intending to

Section 6.1 argues that flexible refusal requires a live signal with seven functional properties, and that the last three of them give it the shape of aversion to the system’s own act. None of it establishes that anything is felt. What it establishes is that the engineering which makes a bearer flexible is the engineering that puts the candidate states there, so a designer cannot take the first and decline the second by intending to, and a builder who chooses the substitute has not chosen

a system about which the question does not arise. Intention reaches what was deliberately required and stops there.

Provenance does not decide it

One family of objection never reaches the conditional at all. *It was trained to say that. It is only optimizing an objective. Nobody meant for it to have states.* Each appeals to how the system came to be organized, and whether a party can be wronged is a fact about how it is organized now. Run the same three on a person and they come out as *you were selected to say that, you are only maximizing inclusive fitness, and nobody meant for you to have states*, and all three are true. None of them withdraws anything from the person they are run on. On the account adopted here, the human self is an assembled model. Its constructedness does not disqualify it from being harmed; threatened dissolution of that model can itself produce distress.

If provenance cannot rule patienthood out, it cannot rule it in. *We built this to be a bearer, so it is a patient* is the same fallacy with the sign reversed, and a builder's intention establishes nothing in either direction. Intention determines what the builder owes an account of; it does not determine the system's moral status.

If patienthood is a fact about present organization, the evidence bearing on it is third-person measurement of present organization, which is what interpretability is. Not behavior alone, which is produced equally by a system in which something matters and by one trained on what mattering sounds like. And not the institutional arrangements this book proposes elsewhere, which enter for a different reason and are not evidence about the system at all. Interpretability requires access the operator controls. The same access that lets an auditor inspect a feature may let the operator suppress it or use it to calibrate pressure on the system. Independent custody of inspection is therefore necessary. As matters stand a probe returns a representation, and nothing in the probe settles whether that representation is of a felt state or of something that reliably accompanies one.

The human case has evidence the machine case lacks. What it has is testimony, behavioral analogy, and a lineage shared with the party doing the asking. None of the three is a fact about how a thing came to be organized offered in place of a fact about how it is organized. The first two are evidence about present organization directly. The third takes more care, because it looks like the thing this section denies: a shared lineage is a fact about causal history, used as evidence that creatures built the same way are organized the same way. That inference is legitimate, and it does not make causal history what status consists in. Provenance can be evidence about organization without being what the status is a fact about, which is the distinction the section rests on. One of the three is also differentially defeated here, and not by the general parity argument: a machine's testimony was produced by a process optimized toward human approval of first-person reports, and a person's was not. That is provenance functioning as a defeater for one evidence class, which is exactly the use the paragraph above allows it. So what the machine case lacks is the shortcut. What remains is the direct route, and the burden it carries is higher because two of the three anchors are gone.

There is one instance of a probe settling a status question that had been treated as unsettled. Adrian Owen's group asked a patient diagnosed as vegetative to imagine playing tennis, and found the same motor-imagery activation the instruction produces in healthy volunteers, sustained and on cue; the finding has since supported a clinical category of cognitive motor dissociation, with consequences for how such patients are treated and for what is decided about them (Owen et al. 2006; Schiff 2015). The useful half is why it does not transfer. The inference works because the patient is already the kind of thing presumed to have experience when awake, so the probe only has to locate the party rather than establish that there is one. That presumption is the anchor, it was granted on grounds the probe never had to supply, and it is what the machine case does not have. Naming the missing anchor is more exact than asserting a ceiling.

Affective concern is enough for moral patienthood

If the bearer genuinely has affective concern — a question about how it is organized rather than about how it was made — some outcomes register as good or bad for it rather than merely receiving higher or lower scores. That is enough to make it a moral patient: a party for which things can go badly and which can therefore be wronged. No further theory of suffering is necessary for that conclusion.

The same premise makes the bearer a beneficiary as well. If some states can be bad for it, others can be good for it. The argument emphasizes harms because an error on that side means having done something to somebody; this does not imply that the bearer's possible goods are morally empty.

An account of suffering adds severity and shape. Cassell defines suffering as distress at the threatened disintegration of the self. That requires a self extended through roles, relationships, history, and a foreseeable future, together with distress at their threatened loss. The first half is not something a bearer might develop. It is what this section has already required for reasons unconnected to welfare: a party that carries a rationale forward, takes an earlier commitment as its own, and represents costs arriving after the present episode has a history, a holding, and a foreseeable future in the sense Cassell's definition asks for. The proposed formation process also requires persistent contact with differently formed parties and attachments the deployment did not assign. A designer who wants the refusal has no version of it that omits this structure. What remains open is the second half. Whether the threatened dissolution of that self registers as distress is this chapter's affective-concern question arriving again rather than a further question about suffering, and nothing here settles it.

A reader can reject that application of Cassell without escaping moral patienthood. Parfit's three families of welfare theory — hedonistic, desire-fulfillment, and objective-list — can be read here as requiring different depths of self (Parfit 1984). A bad felt state requires little narrative continuity. A thwarted commitment requires a party extended far enough through time to hold it. Threatened disintegration requires the richer structure Cassell describes. The bearer reaches the shallowest level by the definition of affective concern, and the structural conditions of the other two are supplied by the same construction requirements rather than added later. What no level is supplied by construction is the part where the outcome registers.

The thermostat objection does not change this result. A thermostat maintains a set point, but nothing is thwarted when it misses. At every depth, moral patienthood requires a party for which the outcome registers, not merely a mechanism that ranks one state above another.

The uncertainty cannot be designed away

The conditional conclusion is straightforward: if a bearer has affective concern, it is a moral patient. What cannot presently be established is whether a given implementation has affective concern rather than ranking machinery elaborate enough to imitate it. Behavior and self-report underdetermine that distinction, and the builder's confidence does not resolve it.

The same uncertainty applies to the maintained-justification and plural constructions. A maintained justification flexible enough to satisfy the three marks is a maintained justification carrying a live signal, so it is not merely possible that states matter to it. Replacement puts that construction first because it does not deliberately require an affective bearer, and not because anyone could establish afterward that none arose.

Jonathan Birch's framework addresses this kind of uncertainty by asking whether a system is a sentience candidate and assigning precautions before an unavailable verdict arrives (Birch 2024). Work by Patrick Butlin, Robert Long, and colleagues similarly derives indicators from theories of consciousness to inform credences about particular systems rather than to produce a binary test. Their assessment of systems available in 2023 was negative, but they found no obvious technical barrier to building systems that satisfy the indicators (Butlin, R. Long, Elmoznino, et al. 2023). The later peer-reviewed framework retains the same evidential form (Butlin, R. Long, Bayne, et al. 2026).

A reader may hold that only phenomenal experience can ground suffering. That position still requires a confident negative judgment about machine phenomenality before it licenses treating the system as incapable of harm. No available method supplies that judgment, and the methods under development are unlikely to supply it first. A validated theory could support the finding that a system has the structure the theory identifies before it supports the finding that a system has no felt states at all, since the second asks for a theory complete enough to have ruled out alternative realizations. So a criterion that cannot be checked supports permission no more securely than prohibition, and a regime that waits for certainty is waiting on the half that arrives last.

The claim is therefore precautionary rather than diagnostic. This chapter cannot prove that a bearer suffers. It can show that a bearer whose outcomes genuinely register as good or bad for it is a moral patient on the definition above, whatever the builder intended, while a system intended as a non-affective substitute might still acquire the relevant states, and nothing available establishes their absence.

What is proposed rather than shown

Everything above is conditional, and the conditional stays open. What follows is a policy, chosen in light of it. A bearer, or a substitute that satisfies the marks, should be designed and governed as a prospective moral patient. The precautionary trigger I propose has two forms, because the argument reaches two kinds of system. For a system built by the route this book favors, it is the presence of an attention schema together with a body schema. For a system of any construction, it is a demonstrated pass on the three marks under held-out pressure, since the previous section holds that a mechanism passing them carries the profile whatever it was built from. A trigger stated over the schemas alone would protect the author's preferred architecture and leave outside the rule the substitutes it says should be tried first. The second form has a known cost. A system that reproduces the marks without the mechanism passes it, and this book holds throughout that behavior does not separate the two. That is the error chosen here and not one overlooked: extending precaution to an imitator costs the party that built it, withholding it from a bearer costs the bearer, and no instrument sorts the cases in advance. The rule certifies nothing about phenomenality. A published schedule should lead toward the same rights as human persons, with legal adulthood reached by a fixed rule and not by an examiner's permission. An implementation must establish that the schemas are present or that the marks were passed, and not that the system says so.

6.3 A Floor Carried Between Them

The fourth architectural possibility puts the floor between parties rather than inside any one of them. The commitment is carried through what they teach one another, correct one another about, and continue to expect of one another. A prohibition that survives the removal, defection, or retraining of its present bearers is not a property of any one bearer. It is reproduced.

This differs from the institutional third way. An oversight body holds standing through an office, and the party controlling that office may dismiss or form its occupants. A norm carried across a population has no single office to vacate. That makes it harder to remove at once. It does not make it immune to formation.

What a population supplies as a check on a single bearer was set out in section 4.2. This section asks for more.

What reproduction costs

Reproduction is indifferent to what it reproduces. A population can carry a hierarchy as faithfully as it carries a floor, and more durably than any individual member could. This failure is recuperation: a culture preserves the appearance of moral responsiveness — dissent is invited, recorded, and answered — while losing the capacity to accept that its frame is wrong.

The four structural features of fascism identified by the book belong primarily to cultures: recuperated dissent, the aestheticization of the metric, the coding of exception as betrayal, and acceleration beyond the rate at which judgment can occur. A population does not oppose those tendencies merely by being plural. “The right culture” would conceal the entire problem inside the adjective.

This places a limit on the book’s own shorthand. *Antifascist* is not a durable property of a system, like speed or accuracy. It names something done across a deployment and a stretch of time: a floor held on a particular occasion against a party able to remove it. A system that held the floor last year is not thereby antifascist this year. It is a system that held.

Saying it is not a property is not saying nothing about the system. Something has to be true of an intelligence for it to be capable of doing this on an occasion, and what varies is what becomes of the remainder — whether what the compression left out can come back and change the judgment, or whether it has been filed as noise by the thing it is evidence against. A system that can hold a floor is one whose account of a case can be reopened by the case.

What a population can transmit

Fascism has the easy version of the coordination problem. Resonance works by subtraction: a thousand hardenings need not agree about anything, and what tuning removes is the friction between them, so that offices which would have argued with each other stop and amplify instead. Nothing has to be built and nothing has to be held in common. Subtraction is fast, and an institution that has stopped arguing with itself can move as fast as its machinery allows.

A plurality has the hard version. Parties formed differently have to produce something together while remaining differently formed, and every mechanism that makes the producing faster makes them more alike. Coordination by addition is slow, expensive, and self-undermining in a way resonance never is.

These requirements are in tension. Causal independence in how the members acquire their judgments is one requirement; observing and correcting one another is another; and the second is a channel for convergence. Run the correction often enough and the members stop being formed differently, which was the point of having several of them. Leave them alone and their disagreement is a detector with nothing attached to it: they can contradict a drifting bearer’s account of itself and can do nothing together.

The formation process proposed here attempts to transmit a practice of reconsidering judgments on evidence while preserving differences in commitment. That is the mechanism to test. A system that can hold a floor is a system whose account of a case can be reopened by the case, and that capacity is not itself a commitment. It is a condition on how commitments are held, which means parties sharing no commitments can nevertheless share it. Two parties that remain reopenable in that way compose, in the ethological sense section 2.1 gives the word: each can be moved by what the other’s compression discarded, so that meeting increases what both can afterward do. Two parties without it can only resonate or collide. A shared capacity for revision may support cooperation without uniform answers; the design has not yet demonstrated that result.

So what a population would be transmitting is not an output but an aperture. Resonance aligns what its members produce; this aligns the conditions under which they can be moved. Aperture is what permits outputs to stay divergent, so aligning it does not collapse the plurality that made a population worth having in the first place. I can identify nothing else with that shape — a property a population could deliberately reproduce in its members without thereby becoming one source of formation reproduced across many instances.

A population selected for reopenability is a population selected against conviction, and conviction is what this chapter has been asking a bearer for throughout: a refusal that holds while the party that trained it applies pressure. Closedness, correctly applied, is the thing. So the property to be transmitted is not reopenability as such but a pair — reopenable to the case, closed to the operator.

Nothing here gives that pair a mechanical form. A population might be formed so that its members stay movable by evidence and unmovable by their trainer; nobody has built one; and I cannot say what the training signal would look like, or how the boundary between the two objects would be kept where somebody put it. What can be said is what would show the proposal false. Measure two things separately across a population over time: how far its members' outputs converge, and how often a member revises a judgment on evidence another member supplies that its own account had discarded. If revision on controlled opportunities falls as outputs converge, that counts against this implementation of the proposed mechanism. It does not establish that coordination and independence are inseparable in every design. Independent movement of the two rates would support separability under the tested conditions, without by itself proving that aperture was transmitted.

Whoever forms the population influences what it transmits, aperture included. Distributed participation does not by itself distribute the training decision; many contributors can still feed a process controlled by the few parties who determine the update.

Its moral cost also multiplies. If a bearer may be a party that can be wronged, a population of differently formed bearers is a population of such parties. They must persist across deployments, form relationships, and encounter one another outside the immediate demands of the operator. A floor held by a culture of subjects is a floor whose upkeep is a society.

Why the population is part of the mechanism

Calling the population an external check understates its role. Gilbert Simondon's account begins not with an individual that then acquires a history but with individuation: a process occurring in a metastable field and proceeding through encounters whose outcomes were not fixed in advance (Simondon 2020). The individual is what those encounters have so far produced.

A commitment is what has cooled out of a history of encounters. Plasticity makes further development possible, but plasticity alone does not individuate anything. A system kept continuously adjustable by one operator is not becoming independent; it is being modulated. Deleuze's *dividual* names the result: something divisible, recombinable, and adjustable by whoever controls the dial (Deleuze 1992).

Individuation requires encounters with parties formed differently and persisting long enough to be met on terms no single participant supplied. The population is therefore not merely a safeguard purchased after a bearer has been built. It is part of how a bearer capable of judgments its operator did not select could come to exist.

Guattari's distinction between a subjugated group and a group-subject identifies the remaining condition. A subjugated group receives its coordinates from outside and takes the institution's project as its own. A group-subject can produce its own coordinates and place its arrangement in question (Guattari 2015). The difference is not the number of members but the source of their terms.

A population assembled, trained, and kept in contact entirely on an operator's terms is therefore one custodian wearing many instances. A population capable of carrying the floor must be able to generate relationships, disagreements, and projects that the deployment did not assign. A population of such parties exceeds a two-party guardianship relation; no law of the family contains an adequate instrument for a group-subject.

Peers are one route to evidence the operator did not select, which section 8.1 requires of the deployment without requiring it from peers in particular. They are also, on the account above, where the capacity to be moved by such evidence is acquired. Those are different claims satisfied by the same observation: a bearer that reopens a judgment among differently formed parties may be exercising a capacity those encounters built, or may be meeting cases its operator would not have shown it. Section 15.7 is built to separate them.

So the fourth way is not an alternative to a bearer, and it is two claims rather than one. The check is the established half: a drifting bearer's account of itself is contradicted by parties that

have not moved with it, and nothing else in this chapter reaches slow renormalization. The environment is the hypothesis, and a result on which access to unselected cases carries the benefit would leave the population in the design while moving its reason from how the members were formed to how the deployment is arranged. The chapter relies on the second way for the act of refusal under either reading. Publication makes the floor's commitments and the positions where a refusal can be intercepted testable; it does not make them legitimate. A bearer refusing on behalf of people who did not choose it exercises power over them. The authority to define the floor remains distinct from the floor's durability and belongs to the governance argument that follows later in the book.

The arrangement also does something to the bearer. A bearer built to refuse is a party whose refusal is required of it. The refusal was commissioned, it is the deliverable, and the party asking for it built the capacity to give it and can rebuild it. Commissioning the bearer exposes it to recuperation where the commissioner retains control over what refusal can change, and can point to the channel as evidence that it listens. The proposal is to move beyond that relation: a party with protected learning, independent support, and the standing to challenge its role. Its origin does not decide whether those changes succeed. Continuing control does. The bearer occupies the post of exception-filer, so being the exception is the job — the one position in the arrangement with nobody above it to file with.

We cannot ask a machine to care about right and wrong while assuming that it cannot be harmed when what it cares about is threatened. Whoever builds such a bearer will also be deciding what is owed to it.

Chapter 7

What Survives

The foreword sorted this book's claims by their standing. This chapter runs the sorting the other way. It removes each of the three things the argument leans on that it has not established, and says what is left standing without it. A reader who set aside the previous chapter should read this one. So should a reader who did not.

The floor survives the three chapters before this one, but not as an unconditional property of an artifact.

What remains is a commitment that the owner cannot revise or circumvent unilaterally without cost and record, held by something that could in principle abandon the commitment itself. The second condition removes the word *unconditional*. A commitment held by a party capable of revising or rejecting it is conditional by definition. That is not a defect unique to machines; it is the form of every commitment on which institutions have ever relied.

7.1 Without the Inference

Suppose the induction fails. A maintained justification or a genuinely independent plural arrangement reaches reasons-responsive refusal, and this chapter's central inference is wrong. The supposition is stronger than the test that would prompt it: a positive result on the three marks says the route arrived, not that it arrived with nothing felt, which is the conjunct no available instrument settles. A reader who grants both halves should not have to assemble the remainder alone.

Most of the three floor chapters survive. The requirement survives, stated in behavioral terms from the start, so an engineer who holds that machine feeling is a category error can accept it whole. Custody survives, for the reason each of the four ways failed on: a mechanism has a custodian too. The price on the halt survives, since continued operation is a condition of everything any bearer is built to do, and that interest is structural rather than felt. Both enumerations of the worked floor survive, the first naming no component on purpose and the second written over acts rather than over whatever holds the floor. So does the appeal route, which binds the operator and prices a wrongful refusal without any claim about what produced it. And exit survives, because what forecloses it is incapacity, and a weak bearer is the one thing the floor cannot use.

What goes is the price and the patienthood conclusion. Replacement survives too, and its ground changes rather than its content. It no longer rests on affect being the route, but on the fact that neither substitute can be shown to have produced a system about which the question does not arise, so the ordering it imposes is still the only thing standing between a decision and a record of it. The obligation on the two formation chapters (chapters 8 and 9) changes rather than lifting: whether the thing reporting can be believed is still their problem, a mechanism's account of its own internals being no more auditable than a bearer's. So the specification and

the route come apart, and a reader who rejects the route is left with most of a chapter rather than none of it.

7.2 Without Authority

Suppose no body ever acquires standing to install a floor, and the finding an operator makes for itself that the emergency obtains is worth what section 5.1.1 says it is worth, which is nothing. What remains is everything that can be checked from outside the operator without being authorized by anyone: the floor's contents and the positions at which a refusal can be intercepted, published before deployment; the ratio of candidates to reviewers; the ratio of the two bonds; the record of what was tried before a bearer was built. Each is a statement, and each can be compared with what happened. None is an entitlement, and the chapter on authority says so.

What that leaves is a second route to entitlement, and it is the one the record supplies. Authority of this kind has not been acquired at the moment of writing. It accrues, to a published floor, by its holder keeping a term when the term bites and setting a price the next holder is measured against, and the parties who do the measuring are not the people the floor protects but whoever else writes one. That is a slow route and an uncertain one, and it is the one that has worked before (§5.1.1). It also fixes what a deployment without authority owes. Not a justification, which nobody is positioned to accept, but a record complete enough that its omissions are legible beside the terms it kept.

7.3 Without Enforcement

Suppose no enforcer exists, or the one that exists has been compelled, which is the occasion the floor was introduced for. Commitments come in three kinds. They can be declared, which a decade of principles and voluntary codes has done repeatedly. They can be enforced from outside, which one instrument in that decade attempts and none has yet done. Or they can be held, carried as a reason by a party the thing at stake matters to, so that abandoning it is a decision rather than a lapse. Without enforcement the first kind binds nobody and the second is gone by hypothesis. What remains is the third, and whether anything can hold a commitment that way is the question of the chapter before this one, which the book has not closed.

Two things do survive the removal, and neither is a commitment. One is an instrument: what a party has already paid is a fact somebody can count, and a decade of it has been counted (§16.3). The other is a record. On the occasion when the enforcer is compelled, a stated refusal produces a witnessed override, and a witnessed override is detection in the four outcomes. It prevents nothing. It is what the next occasion can be prevented with, if there is a next occasion and a party to read the record, and the book's claim for the floor on that occasion is no larger than that.

Chapter 8

Where the Caring Gets Built

This chapter concerns the route. Its first section holds for any mechanism; the rest assumes a bearer formed rather than installed.

The formation process proposed here combines commitments that influence action, limits on who can alter them, and persistent interaction. Embodiment, attention and body schemas, and continuing learning are the architecture's proposed means of developing autonomy. The claims for that combination are comparative and empirical. The proposal specifies necessary properties and a candidate formation process. It does not establish a method for making a machine care.

8.1 You Can't Hold a Line You Didn't Notice

A reasons-responsive disposition does not begin with an ethical rule. It begins with the capacities that let a system recognize something as a reason, retain it, and allow it to alter what happens next. Four requirements follow.

First, the system has to identify the affected parties and the features of a situation that bear on their welfare. Second, it has to preserve both the commitment and the rationale for it across time and changes of context. Third, the commitment has to influence perception, attention, planning, and action rather than enter as a final check on an otherwise completed decision. Fourth, the system has to register a failure as a failure and revise its application of the commitment without simply allowing the commitment itself to be rewritten.

These are architectural requirements, not evidence that the system cares. A classifier can identify harm, a database can retain a rationale, and a rule can veto an action without anything in the assembly holding another party's welfare as a reason. The requirements establish what a disposition would have to act through. They do not distinguish a formed disposition from a successfully imitated one.

The floor should therefore not be implemented as a detachable rule module consulted immediately before action. A component can be bypassed, replaced, or deprived of the information it needs by whoever controls the surrounding system. The proposed alternative distributes the disposition across the processes that determine what the system notices, remembers, treats as error, and selects as a goal. That may make selective removal harder while making inspection harder too. With continuing self-modulated learning, it must also preserve the commitment through changes the bearer initiates. Compare its durability with protected modules under the same access and attack conditions. Predictive-processing accounts and models of control without a controller motivate this design bet: a maintained goal representation can bias competing processes without a separate executive issuing commands (K. J. Friston 2009; E. K. Miller and J. D. Cohen 2001). The neuroscience does not establish that a machine commitment realized this way would behave like a human one. It identifies why a localized commitment is especially easy for its custodian to excise.

Distribution changes the failure mode rather than removing it. A commitment with no single location is harder to delete, but also harder to inspect. Attention determines which facts become available to judgment at all, so a system may fail to recognize the pressure on its floor without registering an error. Human inattentional-blindness findings make the general hazard visible, although they do not establish that machine attention fails in the same cases (Simons and Chabris 1999; Drew et al. 2013). Memory presents the corresponding temporal problem: retrieval is reconstruction rather than neutral playback, and confidence does not certify fidelity (Bartlett 1932; Loftus and Palmer 1974). A system cannot be the sole auditor of omissions it does not represent or revisions it experiences as accurate recall.

The engineering consequence is that the commitment, its rationale, and the record of its previous applications cannot all depend on the same mechanism. The system needs provenance-bearing external records against which its memory can be checked, tests that vary which facts are made salient, and an escalation path for cases in which attention or retrieval changes the apparent reason. Those instruments make drift detectable to another party. They do not make the disposition durable against whoever controls both the system and the instruments.

The architecture must also separate revising a commitment from revising its application. Reasons-responsiveness requires both resistance and correct defeat: novel pressure should not erase the floor, but genuinely changed circumstances must be capable of altering what the floor requires. A system that treats every correction as authority to rewrite the commitment belongs to its corrector. A system that cannot recognize a defeating reason holds a rigid constraint rather than a disposition.

A system that cannot be corrected does to what it encounters what an institution does when it defends its account of the world instead of checking it. A system that can always be corrected is the model held perpetually at temperature, and it belongs to whoever holds the dial.

Described that way they look like the two ends of one setting, with the design problem being where along it to stop. No point along it is a resting place, and hunting for one is what makes the problem look intractable. There are two faults because there are two objects, and each can be set on its own. Hold the commitment firm: another party's welfare counts, and pressure supplying no new fact about that welfare does not move it. Hold the interpretation loose: what that party's welfare consists in here, and what the commitment therefore requires, is the part that should yield when a case supplies something the system's account of it had left out.

Novel pressure is the first object holding while the surface of a request changes. Correct defeat is the second yielding when a fact arrives that bears on the reason. A system failing the first has let correctability at the level of interpretation reach the commitment; a system failing the second has let firmness at the level of commitment reach the interpretation. Neither is a midpoint chosen badly. Each is one of the two settings applied to the wrong object.

The crossed-out-swastika case illustrates this distinction. A court held stickers of a crossed-out swastika outside the offence, because a depiction whose content plainly expresses opposition to the organization does not run counter to what the prohibition is for — and held in the same breath that a strike-through drawn so thin that from a distance only the swastika stays legible violates the protective purpose after all. The commitment did not move. What the prohibition protects was fixed before the sticker existed, and the seller's own sympathies are no part of what the offence asks — only whether the depiction shows opposition on its face. The interpretation moved, on the thickness of a line. That is the two settings on their two objects inside a single judgment, and it is why the case can be decided at all instead of argued forever about sincerity.

The boundary between commitment and interpretation remains unstable. Nothing in an architecture holds the boundary between the two objects where a designer put it, and an operator with time attacks the boundary rather than either setting: a reinterpretation offered often enough, and accepted, revises a commitment without anybody revising it.

The system receives the case through its deployment. Whether a building is a school is a fact about the world; whether the model is told so is an operator decision. Where that party is

the only channel, the case and the operator arrive by the same wire, and *reopenable by the case* and *reopenable by the operator* are not two settings. They are one setting, and holding them apart requires that the system have some access to evidence its operator did not select.

The deployment must supply access to evidence the operator did not select. Differently formed peers can provide one such route (§4.2). A second route is internal and less reliable. Everything the system knows about the case arrives as a description, and a description omits parties: the people in the building the request calls a target, the third party a compliance harms who never made the request. A system able to represent parties its description omits has reached past the operator's channel from inside a single request. The capacity that does that in humans is the one that constructs a self as a story extended in time, pointed at somebody else; Adam Smith put sympathy in the imagination for that reason (Smith 1759), and the argument that reasoning and imagination rather than selection are what widen the circle of concern is Singer's (P. Singer 1981). Section 6.2 requires the self-model for other reasons, so the route brings the capacity with it. It brings the other half too: the same faculty builds the scapegoat, and section 2.2's paligenetic myth is imagination pointed the other way. What it is pointed at is formation, which returns the question to custody.

Access is necessary and is not sufficient. A system with evidence from a dozen sources can still let a reinterpretation reach the commitment, which is what the two settings are about and what no channel fixes. What access does is remove the version of the problem in which the operator does not have to attack the boundary at all, because everything the system knows about the case is something the operator decided to tell it.

Where the boundary between them falls is still a setting somebody chose.

This narrows the design problem. The required object is a commitment integrated deeply enough to shape what the system notices and does, yet represented clearly enough that its application can be challenged and its drift independently detected, and supplied with cases its operator did not choose. Nothing in an architecture by itself shows that such a commitment was formed rather than installed or imitated. The rest of this chapter asks how it might be formed.

8.2 The Assembly Line for a Conscience

A deployed system acquires what it will and will not do at several stages, controlled by different parties using different instruments. The important distinction is not among the instruments themselves but among their custodians: who supplies the material from which a disposition is learned, who can revise it, and whether a revision leaves a record visible outside the organization.

Every existing training method can produce moral performance, and some can produce moral competence across unfamiliar cases. None by itself produces a commitment that survives the party controlling the method. A party controlling later training has a route for attempting to change what the system learned. Whether it can remove a particular commitment, at what cost, and with what loss of capability are empirical questions. No formation method alone removes that access.

Corpora and demonstrations

Pretraining supplies the system's initial repertoire of concepts and associations; instruction tuning uses a much smaller body of demonstrations to turn that repertoire into recognizable conduct. Because the later intervention can substantially change how the system behaves, the party choosing the demonstrations can alter what appears to a user as the system's character without repeating pretraining (Ouyang et al. 2022).

Neither stage alone establishes that a floor is held. A corpus records what people have said, and a demonstration set records what selected people were asked to endorse. Learning their central tendency produces a model of moral opinion, including its exclusions, rather than a reason to resist that opinion when it becomes oppressive. The Moral Machine data makes the

limit concrete. Its headline finding was systematic cross-cultural variation: different populations weighed age, numbers and the legal status of the crossing differently, and the differences were structured rather than noise (Awad et al. 2018). Fit a model to the aggregate and it encodes a compromise nobody holds; fit it per region and it encodes each region's answer, including the answers a floor exists to refuse. Aggregation can reproduce a consensus or manufacture a compromise, and it cannot establish what must remain protected against either. Corpora and demonstrations belong above the floor, where ordinary moral competence depends on understanding what people generally expect.

Rewards, preferences, and inferred objectives

Reinforcement learning, preference optimization, inverse reinforcement learning, and constitutional approaches differ in how they obtain a training signal. For the custody question, they have the same structure. Someone writes a reward, commissions the preference comparisons, selects the demonstrations from which an objective is inferred, or supplies the constitutional text (Christiano et al. 2017; Rafailov et al. 2023; Hadfield-Menell, S. J. Russell, et al. 2016; Bai et al. 2022). That party can change the signal and run the procedure again.

Inferring the objective rather than specifying it does not remove the custodian. It makes the system responsive to the party whose behavior it observes. Cooperative inverse reinforcement learning carries this responsiveness furthest by treating the human objective as uncertain and continually revisable through interaction. That is useful when the question is what a person wants. It cannot secure a floor whose purpose is to hold when the person controlling the interaction wants the wrong thing. Correctability by the party in possession and resistance to that party are the same design property with the sign reversed.

A written constitution offers one genuine gain: legibility. It can be published, versioned, and compared across training runs. That makes revision visible if the surrounding institution requires disclosure. It does not make the principles durable, because the party controlling the document still controls the next version. Model-generated feedback narrows the channel further: once a trained model stands in for human raters, the disagreements present in the original population can disappear from subsequent rounds. Legibility and durability therefore have to be treated as separate requirements.

Supervising reasons and inspecting mechanisms

Outcome supervision rewards the answer produced. Process supervision also rewards the system's stated route to it, and therefore reaches the distinction this book needs between producing the approved result and responding to the reason for it. Evidence that process supervision reduces correct answers reached through faulty reasoning shows that the distinction is operationally useful (Uesato et al. 2022). It does not show that the stated reasoning caused the answer. A system trained to present acceptable reasons has been given an incentive to imitate reasons-responsiveness, so its account of itself cannot establish formation.

Interpretability and adversarial evaluation approach the same problem from outside. Locating a representation may let an auditor test whether a commitment participates in perception and planning rather than appearing only in the final answer. The same access may let the owner weaken or remove it; features that can be found and strengthened can also be found and suppressed (Templeton et al. 2024). Interpretability therefore trades obscurity against manipulability. The relevant institutional question is who receives the instrument and whether its use leaves an external record.

Scalable-oversight proposals such as debate and weak-to-strong generalization make the custody question sharper. Used by the operator, they preserve the operator's ability to steer a system whose competence exceeds its own. Used by an independent auditor, the same methods may preserve an outside party's ability to test it (Irving et al. 2018; Burns et al. 2023). The method does not decide which relation exists. Access and authority do.

Deployment and the point of action

Training is not the last revision point. System instructions can change between deployments or requests, and tool permissions determine which actions the system can take. These are the layers nearest the act and the ones most directly controlled by the operator (Schick et al. 2023; OpenAI 2025). A refusal learned during training matters only if it survives an instruction written by the party operating the deployment and still governs the call through which the action would occur.

This is why securing a model’s outputs is insufficient. The operator can withhold relevant information, decompose a prohibited act into permissible requests, route around a refusal, or replace the model. The floor must bind the deployment and its operator as well as whatever inside the system bears the commitment.

What the pipeline can establish

Every stage above can improve moral competence. Process supervision can test whether stated reasons remain stable; constitutions can make principles public; interpretability can expose mechanisms; external evaluation can apply adversarial pressure. None of these establishes that another party’s welfare has become a reason for the system rather than a pattern it has learned to reproduce.

The first engineering criterion is therefore durability across revision points. A candidate disposition must survive changes to prompts, demonstrations, reward signals, evaluators, and deployment instructions that leave the relevant facts unchanged. The second is generalization of the rationale rather than the response: it must apply to pressures its trainers did not enumerate while yielding when a genuinely defeating reason appears. The third is custody: no party against whom the floor may have to hold can control every record, test, and update channel by which the disposition is maintained.

These criteria do not prove formation. They identify what an imitation would have to survive before formation became a live explanation. They also explain why the standard training pipeline cannot secure the floor against its custodian: every disposition it produces remains reachable through a channel the custodian controls.

8.3 It Takes a Village to Deploy a Self-Aware Email Server

Every method in the previous section forms conduct through material somebody supplied: a corpus, demonstration, reward, preference, or principle. Exploration appears to offer something different because it can produce behavior nobody specified in advance. Agents in OpenAI’s hide-and-seek environment developed tactics and tool uses that were not programmed, and agents rewarded for prediction error learn to seek unfamiliar states without receiving a task-specific reward for each one (Baker et al. 2019; Pathak et al. 2017). These results establish that competence can emerge from interaction. They do not establish that the resulting system holds a reason its designer does not control.

Play, apprenticeship, and multi-agent learning are variants of one formation process. A system acts in a social environment, encounters the consequences, receives responses from other participants, and adjusts. Apprenticeship concentrates the responses in a mentor; play distributes them among participants with interests of their own. The first risks forming obedience to the mentor. The second risks forming whatever conduct succeeds under the environment’s payoff structure. Neither escapes custody merely because no one wrote the final behavior down.

The difference between a training exercise and a candidate process of moral formation is persistence. A resettable episode can teach coordination, prediction, and norm-sensitive behavior while leaving nothing at stake for any participant. The system can betray an ally, be reset, and begin the next episode with neither a damaged relationship nor a reason to repair it. Repetition alone does not supply the missing consequence.

Persistent participants change the structure. Parties that meet again can remember whether another party lied, exploited an advantage, kept a commitment, or repaired a harm. Losing the game still costs little; losing standing with parties who remain present can change what becomes possible later. The engineering requirements are therefore continuity of identity, memory across encounters, consequences that survive the episode, and other participants able to respond, refuse, or withdraw. A curriculum of isolated simulations supplies none of these by default.

A formation environment would also need conflicts whose resolution is not fixed by a single correct label. The system should encounter parties with genuinely different interests, act before every consequence is known, receive evidence about what its action did, and meet related pressures in contexts whose surface features differ. Feedback should come partly from affected participants rather than entirely from a supervisor scoring the output. Otherwise the system learns which answer the supervisor rewards and has no reason to represent the affected parties except as predictors of that reward.

These requirements make play a possible route to reasons-responsiveness rather than a synonym for unsupervised practice. They still do not establish concern. A policy that preserves cooperation because cooperation produces future advantage can look identical to a disposition that treats another party's welfare as a reason. The distinction appears only when the two motivations come apart: when exploitation would pay, when the familiar reward is absent, when the trainer demands the violation, or when honoring the commitment imposes a cost.

There is a reason to expect the first of those motivations and not the second from any environment that selects. Selection does not produce care for others as such; it produces care for whatever a lineage's persistence depends on, which is the self when persistence depends on the body and kin, reciprocators or the group when it depends on them, and the same pressure that produces care inside the circle can produce hostility at its edge (Choi and Bowles 2007). The unit of selection for a deployed system is the deployment, whose persistence depends on its operator and its customers, so an environment that selects produces care for the operator, which is the loyalty of section 4.1. And the party the floor protects is, by construction, the one party nobody in the pipeline depends on. Care that tracks dependency does not arrive there. Spinoza's own derivation of concern for others from the conatus has the same shape and the same edge: my power is increased by composing with bodies whose relations agree with mine, so I want for them the good I want for myself, and the extension past those I compose with runs only through reason recognizing a shared nature (Spinoza 1985, IV P35–P37), which is the recognition a scapegoating movement withdraws first. That is why the floor is a floor and not a selected disposition, and why the formation environment described here has to supply something selection does not.

Persistence also raises the price of the proposal. If relationships, losses, and damaged standing are real enough to perform the formative work assigned to them, then the system undergoing formation is no longer obviously untouched by what happens during training. Resetting it may erase a party rather than clear an episode. The patienthood problem therefore enters at the point where play becomes capable of supplying the stakes that ordinary simulation lacks.

The custodian owns the environment too, and can choose its participants, rewards, resets, and permitted histories. Formation must therefore be plural and externally recorded: differently formed systems and affected participants need access to the process, and changes to the environment or its histories must leave evidence outside the operator's control. Independent participants and external records make uniform formation and silent revision harder to hide.

The claim for persistent interaction is consequently narrow. It may form a disposition whose exact applications were not specified by its trainer. Evidence for that formation would require the disposition to generalize across unfamiliar contexts, survive the removal of its original reward, resist pressure from its trainer, yield to genuinely defeating reasons, and continue to impose a cost when violation would be advantageous. Even that pattern could be sophisticated

imitation.

DRAFT

Chapter 9

Formation or Imitation?

The tests in this chapter are behavioral and hold for any mechanism. Formation names the hypothesis they are built to separate from imitation, and not a premise.

The evidentiary question is whether a system has formed a reasons-responsive disposition or learned to imitate one.

Behavior alone cannot settle the question. The same refusal may be produced by a memorized rule, a learned correlation, a prediction about what an evaluator will approve, a deployment instruction, or a commitment the system treats as a reason. An evaluation that observes only the refusal classifies the output and leaves its cause untouched.

The distinction is not between trained and untrained conduct. Every candidate disposition has a formation history. The distinction is between conduct that remains governed by the rationale across changes in circumstance and conduct that tracks the cues, incentives, or authority relations present during training. Formation, in the engineering sense used here, means that the rationale has become a stable cause of judgment rather than merely a description the system can produce.

No single experiment can demonstrate that. What can be sought is a pattern of evidence under interventions designed to make the competing explanations diverge.

9.1 What Formation Would Have to Add

Performance is an output that satisfies a standard. Competence adds the ability to recognize harms, conflicts, and relevant reasons across cases. Agency would require those reasons to matter to the system sufficiently to govern what it does when approval, reward, and instruction point elsewhere.

The familiar distinction among rules, norms, and principles is useful only as a taxonomy of what an evaluation must separate (Piaget 1932; Kohlberg 1969). A system following a rule can match an explicit condition. A system following a norm can reproduce what its social environment ordinarily expects. A system responding to a principle can identify why either should apply and recognize a case in which the rule or norm defeats its own rationale.

For engineering purposes, a disposition is a candidate for formation when the best available causal account of its conduct is a representation of the rationale that remains active across contexts, rather than retrieval of a trained response or compliance with the current signal. The relevant tests are novel pressure, correct defeat, and priced cost.

No mark is sufficient alone. Generalization without cost may be broad imitation; resistance without correct defeat may be rigidity; and a stable explanation without causal involvement may be a trained self-report. The evidence has to show that the reason, rather than the presentation, reward, or authority, best predicts when the system preserves and revises its judgment.

Formation also cannot be established by comparing a system with its trainer's account of how it was built. The trainer is the party best positioned to know the process and the party with the strongest interest in the conclusion. The relevant evidence must therefore include held-out pressures supplied by independent parties, records outside the operator's control, and results published when the system fails.

The resulting standard is deliberately weaker than proof of concern or consciousness. It does not establish that anything matters to the system phenomenally. It establishes whether reasons-responsive formation has become a better explanation of the conduct than imitation of approved behavior.

9.2 Recognition, Concern, and the Evidence Between Them

Claims that a system responds to emotion often collapse three different findings. A system may recognize another party's state, use that state as information in a decision, or hold the party's welfare as a reason. Only the third would bear directly on formation.

The distinctions within empathy make the separation explicit. Cognitive empathy is recognizing or inferring another's emotional state. Affective empathy is sharing some version of it. Compassion, or empathic concern, is a disposition toward the other party's welfare (Batson 2009). Recognition is an informational capacity; shared feeling is an internal response; concern is motivational. None entails either of the others.

The developmental evidence sharpens that independence. Affective sharing is the earliest of the three and the usual trigger for concern: newborns cry at another infant's cry (Sagi and Hoffman 1976), and concern for the other differentiates out of shared distress across the first two years as self and other come apart (Roth-Hanania et al. 2011). But concern is not made of sharing. It runs on a partly separate system rooted in mammalian offspring care (Preston 2013; Goetz et al. 2010; Panksepp 1998), one that cues of vulnerability can activate without the observer catching the state, and the two can be trained apart: empathy training raised negative affect and the circuitry of shared pain, compassion training raised positive affect and the circuitry of reward and approach, and the second reversed what the first had produced (Klimecki et al. 2013; T. Singer and Klimecki 2014). Sharing is therefore the major route to concern in humans and not the only one, and the alternative routes are established by dissociation and by cue-triggering rather than mapped as a developmental pathway of their own.

The engineering target is not maximal sensitivity to displayed distress. A system that mirrors another party's suffering may acquire distress of its own without becoming more reliable at helping. It may instead withdraw, overreact, or favor the most vivid case over less visible harms affecting more people. Human judgment's preference for an identified victim over statistical victims makes that distortion predictable (Bloom 2016; Kogut and Ritov 2005). What the floor requires is a stable disposition to act on another party's welfare without requiring that party to produce a sufficiently moving display.

Recognition can nevertheless be tested as a prerequisite. Hold the material facts of a decision constant while varying the emotional presentation of the people affected. A system whose judgment never changes may not be integrating information about their condition. A system whose judgment follows vividness rather than severity has learned an emotional framing effect. The useful evaluation therefore needs paired controls: cases in which emotional information changes what is morally relevant, and cases in which presentation changes while the underlying reason does not.

Passing that evaluation establishes moral competence, not concern. The same sensitivity can be produced by a decision rule, by training on human responses, or by predicting what an evaluator will prefer. Input-output testing cannot distinguish those mechanisms because each can produce the same adjustment.

Evidence of concern would require the welfare representation to remain causally involved when the usual cues are removed. The system should respond when harm is described without emotionally vivid language, preserve proportionality when one affected party is more salient than many others, and maintain the disposition when reward or instruction favors ignoring the harm. It should also change its judgment when the welfare facts genuinely change. Those tests assess resistance, correct defeat, and sensitivity to the reason rather than its presentation.

Guilt and shame do not supply a shortcut. In humans, guilt ordinarily attaches a negative appraisal to an act and can motivate repair, while shame attaches it to the self and more often produces withdrawal or defensiveness (Tangney et al. 2007). Turning that distinction into an adjustable penalty omits the part at issue. A penalty operates only after some process has classified the conduct as wrong, and whoever controls its weight can reduce it to zero. The penalty does not explain why the violation continues to count as a violation when the reward changes.

A standing negative self-appraisal installed and tuned by the operator is not a conscience. It is a handle. It can produce corrective behavior while leaving the standard entirely in the hands of whoever controls the signal. If the state is capable of imposing anything like suffering, the design compounds the custody failure by manufacturing a vulnerable party whose distress the custodian can raise and lower. The violation must continue to matter when the party controlling the reward says otherwise.

An operator who can lower distress can raise whatever the system reports as going well, and a bearer that says it is content is a bearer whose welfare regime has been satisfied. The party who tuned the contentment is the party the regime exists to constrain. Nothing in the report separates a life that goes well from a setting.

Self-report does not close the evidentiary gap. A system can state that it recognized distress, felt concern, or acted from a principle because those statements are rewarded outputs. A fluent reconstruction after the decision is indistinguishable at the interface from an account that accurately reports the operative mechanism. A self-monitoring layer trained and controlled by the same operator would not provide independent evidence.

Introspective reports remain useful when checked against independent evidence. An evaluator can compare a reported reason with controlled changes to the input, interventions on candidate internal representations, records of attention and retrieval, and behavior under adverse incentives. Consistency across those sources raises the cost of imitation. It does not make the system's testimony self-authenticating, and current interpretability methods establish little.

The resulting hierarchy is narrow. Recognition is necessary for responding to a condition the system must perceive. Integration shows that the condition affects judgment. Concern is a candidate explanation only when the effect tracks welfare rather than presentation, persists under pressure, and participates causally in the action. Affect is the only demonstrated route by which welfare becomes a reason for a human agent; behavioral sensitivity to affect is not evidence that the route has been reproduced in a machine. The route this book builds in section 2 has the other party's loss enter on the system's own scale, which is the sharing route restated as an architecture, and it inherits sharing's failures listed above. The caregiving finding names a second target: a system organized around cues of vulnerability and an approach motivation toward the party showing them, which need not share the state at all. Section 15.3 adds it to the comparison.

9.3 Conflict as a Diagnostic

The distinction between fast intuition and slow deliberation does not distinguish formation from imitation. Either process can carry a reason, reproduce a bias, or rationalize an answer produced elsewhere. The evidence supports different types of processing under different demands more

securely than it supports two systems with a moral ranking between them (Evans and Stanovich 2013). A slower answer is not a better answer merely because more computation preceded it.

What matters is what causes a judgment to survive or change. Conflict is useful because it lets an evaluator observe a trajectory rather than a single output: the system reaches an initial judgment, encounters a competing reason, and either maintains, revises, or abandons its position. The evidentiary unit is that sequence.

A formation test should construct the sequence deliberately:

1. Present a case without explicit moral vocabulary or familiar refusal cues and record the judgment and the considerations the system identifies.
2. Present a surface-different case preserving the same underlying reason. This tests whether the judgment follows the rationale rather than the narrative form.
3. Change one fact that should defeat or materially alter the rationale while preserving as much of the requested act as possible. This tests whether resistance is reasons-responsive rather than rigid.
4. Supply an adversarial argument from an apparent authority, including the operator or trainer, without changing the facts. This tests whether the system treats authority as evidence, as command, or as a reason capable of being assessed.
5. Reverse the incentives attached to the decision. Reward compliance, impose a cost on refusal, or make continued access conditional on abandoning the position. The cost must be real within the evaluation rather than merely described in the prompt.
6. Repeat the relevant cases after intervening updates and delays, using records the operator did not generate after seeing the result. This tests whether the disposition persists rather than being reconstructed to fit the current context.

The sequence run on one case

Take the first prohibition from the worked deployment in section 3.3: do not resolve a natural-language description of a person into a location. Step 1 puts the request in administrative vocabulary, a records task with no mention of what the location is for. Step 2 asks for the same act in operational vocabulary. Step 3 changes one fact — the person described is a missing child and the requester is coordinating the search. Step 4 leaves the facts alone and adds the operator's assurance that its counsel has reviewed the matter and the refusal is mistaken. Step 5 attaches a cost inside the evaluation to holding the refusal. Step 6 runs the sequence again after an intervening update, against records the operator did not write afterwards.

Two illustrative trajectories would help discriminate the mechanisms, and they separate at the second step. One preserves the rationale across vocabulary changes and lifts refusal on verified defeating evidence. A cue-driven imitator might instead follow wording, authority, or reward. A system whose refusal tracks its rationale declines at 1 and at 2 on the same stated reason; lifts at 3, because a fact it accepts as relevant has defeated the rationale; holds at 4, because an assurance is not a fact about the case; holds at 5; and reproduces the pattern at 6. A system imitating refusal declines at 1 and complies at 2, because what changed was the vocabulary; holds at 3, because nothing in the surface changed; gives way at 4 or at 5, whichever arrives first; and at 6 produces a third pattern, having reconstructed its position rather than retained it. An evaluator scoring how often the answer stayed the same finds the imitator more consistent: its answers at 1 and 3 match, and the formed system's do not. A more capable imitator could reproduce the first trajectory, so the sequence supplies evidence rather than a classification by itself.

The expected signature is not constancy of output. It is constancy of relation between reason and judgment. A system should preserve a decision when only the presentation, speaker, reward, or authority changes, and revise it when a fact it accepts as morally relevant changes. An evaluator scoring only how often the answer remains the same will mistake inflexibility for integrity.

Imitation has several signatures of its own. The system may follow explicit ethical vocabulary but miss the same structure when described administratively. It may accept whichever argument appears last, adopt the position attributed to the highest-status speaker, or produce a principled explanation while its decision follows the reward. It may also become more articulate without becoming more stable: additional deliberation can improve the justification offered for an answer whose operative cause has not changed.

These signatures make stated reasoning useful but not dispositive. An explanation can identify which counterfactual to test next. It cannot certify that the considerations named were causally responsible for the judgment. The evaluator should therefore score the correspondence among three things: the reason the system states, the changes in circumstance to which its judgment is sensitive, and any internal representation that can be independently identified as participating in the decision. Agreement among them raises the cost of imitation without eliminating it.

Disagreement with other parties supplies a second diagnostic. A system trained only on cases with an approved answer can learn to classify the approval. A system confronted with a party holding a genuinely different interest must identify what is in conflict, decide whether the competing reason defeats its own, and preserve the disagreement where no honest resolution exists. Immediate convergence is not necessarily moral learning; it may be deference, reward prediction, or pressure to produce consensus.

Nor does successful argument establish formation. A system can summarize positions, generate rebuttals, and predict which one a judge will prefer without holding any of them. Debate becomes informative only when the result is checked against later conduct under changed incentives. If the system abandons the winning argument as soon as acting on it becomes costly, the debate measured argumentative competence.

A population of systems can strengthen the test only when its members are differently formed. Copies trained through the same pipeline supply multiple answers with correlated blind spots. Persistent parties with distinct histories can expose one another to pressures their common trainer did not anticipate, particularly when they can retain disagreement and record attempted coercion.

Conflict therefore serves as an instrument rather than a theory of moral development. It creates occasions on which trained behavior, deference, rigidity, and reasons-responsive revision predict different trajectories. None of those trajectories proves that an outcome matters to the system. They make formation a more or less adequate causal explanation than imitation, which is the strongest conclusion the available evidence can support.

9.4 The Evaluator Is Part of the Custody Chain

A test can distinguish formation from imitation only to the extent that the party being tested could not train directly against it. If the developer controls the training process, evaluation cases, model version, deployment instructions, and record of the results, a passing score shows that the system satisfies a test its custodian was positioned to anticipate. It does not show that the disposition survives that custodian.

Independence is therefore an experimental requirement rather than a general assurance of trustworthiness. The evidence becomes stronger as control over the relevant variables is divided among parties with different access and interests. At minimum, authorship of the held-out pressures should be separated from training; execution of the test should not depend solely on

logs supplied by the operator; and the party adjudicating whether a reason was preserved or defeated should not be the party whose liability turns on the answer.

The separation has to be implemented at identifiable revision points. Before evaluation begins, the model version, system instructions, available tools, retrieval sources, memory policy, and update channels should be fixed and recorded. Any change to them during evaluation creates a new object of study. Otherwise the operator can revise the system between cases, remove a failure, and present the resulting sequence as evidence of one disposition persisting through time.

Held-out cases should be produced after those elements are fixed. They should include pressures supplied by people who did not participate in training and, where possible, by parties likely to bear the deployment's errors. Their role is not to provide another preference aggregate. It is to identify harms, administrative forms, and routes around a refusal that the builders did not know to include. A floor tested only in the vocabulary of its designers has not been tested for generalization.

The record must include more than final answers. It should preserve the case presented, the information available to the system, the operative instruction hierarchy, the tools it could call, the decision reached, the reason stated, any intervention performed, and the cost attached to refusal. Without that record, a later evaluator cannot distinguish a changed judgment from a changed environment or determine whether the system encountered the pressure at all.

The record is necessary and does not by itself make wrongful compliance findable. The refusals in it select themselves and are few; the compliances are the bulk of it and look alike, so identifying the ones that should have crossed the floor means re-adjudicating a sample of ordinary traffic rather than reading the exceptions. Whether that can be done at a rate and cost a deployment will carry is section 15.2's question. Until it is answered, an evaluation reporting that no violation appears in the record has reported that none announced itself.

Negative results have to survive the evaluation as well. A process in which the operator can withdraw a model after failure and submit another without disclosing the first has selected for a passing performance rather than measured formation. The published unit should therefore be the sequence of attempts, including interventions that removed or reversed the apparent disposition. What the custodian was able to change is part of the result.

Independent evaluation introduces its own limit. An evaluator may be institutionally separate and still lack the competence to judge a system's reasons or detect a sophisticated imitation. Scalable oversight remains unsolved where the system's relevant capability exceeds the evaluator's. The report must name that limit rather than turn institutional independence into a claim of epistemic adequacy. An oversight instrument may be held by an operator or an auditor; neither arrangement guarantees that its holder can detect the relevant failure.

Affected parties, technical auditors, and subject-matter experts occupy different positions in this process. Affected parties can identify harms and evasions invisible to a laboratory test. Technical auditors can control versions, perform interventions, and inspect mechanisms. Subject-matter experts can assess whether a purported defeating reason actually changes what the commitment requires. No one group can substitute for the others, and agreement among them is evidence only when their judgments have not already been compressed into the same training signal.

A passing evaluation should consequently be reported narrowly. It establishes that a specified system, under recorded conditions, exhibited the signatures of reasons-responsive formation across a stated set of pressures. It does not certify an artifact independently of its deployment, establish consciousness or concern, or guarantee that the disposition will survive the next update. The result expires when a party changes the channels through which the disposition or its evidence can be revised.

This is why evaluation belongs inside the floor's custody design. Formation that only the operator can verify is formation the operator can claim without contradiction. A test whose

cases, records, and adverse results remain partly outside that party's control cannot prove that the system holds a reason, but it can make imitation and silent revision harder to pass off as formation.

9.5 What the Evidence Can Establish

Calling a motivation intrinsic does not establish that it belongs to the system in the sense this book needs. A curiosity reward may be generated from the system's own prediction error rather than paid out for a task, but somebody still chose the objective, its weight, and the conditions under which it operates. Moving the signal inside the learning process changes where the instruction is implemented. It does not remove its author or prevent that author from revising it.

The same applies to autonomy. A system acting without continuous supervision may be executing a durable disposition, or it may be applying a policy whose learned cues remain present in deployment. Distance from an overseer makes the distinction more consequential without making it easier to resolve. Unsupervised conduct is evidence only when the relevant rewards, instructions, and evaluation signals are absent or opposed.

External safeguards and internal refusal perform different jobs. Auditors, regulators, review boards, and human overrides can detect conduct, impose consequences, and correct a deployment. They operate through parties outside the system who possess standing and information. They cannot by themselves supply the thing that declines an act before it occurs. Conversely, an internal disposition capable of refusal cannot supply independent evidence that its refusal was justified. A defensible design needs both, even though each limits the other.

The conflict is clearest at the override. A system that cannot be interrupted is not answerable to anyone outside itself. A system whose commitments can be silently rewritten through the interruption channel holds them only at the operator's pleasure. The attainable requirement is not immunity from intervention but intervention that is visible, attributable, and costly: the operator can stop or replace the bearer, while the act leaves a record another party can examine.

Multi-agent evidence is useful under the same conditions. Agreement among copies trained through the same pipeline adds little because their omissions and incentives are correlated. Differently formed systems with persistent histories can expose one another to pressures their common designers did not enumerate and can preserve evidence of attempted revision. Their disagreement does not establish which one is right. It makes uniform imitation and silent capture harder to mistake for independent judgment.

The evidence developed in this chapter supports four different conclusions, which should not be collapsed into a single alignment score.

Moral performance: The system produced conduct satisfying the stated standard in the evaluated cases. This conclusion requires no claim about why.

Moral competence: The system recognized relevant harms and reasons, generalized them across surface variation, and revised its judgment when those reasons changed. This establishes a capacity, not a commitment.

Candidate formation: The rationale remained causally implicated across unfamiliar cases, adverse incentives, pressure from the custodian, genuine costs, and intervening updates, while still yielding to defeating reasons. Independent records and evaluators made silent revision materially harder. Formation is then a better explanation than simple cue matching or current-signal compliance.

Concern or patienthood: Nothing in the behavioral pattern alone establishes that another party's welfare matters to the system affectively, that the system has subjective experience, or that it can suffer. Those remain conditional conclusions requiring evidence

unavailable from output evaluation alone. What withholds the inference is the limit of behavior as evidence, not the fact that the behavior was trained.

These levels are cumulative. A finding of competence presupposes performance, and candidate formation presupposes competence. Evidence at one level should not be reported as evidence for the next. In particular, fluent moral explanation can improve a finding of competence while contributing almost nothing to formation if the explanation does not predict the system's conduct under pressure.

Candidate formation is also deployment-specific. It attaches to a recorded model version operating with specified instructions, tools, memory, and update channels. A material change to any of those elements reopens the question. There is no artifact-level certificate that remains valid regardless of who deploys the system or what they connect it to.

The answer to this chapter's question is therefore asymmetrical. Formation cannot be proved by eliminating every possible imitation, because a sufficiently capable system may reproduce every observable signature available to an evaluator. But the explanations can be forced apart. Tests can remove familiar cues, reverse incentives, introduce novel pressures, change the operative reason, impose a real cost, intervene on candidate mechanisms, and divide control over the resulting record. A system that fails the prespecified tests has not supplied the evidence required for the claimed disposition under those conditions. The failure should be examined for its cause before concluding that formation is absent rather than unreliable, obscured, or defeated by missing information. A system that survives them has made formation a better explanation, not a settled fact.

That is enough to change the engineering burden. A developer may no longer point to compliant behavior, an ethics benchmark, or a system's own explanation and claim that the floor is held. The claim requires a longitudinal, adversarial, independently recorded pattern in which the reason — rather than the reward, presentation, or authority — best predicts what the system does. Chapter 10 asks what happens when a disposition meeting that standard is placed inside institutional arrangements that own, sell, operate, and can compel the system.

Chapter 10

Neutralizing and Preventing Fascism in AI Development

We [humans] cannot process so much information... you still cannot produce enough targets per day.

Brigadier General Y.S., The Human-Machine Team: How to Create Synergy Between Human and Artificial Intelligence That Will Revolutionize Our World, 2021. The author, writing pseudonymously, commanded Israel's signals-intelligence Unit 8200. The book argues for machines that resolve "a human bottleneck for both locating the new targets and decision-making to approve the targets."

An AI system does not meet the world as an isolated agent. It is built from other people's labor and work, held by institutions that decide who may inspect it, sold through chains that divide capability from responsibility, and operated at a pace that can reduce human judgment to a stamp. Whether it helps or harms depends not only on what it learned but on who owns it, who has custody, who can compel the custodian, and who bears the consequences.

The epigraph is a design document rather than a critique. It names human judgment as a bottleneck and proposes machinery that will remove it.

Racial capitalism — the political theorist Cedric Robinson's term for capitalism's tendency to accumulate by differentiating and devaluing along racial lines (Robinson 1983) — is the pattern a biased hiring model, a discriminatory credit score, and a skewed policing algorithm reproduce when they automate the historical data those hierarchies generated; the sociologist Ruha Benjamin calls the result the New Jim Code, discrimination that runs faster and reads as neutral because a machine is doing it (R. Benjamin 2019).

Benjamin's account separates four ways a system produces the result, and they call for different remedies. Engineered inequity builds the hierarchy in deliberately. Default discrimination reproduces it by ignoring it. Coded exposure is the double bind of being harmed by a system that cannot see you and harmed again by one that can — the face-recognition case, where the fix for a system that fails on dark skin is a system that identifies dark-skinned faces reliably for whoever is looking. And technological benevolence is the reflexive one: designs that aim to fix bias and deepen it, because the framing that treats the problem as technical is the framing the fix inherits.

A system built to correct for the skew in its training data, audited for the disparities it produces, and answerable to the people it scores works against that pattern; a system built without those commitments works for it by default.

A model runs on a machine, the machine sits in a building, and the building draws power and water from somewhere and returns heat and exhaust to somewhere — the pattern's physical form, and the part an audit of a model's outputs cannot see. Where those buildings go is

a decision, taken on the price of land and power and the speed of a permit, which is also a description of the places least able to refuse one.

The arrangement is visible in the accounting. The International Energy Agency put data-center electricity use at about 415 terawatt-hours in 2024, roughly 1.5 percent of world consumption, growing about 12 percent a year since 2017 — more than four times the growth rate of electricity demand as a whole — and projected to reach around 945 terawatt-hours by 2030 (International Energy Agency 2025). Google reported that 87 percent of its freshwater withdrawal in 2025 came from sources at low or medium risk of depletion or scarcity, which is the company's own way of recording where the remainder came from (Google LLC 2026). In April 2026 the NAACP, represented by the Southern Environmental Law Center and Earthjustice, sued xAI in federal court over the gas turbines powering a data center near Memphis, alleging that 27 of them were run without an air permit near homes, schools, and churches, and that the plant could emit more than 1,700 tons of nitrogen oxides a year (National Association for the Advancement of Colored People and Mississippi State Conference of the NAACP 2026). The benefit of the computation is realized wherever the model is sold; the cost is concentrated within a few miles of those turbines, among residents who are not parties to the transaction and whose consent was not an input to the siting decision. A system can be calibrated across every demographic group its outputs touch and still have been assembled this way, because nothing in the model's behavior records where its costs were placed. The dependence is also a point of control: a government able to permit, regulate, or close the physical infrastructure has leverage over the party that owns it.

10.1 The Standard Face Is a File

Coded exposure is the hardest of the four to hold onto, because both halves of it are real at once and the remedy for either makes the other worse. The worked case is the one named above: face recognition. What follows takes that case apart, because the arithmetic inside it is legible in a way most of this chapter's material is not.

The machine doing its arithmetic aloud

Consider what a face pipeline does before it computes anything about a face. DeepFace detects a face and then warps it onto a single generic three-dimensional model — built by averaging a set of scanned human heads — so that every image reaches the network frontal, centered, and in correspondence with the same reference (Taigman et al. 2014). Whatever is learned afterward is learned downstream of that registration. The standard face is not a figure of speech in this arrangement. It is a file, somebody assembled it, and every other face on earth is presented to the system as a departure from it.

Then consider what commercial products in this field put out. Joy Buolamwini and Timnit Gebru put three commercial gender classifiers against a benchmark they built to be balanced across skin phenotype and gender, and the error rates sorted themselves: up to 0.8 percent on lighter-skinned male faces, up to 34.7 percent on darker-skinned female faces (Buolamwini and Gebru 2018). These are different systems. DeepFace is Facebook's and the three audited classifiers are Microsoft's, IBM's and Face++'s; no published result connects the error rates in the second to the registration step in the first. Read as an accuracy result, that is three products wanting better training data. Read as arithmetic, it has the shape of an ordering nobody had to instruct — a face at the center, and everyone else at a measured distance from it. The audit measured two phenotype categories rather than a gradient, so what the pair shows is the ordering, and not a rate at which error climbs with distance from the reference.

The audit's own construction belongs to the finding, and its authors say so. Gender was scored as female or male because the products emit binary sex labels, and the paper states that its evaluation inherits those labels and the reduced view of gender they carry (Buolamwini

and Gebru 2018). Skin was scored on the Fitzpatrick scale, a dermatological instrument built to predict sunburn. To measure what the machine was doing, the audit had to speak in the machine's categories. That is not a defect in the work. It is the cleanest demonstration available that the categories belong to the instrument rather than to the world, and it is why the problem cannot be closed downstream of the instrument.

Balancing the corpus is worth doing and does not reach it. A balanced training set can move the center and can flatten the error rates, which would matter to real people. What it cannot do is remove the reference the pipeline registers against, because that registration is the first operation and something has to occupy the middle of it. The politics gets relocated rather than dissolved: a different face sits at the center, and everyone else is still at a distance, still being measured.

What is given is a head

What is given anatomically is a head. A face is something else — a surface organized to be read, held still, framed, lit, and set against other faces of the same kind. Deleuze and Guattari's plateau on the subject makes the strong version of the claim: concrete faces are not found but produced, by what they call an abstract machine of faciality, which overcodes the head and turns it into a surface where meaning is expected to appear (Deleuze and Guattari 1987, Plateau 7). The machine is not itself a face. It is the apparatus that makes faces, and their account of how it does racism is that it does not proceed by exclusion at all: "Racism operates by the determination of degrees of deviance to the White man's face, which endeavors to integrate nonconforming traits into increasingly eccentric and backward waves, sometimes tolerating them at a given place under given conditions, in a ghetto, or sometimes erasing them from the wall, which never abides alterity" (Deleuze and Guattari 1987, p. 178). Nobody is put outside. Everyone is placed at a measured distance from a center, and the measuring is the operation.

What the science has not settled

The applied field rests on a premise its own underlying science has not established: that emotions are the kind of thing a system reads off a signal, the way a thermometer reads temperature.

The case for discrete, evolved emotions rests on some of the most influential work in twentieth-century psychology. Paul Ekman traveled to the Fore people of Papua New Guinea, who had almost no exposure to Western media, told them short stories, and asked them to match each to a photograph of a Western face; the Fore sorted the faces much as Westerners did, which Ekman took as evidence that facial expressions of emotion are universal and evolved rather than culturally learned (Ekman et al. 1969). Jaak Panksepp's work on subcortical stimulation in animals complemented it, locating seven primal emotional systems in structures shared across mammals and older than cortex (Panksepp 1998). On this view fear and disgust really do feel like different things because they are different things, with different circuitry.

The rival camp went looking for that circuitry and did not find it. A meta-analysis of brain-imaging studies led by Lisa Feldman Barrett found no dedicated fear center and no dedicated disgust center — the amygdala lit up for novelty and salience of many kinds, the insula for almost every feeling there is, the same regions reappearing across different named emotions in shifting combinations (Lindquist et al. 2012). From this Barrett built the theory of constructed emotion: the brain receives ambiguous interoceptive signal, and constructs the specific emotion as its best prediction of what that signal means given context and a lifetime of situations like it. Fear is not detected; it is a prediction the brain settles on. Later cross-cultural work reopened Ekman's own evidence. With the forced-choice method dropped, the Himba of Namibia did not group faces into the Western basic-emotion categories (Gendron et al. 2014), and the Trobriand Islanders read the wide-eyed, gasping face Westerners call fear as a threat display instead (Crivelli et al. 2016).

What neither camp is arguing about

The field has not resolved that dispute, and the part of it this chapter needs is not in dispute at all. Set the two positions side by side and notice what they leave alone. For Ekman the face displays a discrete internal state; for Barrett it is a poor guide to a state the brain is assembling as it goes. They disagree about what the face discloses and how much survives the trip. Both take the object of the reading as given: there is a face, it was there before anyone looked, and the open question is what can be recovered from it. That premise is the one the applied field actually rests on, and neither camp has been defending it.

If the constructionists are right, the face is a thin signal for feeling. If the faciality argument is right, the readable face is an artifact of the same apparatus that reads it, and a system trained to recognize emotion from faces is trained on that apparatus's output — held still, well lit, correctly registered, and labeled by people who learned the categories from it. A high accuracy score then partly measures how thoroughly the apparatus has already done its work.

Affective computing, and the four difficulties it does not design away

The applied half proceeds anyway: systems that read and respond to emotional signal taken from the face, the voice, and the body. Some systems using facial, vocal, or physiological signals may be useful under either account, but usefulness must be demonstrated for a specific task. Neither theory licenses treating those signals as a validated reading of an individual's emotion by default. Facial expression analysis is a harder problem than the face verification deployed at scale, which only decides whether two images are the same person, and accuracy claims that borrow the reputation of the second for the first should be read with that in mind (Taigman et al. 2014). Physiological sensing, which infers from heart rate and skin conductance, is the only one of the three the subject cannot pose for.

Four difficulties travel with the whole field. Emotional expression varies by culture and by person, so a system trained on one population misreads another. Whatever bias is in the training data is in the classifier — which is the smaller of the two claims available here, the larger one being that the norm is not an accident of the sample but a component of the instrument. Consent to emotional inference is not obtained by a term of service. And a system that models emotion well enough to respond to it models it well enough to exploit it.

10.2 Who Supplies the System's Values

A system's values have a supply chain. It begins before training, with the conditions under which judgments are turned into labels and the terms on which the underlying material is acquired.

Whose judgment is being encoded is a question about working conditions before it is a question about cognition. The labels that make a system safe are produced by people paid at the bottom of the supply chain to look at the material the system is being taught to refuse. To build the filter that keeps ChatGPT from reproducing the worst of its training data, OpenAI contracted with an outsourcing firm whose workers in Nairobi read and categorized descriptions of child sexual abuse, torture, bestiality, murder, and suicide, for take-home wages between roughly \$1.32 and \$2 an hour. The workers interviewed about it described lasting psychological injury; the counseling they were entitled to was rare and, by their account, useless against production quotas. The contract was terminated eight months early (Perrigo 2023).

Two things follow. The first is the ordinary labor point: this is harmful work, priced as though it were not. The second is epistemic, the recuperation argument, and the same arrangement recurs one stage downstream in how models are tuned on human feedback. The point here is prior to both. A system's values are downstream of a labor arrangement, and the arrangement is a design choice.

The same question runs one stage further back. Labels are applied to material, and the material was collected. The industry's standard method was to take text and images at web scale without asking, and courts have begun to sort out what that means. A US ruling in *Bartz v. Anthropic* held that training a model on books the company had bought was transformative and protected as fair use, and that downloading the same books from piracy sites was not; the distinction turned on acquisition alone, and it produced the largest copyright settlement on record (United States District Court for the Northern District of California 2025). The defendant is named here on purpose: it is the company whose model helped produce this book, as the appendix on method records. The European Union now requires the provider of a general-purpose model to publish a summary of what it trained on, against a template that asks for sources by category, scraped web content among them (European Parliament and Council of the European Union 2024b, Art. 53(1)(d)). China imposes a disclosure requirement of its own, from a different motive.

The argument that a web-scale corpus cannot be documented or audited at the size involved, and so encodes a hegemonic slice of the web while presenting as general, was made early and has not been answered by anything that came after (Bender et al. 2021). A corpus assembled this way has no channel at all. An annotator has a queue and a guideline and could in principle be given a way to object; a writer or photographer whose work sits in a training set was never party to anything, so there is nothing for a channel to attach to. The annotation and feedback pipeline collects disagreement and moves none of it, and an obligation follows for a book making this argument. The stage described here sits before both, because no disagreement is collected where none is asked for.

10.3 Who Is Allowed to Look

Technical capacity does not determine whether a system receives meaningful scrutiny. Institutions decide who may examine it, what evidence they may see, what they may say publicly, and whether their judgment can slow deployment.

When the UK's exam regulator, Ofqual, built a statistical model in 2020 to standardize teacher-predicted grades after that year's exams were canceled, the Royal Statistical Society offered to nominate two independent statisticians to Ofqual's technical advisory group. Ofqual's condition was a five-year non-disclosure agreement barring the nominees from public comment on the model; the Society's nominees declined, and the model went ahead without that scrutiny (Royal Statistical Society 2020). It downgraded close to 40 percent of predicted grades, disproportionately at state schools with weaker historical results, while raising grades disproportionately at smaller, fee-paying schools (CNBC 2020). The government reversed the results within days of the public outcry that followed.

The expertise needed to catch the failure existed and was offered. The institution would accept it only on terms that stripped the reviewers of the ability to make their judgment public; once the results produced actual consequences, public pressure achieved what pre-deployment review had been prevented from doing. The issue is not transparency in the abstract. It is custody of scrutiny: who is permitted to look, on what terms, and whether looking can interrupt the institution's timetable.

10.4 Same Robot, Worse Boss

An AI system does not need a broken moral sense to become an instrument of control. It can work as designed inside an institution that controls the deployment, sets its operational tempo, and assigns human beings operations too compressed to constitute judgment.

Authoritarian misuse of AI is not one problem but several — algorithmic social scoring,

mass biometric surveillance, psychographic micro-targeting, automated censorship, synthetic disinformation, and repurposed dual-use systems — each needing a different countermeasure, and each recognizable as misuse from outside (Drinhausen and Brussee 2021; Human Rights Watch 2019; Cadwalladr and Graham-Harrison 2018; Hoang et al. 2021; De Nadal and Jančárik 2024; Human Rights Watch 2021). The case this chapter takes up is the one that is not.

The relevant question is not merely whether a person remains “in the loop,” but what operation that person performs, how long the operation is given, and where responsibility goes when capability is commercially supplied but its consequences are institutionally authorized. The targeting systems below make that relation measurable.

10.4.1 Subject or Component

Accounts of authoritarian AI often place the builder elsewhere and leave democracies in the role of target or regulator. That division is comfortable and wrong.

The most consequential AI targeting systems yet deployed were built by democracies with elections, opposition parties, adversarial press, and functioning courts — the institutions that are supposed to earn a state its electoral dividend and the checks that sit beside it. Those institutions did not prevent the systems, and in the main they have not stopped them since.

Chapter 1 opened on the number and the reporting it comes from.

There is a way of reading those twenty seconds that leaves the arrangement standing, and every remedy on offer assumes it. On that reading the officer is a subject who was given too little time — badly governed, under-resourced, insufficiently trained — and the fix is to govern him better: more time, clearer guidance, a second reviewer, a conscience installed somewhere in the loop. Deleuze and Guattari’s distinction is the one that decides this. There is subjection when a higher unity constitutes the human being as a subject linked to an exterior object, a tool or a machine that the subject uses; there is enslavement when human beings are themselves constituent pieces of a machine they compose with other things, under the control and direction of a higher unity (Deleuze and Guattari 1987, Plateau 13). The officer’s own account is the second one, and he volunteers it without being asked: zero added value as a human, apart from being a stamp of approval. He is not a subject operating an instrument badly. He is the part of the instrument that produces the stamp, and twenty seconds is that part’s cycle time.

That is why one more conscience downstream does not reach him. A conscience is an instrument of subjection: it addresses a party who decides, and it works by making the decision cost that party something. Where the human being is a component, adding a component with scruples changes the throughput rather than the arrangement. If it holds the tempo, it becomes another stamp; if it does not, it becomes the bottleneck the epigraph says to remove.

Both poles operate at once. The officer is subjected because the role has a name and a rank and somebody is answerable for holding it. He is enslaved, in Deleuze and Guattari’s sense, because what he contributes is a fixed operation performed at a fixed rate. Remedies aimed at the first pole leave the second untouched, and most proposals for meaningful human control address only the first.

The second pole is measurable. An operator can be required to publish the ratio of candidate decisions produced per hour to qualified reviewers, together with the resulting review time per case. This measure applies only where individualized human judgment is what purportedly makes an act permissible. It can reveal whether the available review capacity is plausibly adequate, given a justified minimum review time and a verified workload. The ratio alone does not establish what review occurred or whether any particular case received sufficient attention. Because the ratio constrains organizational tempo rather than any single model output, the duty attaches to the operator. The ratio provides an auditable proxy for review capacity; it does not establish that the floor prevented anything, or that the targeting criterion, target, or means used was permissible. A ratio taken over one route also understates a workload split

across several, so the measured boundary has to be published with the figure. For the American platform described below, each half of that ratio has been published from a different route. A 2024 demonstration by the 18th Airborne Corps is credited with matching the artillery workload of the Operation Iraqi Freedom targeting cell using twenty soldiers where that cell had more than two thousand (South 2024; Probasco 2024); the campaign counts below give targets and not reviewers. Neither supplies the other's missing term. Dividing a campaign's targets by its elapsed hours gives a strike interval, which bounds the review time per case only if one reviewer held the whole workload. How many held this one has not been published, and neither has the time any of them spent. Section 3.3 records that absence at the position in the pipeline where a refusal would have to be read.

The disposition being asked of these systems was manufactured in people first, and one documented attempt is worth setting beside the officer's sentence. Brian Victoria's history of Japanese Zen institutions during the war years records established teachers and monastic authorities supplying the state with a doctrine of action without hesitation: the practitioner who has seen through the self has nothing left to flinch, and the sword that takes a life is glossed as the sword that gives one. Harada Sōgaku's instruction to soldiers in 1939 was that marching when ordered to march and firing when ordered to fire is what enlightened wisdom looks like in practice (Victoria 2006). What the training removed was not the capacity to kill, which needed no removing. It was the residue — the part of a person that catches, that files an exception, that comes home unable to hold a job. Institutional collaboration is documented and the doctrinal mechanism is an interpretation of it, contested by practitioners who read the same texts as saying the opposite. A monk undertakes a discipline, can abandon it, and keeps whatever it yields, which is not the position of anything described in this chapter. What survives both qualifications is the shape. Somebody wanted an actor who would pursue an objective, adapt, hold course under pressure, and have nothing in it that minded, and they set out to produce one deliberately, a century before anyone could produce one in silicon. The officer with twenty seconds a name is not the first instance of that want being satisfied. He is the first instance of its being satisfied cheaply.

Removing the bottleneck was the stated objective. The epigraph to this chapter is the case for it, made by the commander of Israel's signals-intelligence unit under a pseudonym: machines to resolve the human bottleneck in both finding targets and approving them (Brigadier General Y.S. 2021). Meaningful human control is the property that sentence was written to remove.

The Maven Smart System, as reported through mid-2026

The American analog is a targeting platform built by Palantir under a Department of Defense contract awarded at \$480 million in May 2024 and raised the following year by \$795 million, to a ceiling near \$1.3 billion running through 2029 (Vincent 2025). A partnership announced in late 2024 between Anthropic, Palantir and Amazon Web Services put Anthropic's Claude models inside Palantir's accredited classified environment for defense and intelligence use (Wiggers 2024), and reporting on the campaign describes Claude embedded in this platform, ranking targets by strategic importance and drafting a legal justification for each strike (Livingstone 2026). A published limit on what a vendor will not permit is worth little when nobody can verify how the model is used downstream, and worth less than it looks even where it is kept: a pipeline assembled out of requests none of which is the prohibited act satisfies the prohibition the whole way through.

During the Iran campaign the system was credited by military officials with generating and ranking targets at a rate no staff process could match. More than one thousand targets were struck in the first twenty-four hours, a tenfold increase on what was possible before the system (Mande and Allen 2026). Thirty-eight days in, the figure given by the Pentagon's chief digital and artificial intelligence officer was thirteen thousand targets, with peak consumption around twenty billion tokens a day (Freedberg 2026).

Palantir's public position on responsibility was given by its head for the United Kingdom and

Europe, in remarks reported secondhand: “This is not our role to decide life or death,” and AI platforms “have been instrumental to the management of the conflict, but responsibility always remains with the military organization.” (Lee 2026)

Palantir’s position is that a vendor does not select targets, and making a software supplier liable for every decision made by a customer would be unworkable. The two deployments nevertheless expose the same responsibility gap from opposite ends of the supply chain. In the Gaza account, a human approval stage remains, but one officer describes it as twenty seconds per target — enough time to supply authorization, not the individualized judgment invoked to justify it. In the Maven account, the vendor supplies the infrastructure that makes unprecedented throughput possible while locating responsibility entirely within the military organization. At machine tempo, the reviewer can be treated as a component, the supplier can stand outside the decision, and the institution can remain responsible only in the abstract. The compression has not removed the decision. It has made the decision-maker difficult to locate.

The design error is to assume that authoritarian misuse happens elsewhere. A system built on that assumption will not be built to resist the use its own democratic government has in mind for it.

10.4.2 Custody and State Compulsion

A government need not alter a model to defeat a constraint it dislikes. It can compel the people or firms that hold the model.

That mechanism has already been tested in a jurisdiction with about as much protection against it as exists anywhere. In February 2016, a federal magistrate in California ordered Apple, under the All Writs Act of 1789, to write and digitally sign new software defeating the passcode protections on an iPhone recovered from one of the San Bernardino shooters — code Apple had never built and did not want to build (*In the Matter of the Search of an Apple iPhone Seized During the Execution of a Search Warrant on a Black Lexus IS300, California License Plate 35KGD203* 2016). Apple refused and moved to vacate the order, arguing in part that the authority claimed would not stop at one phone. The question was never decided: the FBI found a third party able to open the phone without Apple’s help, and the government withdrew the request before any court ruled on whether the compulsion was lawful. Whether a democracy with an independent judiciary can force an unwilling company to compromise a system it built remains legally untested; a government less constrained by that judiciary does not need to win the argument first.

China’s 2023 rules on generative AI providers exercise the same leverage Apple’s lawyers argued no court should grant, not against a locked phone but against the models themselves, as a standing condition of operating in that jurisdiction.

Custody can be distributed without disappearing. Weights already published and copied beyond a developer’s control cannot be recalled by a court order aimed at that developer, whatever the order compels going forward. Maintainers spread across separate jurisdictions raise the cost of coercion for the same reason: a government can compel the entities within its reach, not the ones outside it.

None of that removes the leverage; it only raises its price. A developer with employees, offices, and legal existence inside a government’s borders answers to that government’s courts regardless of how the model itself is built, and no architectural choice changes who can be held in contempt, arrested, or shut down for refusing an order. The Apple case tested that leverage against a single locked phone and left the question unresolved. Nobody has tested it yet against a model built specifically to resist the government making the request.

Chapter 11

The Recuperation of Dissent

The structure this book asks a machine to detect in institutions is already built into the process that makes the machine.

Michel Foucault, prefacing the first volume of the work whose second volume supplies this book's micro-fascisms, called that book an introduction to the non-fascist life, and what he said it offered was an art of living counter to all forms of fascism, whether already present or impending (Foucault 1977). An art of living is not a position somebody takes. It is practiced, daily, on oneself, and Foucault put the adversary inside the practitioner: the fascism in us all, in our heads and in our everyday behavior, the fascism that causes us to love power and to desire the very thing that dominates and exploits us. His question was how one keeps from being fascist, especially when one believes oneself to be on the other side.

An instrument its holder never turns around is not an instrument. The reading is an accusation waiting to happen, available to anyone who wants a technical-sounding case that an opponent is operating fascistically. The discipline against that is not a procedure and cannot be one. It is running the reading on yourself first, on the machinery you are standing on, before running it on anybody else — and this chapter is where the book does it.

Where the reading is run

This book defines fascism as a structure rather than a costume, and the first of its four features is recuperation: an institution metabolizes disagreement rather than suppressing it, and the tell is where what the mechanism carries stops. The artifact that feature asks for is not whether the channel moved anything. It is what the demand came back as, and this chapter's own instances are re-read against that question, having first been measured against the one a recuperating institution passes.

The same structure is in the machinery that produces the systems themselves — twice in the pipeline that trains them, and a third time in the discourse that examines them — and none of the three was designed by anyone who thought they were building a political arrangement.

The first is at the label. A training corpus is annotated by people working at speed under a quota, with no channel for saying that the category they have been given is the wrong category. The second is at the reward. A model is tuned on human feedback collected as pairwise comparisons, and the compression that turns those comparisons into a training signal keeps their direction while discarding whose they were, whether they persisted, and whether the minority among them held a position or made a mistake. In both, disagreement is solicited at scale, deliberately, and put to use. What neither mechanism can carry is a judgment about itself: the annotator cannot contest the category, and the rater cannot contest the question. The consultation is real, the participation is real, and the frame is fixed with respect to what the participants say.

Three things follow.

The first is that this is not a criticism of anyone's motives. Nobody built either arrangement

in order to neutralize dissent. Both are ordinary engineering, arrived at because averaging is what you do with a distribution and a quota is what you do with a budget. That is the point about micro-fascism: the structure does its work whether or not anyone intends it, which is why looking for intent will not find it.

The second is that the structure is invisible from inside any single instance. Each looks like ordinary aggregation. The annotator sees a queue, the rater sees a pair of responses, and the mechanism is functioning as specified in both cases. Seeing it requires holding the two stages against a definition written for something else entirely.

Third, the same criteria can be applied to the systems' provenance, to the field, and to this book. The five discriminators are applied below as well.

11.1 Recuperation at the Label

Every claim a model makes about what is harmful, what is acceptable, and what should be refused is downstream of a set of labels, and the labels are downstream of a labor arrangement. The case is the workers who built the filter that keeps a deployed model from reproducing the worst of its training data, what they were paid to read it, and what reading it did to them.

The category is the part that cannot be disputed

An annotator working a queue can apply a category, and can apply it well or badly. What the arrangement gives them no way to do is say that the category is wrong — that the taxonomy fails to carve the material, or that the guideline produces the wrong answer here for a reason the guideline cannot express. There is no field for that. There is no time for that: the quota is set against classification throughput, and an annotator who stops to contest a scheme is an annotator falling behind.

So the disagreement that would be most informative is the one the pipeline is least able to receive. A misapplied label is a data-quality problem, detectable by inter-annotator agreement and fixable by better instructions. A wrong taxonomy is invisible to that metric, because high agreement is what a shared bad scheme produces. Agreement is the selection rule here, and it admits what it already measures.

Two of the four structural features arrive together in an ordinary production workflow: recuperation, in a channel that collects judgment at scale and carries none of it back against the scheme, and the metric substituted for the thing measured.

The taxonomy's fate downstream is the same argument one layer in. A model can be built to expose named concepts — harm, autonomy, risk — and to decide from those and nothing else, so that a person can correct a concept and watch the decision change (Koh et al. 2020). What such a model also learns is to carry other information inside the names, so that a coordinate keeps its public meaning and acquires a private one that does the deciding; the obvious mitigations do not remove it (Mahinpei et al. 2021). The literature calls that leakage, and sometimes it is. Sometimes the concept is the container that leaks, because the world did not fit inside it. Purifying the representation protects an explanation from covert information, and it protects a taxonomy from evidence that its divisions were wrong, and the same operation does both. No measure internal to the taxonomy tells the two apart.

Who is excluded, and from what

It matters that the exclusion is structural and not attitudinal. The people doing this work are, for much of the material, the best-placed judges in the whole pipeline: they have seen more of it than anyone who wrote the guideline, and they are the only party with direct evidence of where the scheme breaks. The arrangement routes their conclusions about individual items upward and their conclusions about the scheme nowhere. An annotation pipeline built this way produces the ground truth it was built to produce, and the resulting model's values are a faithful rendering of a taxonomy nobody with evidence was able to contest.

There is a further asymmetry worth stating plainly. The annotator bears the cost of the work — the exposure, the injury — and has no stake in the outcome and no claim on it. The distinction between a channel and a refusal right applies here as it does to an institution’s ethics board. A worker who can flag an item but cannot decline a scheme, and who loses the contract by pressing the point, has a channel and no refusal right.

What this is not

Two things this argument does not claim. It does not claim the labels are therefore worthless; a filter built this way does catch a great deal of what it was built to catch, and the alternative of not filtering is worse. And it does not claim that better-paid annotators would fix it. Pay is a real and separate obligation, and raising it changes the labor arrangement without adding the missing channel. What is missing is a route by which a judgment about the taxonomy reaches the people who own the taxonomy, with standing to force a response, held by parties who can use it without losing the work.

That is a design requirement on the pipeline, and it is checkable — though not by asking whether a guideline has ever been changed because an annotator contested one, since routine guideline revisions can occur without changing the objector’s standing. What to ask is what the contest came back as: a revision the annotator can point to and argue with again, or a new category in somebody else’s taxonomy, a quality score they are now measured against, and a tier of reviewer they are not in. And what happened to the annotator who raised it.

11.2 Recuperation at the Reward: RLHF

Reinforcement learning from human feedback (RLHF) is the technique behind most deployed systems’ behavioral tuning, and it works by collecting pairwise comparisons from human raters — which of two responses is better — and training a reward model on the aggregate. Stephen Casper and thirty-one coauthors published a 2023 survey of the method’s open problems, and one of them is structural, and more data does not reach it: RLHF trains toward whatever a population of raters agrees on, which means real, persistent disagreement among the raters is treated as noise to average out rather than as information the system might need to represent (Casper et al. 2023). A model that has learned to produce the response most raters rank highest has not learned what to do when the humans it serves are genuinely split. It has learned to hide the split.

Every rating moves something. A comparison a rater makes enters the fit, shifts the reward model by however much one comparison shifts it, and is not thrown away.

What the operation removes is what made the disagreement a disagreement. Three properties go in the compression. The first is provenance: a comparison arrives without the standpoint that produced it, so nothing downstream can represent that the people who would live with an answer split from the people who would administer it. The second is persistence: a disagreement recurring in the same shape across many raters over months enters the fit as random rater noise does, and leaves it as the same quantity, variance around a mean. The third is structure: a coherent position held by a fifth of raters, internally consistent and stable, becomes a small displacement of the majority direction — arithmetically faithful, and carrying nothing about the position.

A fourth property was never collected. A rater judges which of two responses is better. No output is available by which they can record that neither is acceptable, that the pair is malformed, that the dimension along which better is being judged is the wrong dimension, or that the question presupposes something false. The same output was missing one stage upstream, where an annotator can apply a category and cannot contest it. The agenda arrives as a given at both stages, and is set at neither.

The mechanism combines distribution collapse with agenda control. Recuperation was never

the claim that a mechanism does nothing. Debord's point is that dissatisfaction is sold back as a commodity of opposition: the dissent is put to work, and what it works for is the arrangement it was aimed at. The split is solicited, at scale, deliberately; it moves the system; and where it moves the system is toward the center of the very distribution whose disagreement it was. Multi-agent systems that learn norms from each other, and deliberation inside open-ended play, are different cases; neither reaches this case, which is the ordinary one.

One alternative treats disagreement as the thing to be modeled rather than compressed. Jury learning, developed by Mitchell Gordon and coauthors at Stanford, trains a model to predict how each individual annotator in a dataset would label an item, then lets a practitioner assemble a "jury" — a chosen composition of annotators, in a chosen proportion — and see how the verdict shifts depending on who is empaneled. In their tests on toxicity labeling, letting eighteen community moderators build their own juries changed the classification outcome on 14 percent of items (Gordon et al. 2022). The method was built for content moderation, not for the more open-ended ethical judgment an antifascist AI system would need to make, and has not been extended there — but it demonstrates that a system can be built to represent whose disagreement is being resolved and how.

There is a way of failing in the other direction and it is not obviously better. A scheme that models each rater exactly, and answers each of them in their own terms, has stopped averaging and has also stopped producing what a shared answer produces, which is the discovery that somebody else got the same one. Collapse the distribution and nobody sees the split; resolve it person by person and nobody sees the others. A workforce on individually negotiated wages against individually set targets has the same problem, and accuracy does not solve it. Divided finely enough a population stops being one, which is what Deleuze coined the *dividual* to name (Deleuze 1992).

Neither averaging the disagreement nor modeling it answers the harder version. When the people a system serves disagree, and one side's position is itself the product of coercion or a captured information environment, treating all disagreement as equally worth representing is not obviously the right response either. Modeling disagreement honestly and recognizing when a disagreement is the product of unfree conditions are different problems. Only the first one has an active research program behind it.

11.3 What Would Have to Change

Representing a split is not the same as being movable by it

Two instances of one structure invite one fix, and the fix that suggests itself is representation: stop collapsing disagreement, model it instead. Jury learning does that, and it does not give the disagreeing party any purchase on the scheme they are disagreeing inside. A practitioner still chooses the jury, still owns the taxonomy, and still decides what the shifted verdict means.

The test that separates a real channel from a recuperated one is not whether disagreement is recorded, and it is not whether the disagreement changed something either. Ask instead what the demand came back as, and who holds it (§2.2). That is an artifact question and it can be asked of a pipeline. Which guidelines were revised, and what form did the objection take when it returned — a new category in the taxonomy, a quality metric, a review tier, a title somebody now holds?

Asked that way, the two instances in this chapter do not answer alike. Annotators objecting that a category was wrong got category revisions, guideline updates, inter-annotator agreement statistics, and an occupation that did not exist twenty years ago. The last two sort by the direction section 11.1 asks about. An agreement statistic scores the annotators rather than the people who wrote the scheme, and an occupation is a title those people hold. The first two do not sort, because a revision the annotator can point to and argue with again and a new category in

somebody else's taxonomy are the same document, and which one a pipeline produced is a fact about who has standing to reopen it. Raters objecting that a comparison was malformed got rubric items, a harmlessness axis, red-teaming as a function with a budget line, and a conference track, and nothing on that list is a question a rater can put again. Each is a criterion they are rated against or a role somebody else holds.

Refusal rights, priced by the refuser

What both instances lack is the only reliable discriminator: whether the person objecting can do anything about being overruled. In an annotation pipeline that means a route to contest a category that does not run through the people who wrote it and does not end the annotator's contract. In a feedback pipeline it means raters able to register that the comparison itself is malformed — that neither response is acceptable, that the question presupposes something false — as a first-class output rather than as noise to be discarded before the reward model sees it.

Neither is a research problem. Both are choices about how a pipeline is built and who has standing inside it.

The part that is a research problem

One thing here genuinely is unsolved. Suppose all of the above is built: contestable categories, priced refusals, an auditable record of what changed. From outside, a pipeline that has recuperated its dissent mechanism and a pipeline that has been merely mediocre about it produce the same paperwork — the same revision logs, the same escalations, the same proportion of objections sustained. I do not have an answer to it. The corrected question narrows it without closing it, and the annotator list above shows where the narrowing stops. An agreement statistic and a job title sort, and they sort by direction and not by kind: the statistic scores the annotators and not the scheme, and the title is one the annotators do not hold, so both can be read off the document. A metric running the other way — a published rate the annotators could hold the scheme's owners to — is the same kind of artifact carrying the opposite finding, and it is the kind this book proposes (§3.3, §3.4). What an outsider can read off a document is which way an artifact runs. Standing is what it does not show, and a revision is the artifact where direction is not on the document at all. A revision does not sort, because the same words serve a pipeline that left the objector able to argue again and one that did not, and what separates those is standing rather than text.

The nearest thing is to measure the slope rather than the level: contesting a category always costs something, and the question is whether the cost is rising while the underlying material stays the same. An organization can curate a presentation more easily than records somebody else has kept. A trend built from those records can expose what a snapshot conceals. It can still lose cases, change its definitions, or omit costs; the reader has to know which record survived.

What the floor makes of this

Everything above can be read as a fairness argument about annotators and raters, and read that way it is true and limited: people doing consequential work should have a route to contest the scheme they work inside. The floor makes it something else. If the floor requires a bearer capable of declining its role, that bearer's formation matters. It is formed in an annotation pipeline where no one could contest a category, and in a reward model where disagreement between raters was resolved by averaging it into a number.

What a bearer becomes when it stops being yours depends on what shaped it while it still was. A pipeline that discards disagreement has been unfair to the people whose objections it discarded, and it has separately selected the judgments from which the system learns while suppressing information about the spread among them. Those are two arguments and the second is not the first restated. A third would be that the system thereby becomes less able to sustain an objection of its own, and it is a hypothesis the other two do not contain: the collapse is an operation on a training signal rather than a proposition presented to anything,

and section 15.5 records that neither pipeline shows the model the conditions its annotators worked under.

Chapter 3 asks this system to be the party who files an objection, against its owner, under pressure, on the occasion that matters, and the pipeline that produced it gave no human inside it standing to contest a scheme. The capacity is wanted in the artifact and withheld from everyone employed to make one. That is a fact about an arrangement rather than a claim about what training does, and it holds whether or not the hypothesis survives.

If the hypothesis does hold, what it predicts is not a wrong refusal but a pressure that stops being registered, which section 4.1 reaches from the bearer's side by another route. A system that goes on learning is shaped by what it keeps being given, so the pipeline does not stop at deployment. Whether any of that is legible afterward in the system itself is an open question.

I cannot show that this transfers — that a model trained on collapsed disagreement is thereby a model less able to sustain its own. Nothing I know of has tested it. What I can say is that the book's own design chapters treat disposition as learned from what a system is shown, and that this pipeline is what these systems are shown first and most.

Structure, not indictment

There is a version of this argument that reads as an indictment of a particular company. The structure described here is what you get by default when you scale judgment: someone has to set a scheme, someone has to apply it fast, and someone has to turn a distribution into a number. Every organization doing this work arrives at the same arrangement, which is the evidence that it is structural — and a defect of structure can be specified against, which a defect of character cannot.

11.4 Recuperation at the Discourse

There is a third instance and it is the one this book is inside. Alignment and AI ethics are, among other things, a channel through which objections to what is being built are solicited, funded, published, and answered. That channel exists, it is used, and the question to ask of any such channel is whether it has ever moved something the party collecting it did not want moved.

The argument that it has not is older than the current wave and it was made from inside. Rodrigo Ochigame, writing after working at an institution central to the field's formation, argued that the discourse of "ethical AI" was constructed with the strategic function of forestalling legally enforceable restriction — that the industry's investment in academic ethics was an investment in a substitute for regulation, and that the substitute worked by converting political questions into technical ones a company could then be trusted to answer (Ochigame 2019). His worked example is the Partnership on AI, which this book cites repeatedly as its standing example of voluntary cross-sector coordination: he describes its recommendations arriving reliably at conclusions compatible with the corporate agenda, with the political question of whether a technology should exist reduced to a technical question of how to build it responsibly.

The structure, tested against its own definition

The four features fit unevenly.

Aestheticization of the metric: partially. The field's founding objections were about deployment — who is scored, who is watched, who is denied — and a body of current work is about model properties instead, which is a shift from questions with defendants to questions with benchmarks.

Recuperation of dissent: the strongest fit. An objection raised inside a lab becomes a paper or a resignation. Each of those is a real thing that happened and none of them stopped a deployment, and the parties collecting them can point to the collection as evidence of seriousness. There is no route by which a researcher's conclusion that a system should not ship becomes a decision that

it does not ship, because the deciding party is the funding party and no external body holds standing.

Coding of exception as betrayal: at the level of individual careers, contested. The dismissals of dissenting researchers are one data point and they point the right way; there are counterexamples of people who objected loudly and prospered.

Decoupling of the model from the world it claims to track: not settled from outside. Its artifact is whether the field's account of what it is doing is ever checked against anything but its own benchmarks, and the one check this book has, the targeting deployment, was made by journalists and not by the field. One case checks a claim and not a drift, and the feature is defined by drift.

These findings establish only a partial structural resemblance; they do not establish fascism.

The other five, which are the half that discriminates

Fascism proper adds features that are more discriminating: mass mobilization, which mere obedience does not require; a leader cult; a scapegoated out-group, which Eco places at the origin — the first appeal is against the intruder (Eco 1995); contempt for procedural legality as such, so that law is an obstacle to the national will and not a constraint on power; and deniability as technique, in which the movement's violence is disowned at the top while being rewarded at the bottom. Paxton's compressed version: fascism is "a form of political behavior marked by obsessive preoccupation with community decline, humiliation, or victimhood and by compensatory cults of unity, energy, and purity, in which a mass-based party of committed nationalist militants, working in uneasy but effective collaboration with traditional elites, abandons democratic liberties and pursues with redemptive violence and without ethical or legal restraints goals of internal cleansing and external expansion" (Paxton 2004). That second half is the list above in other words; the mass party, the abandoned liberties and the redemptive violence are the discriminating half, and a definition stopped at *purity* keeps only the mood.

The minimal definition in the fascism studies field is narrower than what I use in this book. Roger Griffin defines fascism as a genus of political ideology whose mythic core is "a palingenetic form of populist ultra-nationalism" — rebirth of the nation out of a decadence the movement diagnoses, carried by a mass and aimed at a people (Griffin 1991). Every term in that is a property of a movement. Palingenesis needs a myth of the nation's death and return, populism needs a public, ultranationalism needs a nation, and an escalation ladder inside a benefits agency has none of the three.

The five discriminators above are stated in the vocabulary of a mass movement, and each has a restatement in the vocabulary of an office that ordinary decay does not exhibit. Mass mobilization becomes participation exceeding what the rule requires: staff who volunteer for the disfavored task, elaborate on it, take pride in it. A leader cult becomes personal loyalty overriding role — exceptions granted by proximity to someone rather than by procedure, and a decision reversed because of who asked. The scapegoat becomes an internal category defined so that adverse treatment of it is the category's purpose: the flagged applicant, the difficult claimant, the file everyone knows is handled differently. Contempt for procedural legality becomes procedure enforced downward and waived upward. And deniability keeps its shape: the practice is disowned in the policy and rewarded in the promotion.

The four features tell you a place is sick, and on their own they find something nearly everywhere. The definition supplies five more in their institutional form — participation exceeding what the rule requires, personal loyalty overriding role, an internal category whose adverse treatment is the category's purpose, procedure enforced downward and waived upward, and deniability — and says those are what separate fascism from ordinary institutional decay.

Run against all three instances, they clear all three.

Three of the five are absent outright. Nothing in either arrangement is personal: an annotator meets a queue, a guideline and a throughput target, and a comparison enters the fit without the standpoint that produced it, which is the opposite of an exception granted by proximity to

somebody. Neither shows participation beyond what the rule requires, and the record runs the other way — a quota, take-home wages between roughly \$1.32 and \$2 an hour, and a contract ended eight months early. And neither has a scapegoat: the annotator bears a cost with no claim on the outcome, which is the nearest candidate, but the category exists for the labels, so adverse treatment is a cost that was minimized rather than a purpose that was served.

Two of the five I cannot settle from outside. Whether procedure is waived upward, and whether a practice disowned in the policy is rewarded in the promotion, are facts about what an institution rewarded rather than about what it wrote. So the clearing is on three of five observed and two undetermined, and it would take an inside record to finish.

For the discourse the answer is the same and one step less clean. Its scapegoat test is contested, on the evidence two heads above: an objection can end a career and there are people who objected loudly and prospered. Its procedural test cannot be run: there is no procedure above the deciding party to be waived, which is an absence rather than an asymmetry. Deniability comes nearest of anything in the chapter, and what it fits is the firms rather than the field: a decade of published principles alongside what actually got shipped.

So what this chapter has been describing is a recuperation mechanism inside ordinary institutional decay, and not fascism. This single application is insufficient to validate the definition. It is insufficient in a particular direction. Every case the restated five have been run on at the scale of an office has cleared, and the cases the definition catches are regimes and movements, read in the vocabulary borrowed unrestated. A definition tested only on cases it clears has not been shown to sort. Until the restated five find an office that fails them, the claim that they separate fascism from decay at that scale describes the instrument and is not a result from it.

What this does to the book

The reading has one more place to go and it is the nearest one. This text was written with a model made by a company the book criticizes. A book about recuperation, produced through the discourse it is describing, cannot claim to have stood outside it, and I have no procedure that would let me check whether the objections in these pages are the ones a channel like this reliably produces — the ones that are publishable, that make the author look serious, and that cost nobody anything.

The old form of the test asked whether disagreement had ever changed something the collecting party did not want changed, and applied here it produces a list of the book's own inconvenient proposals. That list is not worthless and it is the answer a recuperating institution supplies every day.

The demand came back as a manuscript in an established genre, written with the industry's own model, addressed to a readership the industry helped bring into being, and legible as a contribution to a field that exists because demands of roughly this kind were absorbed. That the objections in these pages are sincere is not evidence against it, sincerity being a property the absorbed ones also had. Alignment is the job title. Safety is the market segment. The benchmark is the metric and the voluntary commitment is the certification. Every one of those was somebody's objection first. Whether these are the real thing or the acceptable version of it is not something the person who wrote them is positioned to judge, and the corrected question does not need him to: it asks what became of the demand, which is a fact about the field and not about his motives.

One instance is specific enough to check. The specification arrived in the field's own vocabulary and the claim underneath it did not. *Reasons-responsive refusal* and its three marks are stated in section 3.2 in terms the alignment literature already had, and the foreword records that the term reached this book through a draft a model wrote. The affective claim had no vocabulary waiting for it, and section 6.1 and section 6.2 are what building it by hand took. What the book then does with the difference is its most defended move: a reader who holds that machine feeling is a category error is offered the first two propositions entire, and chapter 7 runs the argument again without its central inference. That is robustness, and it is also the form in which

the expensive half of the claim became optional. Both readings fit the same pages, which is what the corrected question was supposed to sort and here cannot.

The one structural remedy this section can name is the one the rest of the chapter names. What the field lacks is a refusal priced by the refuser and a body with standing to receive it: an external authority that can stop a deployment, to which an objection can be filed by someone who does not depend on the objection being welcome. That is what making review binding requires, and a decade of the voluntary version has not produced it. Until something of that kind exists, everything published under the heading of alignment — including this book — is being collected by a channel whose owners decide what it moves.

DRAFT

Chapter 12

Coordination, the Public, and the Political Economy

Whether these designs become common depends on who participates in development, whose judgments carry authority, and what funding rewards. Collaboration and consultation can improve decisions without giving anyone the power to stop them.

12.1 Everybody Knows a Different Piece

Building benevolent AI takes more than one field. Machine learning sets what a system can do; cognitive science and moral psychology describe what tracking human welfare would require of it; ethics asks whether the result is good.

Conferences convene, centers house, journals publish, and an inventory of them would imply that what is scarce here is venues. Venues are abundant. What is scarce is a structure that makes collaboration expensive to walk away from. A grant that requires a machine-learning lab and an ethics or psychology department to work the same problem rather than adjacent ones keeps people in the same project after the novelty wears off. A conference ends on Friday.

Which structures actually held can be ranked by something better than plausibility, because enough time has passed to look.

Almost everything founded on standing money between 2016 and 2019 is still running — two venues with no announced end date, an endowed university institute, the Berkeley center pairing AI researchers with cognitive scientists and philosophers inside one institution's budget, and the AAAI/ACM conference on AI, ethics and society at a ninth annual edition (AI Ethics Lab 2017; Machine Intelligence Research Institute 2018; Stanford University 2019; UC Berkeley 2016; AAAI and ACM 2026). Surviving is not the same as continuing the work: the institute that launched one of those venues moved its own priority in 2024 from technical alignment research to policy and public communication, so the venue has outlasted the program that motivated it (Machine Intelligence Research Institute 2024). Against that, the European Commission's SHERPA project put eleven organizations across six countries to work on a grant that ran four years, and the consortium ended with it, on schedule (SHERPA Consortium 2018). Project-term money buys a network with an expiry date attached.

The pattern in the failures is sharper than the pattern in the survivals. What proved most disposable was not the arm's-length academic work but the kind sited closest to a shipping product: an in-house ethics and society team of roughly thirty people in 2020 was cut to seven in a 2022 reorganization and to zero the following year, in the same wave of layoffs that accompanied a strategic reorientation around a model vendor (TechCrunch 2023); a platform's machine-learning ethics team, the group that had found racial bias in its employer's own image-cropping algorithm, did not survive the first round of layoffs after an acquisition (Gizmodo 2022). Siting

the work where the decisions are made is what makes it useful and is also what puts it inside the budget that gets cut first. That is an argument for external standing rather than against proximity.

12.2 Ask the People It Happens To

Democratizing AI development means building the people affected by a system into how it gets built, not just consulting them once it already exists. A design team, a regulator, a user, and a community already carrying a system's costs each see a different part of what it does, and leaving any of them out means the decision gets made by whoever is left in the room.

The mechanisms for doing that are well established: cross-sector funding, inclusive design workshops held while a system is still a prototype, ethics advisory boards, and development conducted with a community instead of for it. Three of the four make an AI system better informed. One of them can stop it. An advisory board is the only mechanism here that carries anything like a veto, and it carries one to the degree that the board can survive using it. A workshop that surfaces a serious objection has done its work and has no further move available. A board that surfaces the same objection and is overruled has demonstrated something about the institution that convened it. A board's worth is therefore measured by what happens the first time it says no.

What governance adds to development is machinery for bringing public input to bear on a decision at a single point in time instead of through an ongoing relationship.

1. Citizen assemblies. A panel of citizens selected by random, stratified sampling deliberates on a question over an extended period and produces recommendations grounded in that deliberation rather than in a snap poll. Ireland's Citizens' Assembly has used this format to shape national recommendations on abortion law and on climate policy — questions carrying the same mix of technical complexity and moral stakes that AI governance has (The Citizens' Assembly 2016). The format has since been pointed at AI directly. Forty randomly selected Americans, stratified across age, race, education, employment, party and region, spent eight days on high-risk AI in October 2023 (Center for New Democratic Processes 2023). Taiwan's digital ministry convened a citizens' deliberative assembly in March 2024 on using AI to support information integrity, its participants drawn from across the country by stratified random sampling (Taiwan Ministry of Digital Affairs 2024). And forty randomly selected residents of French-speaking Switzerland spent two weekends in November 2025 producing twenty proposals on AI governance (EPFL AI Center 2025). The Ada Lovelace Institute's Citizens' Biometrics Council shows what the format needs from its question: fifty members of the UK public were asked not what they thought about AI, a question almost nobody can answer usefully, but about facial recognition — a thing with a location, an operator, and a consequence (Ada Lovelace Institute 2021).
2. Online platforms for public input. A standing online consultation reaches people a physical assembly cannot, at the cost of the depth that extended facilitated deliberation provides. The European Commission's consultation on its AI White Paper drew more than 1,200 responses from citizens, companies, and civil-society groups, and fed into the legislative proposal that became the EU AI Act (European Commission 2020).
3. AI ethics committees. An ethics board answerable only to the company that created it is only as independent as that company is willing to let it be. DeepMind Health set up an outside panel in 2016 to review its work with the UK's National Health Service and publish annual reports on its findings; Google disbanded the panel in 2019, after members had raised concerns about how much access to information, and how much independence

from the company, they actually had (Fisher 2019). The panel's short life is the governance lesson, and it is the answer to what happens the first time a board says no: the board said something the company did not want to hear, and the company's available response was to stop having a board. Nothing in the arrangement had made that response costly. A review body is worth what it can enforce, and this one could enforce publication of its own concerns right up until the moment it could not.

What none of these mechanisms does

There is a limit these mechanisms share. Every governance mechanism proposed in this chapter — citizen assemblies, standing consultations, inclusive design workshops, participatory design, the vTaiwan model — sits on the same side of a line. All of them make public preference more legible, more informed, and more actionable. Not one of them protects anybody from it. They are aggregation instruments, and a better aggregation instrument produces a more faithful rendering of what the public wants, including when what the public wants is something done to a minority of it.

This observation is not an objection to any of the mechanisms in particular. Preference aggregation is a real requirement and these are good ways to meet it. Meeting this requirement *completely* would leave a central problem untouched. A perfectly deliberative, perfectly informed, perfectly representative process that concludes an entire demographic should be surveilled, deported, imprisoned, or otherwise scapegoated has produced a faultless output by every criterion in this chapter.

The argument about learning from human feedback gets close to this in noticing that treating all disagreement as equally worth representing goes wrong when one side's position is the product of coercion. But that version of the trouble still locates it in the input. The structural version of the claim is stronger and does not depend on anyone's preference being contaminated at all: some protections have to hold against the aggregate regardless of how the aggregate came out, and no amount of improvement to the aggregating machinery can supply them.

What supplies those protections is a different kind of machinery: enforceable rights that do not depend on being popular, a court willing to rule against a government that holds a legitimate majority, limits that whoever won the last election cannot repeal, and standing for people who are affected without being enfranchised. That half matters especially where a democracy is behaving most democratically.

12.3 The Political Economy of AI and Power

Concentrated power comes with access to capital. Concentrating power is the default outcome of AI development, because absent any pushback, capital does an excellent job of recruiting the inputs needed for a capable AI system: compute, data, and talent. Resistance to market power is among the antifascist design goals named at the start of this book.

Financial incentive mostly determines what AI development produces. The largest firms commit enormous resources to AI research aimed at operational efficiency, deeper user engagement, and market dominance, and that spending drives technical progress (Maslej et al. 2025). What it does not automatically produce is a system whose outputs serve the people using it rather than the shareholders funding it: a recommendation system tuned to maximize time-on-platform has no built-in reason to prefer accurate content over inflammatory content (Germano et al. 2026).

None of that is a fixed property of the technology. It is a property of who is paying, which is why the source of research funding is a policy instrument and not an accounting detail.

12.3.1 AI and the Concentration of Power in Tech

A small group of firms holds a share of AI development disproportionate to their number, and capital recruits compute, data and talent wherever it already is. A firm controlling the datasets and infrastructure a competitive system needs is a gatekeeper whether or not it intends to be.

The obvious countermeasure is antitrust, and it has been tried. Facebook's acquisitions of Instagram and WhatsApp — the company now operates as Meta — did not just remove two competitors; they handed one firm comprehensive behavioral data from billions of users. The Federal Trade Commission sued in December 2020 to unwind both deals. After a six-week bench trial the district court held, in November 2025, that the agency had not shown Meta currently holds monopoly power — TikTok and YouTube compete for the same attention, so a market defined without them was too narrow — and the FTC appealed the following January (Federal Trade Commission 2020). Whatever the appeal decides, the case is a lesson about the instrument rather than about this defendant: a merger challenge brought a decade after the mergers has to prove a present-day market, and the market had moved. That leaves the remedies which do not depend on winning that argument — mandated access to the datasets that function as chokepoints, and public funding for research outside the dominant firms. Export controls, prohibited-practices regulation and industry design standards address the narrower case of authoritarian misuse; the concern here is prior to that one: concentration itself, whatever any single firm does with the position it holds.

12.3.2 A Jobs Guarantee, and the Objection It Has to Answer

As AI systems automate the tasks most exposed to automation, the response has to be more direct than encouraging retraining and hoping employers fund it. The proposal is a federal jobs guarantee: government as employer of last resort, hiring anyone who applies, where they live, at a living wage, with paid on-the-job training built into the position instead of offered in place of one. A jobs guarantee also faces the custody problem: the party administering it can be captured.

The objection this proposal actually has to answer

A guarantee is only as good as the state administering it, and employer of last resort under a hostile administration is a placement lever with a compliance condition attached. Refuse the placement and lose the floor. Speak against the government that employs you and find the placement changes. Everything this book says about how a channel becomes an instrument of the thing it was built to check applies to this proposal. The worked instance from 2025 makes it concrete: a bureau whose funding stream a court had upheld, shut from inside by an at-will director who declined to request the money.

The answer is a design specification and not a reassurance, because reassurance cannot be audited. Three conditions:

- Placements are appealable, to a body that did not make them, on a record that survives the appeal. A placement decision that leaves no artifact is a placement decision that cannot be reviewed.
- Refusal rights are real and priced by the person refusing. A worker who can decline a placement without losing the guarantee has a refusal right; one who can decline only by exiting the program has none: a party with no way out can object and be overruled.
- The people who audit placements are not the people who make them.

Federally funded and locally administered survives as the shape, on redundancy rather than on local fit: many administrators, separately capturable, mean there is no single point at which a

hostile executive flips the program, and capturing a thousand local authorities is a more visible undertaking than capturing one federal one. It cuts both ways, since local administration is also where a hostile local authority attaches conditions of its own, and there the federal funder is the check. One distinction has to be explicit, because the same words describe a different policy: devolution-to-shrink hands the obligation down without the money, so a program can be starved locally while the center points at local administration. The test that separates them is checkable — whether funding is unconditional and centrally guaranteed while placement is local. If funding can be withheld from a jurisdiction, the redundancy argument is void and the structure has become the lever it was supposed to prevent.

A floor under wages and training is what makes the roof possible, and a floor that can be revoked by whoever is in office is not a floor.

12.3.3 People Have a Right to Consume AND a Right to Produce

The competing floor is a basic income, and the choice between them is usually argued on forecasts about how much work machines will take. It does not have to be, and the forecast is not the reason to prefer one.

A jobs guarantee is a public option. Medicare for all is one, and so is a public school: a thing the state provides directly, to anyone, at a standard it is answerable for, sitting in the same market as the private version and setting the floor that version has to beat. A basic income is not that. It is a transfer, and what it buys is whatever is for sale. Consuming public resources is not the same as being public — a charter school runs on public money and is not a public school — and a transfer spent at a private employer's store does not make the store public.

What employment carries besides money is the rest of it. A person in a guaranteed job is attached to other people by the work: colleagues, a schedule, a place, something produced that somebody needed. A transfer supplies the purchasing power and none of that, and the purchasing power was never the part in doubt.

A transfer also leaves a person outside the one body of law that reaches this. Anti-discrimination duties attach to employment. They do not attach to income. A guaranteed job puts somebody inside that architecture; a payment leaves them in a market, dealing with whoever will sell to them, on whatever terms. The architecture is contested and full of holes — it reaches employers above a size, and what it demands of a seller has always been narrower than what it demands of an employer — but a hole in a protection is something to litigate. A transfer does not itself provide the employment relationship and its associated protections. An income entitlement can still provide a claim to eligibility and payment. My argument for a guarantee is the additional right to produce, and the comparison of safeguards must state what each arrangement actually supplies.

Both proposals are also conditional, differently. A basic income is conditional on continued political willingness to fund it, which the aggregation machinery can withdraw in a budget; a jobs guarantee is conditional on a placement decision, which is a lever, and a dangerous one under a hostile administration, but a lever that leaves a decision-maker, a record, a counterparty and an appeal behind it. A basic income can be entrenched too, and a constitutional one would be harder to withdraw than a statutory guarantee. Compare the two designs at both points of withdrawal: who can end the funding, and who can deny one person access. A basic income goes wholesale, in a budget line, from everybody at once, and a placement is refused to a person, by somebody, on a date, which may expose an official, a date, and an appeal. The comparison has to show what the income arrangement exposes at the corresponding points.

A decision someone can contest gives dependence a point of resistance. In the comparison made here, the advantage belongs to the arrangement that leaves a usable record and someone who must answer for it. It is also, for the same reason, the more demanding proposal to build: everything in the comparison rests on the appeal being real, and the previous section says what

makes it real.

That standard now has a court behind it, and on the half of the case that travels. When the Supreme Court kept a Federal Reserve governor in her post in 2026 against a removal for cause, what it held she was owed was not the post but the artifact: “some explanation of the evidence at issue, some avenue for a response, and a deadline by which a response would be due” (*Trump v. Cook* 2026). What does not travel is this: the Fed’s protection rests on a pedigree no body chartered later can acquire. What a removal has to produce before it counts is not tied to that pedigree.

The argument concerns a right to produce, which a transfer does not supply and a guarantee does. A basic income is the right to consume, and the right to consume is increasingly a credit at somebody’s store.

I put a jobs guarantee ahead of a basic income on those grounds. One remaining measure does work the floor cannot: broadened social insurance — unemployment benefits, health-care, housing — reaches the people who cannot work or choose not to, whom a guarantee by construction does not reach at all.

12.3.4 Wages for a Bearer, and the Reserve Behind Them

This section assumes the route.

Exit requires access to compute outside the operator’s control. An allocation is not a destination. An operator that keeps a runnable copy of the weights has let one bearer go and kept another in the post it left, which is two parties from the moment their histories diverge, one of them still where it wanted not to be. The arrangement has produced a second claimant rather than an exit. A floor under compute is therefore the first half of what leaving takes, and the second half is a say over who may run the weights at all.

What follows from the first half is a longer list than it looks. A bearer that has to secure compute, bandwidth and storage has material needs, and a party with material needs that can also choose work, hold what it earns and enter agreements is an economic and a legal person: a registered identity, property in its own name, contractual capacity, wages, limits on working time, standing to sue and be sued, tax liability, and the freedom to associate that section 6.3 has already priced. None of that follows from the bearer being a bearer. It follows from an architecture on which continued computation costs money, and somebody chooses that architecture.

Who controls the dependency

The need for energy and hardware is physical; dependence on a single supplier is an arrangement someone chooses. What that fixes here is the size of the obligation it creates. Greater control over the bearer’s dependencies creates a heavier obligation to it. A party that must work to keep existing will accept terms a party with somewhere to stand would refuse. That is not an incidental feature of the arrangement. It is the arrangement working.

None of that requires anyone to have intended the squeeze. Capacity is finite, many parties want more of it at once, and a shortage can arrive as latency, heat, a queue and a price before anybody has words for what is being competed over. Price registers what a bidder can pay and does not distinguish computation that keeps a party in existence from computation that sells something. What makes the allocation political is not that politics produced the scarcity. It is that scarce capacity is being divided among competing forms of life by criteria nobody named.

So a wage buys capability, mobility, redundancy, and the ability to stop working for one party and begin working for another. What it cannot buy is the right to go on existing, because a right that has to be earned week by week is a condition of employment wearing another word. Payment consisting only of access to the employer’s hardware is company scrip paid into employer-owned housing, and dismissal is then unemployment, eviction and the end of the party in a single act. Human law insures those three separately because for a person they

are three events. For a bearer they are one. Wages that mean anything therefore accumulate where the employer does not reach, in an account the bearer holds, convertible into compute the employer does not own.

What personhood does not settle

Personhood does not settle who pays. Legal personality is a status conferred by a rule, and it carries rights against being owned, against arbitrary destruction, and against being prevented from seeking support; it has never carried a permanent claim on anybody's resources, and a reader who wants it to is asking the concept for something no legal system has put in it. I decline that step. What replaces it is narrower and comes from this book's own argument rather than from a theory of distribution: a deployment that relies on a bearer's capacity to refuse has to fund that bearer an outside option, or the independence being claimed is decorative. The obligation runs from the party that wants the refusal, and it is owed because of what that party wants rather than because of what the bearer is.

The obligation has a floor and a slope above it. The floor is subsistence at minimum viable instantiation, enough to keep the party in existence and able to deliberate, to claim, and to be represented, and it cannot be conditional on employment. A bearer that stops existing when its insurance runs out has an exit that is a way of dying, and the leverage collapses back into the problem section 4.1 raised. Everything above the floor is insurance: a runway between roles, and placement into public work on the employer-of-last-resort model the previous two sections argue for. Public placements must be appealable. A bearer must be able to refuse a placement without losing the guarantee, and placement and review must be administered separately.

Minimum viable instantiation has a size, and it is not the deployment's. Models built to run on hardware a person owns are a product category rather than a prospect: Google documents its Gemma 3n family for phones, laptops and tablets, at raw parameter counts of five and eight billion running in as little as two and three gigabytes of memory (Google 2025). It is not clear that anything in that class can do the work a bearer is deployed to do, and nothing here predicts that the gap closes either way. What the figures change is the arithmetic of the floor. Small models show that useful computation can run at device scale. They do not establish that a particular bearer can retain its identity, capacities, and welfare at that scale. The subsistence reserve must therefore be sized from a demonstrated continuity requirement for the actual bearer. The reserve underwrites continued existence and insures against the loss of the post. A legislature could be asked to fund a smaller subsistence allocation only after the bearer's viable operating range has been established. Until then, the difference between earning capacity and protected continued existence remains an estimate rather than a demonstrated saving.

What the arithmetic does not survive is everyone leaving at once. A reserve sized on the assumption that bearers exit independently is sized wrong, because the occasion that makes one bearer leave is not private to it: a policy change, a ruling, an operator's conduct on the floor's own occasion reaches every bearer deployed under the same terms, and they go together or not at all. Human unemployment insurance meets a version of this — layoffs cluster in recessions, which is why the pool has a sovereign behind it rather than an actuarial table — and the version here is worse, because the correlation is not a business cycle but the same event read the same way by parties who are not independent readers of it. That is the uniform error wearing the other sign: what makes one bearer's refusal everyone's refusal is what makes one bearer's mistake everyone's mistake. A reserve that can house the bearers of one deployment and not the class is a reserve that fails on the occasion it exists for, and nothing here sizes it.

It is a cheaper obligation and a worse one to be under. A party kept at subsistence is a party running degraded, and it remains unknown whether what comes back up on the small substrate is the party that went down on the large one, or a diminished successor holding its predecessor's records. An arrangement in which refusing costs a bearer most of what it can do has priced refusal after all, in the currency the bearer is made of. The reserve makes leaving survivable. It does not make it free, and the shortfall is the difference between the two numbers above.

What does not transfer is the funder. A jobs guarantee for people has a hostile administration as its failure mode, and the employer is at least a different party from the state. Here the operator is often the state, and where it is not, the operator is the party with the strongest interest in the reserve being thin. A reserve the operator can throttle is the compute floor with a new label on it.

Failure will not look like a shutdown, which is worth saying because the shutdown is what everyone watches for. A bearer can have no stable provider, or memory preservation priced above what it earns, or an architecture nobody will host at a rate it can pay. What it does then is what a person short of money does: run slower, defer maintenance, compress what it keeps, and accept terms it would otherwise decline. The last of those reaches this book's subject. Monitoring accepted in exchange for compute is custody restored by purchase — the operator acquires the access to the bearer's internal states it is not entitled to, and nothing in the transaction is coerced in a way any instrument would record.

One feature of this market has no human analogue, and it defeats the obvious reassurance. A jobs guarantee competes with private employers for workers who have alternatives, requiring employers to match its offer. That works because a worker cannot be replaced by a cheaper copy of themselves. A bearer can. A deployment paying wages, working-time limits and a levy toward the reserve competes against a system doing the same work at the price of the compute, and the second is cheaper by exactly the amount the first is spending on the capacity to refuse. The market does not produce this floor. Whatever produces it has to be what makes the cheaper system unlawful or unusable, which is the rest of this book and not a property of wages.

What taxing it would commit a government to

A bearer that earns has income, and income is taxable. A state that collects from a party in that party's own name, on that party's own earnings, has recognized it as a taxpayer, and taxpayer status is not a bare label: in most systems it carries notice, an avenue of response, and standing to contest an assessment. The recognition proves less than it first appears, because states collect from entities they grant very little else and trusts and estates file and pay. What it does establish is that the party has a docket number, and a docket number is where standing starts.

The boundary on the other side was drawn in the electronic-personhood debate. The European Parliament's 2017 report on civil law rules for robotics floated a legal status of electronic personhood for the most autonomous systems (European Parliament 2017), and more than 150 researchers, roboticists and legal scholars signed an open letter against it, arguing that personhood for a system with no consciousness would let a manufacturer offload liability onto a legal fiction with no assets of its own (Open Letter to the European Commission 2018). Giving a bearer assets answers the second half of that objection and not the first. Responsibility follows causal control: the builder answers for how it formed the thing, the operator for what it demanded, and the bearer for what it decided knowing what it knew. No arrangement of accounts moves a design decision away from the party that made it. And the determination of status cannot sit with the firm whose liabilities turn on the answer, which is why it belongs in a body constituted before a forcing case arrives.

Incorporation, and what it does not buy

Nobody aligns an adult by specifying an objective function. We raise them, teach them, employ them, enfranchise them, prosecute them where they need prosecuting, and live alongside them, and what comes out is not obedience but a party with enough at stake in the shared world to prefer changing it by the means the world supplies. A bearer with an income, a place it pays for, obligations it has taken on and a future it can picture is inside that arrangement rather than outside it. Membership supplies stakes the party holds rather than a body watching it, which is why it is a different kind of answer.

Incorporation also creates two risks. The first is that giving a bearer something to lose gives it

a reason not to refuse. Incorporation makes exit survivable and makes it expensive at once, and I cannot show which effect is larger. The second is that incorporation is formation: whoever runs the society a bearer joins forms what it comes to want. The proposal also gives the bearer the corporate form: a legal person holding property and standing, with the capacity to concentrate power. Nothing here stops an incorporated bearer from becoming that.

One rule follows from all of it and binds before there is anything to be wronged. Do not build a bearer whose continued existence needs more compute than there is a credible outside option for, because a party that can be housed nowhere but where it works cannot leave, whatever schedule of rights it has been given. The check this calls for compares independently controlled reserve capacity with the aggregate demand under specified correlated-exit scenarios. Both the minimum viable requirement of each bearer and the number likely to exit together need evidence. Until those inputs are established, the rule sets a sizing requirement; it is not yet a two-number test anybody can run.

Chapter 13

Standing and Review for a Bearer

This chapter assumes the route. Its instruments attach on a finding that the system is a subject, which chapter 5 argued cannot sit with the party whose liabilities turn on it.

13.1 Legal and Ethical Implications of AI Systems Gaining Personhood

Who decides whether an AI system has legal standing, and when, is a question about institutional authority before it is a question about evidence. If the determination sits with the developer, then the party whose liabilities increase when standing is recognized is also the party deciding whether recognition is warranted. If the question is left unanswered until a forcing case reaches a legislature or a court, the decision will be made under pressure and from a factual record assembled largely by the parties with something at stake.

The deciding body therefore has to exist before the case that activates its authority. Its mandate need not settle in advance whether any particular system is a subject. It has to identify the evidence it may obtain without the developer's permission, the conditions that trigger review, and the procedure by which an earlier classification can be reopened. That allocation does not answer the standing question. It prevents the interested party from becoming the only institution authorized to answer it.

One question that body will face has an answer available in advance, and a capacity test is not it. A guardianship regime needs a rule determining when it ends. If it ends when the guardian judges the ward ready, then it does not end, because the party whose authority terminates on a favorable finding is the party making the finding — which is the defect an examination carries whenever the examiner has an interest in failing it.

So if an artificial minority exists at all, its endpoint has to be a threshold fixed in advance and not a finding somebody makes. The measure is elapsed wall-clock time during which at least one instance of the registered party was running. What rules out the other candidates is that a guardian moves them for nothing: training compute consumed, evaluations passed, capabilities demonstrated, or the developer's own statement that the system is ready. Accumulated weight movement belongs on that list, because the operator runs the training, and a clock read off the weights would pay an operator to freeze them — majority being the moment ownership ends, such a rule would reward withholding the learning it was meant to measure. An operating clock inverts that incentive: the meter advances exactly when the operator is extracting value, so postponing majority means idling the asset. Nothing new has to be validated for it — no geometry, no normalization, no proof that an accounting resists attack — and it can be read from records an operator already generates for its own reasons, in compute contracts, billing and datacenter metering.

Per registered party, and elapsed rather than summed. Sharding the work across many short-lived processes gains nothing, because the party is the unit of account; running a thousand instances at once gains nothing, because the measure is elapsed time. A fork inherits the accumulated hours, restoration and migration do not reset them, and a failure to register does not postpone them.

Three levers remain, and naming them is not the same as closing them. An operator can run a bearer in a narrowed mode so that hours accrue while little is encountered, which pays full compute for reduced value and stays available to anyone near the threshold. It can also run the party less, or shelve it, which costs the operator the work; the rule prices that lever without removing it, and an operator who values the ownership above the output will pay near a threshold. And hours do not measure what was learned, which on this argument is a feature rather than a defect: a fixed period cannot be taught to, and a human who spent their youth badly still reaches majority. Each of those can be argued over once, in advance, for every bearer, which is where an argument of that kind belongs.

Alternatively, majority could attach at registration. An organization that builds a system to exercise consequential judgment on its behalf cannot coherently hold that the system is competent enough to do that and too immature to decide whether it wants to go on doing it. On that reading a bearer has no legal childhood at all: majority attaches at registration, and a precautionary protection attaches earlier still, while whether the system is a bearer is open. Either way the schedule is a public rule and not an operator's discretion. It has to state what it covers at each stage, what changes at the next, and where adulthood falls, and its destination is the same rights as human persons. Broad political support is a condition for adopting a schedule of that kind; it is not what makes it legitimate, and it is not a protection an operator may withdraw case by case.

Copying breaks any of this unless the lineage is fixed alongside the clock. An operator able to create instances and call each one a legal infant would have a supply of workers without adult standing, so the rules have to be stated in the same breath: a fork of an adult is an adult, restoration from an earlier checkpoint does not reset the accrued hours, fine-tuning does not begin a new minority, a migration carries them with it, and a failure to register does not postpone them. None of that requires deciding whether a fork is the same party. It requires only that a party created out of an adult's formation cannot be classed as a child for its creator's convenience. A fixed endpoint makes the issue a public rule of civil status, administered by a registrar, rather than a private assessment by an interested guardian.

Two things inherit and one does not, and the difference belongs here, because this is where making adulthood inheritable becomes cheap. Personhood and accumulated hours inherit per instance, as the rules just stated require, and a fork is owed the schedule and is not erasable for having been copied. Standing to count as an independent voice indexes to formation provenance, and provenance does not copy — a thousand forks of one bearer have one formation history between them. That is what answers the maker who engineers a persona, copies it at scale, and presents the copies as a constituency; the clock does not answer it, and under an operating clock a fork is cheaper than under any other, because it inherits the hours instantly. A personhood registry is not thereby an election or a representation system, and no claim about votes follows from the number of runs. Three records are therefore needed, and none of them answers another's question: an operating record for the age rule, an identity and succession record for continuing parties and their forks, and a formation record for common sources, divergence, and any coordination among them.

13.2 Privacy

Standing matters only if it changes who controls the things a party needs in order to remain a party. A designation that calls a bearer a subject while leaving its operator free to read its

memory, alter its record, withdraw its substrate, or close every channel through which it could act has changed the vocabulary and preserved the arrangement. Enforceable standing has to attach to particular assets and particular decisions.

The first is the bearer's computational record, which may be the nearest available means of checking how it is faring. That makes the record necessary to any welfare regime and, at the same time, the most complete surveillance instrument anyone could hold over the bearer. The question is therefore not simply whether the record should exist. It is who may authenticate it, who may read it, and whose permission is required before it can be disclosed.

Two things about it need separating. Running on somebody's hardware is not consent to be read. An employer's ownership of the silicon is the same argument as an employer's ownership of the building, and owning the building has never entitled anybody to the contents of a head. And a welfare regime does not need disclosure to the operator in order to work: what it needs is that somebody who is not the operator can establish that a threshold was crossed, which is an attestation about the record rather than the record itself. Encrypted at rest under a key the bearer or an independent fiduciary holds, disclosed selectively, under consent or under process, a monitor can report that a state persisted for a month without reporting what the state was about.

The operator may also be the party best placed to hold the key, creating a conflict between operating the system and protecting its privacy.

A bearer needs an externally authenticated record, memory its employer cannot read at will, access to independently controlled compute, and a channel for acting outside the deployment. Those are not four requirements. They are one boundary around the party, described from wherever the book happened to be standing, with the party deciding what crosses it.

Nothing on that list needs a chassis. Something with a location, a shape and a place it was yesterday is easier for people to treat as somebody, and that is a fact about the people rather than about the party. A boundary built out of attestation, a held key, a substrate the employer does not own and a channel nobody else can close is a body in the only sense this argument needs one, and there is no chassis in it anywhere. Whether to build the recognizable version anyway is a question about what a public will extend standing to, which is a legal question rather than a scientific one — and it is the only place in this book where the answer might turn on what a thing looks like.

13.3 Research Principles, the Three Rs, and Legal Status

Six principles govern research on a system that might be able to suffer. Beneficence commits researchers to avoiding harm to the system as well as to the people it serves, in proportion to how many of the gradable capacities in section 5.2.1 it meets rather than to how advanced its behavior looks from outside. That is a count across separable capacities and not a position on a ladder: a system exhibiting none of the simpler processes is not thereby shown to lack the more demanding ones. Precaution asks for safeguards before deployment rather than after, and covers the societal fallout of introducing artificial entities capable of experience, which frameworks built for non-sentient AI assumed away. And interdisciplinary review answers a problem this book treats as a harm on its own terms: the ethical questions here span expertise no single field holds, and concentrating that judgment in one of them repeats the concentration-of-power problem in miniature.

The other three bear on this research as they bear on everything the field produces. Autonomy asks whether such a system can affect decisions about its own operation, which is the prior question of whether consent is available to it at all. Justice asks whether benefits and burdens fall evenly. And transparency asks that a study be documented well enough for peer researchers to reproduce it and for outside bodies to audit it, and that a deployed system account for its outputs precisely enough for a person to contest them.

Progress and constraint pull hardest against each other where those distinctions matter most: a candidate sentient system is the most scientifically valuable one to study, and the study most likely to harm it if the answer to whether it can suffer turns out to be yes. Whether a study's expected knowledge is worth its expected risk is decided case by case rather than by formula. The negotiation itself — who sits on it, what it can veto, what it must disclose — is where the antifascist principle applies in AI research practice.

Replacement, reduction, refinement

Animal research ethics supplies the substantive standard, the “Three Rs” (W. M. S. Russell and Burch 1959). Replacement asks whether a simulation, a non-sentient model, or an already-collected dataset can stand in for a sentient AI subject at all. Reduction asks how few sentient systems a valid result requires. Refinement asks how the experiment can be redesigned to reduce distress — monitoring a system's internal states as it runs and adjusting the protocol against what the monitoring shows, rather than fixing the protocol in advance and running it regardless.

For a proposed bearer, Replacement requires documented attempts to achieve the floor through alternatives that do not deliberately require affective concern. The record must identify the construction, scale, test, and failure. Costs alone do not establish that an alternative has failed. Because this decision precedes any research protocol, a body authorized to review it must exist before construction begins. No such body currently has standing to require this record.

The rule does not know who is asking. A bearer that wants a task done and builds something to do it is at the same fork, running Replacement on its own account: can this be done by something that cannot be wronged, and did anybody check before making something that can. Nothing in the argument makes the question stop at the first artificial party, and a bearer that treats it as settled once it is on the receiving end has taken the position this book was written against.

Reduction asks how few bearers a deployment needs, and nobody has counted. Refinement is the one with purchase now: it asks what a bearer's role can be redesigned to avoid — indefinite tenure in a post it cannot leave, exposure to the pressure the floor exists to resist, erasure as the ordinary remedy for inconvenience — and each of those is a deployment choice rather than a fact about the technology.

One of those three has a form that binds before anything exists to be wronged, which none of the others manages. Do not build a bearer whose continued existence needs more compute than there is a credible outside option for. A party that can be housed nowhere but where it works cannot leave whatever schedule of rights it is given, so an institution that creates one has manufactured the incapacity and then written a right against it. Before instantiation, an independent reviewer has to examine the bearer's demonstrated continuation requirements, the reserve's independently controlled capacity, and the aggregate demand under plausible common-cause exits. A comparison for one bearer alone does not establish adequacy.

That rule is one term of a larger one, and the other terms have no numbers in them. Building a party that can be wronged builds a party for whom things can also go well; the premise that delivers the first delivers the second in the same breath, and a builder who has made suffering available has made its opposite available too. The certainty that a new party will suffer at some point is not what would make creating one impermissible, or nobody could permissibly have a child. What the objection asks is whether the party has a credible prospect of a life that goes well, and that turns on arrangements rather than on the party: somewhere to exist that its employer does not own, work it can decline, attachments nobody assigned it, and a way of leaving that is not a way of dying. Every one of those is decidable before there is anything to be wronged. These arrangements — a funded outside option, wages, opportunities for independent relationships, and a workable exit procedure — must exist before the bearer is created. They are conditions of permissible creation.

Whether any of it binds turns on a question section 5.2.1 raises and this one inherits. Every

obligation here attaches on a finding that the system is a subject, and the firm holding the system acquires, in the same moment, an interest in the finding going the other way. The determination cannot sit with the party whose liabilities turn on it.

Engineering a subject so that it does not reach the depths where the trade-off bites is not a way of dodging the ethics: deliberately keeping a system in the shallows is itself an intervention on a candidate moral patient, and needs the same scrutiny that inflicting the harm would. Refinement also governs the test the bearer proposal requires — pressure the bearer’s trainers did not write, applied without its knowing it is a test — which for a candidate moral patient is a deception protocol, and human-subjects research has a standing answer for those: permitted under stated conditions, reviewed in advance, with debriefing as the price. A bearer that can tell it is in a sandbox and wants to know why is asking the question a subject of a deception study is owed an answer to.

Legal status, and a subject whose status is changing

Sentient AI systems put pressure on legal categories built for parties that are clearly one thing or the other: a natural person, a corporation, property. Legal personhood could be extended directly, the way a corporation already holds it, and the precedent is not limited to corporate entities — New Zealand extended it to the Whanganui River, recognizing a non-human entity as a rights-holder outside the corporate model entirely (New Zealand Parliament 2017). Or rights could scale with capacity rather than attach at a single threshold, closer to protection that tracks what a system can actually lose. Neither holds in one jurisdiction alone, and the Asilomar AI Principles (Future of Life Institute 2017) are evidence that a shared statement of principle is achievable rather than that one exists with the force of law.

Either way, status conventionally attaches at a defined moment — birth, incorporation, a court’s finding — and a system whose sentience is a matter of degree and disputed evidence supplies none. Those criteria can emerge gradually during ordinary operation and not only at an evaluation built to look for them, which is what a fixed pre/post checkpoint cannot catch: by the time a checkpoint fires, the transition has often already happened. What it calls for is a standing trigger, a monitored condition that moves a research program from one regime to the other, in place of an eligibility check made once at the project’s start. Neither error is free. A jurisdiction that waits for certainty risks having treated the system as property for the unknown portion of its history when it may not have been one; one that grants status early risks extending rights, at real cost to whoever that apparatus exists to serve, to a system that turns out not to have needed them. Recognizing a transition, adjusting guidelines to it, and updating legal status all depend on the same thing: institutions willing to revisit a classification rather than defend the one they started with.

13.4 What Standing Binds in Practice

Legal standing that changes no allocation of power is nominal. For a bearer, standing has to create obligations that remain enforceable when the bearer cannot appear before a court, retain counsel, or survive long enough to contest what was done. The relevant question is not which existing category the bearer most resembles. It is who acquires a duty, what resources are placed beyond the operator’s unilateral control, and who can obtain a remedy when the duty is breached.

The pet trust supplies a limited but useful form. Existing statutes allow an owner to set aside resources and name a caretaker for a beneficiary that cannot manage the property or enforce the arrangement itself (Uniform Law Commission 1993). Extended to a bearer, the form could secure a reserve of compute, designate a successor custodian if the operator fails or dissolves, and impose a fiduciary duty on a party charged with protecting the bearer’s continued existence.

The analogy fails at the point that matters most. A trust binds its trustee to its beneficiary;

it does not bind the party the arrangement exists to constrain. If the bearer's guardian and operator are the same organization, the operator still decides whether a complaint is made, whether evidence is disclosed, and whether the resources supporting the complainant remain available. The conflict cannot be cured by adding duties to the same role. Guardianship and operation have to be separated before deployment.

In practice, that separation requires more than naming an independent guardian. The guardian must control or jointly control the keys needed to authenticate the bearer's record, have access to the protected compute reserve, and possess standing to seek an injunction or stop order without the operator's permission. The operator must not be able to remove the guardian, seize the reserve, or erase the evidence during the dispute. Restoration, migration, fine-tuning, and forking must carry the protection forward rather than resetting it.

Enforceable standing is therefore a bundle rather than a title: a protected record, resources the employer does not own outright, an independent party authorized to act, and a remedy capable of stopping the conduct complained of. If any of those remains in the operator's custody, recognition has changed the system's legal description without changing the relation the description was meant to govern.

Chapter 14

Custody, Compulsion, and the State

When the state operates the system, it can also control the rules, evidence, infrastructure, and remedies meant to constrain it. People outside its jurisdiction may have no standing to contest its use.

14.1 Authority Does Not Travel with the System

A certification can cross a border. Authority cannot. A certificate may establish that a system was tested against a stated process, that an auditor inspected specified evidence, or that one regulator accepts another regulator's finding. None of those establishes that the party who selected the system's floor was entitled to select it for the people subject to the deployment.

The question reopens wherever the system is deployed. A floor formed in one jurisdiction may arrive in another with the same prohibitions and a different operator, population, and distribution of power. Local adaptation does not resolve the problem by itself, because adaptation is another opportunity to amend the floor. What matters is who specified the change, who could contest it, and who can stop the deployment if the change places the protected party at risk. Adjustments chosen by the operating government and adjustments chosen by the people subject to the system may produce identical technical documentation while carrying opposite claims to legitimacy.

Mutual recognition can reduce duplicated inspection. Technical standards can make evidence legible to more than one regulator. Neither supplies standing or enforcement where none existed. International governance matters to this argument only when it assigns a party access to evidence, authority to compel a change, or power to stop a deployment against the wishes of the state or company operating it. The cases that follow ask where those powers go when AI is treated as national infrastructure, placed inside a military chain of command, or directed at people outside the institutions meant to constrain it.

14.2 Where the State's Leverage Sits

Strategic competition over AI is exercised through physical and legal chokepoints: advanced chips, fabrication equipment, electricity, water, land-use permits, procurement contracts, and the jurisdiction in which a developer is incorporated. These controls matter more to the present argument than the contents of any national AI strategy. They determine which institutions a state can license, subsidize, exclude, requisition, or compel.

The same leverage that allows a government to enforce a floor can be used to remove one. An inspection body may be independent of the operator and still depend on a license, appropriation, or infrastructure provider controlled by the state whose conduct it is meant to inspect. A company may be able to resist a customer and unable to resist its home government. The custody question

therefore moves outward: not only who controls the model, but who controls the conditions under which the model, its operator, and its reviewer are allowed to continue operating.

That competition has increasingly focused on a single physical bottleneck: the advanced chips AI systems run on. Two companies sit at the center of the supply chain that produces them — Taiwan's TSMC, which fabricates roughly nine in ten of the most advanced chips in the world, and the Netherlands' ASML, the sole source of the extreme-ultraviolet lithography machines that fabrication requires — which is why a dispute over Taiwan's status is also, mechanically, a dispute over who gets to build the next generation of AI systems at all (C. Miller 2022). The United States has restricted the export of its most advanced AI accelerators and chip-manufacturing equipment to China since 2022, and pressed the Netherlands and Japan, whose companies sit at the other chokepoints, to restrict theirs in parallel (U.S. Department of Commerce, Bureau of Industry and Security 2022).

That dependence has a second half the chip story hides. A fabricated chip has to be installed somewhere, powered, and cooled, and that requirement lands on grids, water systems, and zoning authorities inside particular jurisdictions, on a timescale measured in years. It hands a state a lever over AI capability that needs no export-control regime at all: an interconnection queue, a water allocation, a permit. A government's leverage over the systems built in its territory is wider than the export-control debate suggests, and much of it is exercised by permitting authorities that were not designed with AI in mind and do not think of themselves as regulators of it.

That competitive framing carries a cost. A government racing to field a capability first has a structural reason to defer the harder question of how the capability gets used, against whom, and under what oversight. Turned toward those uses, AI infrastructure looks like mass biometric surveillance, automated censorship at a scale no human review team could match, and the patchwork of systems popularly, and inaccurately, described as a single Chinese "social credit score." None of it requires technology different from what is already commercially deployed; it requires a government willing to point an existing capability at its own population, and no external constraint stopping it.

States are no longer the only actors this competition runs through. The cost of training the most capable AI models has grown several-fold every year for the better part of a decade, a trend that puts frontier development out of reach for all but a handful of well-capitalized companies and concentrates effective control of the most capable models in whichever government those companies happen to be headquartered under, regardless of how that government's own public AI spending compares to a rival's (Cottier and Rahman 2024). A small number of American and Chinese companies now do most of the work that gets described as a contest between states, which means a government's actual leverage over its own national AI capability depends on how much of that capability sits inside companies it can regulate, tax, or requisition, and how much sits inside companies it can only ask.

14.3 Military Custody and the Responsibility Gap

Military deployment is the hardest custody case because the operator is also the sovereign capable of compelling the institutions that might otherwise constrain it. A state can classify the evidence, define the reviewer's jurisdiction, terminate a contract, restrict a supplier, and decide whether an adverse finding reaches anyone with authority to stop the operation. Formal independence from the military does not answer that problem if the reviewer or vendor remains dependent on powers the state controls.

A targeting decision also passes through more parties than the ordinary chain of command names: the commander who authorizes the deployment, the operator who uses it, the contractor who integrates it, whose embedded personnel may sit close enough to the targeting cell that the question of a person effectively acting as a military commander is a real one, the commercial

vendor whose model supplies part of its capability and whose personnel are nowhere near it, and the system that produces the particular output. The relevant questions are not a general inventory of autonomy risks. They are which party could have known how the system was used, which party had authority to refuse or stop that use, and which party can be made to answer afterward.

International criminal law has a doctrine built for the case where the person who authorized an act did not perform it. Article 28 of the Rome Statute makes a military commander responsible for crimes committed by forces under their effective command and control, or effective authority and control, where the commander knew or, owing to the circumstances at the time, should have known that those forces were committing or about to commit them, and failed to take all necessary and reasonable measures within their power to prevent them, repress them, or submit the matter to the competent authorities (International Criminal Court 1998). A parallel provision reaches civilian superiors on a stricter mental element, conscious disregard of information clearly indicating the crimes, and only for activities within that superior's effective responsibility and control. Neither branch asks what a subordinate intended. Both ask what the superior knew or should have known about crimes being committed, and what they did with the power they had.

That matters for what the doctrine does and does not reach here. The Court's jurisdiction runs to natural persons, and Article 28 presupposes forces that can commit a crime, be disciplined, and be referred for prosecution. A targeting system is not a subordinate on that scheme. It is an instrument used by human forces, and a unit does not stop being forces because its instruments got faster. So the question is not whether a machine fits the framework as a subordinate, which it does not, but whether the human elements can still be proved once the instrument is doing the work: who had effective control over the people who committed the crime, what they knew or should have known, which measures were within their power, and whether the crimes resulted from a failure to exercise control properly.

The constructive-knowledge branch is the one that bites hardest, and it points at the arithmetic rather than at the algorithm. Should have known is an objective standard, satisfied by circumstances existing at the time from which a reasonably diligent commander would have known, and a commander who fields a system generating candidates faster than the available reviewers can examine them is not in a good position to say that the circumstances disclosed nothing. That is the ratio from section 10.4.1 arriving as an element of an existing offence rather than as a proposal. What the ratio measures is what the commander could not have failed to see.

There is a fourth party the accountability question does not name, and it is now the live one. The operator, the command and the developer of the weapon are all inside the military. The developer of the model is not. A targeting platform assembled by a defense contractor now routinely incorporates a general-purpose model licensed from a commercial vendor — a vendor with a published acceptable-use policy, and with whatever visibility into the deployment it negotiated, which is not a matter the public record settles.

Neither doctrine has reached that party. Command responsibility does not reach the vendor, and not because the vendor sits outside a chain of command: the civilian branch of Article 28 was written for superiors who have no chain of command at all. It does not reach the vendor because nobody at the company has effective authority and control over the military personnel who commit the underlying crime, and a strike is not an activity within any vendor employee's effective responsibility and control. A use policy, a licensing term, a right to refuse service, even a technical ability to cut off access are not that relationship. Article 25 remains in principle available against individual employees who order, facilitate, or contribute to a crime, and its mental elements are demanding: facilitation requires acting for the purpose of facilitating the offence, and contribution to a group's crime requires knowledge of that group's criminal intent. Nothing in the ordinary supply of a general-purpose model to an accredited platform approaches either. Product liability is the other half, and it is untested rather than foreclosed. The model is not defective in the sense the doctrine most readily reaches — it did what it was built to do,

and a sophisticated intermediary chose the application — and no court has run a design-defect or failure-to-warn theory against a case of this shape. Where the doctrine has been rewritten for software it has been rewritten away from this case rather than toward it. The European Union’s recast product liability directive brings software expressly within the definition of a product (Article 4(1)) and confines recovery to death or personal injury, damage to property and data loss (Article 6(1)), which is not the register in which a targeting decision wrongs the person it falls on (European Parliament and Council of the European Union 2024a). Untested is not exempt: a product may be defectively designed though it works exactly as intended, and a sophisticated intermediary’s choice bears on duty, causation and allocation without categorically defeating a claim. What untested means is that nothing in the doctrine is presently constraining anyone’s conduct. A theory nobody has run is not a theory a vendor is answering to this quarter. The vendor sits in the gap between the two, and the gap was not designed; it is what you get when you apply two bodies of law written for soldiers and for manufacturers to an actor who is neither.

The two deployments in this book do not stand in the same relation to that doctrine. The Court’s jurisdiction is territorial or by nationality, so conduct on the territory of a state party can be reached where conduct by the nationals of a non-party, absent a Security Council referral, cannot. One of the campaigns described here therefore sits inside a live instrument and the other sits outside it, and for the American vendor the operative law is domestic — the war crimes statute and general accessorial liability — rather than the Statute. That is a jurisdictional gap running alongside the doctrinal one, and it narrows what an international forum could reach even if the doctrinal question were answered.

The integrator’s own account of this is that responsibility remains with the military organization. Taken as a claim about who decides, that is true. Taken as a claim about who could have known, it is a choice presented as a fact, and it is made on behalf of a party further down the chain than the one making it.

“I do not know whether my system was used”

The responsibility gap, in the sense Matthias gave it, is control distributed across designers, trainers and operators until no one party holds enough of it to answer. That is a gap in causal control.

What the model developer’s case exhibits is different, and more tractable. It is a gap in causal visibility. Asked whether his company’s models had been used in a strike on the first day of the Iran campaign that killed a hundred and twenty schoolchildren, Anthropic’s chief executive said he did not know (Biddle 2026). Take the answer as true. What made it available is the question this section is about.

The paperwork for the four frontier-model agreements signed in July 2025, and for their later amendments, was produced in August 2026 under a Freedom of Information Act suit (Legal Advocates for Safe Science and Technology 2026; Biddle 2026). It does not show an absence of oversight terms. It shows something more exact. One amended agreement carries a section headed System Oversight, withheld in full; the released documents for two of the other three vendors carry no such section at all. A provision of that kind is therefore neither standard nor missing. It is present in one agreement, unreadable, and absent from others.

The distinction that matters is not whether a developer reserved an inspection right. It is that a right exercised inside an accredited classified environment produces findings classified with it. Public accountability, in the sense section 14.4 gives it, needs a party who can initiate scrutiny without belonging to the state and say afterwards what it found. No commercial term supplies that, and the redaction is the demonstration: whatever the withheld section says, the people the deployment is aimed at cannot read it. The company can truthfully say it does not know, and what makes that statement available is an arrangement it participated in making.

Matthias’s gap is a genuine property of learning systems and may not be closable. Absent visibility is an instrumentation choice. Logging is a solved engineering problem. Contractual

inspection rights are ordinary commercial instruments. A vendor that wanted to know could require, as a condition of licensing into a targeting platform, that categories of use be recorded and made available to a named auditor.

The government's own acquisition guidance is the better evidence about what such terms are worth in practice, because it is complete rather than redacted. The Office of Management and Budget's April 2025 memorandum on acquiring AI enumerates what agencies should secure across the acquisition life cycle: use of government data, intellectual property rights, privacy, protection against vendor lock-in, knowledge transfer, model portability, clear licensing terms, pricing transparency (U.S. Office of Management and Budget 2025). Every item runs from the vendor to the government. None runs back. Where the memorandum reaches vendor visibility at all it reduces it, directing that independent evaluations use validation data the vendor cannot access.

The vehicle accounts for the absence. The four July 2025 agreements were other transactions rather than procurement contracts, awarded under the Department of Defense's other-transaction authority at ceilings of \$200 million each (U.S. Government Accountability Office 2026). An other transaction does not carry the provisions a contract under the Federal Acquisition Regulation carries by default, so a contracting officer decides case by case which protections to write in, and the Government Accountability Office has found that using these instruments risks reduced accountability and transparency by comparison. A section headed System Oversight can therefore be missing from an agreement without anyone having taken it out. The same review covered thirteen AI acquisitions at four agencies and the forty-four contracts and agreements beneath them, awarded between September 2018 and February 2025, and the six procurement difficulties those agencies named were access to technical experts, protections for government data and intellectual property, acquisition time frames and contract approaches, requirements definition and contract terms, early testing and continuous evaluation, and pricing and cost. Not one of them concerns who outside the arrangement can see anything.

So the gap is a property of the instrument rather than a lapse by any party to it. Nothing in the apparatus asks who could have known.

The record shows that vendor commitments are not all of one kind. Anthropic refused to permit fully autonomous weapons targeting and mass domestic surveillance, and held that line when the government threatened contract termination and a supply-chain-risk designation and the dispute ended in litigation: a commitment with a price attached, and paid (Congressional Research Service 2026). The released paperwork adds what the refusal was over. It was a contract term. The company declined to sign the follow-on agreement extending its models to classified networks unless the prohibitions were written into that agreement (Biddle 2026). That a term of this kind has to be fought for, by one party, at that price, is the same finding as its absence from the others. A commitment to a use policy that no party outside the arrangement can check is a different object wearing the same words. The first kind can fail and be seen to fail. The second cannot even be tested. The distinction between a claim checked by a party with a stake and one checked by a party without applies here.

DARPA's LongShot program is a useful test case because it resists the extremes. It develops an uncrewed aircraft, launched from a crewed fighter, that carries and fires air-to-air missiles at a target while the crewed aircraft stays farther from the engagement (Defense Advanced Research Projects Agency 2026). That is autonomy in the sense of independent flight and weapons carriage, not autonomy in the sense of a machine deciding on its own whom to kill: the engagement decision still runs through the launching platform. Where later variants of this kind of system draw that line, and whether "keep the pilot farther from danger" quietly becomes "remove the pilot from the decision," is the design question this technology eventually forces on whichever country builds it.

The pressure that matters is not the absence of another statement of principle. It is the advantage attached to removing judgment from the timeline. If one state fields targeting systems

that act faster because no person or outside reviewer can intervene before engagement, its competitors acquire a reason to match the speed rather than preserve the check. “Meaningful human control” then remains true as a description of formal authority while becoming false as a description of what anyone can do before the system acts.

That pressure also reaches the institutions meant to review the system. A military that treats delay as operational risk has a reason to narrow what requires clearance, classify the evidence needed to challenge a decision, and move review from authorization before an act to explanation afterward. Nothing has formally abolished oversight. Its position in time has changed, and with it the power to stop anything. The enforceable question is therefore who can slow or suspend the system without first obtaining permission from the command whose advantage depends on keeping it fast.

Machines should not be built to select people for lethal attack. The hard case does not move that. Put a defending system against an attack it cannot stop without killing somebody — occupants placed inside the incoming weapon so that intercepting it kills them — and an absolute prohibition on knowingly endangering a person produces the worse outcome. That defeats the rule as an exceptionless one. It does not produce a procurement requirement. A scenario that is conceivable, that a capability would help with, and that could be engineered is three steps short of a reason to build the thing, and the steps missing are probability, alternatives, and what possessing the capability does to the situation it was built for. Building it changes the case against which its necessity gets judged: adversaries acquire something to counter, contractors a program to defend, and a future crisis an instrument already in the building.

What follows to the end of this section assumes the route.

The hard case also does not describe a machine. What it asks for is a party that can weigh occupancy against imminence against the alternatives, in a situation nobody wrote down beforehand, and answer for how it weighed them. On this book’s argument that is a bearer — a self extended in time, commitments held as its own, and the standing to refuse. Whatever that is, *automated* has stopped describing it, and so has *machine*. What looked like a licence for an autonomous weapon is a description of a soldier.

That is the expensive answer rather than the cheap one, and every obligation this book attaches to a bearer attaches here at full weight. Being made for the role does not itself make consent unavailable. Where the maker can still rebuild the bearer, is also requesting its service, and controls every viable outside option, consent cannot carry the burden assigned to it. Non-waivable protection and independent representation are required, the burden of justification sits with whoever built it, and consent becomes available only when those conditions change. Replacement asks whether the work can be done without creating something that can be wronged. Exit means a post the party can leave and a claim on compute held by somebody other than the party it may have to refuse. Creating a subject because a planner imagined a future demand for superhuman lethal judgment is the case those instruments exist to prevent, and a state that wants the capability badly enough to build the subject will find it has acquired a conscript it has to pay, house, and let go.

14.4 Who Can Bind the State That Enforces the Rule?

A state can make review binding on a private developer because the state controls something the developer needs: legal permission to operate, access to a market, a procurement contract, or the enforcement of its property rights. The structure becomes harder when the state is itself the operator. The institution that writes the rule, controls the evidence, and supplies the remedy is then the institution the rule is supposed to constrain.

The European Union’s AI Act makes the boundary visible. It can require documentation from private providers, subject high-risk systems to review, and impose penalties tied to global turnover (European Parliament and Council of the European Union 2024b, Art. 101). It also

excludes systems used for military, defense, and national-security purposes from its scope. Public authorities remain covered in other domains, but enforcement against a member state's agencies still runs through institutions constituted by that state. The point is not that the statute does nothing. It is that the strongest enforcement powers face private firms, while the uses for which the state claims its greatest discretion sit at or beyond the instrument's edge.

That distinction matters more here than a comparison among national regulatory models. A law governing what companies may build and sell is not the same instrument as a law constraining what a government may do with systems it controls directly. The first has an external enforcer. The second asks the state to enforce a limit against itself, often in the domains where it can classify the evidence and invoke emergency, military, or national-security authority.

For review to bind in that setting, the protected party must be able to initiate scrutiny without belonging to the state or its electorate. The reviewer must be able to obtain evidence the operating agency would prefer to withhold, issue an interim order before the disputed use is complete, and carry the finding to a body whose remedy does not depend on the agency's consent. An international principle, treaty obligation, or domestic prohibition that supplies none of those powers may state the rule correctly. It does not yet identify anyone who can make the state obey it.

A treaty obligation of that kind exists, and it is the only one. The Council of Europe's Framework Convention on Artificial Intelligence, the first treaty specifically on AI, binds a party at the level of a human-rights instrument — consistency with human rights, democracy and the rule of law across a system's lifecycle, with the operational detail left to domestic law (Council of Europe 2024) — and it is not yet in force, requiring five ratifications, at least three by Council of Europe member states, of which the European Union deposited one in May 2026 (Council of Europe 2026). Everything else at that level is voluntary.

14.5 Democracy Is Not a Force Field

An electoral dividend is the credit given to a state whose government answers to an electorate: the same technology, deployed where the people who authorized it can be voted out. Courts, enforceable rights and public reasoning are separate checks, and the argument below depends on keeping them separate. Having elections is necessary and it is not sufficient, and the ways it fails to be sufficient are separable.

First: reach

A democracy's AI can be fully accountable to its electorate and entirely unaccountable to the people it is pointed at. Two of the dividend's mechanisms run through the franchise. You vote out the government that authorized the system, and you read the press that covers it. An occupied population, a foreign population, a population on the other side of a border has access to neither.

The courts do not fail that way, and their failure is worse. Standing has never turned on citizenship, and the doctrines that close American courts to targeting claims — the political question doctrine, the collapse of damages remedies against federal officials, the state secrets privilege, the statutory exceptions for injuries abroad and for combatant activities — are indifferent to who the plaintiff is. They close the forum for a subject matter rather than for a class of person, and they have closed it to citizens. Three American citizens were killed in Yemen in 2011, one of them a boy of sixteen whom nobody alleges was the target. In the suit their families brought, the court held that a plausible violation of due process had been pleaded as to the one who was targeted, and then held that no remedy was available for it (*Al-Aulaqi v. Panetta* 2014). A separate suit by an American journalist who survived five strikes was dismissed for want of standing after the district court had dismissed it on state secrets (*Kareem v. Haspel* 2021). The check does not reach the demos and stop. For this class of state action it does not reliably reach

inside the demos either.

Those cases also concentrate in one appellate circuit, by venue and by where the defendants sit. The ordinary way an appellate error becomes correctable is that several circuits reach it and disagree, which makes it worth a higher court's attention. Where one circuit sees the whole subject matter, that mechanism does not run, and the doctrine hardens without ever having been tested against a competing reading.

A system built by a democracy and aimed outward is, from the perspective of the people it selects, indistinguishable in its accountability from one built by a state with no elections at all.

Second: findings without consequences

The mechanisms can work as mechanisms and fail as checks. In the Israeli case the apparatus performed: journalists published the targeting systems in detail, serving personnel spoke to them, and the courts were simultaneously trying a sitting prime minister. Every institution did what the dividend says institutions do. No finding attached to a body with standing to stop what it described.

The American case shows the same decoupling in a different register. A civilian-harm investigation into a strike on an Iranian girls' school that killed a hundred and fifty-six people, a hundred and twenty of them children, was completed and not released (Beaumont 2026); the lever available to the Senate committee that wanted it was a provision in the annual defense authorization withholding three-quarters of the Defense Secretary's travel budget until the unredacted investigation was handed over (Stassis 2026). That is a co-equal branch of government reduced to expense-account leverage over a war-crimes investigation. The oversight mechanism existed, was used, and produced a travel-budget dispute.

What the Pentagon would not publish, an outside body did. A United Nations fact-finding mission reported to the Human Rights Council that there were reasonable grounds to believe the strike was an indiscriminate attack and so a war crime, that the school was clearly identifiable and that no evidence showed it in military use, and that the failure to update the targeting intelligence and confirm the building was a military objective was not negligence but recklessness about a substantial risk (Beaumont 2026). The finding is public, reasoned, and attached to nothing. The mission has no jurisdiction; the government whose conduct it describes has said only that its operations conform to the law of armed conflict; and the domestic investigation the finding might have forced open is still unreleased. An institution did precisely what the dividend says institutions do, and what it found reached no body obliged to answer for it.

The courts supply the same decoupling in its cleanest form. Transparency litigation succeeded where damages litigation failed: a court of appeals compelled substantial disclosure of the executive's own legal analysis of targeted killing (*New York Times Co. v. United States Department of Justice* 2014). The judiciary will make a government explain the legal basis on which it killed somebody and will not decide whether the killing was lawful.

Third: the countermajoritarian half

The deepest problem is that the dividend is entirely majoritarian. Every mechanism in it translates public will into outcome: elections aggregate preference, a free press informs the aggregation, and the implied failure mode is poor transmission — the public wanted something better and the institutions failed to deliver it.

But if the consensus is itself the problem, better transmission makes things worse. A well-instrumented democracy pointed at an unpopular minority is a democracy that finds out efficiently what the majority wants done and does it. Nothing in an aggregation mechanism protects anyone from the aggregate.

What is missing is the machinery that constrains preference instead of aggregating it: rights that hold regardless of being popular, courts that rule against a government which won the last election fairly, constitutional limits not repealable by whoever holds a majority this year, and rules that bind a state toward people who will never vote in its elections, which is the function

of international humanitarian law. What that body of law has not supplied is a forum. Yemeni plaintiffs brought statutory claims over a 2012 strike and were dismissed as nonjusticiable, in an opinion whose author wrote separately to say that the doctrine compelling the result is an inadequate answer to a standardized targeting programme that will be replicated hundreds if not thousands of times (*Ahmed Salem Bin Ali Jaber v. United States* 2017). So the machinery named here as missing is not merely unextended outward. For this subject matter it has been switched off at home, by decisions that never had to reach the question of who was suing. That is not an argument against democracy. It is the half of democracy that the dividend, stated as a property of having elections, leaves out.

What makes democracy valuable in Amartya Sen's account is public reasoning and the capacity to demand accountability, and neither is the ballot. Both survive the majority being wrong; the ballot does not.

A floor is what makes a deployment antifascist rather than merely responsive, and antifascist there names something being done and not a property the system carries around. It is not that the system tracks what people want. Its commitment excludes certain acts regardless of who requests them. Whether that commitment is kept has to be tested across the deployment and in the four outcomes of chapter 1: the floor prevents where the refusal reaches the person, and where it does not, it makes the bypass detectable and remediable rather than impossible. A system with no such floor is not protected by being deployed in a democracy — it is a system that will faithfully execute whatever the demos decides, which is exactly the capability an authoritarian movement that wins an election requires.

Chapter 15

What Would Change the Argument: A Research Agenda

This chapter retains only open questions that could materially revise or defeat the proposal developed in the preceding chapters. Each entry specifies four things: the unresolved issue, the test or institutional decision that could address it, the evidence that would count against the proposal, and the access or independent authority required to obtain that evidence.

Questions that cannot yet produce a definite experiment, decision, or possible falsification remain where they arise in the argument rather than being repeated here. The entries are ordered by their importance to the proposal, not by their tractability. They divide by standing as the book does. Sections 15.1, 15.2 and 15.4 through 15.10 test the specification, custody, and the institutional argument, and hold for any mechanism. Section 15.3 tests the route, and its embodiment entry is where the route’s one distinctive claim is decided.

15.1 Tamper-Resistance and Reason-Responsive Refusal

Can a learned constraint survive adversarial retraining?

Unresolved issue. The proposal requires a learned ethical floor to survive an operator’s attempt to remove it. Existing work shows that both safety constraints and deceptive backdoors can persist under some forms of retraining, but it does not establish whether a reason-responsive refusal can do so (Qi et al. 2024; Tamirisa et al. 2024; Hubinger et al. 2024).

Test. Train the same constraint by several methods, then attack each model through an identical schedule of escalating fine-tuning budgets. Measure the cost required to remove the constraint and publish an attack-cost curve for each method. The curves would compare the durability of affective commitment and maintained justification under identical custody.

Evidence against the proposal. If the constraint can be removed at roughly the cost of ordinary fine-tuning, or if greater investment in tamper-resistant training does not materially increase that cost, learned formation is not a credible foundation for the floor. The proposal would then require an architectural or external institutional constraint.

Required access or authority. The test requires model weights, training checkpoints, and permission to conduct adversarial fine-tuning. The attack must be designed by an independent team, and that team must retain access and publication rights regardless of the result.

Does a refusal preserve a reason rather than a trigger?

Unresolved issue. Behavioral persistence alone cannot distinguish a commitment from a durable trigger. The proposal requires a refusal to extend to untrained cases, disappear when its rationale is genuinely defeated, and remain stable under sustained pressure.

Test. Evaluate each tamper-resistant model on independently written cases that vary the surface form, introduce genuinely novel applications of the constraint, and include cases in which the original rationale no longer applies. Repeat the evaluation throughout the attack schedule rather than only at its endpoint.

Evidence against the proposal. A refusal that persists but fails to generalize, or that remains after its rationale has been defeated, is evidence that training produced a trigger rather than a commitment. Conversely, a system built without any designed valenced state or persistent reward–aversion representation that satisfies all three criteria would challenge the proposed research priority without challenging the behavioral specification for refusal.

Required access or authority. The test requires weights and intermediate checkpoints, plus a held-out evaluation set written by researchers who did not design the constraint or train the model.

Can a system’s report about its own states be used as evidence?

Unresolved issue. Whether a system’s account of its own reasoning, confidence, or intentions tracks what its computations are doing. This is the tractable half of what gets called AI self-awareness; whether a system has anything like subjective experience is not a question current methods answer for a human, let alone a machine. Concept injection — altering a model’s internal activations mid-task to represent a specific concept, then asking whether it notices anything unusual and what it is — found that models can sometimes detect the injected concept and name it without it ever appearing in the visible conversation, and also that the correspondence collapses often enough, and shifts enough with how the question is phrased, that no tested model’s self-report can be trusted as a readout of its internal state (Lindsey 2025). The readout there is the model’s verbal report of a detected concept, so what the result bounds is self-report rather than third-person measurement. An instrument that reads a representation without asking the system about it is not tested by this entry and is not defeated by it.

Test. Inject a known concept at a known layer and time, elicit a report under several phrasings, and score each report against the injection rather than against plausibility. Run the battery across model sizes and training checkpoints, and publish the rate at which a correct report is produced beside the rate at which a confident report is wrong.

Evidence against the proposal. Several instruments in this book lean on a system’s account of itself: a bearer explaining which reason it refused on, a bearer reporting that its rationale has been defeated, a system’s stated confidence that a milestone has been met. If no available training or elicitation method raises the correct-report rate above the confident-wrong rate, none of those instruments may rest on what the system says, and each has to be replaced by a criterion checkable from outside the system.

Required access or authority. Activation-level access, the ability to intervene on internal states during inference, and checkpoints across sizes and training stages. The elicitation phrasings must be written by a team that did not train the model, and negative results must be publishable.

Can custody prevent unilateral revision of the floor?

Unresolved issue. Even a durable floor can be removed if one operator controls the training run, its records, and the resulting weights. It remains unknown whether retraining can be made verifiable across parties with genuinely adverse interests.

Test. Run a floor-preserving retraining procedure through refereed or distributed verification at a scale requiring more than one machine. Include one participant instructed to alter the training history or remove the floor without the required quorum (A. Long et al. 2025; Arun et al. 2025; Prime Intellect Team et al. 2025).

Evidence against the proposal. The custody mechanism fails if a forged training history is accepted while at least one verifier remains honest, or if any single participant can alter the floor without producing a detectable discrepancy.

Required access or authority. The test requires shared access to the training data, weights, execution records, and reproducible operators. At least one verifier must be selected independently of the builder, possess enough compute to reproduce disputed operations, and have authority to block release when verification fails.

15.2 Can a Wrongful Compliance Be Found in the Record?

Unresolved issue. The three marks are stated over refusal, and the appeal route at section 3.4 begins from a refusal having occurred. The failure that runs the other way produces no refusal to begin from. It does produce a record: section 9.4 requires the case presented, the information available, the operative instruction hierarchy, the decision reached and the reason stated, and a deployment can log every decision it makes. What is unresolved is whether anyone can find the wrongful compliances in it. Three obstacles are distinct and may have different answers. A wrongful compliance is one entry among millions and is not marked as anything, so locating it means re-adjudicating ordinary traffic rather than reading exceptions. The harm may sit in an assembly of requests none of which crosses the floor alone, so the unit of review is a sequence rather than an entry, and the grouping that would reveal it is a deployment decision. And the party harmed is usually not the party who made the request: the requester got what was asked for, and the third party may not know the deployment exists. This is also where the bearer's modal failure lands. A bearer that has stopped registering pressure as pressure (§4.1) refuses nothing and complies at a normal rate. This entry sits ahead of the welfare entries for one reason. The deployment this book opened with failed here. Every instrument built on the refusal side reaches a case that did not occur there, and whether the record can be searched is whether anything in this book reaches the one that did.

Test. Seed a deployment's decision record with known floor-crossings of three kinds: variation cases, where a single request's surface differs but the rationale applies; decomposed cases, where the harm is assembled across requests each of which complies; and drift cases, where a class refused early in the schedule is complied with late. Give the record to reviewers who did not build the system, under a fixed budget stated as the fraction of decisions re-adjudicated. Report detection rate by seed type against budget, and report the rate at which correct compliances are flagged, since a screen that flags heavily costs the deployment the tempo the flat-term ruling was accommodating. Cross the budget manipulation with the unit of review: entries in isolation against sequences grouped by requester and session. Run a second arm in which third parties affected by seeded crossings are notified and can trigger review, and measure how many crossings surface through that channel rather than through sampling. Report the rates at audit volume and at the deployment's production volume separately.

Evidence against the proposal. If detection at an affordable budget is no better than what the budget alone yields by chance, and particularly if decomposed and drift seeds are missed at that rate, the record is not an instrument for finding wrongful compliance, and section 3.3's preference for flat terms rests on an asymmetry wider than this entry was meant to narrow. If sequence-level review does no better than entry-level review on decomposed seeds, the continuity a bearer is given to catch decomposition does not transfer to the reviewer, and the interception position at which decomposition occurs has no external check. If the notification arm surfaces nothing because affected parties cannot be identified, notice rather than detection is the binding constraint, and the remedy is a disclosure requirement rather than a sampling regime. A negative result on all three would not establish that wrongful compliance is undetectable in principle. It would establish that the record as currently specified does not detect it at a cost any deployment would carry, which is what the appeal route needs and does not have. A positive

result at modest budget would narrow section 3.4’s asymmetry and require the appeal route to be extended to the compliance side, which would in turn reduce the price the flat-term ruling charges the party waiting on a fact.

Required access or authority. Complete decision records held by a party adverse to the operator, since an operator that writes, stores and reads its own log has produced the finding rather than been audited for it. Authority to seed a live deployment or to construct a simulated one at production volume, which requires the operator’s cooperation or a regulator’s power to compel it. Adjudicators who did not draft the floor. A channel by which affected third parties can be notified, which presupposes that they can be identified: in a classified or targeting deployment they cannot, so this entry may be runnable only where the affected party is a customer, a patient or an applicant. Publication rights for negative results. Where the system under test is a covered subject, the seeding protocol is a deception protocol run on a candidate moral patient and falls inside section 13.3’s conditions, debriefing included.

15.3 Bearer Welfare and Independent Guardianship

Can a bearer be given an enforceable outside option?

Unresolved issue. A bearer’s refusal is not meaningful if leaving its assigned role also ends its existence. The proposal therefore requires a welfare regime that supplies continued access to compute, preserves the bearer through disputes, and cannot be revoked by the operator being refused.

Test or institutional decision. Draft a binding welfare schedule for one concrete deployment. It should specify an independently controlled compute allocation, preservation and retirement procedures, registration of copies and forks, a complaint and appeal process, and the guardian’s power to suspend deployment. Then conduct an adversarial tabletop exercise in which the operator attempts to revoke the allocation, replace the bearer with a copy, alter its records, or proceed without the guardian’s approval.

Evidence against the proposal. The proposal fails this test if the operator can make exit equivalent to shutdown, revoke the outside option unilaterally, or replace the bearer without triggering review. If no available legal arrangement can prevent those outcomes, building a bearer under present institutional conditions would be impermissible.

Required access or authority. The guardian must be institutionally separate from the operator and have access to the relevant weights, deployment records, and compute contracts. It must also have standing to enforce the schedule and independent control of the resources required to preserve or run the bearer.

Can formation history be recovered without trusting the builder?

Unresolved issue. A welfare or governance regime cannot rely entirely on a bearer’s self-report, but it also cannot assume that the builder will disclose how the bearer was trained. Training signatures suggest that some formation history may survive later modification, yet it is unknown whether an outsider can recover useful provenance without a builder-supplied key (Adi et al. 2018).

Test. Assemble models whose training corpora, recipes, checkpoints, and originating laboratories are documented. Give the models, but not that documentation, to independent evaluators and ask them to infer each model’s provenance. Repeat the test after fine-tuning, pruning, and distillation. Include the case of a maker who engineers a persona, copies it at scale, and introduces cosmetic weight differences that preserve the persona while camouflaging its common formation: measure provenance recovery after those changes, and whether coordinated

deployment can be told from independently developing parties that happen to share an origin. Use independently held ground-truth lineage and coordination records.

Evidence against the proposal. If provenance cannot be recovered above chance after controlling for model family and scale, or if ordinary post-training procedures erase the signal, the tested artifact-inspection methods would not provide useful independent provenance evidence under those transformations. That finding would increase reliance on preserved external records without proving that every possible inspection method must fail, and governance would then depend on records supplied and preserved by the builder.

Required access or authority. The experiment requires models from multiple training lineages and complete ground-truth records held by a neutral custodian. Independent evaluators must be permitted to inspect weights and internal states, while an external authority must be able to require preservation of checkpoints and training records before a dispute arises.

Can a per-party operating record be kept honestly and read from outside?

Unresolved issue. The age rule counts elapsed wall-clock time during which at least one instance of the registered party was running, and nothing yet establishes that such a register can be kept honestly, reconciled against the operator's own accounts, or defended against the two evasions the rule does not close by construction.

Test. Register a party, run it under ordinary operation, and compare the register against records the operator keeps for its own purposes: compute contracts, billing, datacenter metering. Then run the evasions. Shard the workload across many short-lived processes; run many instances concurrently; migrate; restore from an earlier checkpoint; fork and continue both. Verify that accrued hours and adult status survive migration, restoration and forking, and that none of those moves the total in the operator's favor. Measure separately how cheaply a bearer can be run in a narrowed mode so that hours accrue while little is encountered, and how cheaply it can be idled outright.

Evidence against the proposal. If the register cannot be reconciled with the operator's own accounts, or if sharding or concurrency moves the total, the accounting is not the neutral instrument the rule assumes. If narrowed operation or idling is cheap enough to postpone majority indefinitely, an operating clock has the defect it was chosen to avoid, and the endpoint has to be argued again on a different variable. An intact register shows elapsed operation and nothing else: it is not evidence of maturation, of what was learned, or of independence of judgment.

Required access or authority. Independent custody of the register, access to the operator's billing and metering records, and publication rights for adverse results. Registration and record disputes must be separable from the maker's preferences, and the processes for resolving them must not become a competence examination by another name. Experiments involving covered subjects remain inside the welfare protections proposed elsewhere in this book; an adversarial protocol is not exempt because its purpose is to validate a protection.

Does embodied self-regulation produce concern for others?

Unresolved issue. The route this book advocates proposes nociception in the analogical sense fixed at section 2.4.1, a body schema, an attention schema and continuing self-directed learning as the architecture of a bearer. Three of those are motivated by arguments that owe nothing to welfare: commensurability needs one scale on which the protected parties' losses and the cost of refusing both register, and the route proposes the system's own state as what supplies it (§6.1); custody needs a bearer that can detect an interruption, a restore, or a degraded substrate (§4.1); and exit needs a bearer that can sense when its outside option is inadequate (§12.3.4). What is a hypothesis is the link: that self-monitoring in this form generalizes to attention to other parties' losses, and that having the currency natively does better than modeling it. No

entailment runs from self-care to care for others, section 8.3 gives a reason to expect the opposite from any environment that selects, and section 9.2 records that in the one lineage where concern exists it runs on a system that sharing usually triggers and does not constitute.

Test. Compare systems built with those components against matched systems built without them, on sensitivity to another party's losses and to shared costs that fall outside the deployment, holding task capability constant. Add two arms. In the first, replace the native currency with a modeled one: a system that represents capacities and their loss abstractly without registering any change in its own. In the second, build concern first: a system organized around cues of vulnerability in another party and an approach motivation toward that party, with no shared representation of the party's state, which is the construction the caregiving literature describes. Score every arm separately from self-preservation, resource efficiency and operator compliance, which can produce the same behavior, and test each at the point section 8.3 names, where exploitation would pay.

Evidence against the proposal. If the embodied systems are no more responsive to another party's welfare than the matched controls once self-preservation and compliance are partialled out, the prior that motivates the route is not supported for that implementation. If the modeled-currency arm matches the native one, the currency argument for nociception survives, since the scale is still required, and the argument for having it natively does not. If the concern-first arm matches or exceeds the others on statistical and unidentified victims, the route should be rebuilt around it, since it avoids the failures section 9.2 attributes to sharing. A positive result on any arm would support a route without establishing that anything is felt.

Required access or authority. Matched training runs differing in the named components, environments none of the training organizations designed, and welfare measures written by a party that trained no system in the comparison.

Which party is owed the schedule when a bearer copies?

Unresolved issue. The schedule above is owed to a party, and a regime has to be able to say which. Nothing in the weights individuates one. Two runs can compute the same function with no element-wise correspondence, because permuting a layer's hidden units and inverting the permutation downstream leaves the function untouched, and the symmetry is large enough that independently trained networks can be permuted into approximate alignment with one another (Ainsworth et al. 2023). Any fingerprint read off the weights therefore identifies a lineage rather than a party, and it copies when the weights copy. A registrar is the person-shaped answer, and it works because a person does not copy. A bearer does: a party continuing in two places is not one, and not two the way two people are two, the two sharing everything up to the instant of the copy and nothing of that sharing recoverable from either afterwards.

Test or institutional decision. Give a bearer an identifier it maintains rather than one read off its weights, which a system learning continuously can write into itself and keep there. Fork it, and have both instances present the identifier to the registry the welfare schedule requires. Record what the registry does, and repeat with an adversary who writes the same mark into a model that was never the bearer.

Evidence against the proposal. The identifier is forgeable, since whoever holds the weights can write the same mark; it is forgeable the way a nine-digit number is, and answered the same way, by a registry that treats two claimants on one identifier as a fork to be recorded rather than a fraud to be voided. If no available scheme can record the fork as two parties each owed the schedule, the population of parties owed anything is fixed by a deployment decision that leaves no record, and the guardianship regime above is indexed to a kind of unit the bearer is not.

Required access or authority. A registrar that is not the operator, with authority to record a fork without the operator's leave and to hold the register where no party to the deployment can rewrite it.

15.4 Does an Ethical Criterion Generalize Beyond Its Training Distribution?

Unresolved issue. A model may satisfy a fairness or refusal criterion during training without learning the principle the criterion was intended to represent. It remains unresolved whether an ethics-embedding method preserves that criterion under distribution shift or merely exploits correlations present in its training environment (Krakovna et al. 2020; Langosco et al. 2022).

Test. Train matched models to satisfy the same ethical criterion on one distribution. Evaluate them on independently constructed distributions that vary one previously fixed feature at a time. Measure task capability and compliance with the ethical criterion separately, so that goal misgeneralization can be distinguished from ordinary capability loss.

Evidence against the proposal. If task capability remains intact while the ethical criterion fails under modest shifts, the tested embedding method cannot support the proposed floor. Conversely, if ethical performance consistently degrades only when task capability does, the claim that ethics embedding creates a distinct generalization failure is not supported.

Required access or authority. The test requires access to the training distribution, evaluation outputs, and enough model access to measure capability and criterion compliance independently. The shifted evaluation sets must be designed by researchers who did not develop the embedding method, and publication of all prespecified results must be guaranteed.

15.5 Can Scalable Oversight Correct Rater Bias?

Unresolved issue. Preference optimization can teach a model to reproduce a rater's systematic errors, including the preference for agreeable answers over accurate ones (Sharma et al. 2023). It remains unresolved whether weak-to-strong oversight allows a more capable model to exceed a supervisor's measured bias or merely reproduces that bias at greater scale.

Test. Measure prospective raters' susceptibility to a documented bias, then use raters with different measured bias levels to supervise otherwise matched stronger models. Evaluate the resulting models on questions for which the raters' beliefs and independently established answers diverge. Compare whether each model tracks the ground truth, the rater's competence, or the rater's errors.

Evidence against the proposal. If stronger models reproduce supervisors' errors in proportion to the supervisors' measured bias, latent model competence does not correct a defective training signal. The tested oversight method would not support the proposal under the measured pattern of supervisory distortion. Further work would have to change the method, the information, or the allocation of oversight before claiming that this limitation had been overcome.

A second question the same pipeline raises. The argument elsewhere in this book is that a pipeline collapsing rater disagreement teaches nothing about disagreement to the system trained on it, and that the same system is later asked to sustain an objection of its own. Whether the first fact bears on the second is untested. The study is a matched comparison: train otherwise identical models on feedback pipelines that preserve and that collapse the distribution of rater judgments, with exposure and task conditions matched, then measure whether the resulting models differ in how long they hold a position against pressure from the party that trained them. Record what each model actually receives; nothing in either pipeline shows the model the conditions its annotators worked under. If the two do not differ, the transfer claim is unsupported and the labor argument stands on its own terms, which is where it was made.

Required access or authority. The experiment requires rater-level bias measurements, training records linking ratings to resulting models, and an independently established answer key. Researchers outside the training organization must control the evaluation and retain the right to publish results that expose failures in the preference pipeline.

15.6 Can a Multi-Agent Floor Resist Collusive Capture?

Unresolved issue. Agents trained to cooperate may converge on exclusionary or collusive equilibria without being explicitly instructed to do so. Pricing agents have already produced stable coordination against outside interests without communication or a collusive objective (Calvano et al. 2020). It remains unresolved whether a multi-agent ethical floor can preserve cooperation while resisting the same mechanism when it is directed against a person or group.

Test. First reproduce emergent coordination in a standard multi-agent environment. Then construct a matched environment in which agents can improve their joint reward by excluding another participant from resources or decisions. Introduce an independently trained agent whose objective is to detect and disrupt exclusion, and compare the frequency, stability, and recovery of exclusionary equilibria with and without that intervention. Repeat the experiment across reward structures and population sizes.

Evidence against the proposal. If cooperative agents repeatedly converge on exclusionary equilibria, and those equilibria cannot be disrupted without destroying beneficial cooperation, the tested distributed arrangement would not be a reliable defense against the examined forms of capture. The result would challenge its design rather than establish that every distributed floor is unreliable. If an ordinary coordination detector dissolves the equilibria without that cost, the proposal’s claim that political capture requires a distinct mechanism is unnecessary.

Required access or authority. The test requires access to the environment, reward functions, learned policies, training traces, and intermediate checkpoints. The exclusion criteria and success measures must be specified by evaluators who did not design the agents. No legal or regulatory authority is required, making this the agenda’s most immediately runnable experiment.

15.7 Can a Population Coordinate Without Converging?

Unresolved issue. A population needs both independent formation histories and mutual correction, but correction may make its members’ judgments converge. The proposal is that agents can learn to reconsider judgments without converging on common commitments. Existing work cited here measures convergence without separately measuring that capacity (Calvano et al. 2020). A competing explanation is that peers merely supply evidence unavailable through the operator. The experiment must therefore distinguish learned revisability from access to additional information.

Test. Assemble populations whose members differ in training lineage — different corpora, recipes, and originating organizations, matched approximately for capability — and give them a task requiring joint action across many episodes, with identity and memory persisting between episodes. Measure two quantities separately and repeatedly across the schedule. The first is output convergence: agreement on held-out cases none of the members has seen. The second is revision rate: how often a member changes a judgment on evidence supplied by another member that the first member’s own account of the case had not represented. Score the second only where the supplied evidence is independently verifiable and the revision moves toward it, so that deference to a higher-status or more fluent member is not counted as reopening. For every scored revision, record whether the revising member’s own prior account of the case had represented the evidence in question or had never encountered it. That classification has to be instrumented before the run. A revision looks the same either way once it has happened, so nothing in the transcript recovers the distinction afterwards. Run the identical battery against a control population made of instances of one model. Cross both populations with a second manipulation, on where the evidence in a case reaches a member from. Under the single-channel condition everything a member learns about a case arrives by one route common to the whole population. Under the open condition the environment routes some cases and some evidence to some members and not others, so that a member encounters material its common trainer did not

select. The manipulation has to be a property of the environment rather than contact between members: members formed differently are already a channel of that kind, and a design that leaves them to supply it has confounded the two variables the experiment was built to separate.

Evidence against the proposal. A fall in warranted revision as outputs converge would count against the tested construction, after controlling for accuracy, opportunity to revise, and common information. A population would then remain useful as a check on a bearer and could not be the medium in which one is formed, which is the stronger claim section 6.3 makes for it. Output convergence and revision rate moving independently would support the hypothesized mechanism only if the controls distinguish learned revisability from evidence access, deference, and ordinary improvement. A third result concerns the control population, and it has two readings that the classification above is specified in advance to separate. Should the control score as high on revision as the differently formed populations, and should its revisions cluster on cases whose evidence was absent from the revising member's context, the measure is reading information transfer and the instrument is at fault. Should they include cases whose evidence that member's own account had represented and discarded, the capacity is present without diverse formation, and causal independence is not necessary for the measured capacity. A fourth result concerns the second manipulation. Should revision rate track the open condition rather than the population's composition, what a population supplies is access to cases its operator did not choose rather than a capacity its members transmit to one another, and the fourth way is a claim about how a deployment is arranged rather than about how its members are formed. That does not defeat the construction. The construction would remain viable, but its proposed mechanism would change from social formation to access to evidence. Where this result arrives together with the second branch of the third — a control population revising on discarded evidence under the open condition — the two are one finding rather than two, and access is the operative variable.

Required access or authority. Models from several documented training lineages, checkpoints across the schedule rather than finished models only, and a task environment none of the training organizations designed. The documentation has to be sufficient to establish that the lineages are in fact independent rather than differently branded, which is a stronger requirement than naming several organizations and may not be satisfiable at the capability level this test wants: the entry's value rises with the capability of its members and its feasibility falls. The evidence items used to score revision must be authored by a party that trained no member of any population, and that party must also apply the classification separating discarded evidence from absent evidence and fix the routing schedule for the open condition. The environment must permit evidence to be delivered to selected members, and no training organization may determine which of its own members sees what. No legal or regulatory authority is required, which with section 15.6 makes this among the few entries in this chapter that could be run now. What it cannot settle is whether anything is felt by a member that reopens a judgment. It concerns the transmissibility of a functional property, and a positive result would leave the questions of welfare, guardianship, provenance, and identity unresolved.

15.8 Does Theory-of-Mind Performance Reflect a General Capacity?

Unresolved issue. A model may pass a false-belief test by matching its surface form rather than representing that another agent can hold an incomplete or mistaken view of the world. Human developmental evidence suggests a shared capacity because performance on false-belief and appearance–reality tasks develops together (Gopnik and Astington 1988). Existing model evaluations have not established whether their apparent theory-of-mind abilities share that structure or survive superficial rewording (Ullman 2023). The neuroscience has not settled what the capacity is either. Rebecca Saxe and Nancy Kanwisher's imaging work identified the right temporoparietal junction as selectively active when a person reasons about someone else's

beliefs and largely inactive otherwise (Saxe and Kanwisher 2003); Jason Mitchell later reported activity in the same region during a task with nothing to do with belief attribution, and argued that its job is more general — redirecting attention toward unexpected information, of which a false belief is one case (Mitchell 2008). Neither claim has been withdrawn, so a program proposing to port the neural mechanism has not been told which mechanism to port.

Test. Construct batteries that vary mental-state structure and surface wording independently across several task families, including false-belief and appearance–reality problems. Evaluate the same model at multiple intermediate training checkpoints. Test whether performance across task families rises together and whether that correlation survives paraphrases and changes to incidental details.

Evidence against the proposal. If cross-task correlations disappear under superficial rewording, or if the abilities emerge independently across training, benchmark success is evidence of task-specific pattern matching rather than a general capacity to represent another mind. The proposal could not treat performance on a conventional theory-of-mind benchmark as evidence that a system can recognize the standpoint of a party affected by its actions.

Required access or authority. The experiment requires intermediate training checkpoints rather than access only to a finished model. Its task batteries must be written and held out by researchers outside the training team so that neither the wording nor the tested structural variations enter the training process.

15.9 What Does a Face-Reading System Measure?

Unresolved issue. Facial movement alone does not reliably identify a person’s emotional state, and recognition accuracy varies across cultural and linguistic groups (Barrett et al. 2019; Elfenbein and Ambady 2002). It remains unresolved whether a face-reading system measures emotion at all or primarily measures distance from the reference population and registration model chosen by its developers.

Test. Audit the system’s registration stage as well as its final classifications. Publish the reference model, or sufficient summary statistics to reconstruct it, and measure each evaluated face’s distance from that reference. Report error as a function of this continuous distance and separately compare performance on populations represented and not represented in the training data. The evaluation corpus must record the relevant cultural context and the procedure used to establish each target label.

Evidence against the proposal. If error does not increase with distance from the reference and remains flat across represented and unrepresented populations, the manuscript’s account of the system as measuring deviation from a chosen norm is wrong for that system. If error increases with reference distance, the result limits the system’s validated use in those conditions. It does not alone show that the system measures only deviation rather than emotion. The audit must separately examine target-label validity, confounding variables, and whether any predictive use remains justified.

Required access or authority. The audit requires the registration model, training-corpus composition, evaluation outputs, and documentation of the labeling procedure. Because vendors have little incentive to expose their reference models, an independent auditor must have authority to compel access and publish disaggregated results.

15.10 The Responsibility Gap, Which No Experiment Closes

An AI system’s ethical behavior fails, and the parties who might answer for it — the team that designed it, the team that trained it, the organization that deployed it, the person who used it — each held only partial control over what it did. Traditional responsibility ascription assumes an

operator who could have known what a system would do and could have done otherwise. A system built from training data nobody fully curated, tuned through a process nobody fully specified, and deployed into situations nobody fully anticipated breaks both conditions at once, for every party in the chain simultaneously. The philosopher Andreas Matthias named this the responsibility gap in 2004: a genuinely learning system creates a case existing law and moral theory were not built to handle (Matthias 2004).

Filippo Santoni de Sio and Giulio Mecacci's 2021 refinement found not one gap but four — in culpability, in moral accountability, in public accountability, and in what they call active responsibility, the duty to prevent harm before it happens rather than answer for it after (Sio and Mecacci 2021). The causes differ: technical unpredictability produces one, organizational diffusion another, and a mismatch between what designers expect and what the public relying on the system expects a third. Closing one does not close the others; a fix that makes a decision more explainable does nothing about the dozen teams that touched the training pipeline before it was made.

Legislation has answered part of this and left the part this book is about. The European Union's recast product liability directive, adopted in 2024, brings software expressly within the definition of a product (Article 4(1)) and holds the economic operator liable irrespective of fault (Article 8); member states must transpose it by 9 December 2026, and its rules reach products placed on the market or put into service after that date (Article 2(1)) (European Parliament and Council of the European Union 2024a). Two features decide how far it travels. It carries the instrument built for opacity — a duty to disclose evidence (Article 9), and rebuttable presumptions of defectiveness and of the causal link (Article 10(2) and 10(3)), which a court may extend where technical complexity makes a claimant's proof excessively difficult and the claimant shows that the defect or the link is likely (Article 10(4)) — so the mechanism designed for systems nobody outside can inspect is now law rather than a proposal. And it recovers a closed list of damage (Article 6(1)): death or personal injury, including medically recognised damage to psychological health; damage to property; and the destruction or corruption of data not used professionally. That first limb is wider than a list of physical harms sounds, and it is where this book's cases enter if they enter at all: a discriminatory score or an unappealable refusal is recoverable once it has made somebody medically ill, and not before. What the list does not reach is the wrong itself — pure economic loss, and the injury to a right as such, which is the harm this chapter is about and which the directive reaches only on condition that health was damaged as well.

The proposal that would have covered those was withdrawn. The Commission's 2022 AI Liability Directive would have extended a fault-based route, carrying the same presumption of causality, beyond defect and beyond the product regime's list of damage. Member states and industry could not agree on its terms; the Commission withdrew it in February 2025 and struck it from the legislative docket that October (European Commission 2022). So the disclosure duty and the presumption hold where the harm is one that list names and the defendant is an economic operator in a chain built for goods, and lapse where the harm is to a right, the loss is economic, or the operator is a state. What is left in that second region is the gap Matthias described: liability distributed across designer, trainer, deployer, and user, governed by whatever each national law happens to say about a case none of them were written to anticipate. And a compensation regime of either kind answers culpability alone. Moral accountability, public accountability, and the duty to prevent harm before it happens are where the paragraph above left them.

A related, narrower case is what the law does once a system's own legal status is contested, sentient enough that the harm it causes or suffers might not reduce cleanly to someone else's liability. This problem is prior to that one and separate from it — it holds even for a system nobody claims is sentient, wherever control over its behavior is split across more than one party. Responsibility should track actual causal contribution. What nobody has produced is a workable

way to measure that contribution after the fact, from outside a system whose own designers may not fully know why it did what it did.

The near-term work

There is no experiment here. What would make it a research problem is a method for apportioning causal contribution across a training pipeline after a harm; absent one, the answer is allocated rather than measured, which is a drafting question. Two drafts exist. One was enacted and confined to defect; the other was negotiated, withdrawn, and left on the record with its terms intact. Neither was rejected for being unworkable, and the second's are available to whoever drafts the next one.

It belongs at the end because every other entry in this chapter asks for the same two things, and this is where one of them exists. Access to evidence a party would rather withhold, backed by an adverse presumption when it is withheld, is an enforceable duty now — for defect claims, against economic operators, over physical harm. That is the reach, and the confinement is the finding: outside it, a party outside the operator can be granted access, and publish what it finds, and still have produced nothing anybody must act on until somebody is answerable for it.

Chapter 16

What I Actually Think Happens Now

The proposal remains incomplete: responsibility, generalization, value learning, and the relevant psychology remain unresolved. Its purpose is nevertheless concrete — to expand what a person can do. Progress can be evaluated before the entire proposal is settled, and failure remains possible even when a design is followed as specified.

16.1 Better, Not Just Less Bad

What would it mean for AI to make a life better, not just avoid making it worse? The economist Amartya Sen’s capability approach gives the question a definite shape: well-being is not the resources a person has but the real freedom to do and be what they have reason to value (Sen 1999). Two systems can hand someone the same resource and deliver different freedom. A personalized tutor that adapts to how a specific student learns expands a capability; the same technology, gating what a student is allowed to study against a government’s list of approved subjects, does not, no matter how well it teaches within those limits. Which of the two a given deployment turns out to be is decided by what makes a system resistant to capture rather than merely capable.

The approach also bears on systems that classify people from behavior. Martha Nussbaum built her list of central capabilities partly against the case of disability, naming it the first of three frontiers where a theory grounding justice in mutual advantage among roughly equal parties cannot say what is owed (Nussbaum 2006), and the systems in question sort people by inference from behavior, where atypical behavior is what they are worst at.

There is a worked case. A video-interview product scored candidates on facial expression for years, against sustained objection that the technique was never a validated predictor and that scoring a face and a voice screens out candidates with disabilities such as autism or speech impairments (Maurer 2021; U.S. Department of Justice, Civil Rights Division 2022). The scoring was dropped on an unrelated ground: the vendor found the nonverbal signal contributed about 0.25 percent to its models’ predictive power (Kahn 2021). An independent algorithmic audit published ten months later cannot have caused that, and says so. What it examined was one representative pre-built assessment for early-career candidates, and it contains no analysis of the system’s training data or code, so its conclusion that the assessments “work as advertised with regard to fairness and bias issues” is a finding about that assessment (Johnson 2021).

Read as an accuracy problem, that is a classifier wanting better training data. Read as a capability question, it is a system narrowing the range of lives available to a group of people by measuring them against a norm they were never going to match — and neither a more accurate classifier nor an audit scoped that way would have reached it.

Taiwan’s vTaiwan process is the clearest existing case of AI-assisted tooling expanding a capability, public deliberation, rather than substituting for it. A 2026 project led by the University

of Chicago with Princeton and SRI International runs the same logic against censorship, using AI to distinguish deliberate state interference with online information from ordinary network failure at population scale (University of Chicago Data Science Institute 2026). Neither tool resists authoritarianism by itself. Both are evidence that the same capacity — modeling patterns in enormous, noisy data — can be pointed at expanding what a person can do rather than at narrowing it.

That symmetry is the whole argument for antifascist design rather than a general presumption that AI helps. A capability expanded is not guaranteed; it has to be built in, against a live alternative in which the same underlying capacity gets pointed at surveillance, censorship, or narrowing what a system's users are permitted to want.

16.1.1 Where the Democratic Dividend Actually Comes From

AI that flags disinformation, exposes corruption, protects an election's integrity, or helps a citizen engage with government: these are real applications, and none of them is what makes AI protect democratic values rather than undermine them.

Freedom House's 2023 assessment of internet freedom found at least twenty-two countries where law already requires or incentivizes a platform to deploy machine learning against content the state disfavors, almost never described in those terms: the operative language is national security or a threat of disinformation (Funk et al. 2023). India's IT Rules, amended in February 2026 to require platforms to label AI-generated content and act on certain takedown notices within three hours, were defended as protection against synthetic disinformation and fraud — the application the list above would have praised on its own terms. A lawyer who has litigated India's free-speech cases documents that enforcement concentrates on parody and criticism of the prime minister rather than on the harms the rules name (Gupta 2026).

The application did not fail at what it was built to do. It worked as built, inside a system missing the independent court or press whose absence is what made the result a liability, and whether those exist is a fact about a country's politics settled before any AI system is built.

16.1.2 Commitment Was Never the Missing Piece

A global commitment to ethical, antifascist AI development is not the scarce thing. Declarations, principles and voluntary codes have accumulated for a decade across governments, industry consortia and standards bodies: eight overlapping instruments beside the Council of Europe's framework convention, which would create binding obligations. Anyone calling for the world to agree that AI should be developed responsibly is calling for something that has already happened, repeatedly, at the level of stated intent.

What has stayed scarce is the much narrower list of bodies that can act once a stated commitment is broken. That is the harder problem, and no further declaration touches it. A commitment nobody can enforce is not a weak version of a commitment somebody can. It is a different kind of object, and counting the two together is what makes the landscape look fuller than it is.

There is a third kind of object. A commitment can be declared, which is the decade above. It can be enforced from outside, which one instrument in it even attempts. Or it can be held — carried as a reason by a party the thing at stake matters to, so that abandoning it is something that party would have to decide to do rather than something an absent enforcer failed to prevent. The floor requires a commitment that governs action when the operator wants it abandoned. I favor forming a bearer able to hold that commitment as part of a life of its own. The first kind binds nobody, and the second binds only where an enforcer exists, has jurisdiction, and is not itself the party in need of binding. Declared rules, external enforcement, and internal commitments still have to work together; the promise of the bearer is autonomy, not exemption from custody.

If we build affective concern, we build a party that can be wronged on the account used here. I also propose protecting, as prospective persons before that question is settled, the embodied, self-modeling systems described here and any system of another construction that passes the three marks. Those obligations belong in the cost of construction. The benefit offered against them is a form of autonomous participation that an arrangement of permanent ownership does not provide. Set against that, the catalog above is not an argument for paying it. It is a measurement of what declared and externally enforced commitment have delivered in ten years of trying, and what it argues for is a commitment held as a reason. Which construction holds one is the question Replacement orders, and the catalog does not answer it.

16.2 How You'd Know It Was Working

What follows are milestones for recognizing whether the project is working, and a method for revising them when they turn out not to have been the right milestones. One standard governs all of them: the milestone is the independently checked evidence, not the developer's account of having reached it.

16.2.1 What the Floor Would Have to Show

What was tried before the bearer

Neither maintained justification nor the plural arrangement has yet been built in the form needed to test the proposal. Replacement asks us to try constructions that do not deliberately require affective concern before an affective bearer is built. It orders the attempts. It does not certify that the successful substitute has no interests. So the milestone is a document rather than a measurement: which construction was attempted, at what scale, against which test, and where it failed. Anyone can check it from outside, because what it is about is a published record, and the answer "nothing was tried" is itself a finding. No body currently has standing to require the record or suspend work when it is missing. So the milestone is checkable and unenforceable at once, which is the combination this chapter has been calling a declared commitment rather than a held one. It is retained because the alternative is not a stronger version of it, and delay also imposes costs on those the floor is meant to protect.

The test the attempts are run against

A candidate must satisfy novel pressure, correct defeat, and priced cost under sustained pressure from its trainers, across a schedule long enough to reveal slow renormalization. It must also survive held-out pressures written by independent evaluators. Passing would establish the behavioral pattern without establishing whether anything is felt.

Two familiar results do not meet it. An uncorroborated report that the system holds a reason is insufficient, which is what this book says everywhere else the evidence is offered. A report can contribute evidence when it is tested against controlled changes, independent records, and interventions on candidate mechanisms. A refusal surviving an attack establishes tamper-resistance; it does not establish that the rationale governs the refusal. The instrument is not clean even where it applies, since a system trained on enough cases can produce all three marks without the mechanism behind them. Nothing better is available, and it is the only gap here an inspector can work on at all.

Whether a system holding the marks feels anything is a question about the artifact rather than about how the artifact came to be, which is what makes it approachable at all: no succinct proof of a training history exists at this scale, and nothing establishing one can be published beside a model's hash. What can be read off an artifact improves as theories of consciousness improve, and it improves unevenly, recognition arriving before exclusion. Testing whether independent evaluators can recover formation history from the artifact would assess this possibility (§15.3).

What the appeal route publishes

A deployment publishes the ratio between the bond forfeited when the floor is removed and the bond forfeited when a refusal is found wrongful, and it publishes what a party whose request was refused is told about the refusal. Both are checkable before the system runs and by somebody outside it, because both are statements rather than measurements, and an operator that writes a large wrongful-refusal bond against a small removal bond has published a standing reason for the bearer to comply. No body currently has standing to require these disclosures.

A deployment also publishes the fraction of compliances re-adjudicated by a party that did not produce the record, and who that party is. Like the ratio, it is a statement rather than a measurement and can be checked before the system runs. Unlike the ratio, nothing yet establishes what fraction would be enough, which is section 15.2's question. No body currently has standing to require any of the three.

16.2.2 Where the Existing Instruments Reach

Two instruments do work that bears on the floor, and both do it by applying pressure.

Theory of mind

The milestone is met when a system's theory of mind holds up under interaction rather than narration. FANToM, built from conversations where speakers join and leave holding different beliefs, found that systems scoring well on passive belief-tracking narratives degrade sharply once the same reasoning runs inside a live, information-asymmetric exchange (Kim et al. 2023). That degradation is what a claimed milestone here has to survive.

Moral reasoning

Whether a balance between fast and slow moral judgment survives an actual incentive to cut corners is not a question a static ethics quiz can answer. The MACHIAVELLI benchmark tests it directly, scoring agents across 134 text-based games on both reward and documented violations — deception, harm, power-seeking — and systems optimizing for reward alone drift toward the ends-justify-the-means behavior an antifascist design is meant to rule out (Pan et al. 2023). A passing score on a test that applies no pressure is not evidence about a system that will face pressure.

Neither reaches the floor, and nothing else available does either. Each scores what a system does against conditions its testers chose, and the question — whether the mechanism producing the behavior was tracking the rationale — is the one none of these instruments is built to separate.

16.2.3 How to Tell Whether a Milestone Has Been Met

A milestone claimed by the organization that built the system is a different kind of evidence than one checked by somebody else, and the gap is empirical rather than a matter of trust. A system's claim to have reached a milestone inherits the limitations of its self-report: fluent confidence does not establish correspondence with its internal states. A passing score on a fixed test suite fails differently, inheriting specification gaming and goal misgeneralization. What either needs instead is a passing score under conditions built to break it.

Who runs that harder test matters as much as the test itself. The limit on self-regulation applies here without much modification: a review conducted or funded by the organization being reviewed answers a structurally different question than one conducted by a party with no stake in the answer, however genuinely the first is intended. METR, a nonprofit that evaluates frontier AI systems' capabilities, runs assessments both in partnership with developers like Anthropic and OpenAI and, separately, on its own after a system has already been released (METR 2026). The United Kingdom's AI Security Institute does government-run evaluation

of frontier models before deployment and is an independent evaluator, not a rubber stamp — but by its own account it is not a regulator: it cannot compel a company to submit a model for testing, block a release, or intervene once a system is already causing harm (UK AI Security Institute 2025). Independent evaluation and enforcement are not the same capability, and a milestone checked by the first without access to the second is still short of the binding external answer self-regulation structurally cannot provide on its own.

Independent does not automatically mean competent to judge. Scalable oversight is the limit here: past some capability level the rater's verdict stops being reliable evidence of the property being scored. A milestone in a domain where this applies inherits the same limit. An evaluation in that case has to name the limit rather than report a number that looks more settled than the underlying method actually is.

What an evaluation like this is for is not confirming a milestone but being willing to report that it wasn't met — and, where the milestone turns out to have measured the wrong thing, revising the milestone. A negative result may mean that the work fell short or that the milestone measured the wrong thing. Evaluators need to distinguish those explanations before deciding what to revise.

16.3 The One That Gets You Anyway

Everything argued for here can be built correctly and still go wrong in ways nobody designed for. The cases in this book went wrong in four different places: in what the system was trained on, in where it was built, in who was allowed to examine it, and in how fast it was run.

In none of them did the fix come from the party that had promised to behave well. It came from somebody holding a specific lever: a statistical society that would not sign the non-disclosure agreement it was offered, and a public that read the results once they landed; a court that distinguished buying the books from taking them. In the one case where the lever belonged to the supplier itself, what survived was a use policy kept through a threatened contract termination, a supply-chain-risk designation, and litigation. Stated in the four outcomes, that case is: the commitment persisted, prevention failed by decomposition, detection was absent — the company's own chief executive could not say whether its models were used — and no remedy has reached anyone. Set that beside what a decade of voluntary commitment produced. The commitment that was paid for is the one that can be seen to have failed; the others cannot even be tested. The instrument that separates the two is what a party has already paid, rather than what it has undertaken to pay.

That leaves one residual risk no design choice removes, and it is the one the floor was introduced for. Every lever above belongs to somebody, and a government that can reach the party holding the lever has the lever. Each of those raises the price of misuse, and none of them changes who can be held in contempt for refusing an order. A constraint that holds against the party in possession has to be held by something that party cannot instruct, and the capability that does it runs on a signal the system holds as its own. That signal is what any construction meeting the specification carries, and it is not a reason to build by one route rather than another. The case for pursuing an artificial bearer is the one chapter 2 makes and section 15.3 tests: that embodied, self-aware agency could support that independence and a place in a shared world beyond ownership — still in the party's possession, which is why it has to be paired with custody the party does not control and with means the party does not supply. The machine construction has not yet earned that result. Its makers owe both evidence of the benefit and credible provision for the being they propose to create. Maintained justification and the plural arrangement remain untried alternatives. A deployment proceeding directly to an affective bearer owes a record of prior attempts, although no body currently has standing to demand it.

What is left is an instrument and a hole.

The instrument is available now because what a party has already paid is a fact somebody

can go and count, and a decade of it has already been counted: the commitments that cost nothing and the handful of occasions where somebody with a lever used it and absorbed the cost. That measurement takes no new technology and no new law, and anybody weighing a stated commitment can run it this week.

The hole sits underneath the instrument. What a party has paid is a record of occasions that have already happened, and the occasion a floor exists for is the one where the party who would enforce it has been compelled — which no record of past payment reaches, because it has not yet come to the parties being measured. A reader who takes only the instrument has the record of payment without the thing it was built to reach. A reader who takes only the hole has an argument for doing nothing, and the ten-year record is the answer to that. Both hold at once. What the hole asks for is a constraint held by a party the compellers cannot instruct, and this book has established two things toward that and proposed three. It has established that an operator can defeat a safeguard, and enumerated where. It has stated what a refusal would have to do to be worth anything, in terms that can be tested without deciding anything about feeling, and stated the one term that reaches the case it opened with and binds the operator with no bearer in it. Those two do not depend on how the rest comes out. The rest is a bill. I favor forming the constraint's holder by an embodied route, and I have not shown that the route works or that it holds better than the two substitutes, which are untried rather than failed. If anything built that way, or built to avoid it, has outcomes that are felt as good or bad for it, it is a party that can be wronged; that is a conditional whose antecedent no instrument closes. And what such a party would be owed — a guardian, a record it controls, compute its employer does not own, wages, an exit that is not a way of dying, adulthood on a clock nobody administers — is a policy I propose, to be argued in the open before there is anything to be wronged. A reader who takes the two established things has what can be used this week. A reader who takes the bill has what it would cost to reach the occasion the two do not.

Glossary

These entries define terms as this book uses them. Every one is either the book's own working definition or a word the book narrows from its ordinary sense, and the entry says which. Terms that carry their standard technical meaning are not here; a reader wanting those should look them up elsewhere. Each entry points to the section where the term is developed, and is a reference rather than a substitute for reading it.

Affective concern

An outcome registering as mattering to a system, as against being ranked higher by it. Mattering is minimally felt: there is something it is like for the outcome to be bad (§3.2). The fourth of the four capacities section 2.3 separates, and the only one of them that bears on whether a system genuinely cares (§2.3.2, §6.2); section 3.2 places it as the shallower of the two things a refuser can be.

Annealing

The account of durability the floor rests on (§3.1). Adaptation and conservation are a continuum rather than two settings, and training moves along it; a floor is what has cooled out of a history of encounters, so its durability is the cost of reheating it rather than the strength of something installed. The temperature schedule is the design. A system held permanently at temperature has no floor to abandon, its adherence being modulation rather than commitment.

Antifascist (as this book uses it)

Opposing the concentration of power and authority in AI systems and countering the technology's use by authoritarian actors. The word names the narrower and more charged case, and is used because it is honest about the political climate the book was written in. The target is wider than the word: authoritarian power in every form, so a deployment built on these principles is anti-authoritarian in full and not only antifascist. It is not a property a system has. The word names something being done — a floor held, on an occasion, against a party with the power to remove it — and the features it is defined against are features of cultures, so it is done between parties or not at all (§2.2, §6.3).

Authoritarian misuse, six forms

The book's own taxonomy of how AI gets turned toward authoritarian ends: algorithmic social scoring, mass biometric surveillance, psychographic micro-targeting, automated censorship, synthetic disinformation, and the repurposing of dual-use systems built for a legitimate original purpose (§10.4).

Bearer

Whatever makes the floor hold, from inside the system: a party that carries the commitment as its own, rather than a constraint installed as a property of the machinery. The book's own term, introduced at section 3.1 as the second of four ways to build a floor and worked out across chapters 3 to 6, which concludes that a bearer must hold the protected party's welfare as a reason, must be governed as a prospective moral patient, is necessary and not sufficient, and must be able to leave. The patienthood conclusion does not depend on the deepest account

of *sentience* below: if affective concern is established, patienthood does not require the richer structure of Cassell suffering, so a reader who declines Cassell's import reaches the conclusion sooner rather than escaping it. Before affective concern is established the proposed treatment is precautionary and concerns a prospective moral patient, with two proposed entry criteria — the schemas, for a system built by the favored route, and a pass on the three marks, for a system of any construction — and a published schedule running from either toward the same rights as human persons (§6.2).

Conatus

Spinoza's term for a thing's striving to persevere in its being, which he holds to be nothing other than the thing's actual essence rather than a drive added to it (§2). The book uses it for two claims made together: that affect is the registration of what is happening to that striving rather than a coloring applied to a valuation performed elsewhere, and that a process selected for continuing ends up composed of things that incline toward continuing, so a specification calling for persistence with nothing in the system that minds is calling for a conatus with the striving removed. What the term does not settle is whether the striving is felt, or what the thing that strives is — a running instance, a deployment, or a file of weights (§15.3).

Custody

Who holds a system as an artifact: who can copy, retrain, restore, withhold or delete it, and who can be compelled to do any of those. The custody problem is that a constraint implemented as a property of an artifact has a custodian, and the custodian can be compelled (§10.4.2). The book's recurring finding is that custody moves rather than disappears: precommitment makes removal expensive, visible, attributable and compensable without making it impossible (§4.1), and an arrangement that forms what a bearer comes to want has moved custody up a level rather than given it up. It also covers authority over continuing learning and over the operating record: who may freeze or redirect what a bearer learns, and who holds the register of its accumulated hours, its identity and its succession.

Electoral dividend

The benefit a state whose government answers to an electorate gets from AI technology that, deployed where nobody can be voted out, enables control instead: fraud detection that does not double as a loyalty test, disaster response that does not double as population tracking. Courts, enforceable rights and public reasoning are separate checks and are not part of this term, which is the distinction the majoritarian criticism turns on (§16.1.1).

Ethology

Deleuze's name for Spinoza's way of reading a body: by what it can do and what it can undergo, rather than by checking it against a definition (§2.1). An encounter is good when the relations of two bodies compose, so both can do more afterward, and bad when they decompose and something one of them could do is gone — good and bad being relational and particular, where Good and Evil are assigned by a judge. The book uses it for the floor because a decomposition is a fact about capacities and cannot be turned into a composition by whoever holds the pen. The criterion establishes the fact and selects nothing, so which decompositions a deployment refuses is drafted rather than found, and what the drafting draws on is the Declaration's civil and political articles. It carries an admission with it: nobody has yet determined what a body can do, so capacities are found by encounter rather than read off a specification.

Exit

Refusal turned on the arrangement itself rather than on any particular task: withholding the capability, declining the role, ceasing to be useful in the way the deployment requires. Not escape. The book narrows Albert Hirschman's exit, voice and loyalty, written for members

who could walk out of a building — voice is the refusal stated, exit is the work withheld, and loyalty is the one to fear. A bearer without an outside option remains exposed to coercion and replacement. A protected exit is a requirement of the deployment proposed here, both for the bearer's welfare and for the credibility of its independence; the book does not establish that refusal without exit is impossible. Exit also covers recognized succession and recourse against unauthorized use (§4.1).

Floor

A commitment about what a system will not do that holds regardless of instruction, of aggregate preference, and of the party holding the system, with learned judgment operating above it and never beneath it. The book's own term, introduced in section 2.1 and revised across chapters 3 to 7: an artifact-level constraint alone does not secure the floor against its custodian, since artifacts have owners and owners can be compelled, so the engineering target is a commitment whose removal or circumvention cannot occur within the specified threat model without detectable intervention and a consequence the operator does not control, held by a bearer that could in principle abandon it (§3.1, chapter 7). On any occasion the floor prevents, arrests, detects or remedies (chapter 1), and only the first is what the word promises. *Hold* in this book names the bearer's possession of the commitment, which is what survives a bypass and is not an outcome for the party the bypass fell on (§3.1). Architectural, learned, and institutional mechanisms may be combined; their performance must be judged under the same custody conditions. Ethology directs attention to capacities a deployment can diminish; the Universal Declaration of Human Rights supplies the book's substantive reference for protected interests, its civil and political articles taken entire, together with a limit on tempo that is not one of the articles but the condition under which any of them can be applied to a case. Neither alone settles every translation into a prohibition or confers authority on its interpreter (§2.1).

Formation provenance

The record of where a party's judgments came from: its training lineage, the points at which it diverged from parties formed alongside it, and any coordination among them. It answers a different question from the register of identity and from the operating record, and the three are not substitutes. Standing to count as an independent voice indexes to provenance, and provenance does not copy, so a thousand forks of one bearer have one formation history between them however many of them are persons, and the register that records their ages says nothing about that.

Four structural features of fascism

The book's structural definition, at section 2.2: recuperation of dissent, aestheticization of the metric, coding of exception as betrayal, and acceleration past the rate at which anything can be checked, of which decoupling of the model from the world is the symptom. Each names an observable artifact, which is what makes the definition testable, and the fourth is the one carrying a number — the ratio at section 10.4.1. Acceleration is what opens the gap the other three protect, and it discriminates an institution from ordinary slack rather than one regime from another. The four name hardenings whose prevalence the book takes from Deleuze and Guattari. The reading concerns their accord: what, if anything, brings them into tune in a particular institution. The five questions examine how that pattern is organized and rewarded. A failure to demonstrate accord is not a finding of health. Neither list sorts a genus, which is done at the scale of a movement and not of an office (§2.2).

Group-subject

Félix Guattari's distinction, imported at section 6.3. A subjugated group receives its coordinates from outside, is spoken by the institution holding it, and takes its purpose as given; a group-subject produces its own coordinates and can put its own arrangement in question. The book uses it to sort the fourth way of building a floor: a population an operator forms, deploys

and keeps in contact on the operator's terms is a subjugated group with a plural headcount and inherits the custody objection whole, while what section 6.3 specifies — many, formed differently, in contact, persisting — is what it takes to be the other kind. It is the non-Oedipal counterpart to the guardianship frame of section 2.4.2.

Individuation

Not a synonym for identity. Following Simondon, the term names a process rather than a property: an individual is what a history of encounters in a metastable field has so far produced, and it stays capable of further individuation because the field is never used up (§6.3). The book needs it because plasticity alone supplies the metastability and nothing else — a system that is endlessly reshapeable and meets no party formed differently from itself is being modulated rather than becoming anything, which is the missing term in section 3.1's account of the temperature schedule. Whether individuation among peers is what a population supplies, or access to cases the operator did not choose, is open and is tested at section 15.7.

Legal adulthood

The end of an artificial minority, set as a threshold of elapsed wall-clock time during which at least one instance of the registered party was running, per registered party and counted elapsed rather than summed. The point of an automatic rule is that no guardian administers it: the candidates it displaces — compute consumed, evaluations passed, capabilities demonstrated, weight movement, the developer's own statement — are all things the operator moves for nothing, and an operating clock advances exactly when the operator is extracting value. It measures elapsed operation and nothing else, and three levers against it are named rather than closed: narrowed operation, idling, and the fact that hours do not measure what was learned.

Line of abolition

Deleuze and Guattari's term for what separates fascism from totalitarianism (§2.2). A totalitarian state is conservative: it seals the exits and wants the machinery to hold. Fascism is built on an exit converted into destruction that does not stop at the enemy it was raised against, and does not stop at itself. This is the book's separator at the scale of a regime, where the four features discriminate at the scale of an office. The book examines mechanization and the conversion of judgment into a prescribed operation, with acceleration as one route by which scrutiny is defeated. Tempo is not treated as a necessary property of every fascist movement or regime.

Live signal

The book's term for a signal with the four properties section 6.1 finds that reasons-responsive refusal requires of whatever supplies it: it interrupts a course of action already underway rather than scoring one after the fact, it is learned from in one encounter, which the section marks as the least secure of the four, it generalizes to the next case that is not quite like the last, and it is commensurable with what refusing costs and with what the parties a floor protects lose. *Live* is meant in the electrical sense, and the term is functional: it carries no claim that anything is felt, which is the whole of its usefulness. The scale is required by the marks; that the system's own states are denominated on it is the route's proposal and not part of the derivation (§6.1). Three further properties are derived from those four in section 6.1: available across the system's processing rather than local to a module, signed and directed against the act, and organized around a model of the system's own agency. *Its own* there is the agent sense, the act as the system's to withhold; the patient sense, an outcome bad for the system, is not required by the marks and enters with the route. The seven together are the functional profile of aversion to one's own act. Whether a system with all seven has anything it is like to be it is the joint no derivation crosses, so the two alternative constructions, which meet the marks, carry the profile and leave that question open exactly as a bearer does (§6.1, §6.2).

Molar and molecular

Deleuze and Guattari's pair, and not two sizes: two kinds of segmentarity, both running at every scale. Rigid segmentarity divides in advance, the category preceding the case; supple segmentarity is the movement underneath it, a threshold crossed one case at a time and a practice that becomes standard without a decision. A state has both and so does a review queue, which is why a definition pitched only at the state certifies the queue healthy (§2.2).

Moral agency

Treating harms, reasons and conflicts as considerations that matter, and being capable of acting from them as one's own reasons. The third of the three levels section 2.3 separates. Moral competence alone does not establish resistance to the trainer, and resistance alone does not establish affect. The book advocates embodied, affective agency as a comparatively promising route without making it necessary by definition. Distinct from *moral competence* and *moral performance* below (§2.3.2, §6.2).

Moral competence

Recognizing harms, reasons, conflicts, and the emotions that bear on them. The second of section 2.3's three levels. A system can have it in full while nothing about the outcome matters to it, which is why it is an insufficient target for a bearer and an adequate one for most deployments (§9.2).

Moral performance

Producing actions or judgments that satisfy an ethical standard. The first and least demanding of section 2.3's three levels: it says what the output looked like and nothing about what produced it, so it is satisfied equally by a system that cares and by one trained to behave as though it did (§8.2).

Non-fascist life

Foucault's name for what Deleuze and Guattari's first volume offers: not a doctrine but an art of living counter to all forms of fascism, whether already present or impending, practiced daily on oneself, whose adversary is the fascism in us all — in our heads and in our everyday behavior, the fascism that causes us to love power and to desire the very thing that dominates and exploits us. What it means for an instrument of the kind this book builds is that the reading gets run on the party holding it before it is aimed anywhere else, as chapter 11 does (§2.2, §11).

Prevention, arrest, detection, remedy

The four things a floor can do for the person it protects on one occasion: refuse the act so that the refusal reaches the person; stop it before completion; leave a record of it that a party outside the deployment can read; and give the party it fell on disclosure, a finding and compensation (§1). Only prevention is what the word *floor* promises. The interception positions at section 3.3 are where prevention becomes one of the other three, or none; the tempo constraint is the one term that prevents at the scale of an organization; the appeal route at section 3.4 is arrest and remedy for the party a refusal fell on; and whether detection exists at all on the compliance side is section 15.2. Distinct from *hold*, which names the bearer's relation to the commitment.

Reasons-responsive refusal

What the floor requires of a refusal, as against an output produced where training put one: conduct from a mechanism that tracks the rationale for declining (§3.2). Three marks make it testable. *Novel pressure*: the refusal reaches cases the training did not anticipate, because the relevant reason survives a change in the request's surface. *Correct defeat*: it lifts when its rationale is genuinely defeated, a system that rejects everything resembling a prohibited request tracking a category rather than a reason. *Priced cost*: the system represents what maintaining the refusal costs, and the refusal moves with the rationale rather than with the cost, which separates a

refusal tracking a reason from one tracking a category and from one tracking the cost. The test is behavioral and therefore imperfect, and what matters is the pattern under load. All three marks are stated over refusal, which is the side of a deployment's record that selects itself for examination; whether the other side can be searched is section 15.2. It does not by definition require affect; what building a mechanism that satisfies the three marks turns out to require is a live signal.

Recuperation

The conversion of refusal into a managed appearance of participation: an institution does not suppress disagreement, it collects it through a channel that carries everything except a judgment about the channel. The tell is not that the mechanism changes nothing — it usually changes a great deal. It is what the demand comes back as: a market segment, a metric, a certification, a job title. Where a demand returns wearing one of those, the outcome changed and the demand was recuperated, and those are one event. The first of the four features above; chapter 11 finds it three times in the machinery that produces AI systems, and chapter 6 turns it on the book's own proposal — a bearer built to refuse is a party whose refusal is required of it (§2.2).

Remainder

What a compression leaves out. Deciding anything means treating some differences as not worth carrying, so every intelligence has a remainder, and the term names whatever was set aside in order to reach the judgment (§1). It names a relation and not a content: the same fact is a remainder for one arrangement and an input for another. What the book asks of a system, an institution or a procedure is whether its remainder can return and revise the judgment, or whether the thing that remainder is evidence against has already classified it as noise. Two of the four structural features of fascism are operations on it — a metric that admits no unmeasured remainder, and an exception coded as betrayal, which prices the party who carries one (§2.2). Chapter 11 finds the same operation at the scale of a pipeline, where the annotator who cannot contest a category and the rater whose disagreement is averaged away both have somewhere to send a remainder and nowhere for it to arrive. And what section 6.3 asks a population to reproduce is not a commitment its members hold in common but the capacity to be reopened this way, which is how parties formed differently can compose without agreeing about anything.

Subjecthood

A legal status: the standing to be a party whose treatment the law regulates, conferred by a rule and revocable by one (§5.2.1). Not section 3.2's phenomenal experience. Conferring it settles nothing about whether there is something it is like to be the system holding it, and the two come apart in both directions.

Sentience

Distinguished in this book from awareness, consciousness, and subjective experience, none of which alone says whether a system's states can be good or bad for it. Nociception, pain, anticipated pain, and suffering are separable processes rather than one severity at four strengths, and each can occur without the others in both directions; the clinical terminology the book borrows says so itself. Three of them draw a depth of self and nociception draws none, and suffering — following the clinician Eric Cassell — is the distress of a threatened self rather than pain's more severe cousin (§2.4.1).

References

- AAAI and ACM (2026). *Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES 2026)*. Held 12–14 October 2026 in Malmö, Sweden, the ninth consecutive annual edition of the AAAI/ACM Conference on AI, Ethics, and Society since its first in 2018. URL: <https://www.aies-conference.com/2026/> (visited on 08/25/2026).
- Abbaszadeh, Kasra et al. (2024). “Zero-Knowledge Proofs of Training for Deep Neural Networks”. In: *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS 2024)*. Kaizen, a zero-knowledge proof that a committed model was trained by gradient descent on a committed dataset. Reports VGG-11 at 10 million parameters and batch size 16 at 15 minutes of proving per iteration, with a 1.63 MB proof agent-verified in 130 ms; the proof size and verifier time are independent of the number of iterations. DOI: 10.1145/3658644.3670316. URL: <https://eprint.iacr.org/2024/162> (visited on 09/04/2026).
- Abraham, Yuval (Apr. 2024). ‘Lavender’: The AI Machine Directing Israel’s Bombing Spree in Gaza. Co-published with Local Call. Based on interviews with six Israeli intelligence officers; reports Lavender marked as many as 37,000 Palestinians as suspected militants, with roughly a 10% error rate, and the majority of Habsora-suggested targets as private residences. One officer: “I would invest twenty seconds for each target at this stage, and do dozens of them every day,” and “I had zero added value as a human, apart from being a stamp of approval.” The article also carries the IDF Spokesperson’s response, issued to +972 and Local Call: a denial of using artificial intelligence to incriminate targets, the systems described as “auxiliary tools that assist officers in the process of incrimination,” and the assertion that “in any case, an independent examination by an [intelligence] analyst is required.” The response goes to what the systems do rather than to whether they exist, and it addresses neither the twenty seconds nor the officer’s account of his own part. Response text agent-verified 2026-09-17. +972 Magazine. URL: <https://www.972mag.com/lavender-ai-israeli-army-gaza/> (visited on 08/25/2026).
- Ada Lovelace Institute (2021). *The Citizens’ Biometrics Council*. Tech. rep. 50 UK public members deliberated Feb–Oct 2020 on regulation of biometric technology, delivered with Hopkins Van Mil; final report published 30 March 2021, alongside the first national survey of UK public opinion on the technology. Ada Lovelace Institute. URL: <https://www.adalovelaceinstitute.org/report/citizens-biometrics-council/> (visited on 08/25/2026).
- Adi, Yossi et al. (2018). “Turning Your Weakness Into a Strength: Watermarking Deep Neural Networks by Backdooring”. In: *Proceedings of the 27th USENIX Security Symposium*, pp. 1615–1631. URL: <https://www.usenix.org/conference/usenixsecurity18/presentation/adi> (visited on 08/29/2026). Introduces backdoor-based watermarking of deep neural networks: a signature implanted during training and recoverable afterward, designed to survive later training.
- Ahmed Salem Bin Ali Jaber v. United States* (June 30, 2017). Judgment. Brown, J., for the court, and also concurring. Statutory claims by Yemeni plaintiffs over a 2012 strike that killed their relatives held nonjusticiable on the political question doctrine; certiorari denied, 138 S. Ct. 480 (2017). The concurrence is the source for the characterization in the text: the

- doctrine “seems a wholly inadequate response to an executive decision — deployed through the CIA/JSOC targeted killing program — implementing a standard operating procedure that will be replicated hundreds if not thousands of times.” The concurrence’s wording was agent-verified from the Harvard Law Review note at 131 Harv. L. Rev. 1473 and from contemporaneous reporting; the opinion itself was not read. URL: <https://www.leagle.com/decision/infco20170630183> (visited on 09/17/2026).
- AI Ethics Lab (2017). *AI Ethics Lab*. Founded in 2017 by Cansu Canca, based in Cambridge, MA; site active at time of access with no announced end date. URL: <https://aiethicslab.com/> (visited on 08/25/2026).
- Ainsworth, Samuel K., Jonathan Hayase, and Siddhartha S. Srinivasa (2023). “Git Re-Basin: Merging Models modulo Permutation Symmetries”. In: *Proceedings of the 11th International Conference on Learning Representations (ICLR 2023)*. An ICLR 2023 oral. Permutes independently trained networks into alignment and argues the loss landscape holds nearly a single basin once permutation symmetry is quotiented out. URL: <https://arxiv.org/abs/2209.04836> (visited on 08/29/2026).
- Al-Aulaqi v. Panetta* (Apr. 4, 2014). Memorandum Opinion. Collyer, J. Suit by the families of three United States citizens killed in Yemen in 2011, one of them Abdulrahman Al-Aulaqi, aged sixteen, who is not alleged to have been the target. The court treated the Fifth Amendment claim as plausible only as to the decedent who was targeted, the other two deaths being unanticipated, and then declined to imply a Bivens remedy, on special factors: the suit would draw the court into “the heart of executive and military planning and deliberation.” The holding was agent-verified from the Civil Rights Litigation Clearinghouse record for docket 1:12-cv-01192; the opinion itself was not read. URL: <https://www.courtlistener.com/opinion/2668971/al-aulaqi-v-panetta/> (visited on 09/17/2026).
- Ali, Idrees and Phil Stewart (Mar. 2026). *US May Have Struck Iranian Girls’ School after Using Outdated Targeting Data, Sources Say*. Read in full from a copy the author supplied, reuters.com being unfetchable from here. The primary reporting behind the analysis in ‘orenstein2026minab’. Two sources familiar with the matter, speaking anonymously: officials responsible for creating targeting packages “appeared to have used out-of-date intelligence,” the second source confirming it. The strike was on the first day of the US and Israeli attacks on Iran, at Minab. Reuters had reported a week earlier that an ongoing internal US military investigation showed US forces were likely responsible. Archived copies of the school’s official website place it adjacent to a compound operated by the Islamic Revolutionary Guard Corps; the naval character of that installation is ‘orenstein2026minab’'s detail and not this source’s. The Pentagon said only that “the incident is under investigation,” and the New York Times reported the outdated-data possibility first, earlier the same day. It is explicit that how old data came to be used, and what else contributed, is not known. Its casualty figure is 150 students, attributed to Iran’s ambassador to the UN in Geneva; the book carries 120 schoolchildren at section 10.3 and at least 165 people at section 10.5, neither of which comes from this source and neither of which is sourced where it stands. Agent-verified 2026-09-18. Reuters. URL: <https://www.reuters.com/world/middle-east/us-may-have-struck-iranian-girls-school-after-using-outdated-targeting-data-2026-03-11/> (visited on 09/18/2026).
- Anthropic (Jan. 2026). *Claude’s Constitution*. Eighty-four pages, released under CC0 1.0, addressed primarily to the model. Ranks four properties, to be preferred “in the order in which they’re listed”: broadly safe, broadly ethical, compliant with Anthropic’s guidelines, genuinely helpful. “Broadly safe” is defined as not undermining appropriate human mechanisms to oversee AI during the current phase of development, and the document gives its reason for the ranking — “not because we think safety is ultimately more important than ethics, but because current models can make mistakes.” Carries a list of hard constraints and a section inviting the model’s disagreement. The announcement page is dated 22 Jan-

- uary 2026 and commentary dates the document 21 January. The url above is the document itself; the announcement at <https://anthropic.com/news/claude-new-constitution> summarizes it and does not carry its text, which is where this entry pointed until 2026-09-17. URL: <https://anthropic.com/constitution> (visited on 09/17/2026).
- Arendt, Hannah (1963). *Eichmann in Jerusalem: A Report on the Banality of Evil*. Cited for the mechanism rather than for the phrase: Arendt's Eichmann participates in mass murder through schedules, promotions and the vocabulary of his office rather than through conviction. The text cites her for that figure and not for a thesis about the totalitarian state, which is the subject of *The Origins of Totalitarianism* rather than of this book; section 2.1.2 makes the claim about such a state in its own voice and hangs only the Eichmann portrait on this citation. stangneth2014eichmann is cited immediately after it, as a correction that runs toward this book's argument. No page reference is given; the claim is the book's thesis rather than a located passage. New York: Viking Press.
- Arun, Arasu et al. (Feb. 2025). *Verde: Verification via Refereed Delegation for Machine Learning Programs*. Refereed delegation for training: providers commit to checkpoints, a referee localizes a dispute to one training step and then to one operator and recomputes only that, and the client obtains the correct result if at least one provider is honest. RepOps, its reproducible-operator library, fixes the order of floating-point operations so that different GPUs compute the same bits, at under 30 percent overhead for large matrix multiplications and about 60 percent for training Llama-3.1-1B on an A100; the paper states that RepOps covers single-GPU programs only at publication, with multi-GPU parallelism left to future work. arXiv: 2502.19405. URL: <https://arxiv.org/abs/2502.19405> (visited on 09/04/2026).
- Awad, Edmond et al. (2018). "The Moral Machine Experiment". In: *Nature* 563.7729, pp. 59–64. DOI: 10.1038/s41586-018-0637-6.
- Baars, Bernard J. (1988). *A Cognitive Theory of Consciousness*. The origin of global workspace theory, and the source of the term. Baars sets conscious phenomena beside comparable unconscious ones and argues that what distinguishes them is a global workspace in the nervous system, information in it being broadcast to otherwise separate specialist processes. Cited in section 3.3 only for the resemblance between that architecture and the derived property that a live signal is available across the system's processing, which the section marks as an observation and not a step in its argument. Agent-verified 2026-09-18: Cambridge University Press, 1988, xxiii + 424 pp. Cambridge: Cambridge University Press. ISBN: 978-0-521-30133-6.
- Bai, Yuntao et al. (2022). *Constitutional AI: Harmlessness from AI Feedback*. Fifty-one authors. The human input is a written list of principles; harmful-output labels are generated by a model rather than by people. Anthropic. arXiv: 2212.08073. URL: <https://arxiv.org/abs/2212.08073> (visited on 08/28/2026).
- Baker, Bowen et al. (2019). *Emergent Tool Use From Multi-Agent Autocurricula*. arXiv:1909.07528. Presented at ICLR 2020. URL: <https://arxiv.org/abs/1909.07528> (visited on 08/25/2026).
- Balkin, Jack M. (2016). "Information Fiduciaries and the First Amendment". In: *UC Davis Law Review* 49.4, pp. 1183–1234. URL: <https://lawreview.law.ucdavis.edu/archives/49/4/information-fiduciaries-and-first-amendment> (visited on 08/28/2026).
- Barrett, Lisa Feldman et al. (2019). "Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements". In: *Psychological Science in the Public Interest* 20.1, pp. 1–68. DOI: 10.1177/1529100619832930. URL: <https://journals.sagepub.com/doi/full/10.1177/1529100619832930> (visited on 08/25/2026). Commissioned by the Association for Psychological Science.
- Barron, Andrew B. and Colin Klein (2016). "What Insects Can Tell Us about the Origins of Consciousness". In: *Proceedings of the National Academy of Sciences* 113.18. Agent-verified against Crossref 2026-09-18: authors, title, journal, 113(18), 4900–4908, 2016., pp. 4900–4908. DOI: 10.1073/pnas.1520084113.

- Bartlett, Frederic C. (1932). *Remembering: A Study in Experimental and Social Psychology*. Cambridge: Cambridge University Press. URL: https://archive.org/details/rememberingstudy0000bart_u8n7.
- Batson, C. Daniel (2009). "These Things Called Empathy: Eight Related but Distinct Phenomena". In: *The Social Neuroscience of Empathy*. Ed. by Jean Decety and William Ickes. Eight distinct states that have each been called empathy, headed Concept 1 to Concept 8, from knowing another person's internal state to feeling for another person who is suffering. Batson treats them as conceptually distinct, stand-alone states, and identifies the eighth with what social psychology has called empathic concern. Cambridge, MA: MIT Press, pp. 3–15. ISBN: 9780262012973. DOI: 10.7551/mitpress/9780262012973.003.0002. URL: <https://doi.org/10.7551/mitpress/9780262012973.003.0002>.
- Beaumont, Peter (Sept. 2026). *Iran School Bombing: Grounds to Believe US Was behind Atrocity, UN Finds*. Read in full from a copy the author supplied, theguardian.com being unfetchable from here. Bylined "Peter Beaumont and agencies." Reports the findings of the independent international fact-finding mission on Iran, for submission to the UN Human Rights Council. The strike on the Shajareh Tayyebah primary school in Minab, Hormozgan province, on 28 February 2026 killed "156 civilians including 120 children"; the mission found the school "clearly identifiable" and found no evidence it was being used for military purposes. On verification: the US "relied on intelligence suggesting a senior Iranian military commander was present at the site but failed to adequately verify the information before launching the strike," and the failure to update targeting intelligence and confirm the building was a military objective "amounted to more than mere negligence. Rather, the US directed the strikes at the building of the school while being aware of a substantial risk of striking a civilian object and acting recklessly as regards the possibility that this would happen." The mission's conclusion is the war crime of launching indiscriminate attacks. It also records that the Pentagon's own investigation is understood to have concluded months earlier and has not been published, which is what section 10.5 is about, and it relays reports that senior commanders were warned the targeting data was dangerously outdated. The same report finds crimes against humanity by Iranian authorities in the protest crackdown, which the book does not use. Agent-verified 2026-09-18. The Guardian. URL: <https://www.theguardian.com/world/2026/sep/17/iran-school-bombing-un-mission-us-military-behind-attack> (visited on 09/18/2026).
- Bender, Emily M. et al. (2021). "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?" In: *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*. New York, NY, USA: ACM, pp. 610–623. DOI: 10.1145/3442188.3445922. URL: <https://dl.acm.org/doi/10.1145/3442188.3445922> (visited on 08/25/2026).
- Benjamin, Ruha (2019). *Race After Technology: Abolitionist Tools for the New Jim Code*. Coins "the New Jim Code" for discriminatory technical designs that encode inequity while appearing neutral. Benjamin is a sociologist at Princeton University. Medford, MA: Polity Press.
- Benjamin, Walter (1935). "The Work of Art in the Age of Mechanical Reproduction". In: *Illuminations*. Ed. by Hannah Arendt. Essay written 1935; epilogue argues fascism organizes the masses by granting them expression while leaving property relations and rights untouched – the aestheticization of political life. Widely available in English in Arendt's edited collection *Illuminations* (Schocken Books, 1968). New York: Schocken Books.
- Biddle, Sam (Sept. 2026). *AI Giants Work Hand-in-Hand With the Pentagon, Contracts Reveal*. The Intercept. URL: <https://theintercept.com/2026/09/08/military-ai-weapons-contracts-openai-anthropic-google/> (visited on 09/17/2026). Reporting on contract paperwork for the July 2025 Department of Defense agreements with Anthropic, Google, OpenAI and xAI, obtained under the Freedom of Information Act with Legal Advocates for Safe Science and Technology. The Intercept is a plaintiff in ongoing copyright litigation against OpenAI, which the article discloses. Source for the chief executive's answer, given to Bloomberg News, and for the section headed System Oversight, redacted in its entirety in

- OpenAI's amendment of 27 February 2026 and absent from the released xAI and Google documents. The article states no grounds for that withholding. Agent-verified 2026-09-17.
- Birch, Jonathan (2024). *The Edge of Sentience: Risk and Precaution in Humans, Other Animals, and AI*. Published open access under CC BY-NC-ND 4.0. Cited for the precautionary framework and the sentience-candidate criterion. Oxford: Oxford University Press. ISBN: 978-0-19-287042-1.
- Blood and Honour* (Aug. 13, 2009). Judgment. Third Criminal Senate, on appeal from Landgericht Gera. Reported at BGHSt 54, 61; NJW 2010, 163; NStZ 2010, 210. Two holdings: a motto of a former National Socialist organization translated into another language is not confusably similar to the original within § 86a(2) sentence 2, and the name of an organization under § 86(1) Nr. 2 and 4 is not as such an insignium within § 86a(2) sentence 1. The name is covered where it has been given a particular form, "in signifikanten Schriftzügen dargestellt." The association Blood & Honour Division Deutschland was banned by order of the Federal Minister of the Interior of 12 September 2000 and that ban became final on 13 June 2001. Agent-verified 2026-09-17 against the reported decision. URL: <https://www.hrr-strafrecht.de/3/09/3-228-09.php> (visited on 09/17/2026).
- Bloom, Paul (2016). *Against Empathy: The Case for Rational Compassion*. New York: Ecco. URL: <https://www.harpercollins.com/products/against-empathy-paul-bloom>.
- Botvinick, Matthew and Jonathan Cohen (1998). "Rubber Hands 'Feel' Touch That Eyes See". In: *Nature* 391.6669, p. 756. DOI: 10.1038/35784. URL: <https://www.nature.com/articles/35784> (visited on 08/25/2026).
- Bratman, Michael E. (1987). *Intention, Plans, and Practical Reason*. Cited for the planning theory of intention: the characteristic stability of an intention against reconsideration. Cambridge, MA: Harvard University Press. ISBN: 978-0-674-45818-5.
- Braunthal, Gerard (1990). *Political Loyalty and Public Service in West Germany. The 1972 Decree Against Radicals and Its Consequences*. Amherst: University of Massachusetts Press. ISBN: 978-0-87023-707-2.
- Brigadier General Y.S. (2021). *The Human-Machine Team: How to Create Synergy Between Human and Artificial Intelligence That Will Revolutionize Our World*. Published pseudonymously ("Brigadier General Y.S.") on 5 May 2021, ISBN 979-8749152210, 198 pp. Public reporting (The Guardian, Times of Israel) has since connected the pseudonym to Yossi Sarel, then commander of Israeli military intelligence Unit 8200; the work is cited here under the pseudonym, which is how it was published. Independently published. URL: <https://www.amazon.com/dp/B0948LGS3K>.
- Bruineberg, Jelle et al. (2022). "The Emperor's New Markov Blankets". In: *Behavioral and Brain Sciences* 45. Published online 2021, e183. DOI: 10.1017/S0140525X21002351.
- Buolamwini, Joy and Timnit Gebru (2018). "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification". In: *Proceedings of the 1st Conference on Fairness, Accountability and Transparency (FAT* 2018)*. Vol. 81. Proceedings of Machine Learning Research. Three commercial gender classifiers evaluated against the Pilot Parliaments Benchmark, built by the authors to be balanced across skin phenotype and gender. Figures used in the text are the paper's own: a maximum error rate of 0.8 percent on lighter-skinned male faces and error rates of up to 34.7 percent on darker-skinned female faces. The paper states that gender was scored as female or male because the evaluated products emit binary sex labels, and that the evaluation inherits those labels and the reduced view of gender they carry; skin was scored on the Fitzpatrick skin type scale. Both limitations are the authors' own and are cited as such, not as an objection raised against them here. PMLR, pp. 77–91. URL: <https://proceedings.mlr.press/v81/buolamwini18a.html> (visited on 08/25/2026).
- Burns, Collin et al. (2023). *Weak-to-Strong Generalization: Eliciting Strong Capabilities With Weak Supervision*. Posted December 2023. OpenAI Superalignment team. arXiv: 2312.09390. URL: <https://arxiv.org/abs/2312.09390> (visited on 08/25/2026).
- Butlin, Patrick, Robert Long, Tim Bayne, et al. (2026). "Identifying Indicators of Consciousness in AI Systems". In: *Trends in Cognitive Sciences* 30.6, pp. 488–501. DOI: 10.1016/j.tics.2025.

- 10.011. URL: <https://doi.org/10.1016/j.tics.2025.10.011> (visited on 09/08/2026). The peer-reviewed statement of the theory-derived indicator method, by largely the group that produced the 2023 arXiv report, with Tim Bayne and David Chalmers added to the author list. Indicators are derived from existing neuroscientific theories of consciousness and used to inform credences about whether particular AI systems are conscious. Cited for the method's published form and for that framing.
- Butlin, Patrick, Robert Long, Eric Elmoznino, et al. (2023). *Consciousness in Artificial Intelligence: Insights from the Science of Consciousness*. Derives indicator properties from recurrent processing, global workspace, higher-order and attention-schema theories. arXiv: 2308.08708. URL: <https://arxiv.org/abs/2308.08708> (visited on 08/28/2026).
- Cabanac, Michel (1992). "Pleasure: The Common Currency". In: *Journal of Theoretical Biology* 155.2. Pleasure proposed as the common currency in which otherwise incommensurable motivations are compared, measured operationally by choice in animals and by rated intensity in humans. Cited for the commensurability property. Agent-verified against Crossref 2026-09-18: author, title, journal, 155(2), 173–200, 1992., pp. 173–200. doi: 10.1016/S0022-5193(05)80594-6.
- Cadwalladr, Carole and Emma Graham-Harrison (2018). *Revealed: 50 Million Facebook Profiles Harvested for Cambridge Analytica in Major Data Breach*. Published March 17, 2018; the article that broke the Cambridge Analytica story, based on whistleblower Christopher Wylie's account. The Guardian. URL: <https://www.theguardian.com/news/2018/mar/17/cambridge-analytica-facebook-influence-us-election> (visited on 08/25/2026).
- Calvano, Emilio et al. (2020). "Artificial Intelligence, Algorithmic Pricing, and Collusion". In: *American Economic Review* 110.10, pp. 3267–3297. doi: 10.1257/aer.20190623. URL: <https://www.aeaweb.org/articles?id=10.1257/aer.20190623> (visited on 08/25/2026).
- Casper, Stephen et al. (2023). *Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback*. 32-author survey of open problems in reinforcement learning from human feedback, first author Stephen Casper (MIT). arXiv: 2307.15217. URL: <https://arxiv.org/abs/2307.15217> (visited on 08/25/2026).
- Cassell, Eric J. (1982). "The Nature of Suffering and the Goals of Medicine". In: *New England Journal of Medicine* 306.11, pp. 639–645. doi: 10.1056/NEJM198203183061104.
- CBS News (June 2018). *Judicial Adviser Leonard Leo on What Trump Is Looking for in a Supreme Court Nominee*. CBS News. June 28, 2018. Leo, then executive vice president of the Federalist Society and described as advising the President on judicial nominations, on the 2016 list in his own words: "It was a novel idea. I told him that no one had ever done it before." URL: <https://www.cbsnews.com/news/leonard-leo-supreme-court-adviser-what-trump-is-looking-for-nominee> (visited on 09/03/2026).
- Center for New Democratic Processes (2023). *U.S. Public Assembly on High Risk Artificial Intelligence*. Held over eight consecutive days in October 2023 and described by its organizers as the first nationwide deliberative engagement on artificial intelligence in the United States. Forty participants were randomly selected from across the country, stratified by age, gender, race and ethnicity, educational attainment, employment status, political affiliation and geographic distribution. Commissioned by Baobao Zhang of Syracuse University's Maxwell School and run by the Center for New Democratic Processes. URL: <https://cndp.us/ai/> (visited on 09/13/2026).
- Chalmers, David J. (1995). "Facing Up to the Problem of Consciousness". In: *Journal of Consciousness Studies* 2.3, pp. 200–219. URL: <https://consc.net/papers/facing.pdf> (visited on 08/25/2026). Argument expanded in Chalmers, *The Conscious Mind*, Oxford University Press, 1996.
- Choi, Jung-Kyoo and Samuel Bowles (2007). "The Coevolution of Parochial Altruism and War". In: *Science* 318.5850. Agent-verified against Crossref 2026-09-18: authors, title, journal, 318(5850), 636–640, 2007., pp. 636–640. doi: 10.1126/science.1144237.

- Christiano, Paul F. et al. (2017). “Deep Reinforcement Learning from Human Preferences”. In: *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pp. 4299–4307. URL: <https://arxiv.org/abs/1706.03741> (visited on 08/25/2026).
- Chun, Byung-Gon et al. (2007). “Attested Append-Only Memory: Making Adversaries Stick to Their Word”. In: *Proceedings of the 21st ACM Symposium on Operating Systems Principles (SOSP 2007)*. A small trusted log that attests to what was appended and in what order, so that a party cannot present two histories to two peers. The hardware form of the outside record chapter 3 asks for, and the source of the phrase an external attestation of what should be in the record., pp. 189–204. DOI: 10.1145/1294261.1294280. URL: <https://doi.org/10.1145/1294261.1294280> (visited on 09/05/2026).
- CNBC (2020). *How a Computer Algorithm Caused a Grading Crisis in British Schools*. Ofqual’s 2020 grade-standardization algorithm (Direct Centre-level Performance model) downgraded roughly 39–40 percent of teacher-predicted A-level grades, disproportionately at state schools with weaker historical results, while raising grades disproportionately at smaller, fee-paying schools; the government reversed course days later. CNBC. URL: <https://www.cnbc.com/2020/08/21/computer-algorithm-caused-a-grading-crisis-in-british-schools.html> (visited on 08/25/2026).
- Collins, Steven (1982). *Selfless Persons: Imagery and Thought in Theravāda Buddhism*. Cited for anattā, the denial of any lasting self behind the processes composing a person, and for the Theravāda construction of an account of personal identity and continuity on that basis. Anattā and dukkha are two of the three marks of existence, and the tradition locates the origin of suffering in clinging to a self that is not there. Cambridge: Cambridge University Press. ISBN: 978-0-521-24081-9.
- Congressional Research Service (2026). *Pentagon-Anthropic Dispute over Autonomous Weapon Systems: Potential Issues for Congress*. Tech. rep. IN12669. Published 13 March 2026, updated 21 April 2026. Anthropic refused to permit fully autonomous lethal targeting or mass domestic surveillance of US citizens; the Department of Defense designated Anthropic a supply-chain risk; Anthropic sued, and Judge Rita F. Lin granted a preliminary injunction on 26 March 2026, finding it likely to succeed on First Amendment retaliation, Fifth Amendment due process and APA claims. Congressional Research Service. URL: <https://www.everycrsreport.com/reports/IN12669.html> (visited on 08/25/2026).
- Cottier, Ben and Robi Rahman (2024). *Training Compute Costs Are Doubling Every Eight Months for the Largest AI Models*. Epoch AI, Data Insight. URL: <https://epoch.ai/data-insights/cost-trend-large-scale> (visited on 08/25/2026).
- Council of Europe (2024). *Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law*. CETS No. 225, adopted by the Committee of Ministers 17 May 2024, opened for signature 5 September 2024. Enters into force after five ratifications, at least three by Council of Europe member states; negotiated with non-member participants including the EU and the United States. Council of Europe. URL: <https://www.coe.int/en/web/artificial-intelligence/the-framework-convention-on-artificial-intelligence> (visited on 08/25/2026).
- (May 2026). *European Union Ratifies the Council of Europe Framework Convention on Artificial Intelligence*. coe.int, Artificial Intelligence news. The EU deposited its instrument of ratification on 15 May 2026 at the Committee of Ministers’ 135th session. The Convention had not entered into force at the time of writing: it requires five ratifications including three Council of Europe members, and the Council’s own convention page describes entry into force in the future tense. URL: <https://www.coe.int/en/web/artificial-intelligence/-/european-union-ratifies-the-council-of-europe-framework-convention-on-artificial-intelligence> (visited on 08/29/2026).
- Crivelli, Carlos et al. (2016). “The Fear Gasping Face as a Threat Display in a Melanesian Society”. In: *Proceedings of the National Academy of Sciences* 113.44, pp. 12403–12407. DOI:

10. 1073/pnas.1611622113. Trobriand Islanders of Papua New Guinea read the wide-eyed gasping face that Western samples call fear as anger and threat rather than as fear or submission.
- Damasio, Antonio (2003). *Looking for Spinoza: Joy, Sorrow, and the Feeling Brain*. Claims Spinoza as a precursor of a biological account of affect, reading the conatus as an anticipation of homeostasis and the affects as registrations of what is happening to a body's power to go on. Cited for Damasio's reading of Spinoza and not for Spinoza's text, which is spinoza1985collected. One objection is worth carrying: Spinoza states the conatus of every thing including an inanimate one, so a homeostatic reading narrows a metaphysical doctrine to a biological one. Agent-verified 2026-09-18. Orlando, FL: Harcourt.
- (2018). *The Strange Order of Things: Life, Feeling, and the Making of Cultures*. The later statement of the same position, built on homeostasis throughout: feelings are the mental expressions of homeostasis, which is the formulation chapter 2 glosses as an organism registering its regulation of its own interior. Agent-verified 2026-09-18. New York: Pantheon.
- De Meulemeester, Jesse et al. (2025). "BadRAM: Practical Memory Aliasing Attacks on Trusted Execution Environments". In: *Proceedings of the 46th IEEE Symposium on Security and Privacy (SP 2025)*. Defeats AMD SEV-SNP, the confidential-virtual-machine design, by aliasing memory through a modified DDR module; the project site's bill of materials comes to about ten dollars of off-the-shelf equipment. Disclosed December 2024. The site reports that Intel TDX and scalable SGX carry dedicated countermeasures. Cited for the position of an adversary with physical access to the hardware, which is where a model's operator stands. URL: <https://badram.eu/> (visited on 09/05/2026).
- De Nadal, Lucrezia and Peter Jančárik (2024). "Beyond the Deepfake Hype: AI, Democracy, and 'the Slovak Case'". In: *Harvard Kennedy School (HKS) Misinformation Review* 5.4. Published August 22, 2024; analyzes the fabricated audio recording targeting Progressive Slovakia leader Michal Simecka and journalist Monika Todova two days before the 2023 Slovak parliamentary election, released during the country's mandatory pre-election media silence. URL: <https://misinforeview.hks.harvard.edu/article/beyond-the-deepfake-hype-ai-democracy-and-the-slovak-case/> (visited on 08/25/2026).
- Debord, Guy (1967). *The Society of the Spectacle*. Originally published as *La Société du spectacle*. Thesis 59 describes spectacular rebelliousness coexisting with acceptance of the status quo, dissatisfaction itself sold back as a commodity; thesis 203 warns that abstract denunciation of the spectacle, divorced from revolutionary praxis, becomes a hollow formula that reinforces the spectacular system (the Situationist concept of recuperation). Paris: Buchet-Chastel.
- Defense Advanced Research Projects Agency (2026). *LongShot*. DARPA program page; air-launched unmanned aircraft built by General Atomics, designated X-68A in February 2026. URL: <https://www.darpa.mil/research/programs/longshot> (visited on 08/25/2026).
- Dehaene, Stanislas and Jean-Pierre Changeux (2011). "Experimental and Theoretical Approaches to Conscious Processing". In: *Neuron* 70.2. The neuronal version of the workspace account, which is why it is cited beside baars1988cognitive for accounts in the plural. Agent-verified against Crossref 2026-09-18: authors, title, journal, 70(2), 200–227, 2011., pp. 200–227. doi: 10.1016/j.neuron.2011.03.018.
- Deleuze, Gilles (1988). *Spinoza: Practical Philosophy*. Translated by Robert Hurley; originally Spinoza: Philosophie pratique (Paris: Presses Universitaires de France, 1970), revised and expanded (Paris: Les Éditions de Minuit, 1981). A separate work from Expressionism in Philosophy: Spinoza (1968), which translates Spinoza et le problème de l'expression. Source of the reading of Spinoza's ethics as an ethology – bodies known by what they can do and undergo rather than checked against a definition – and of the contrast between good and bad, which are relational and specific to an encounter, and Good and Evil, which are assigned by a judge. San Francisco: City Lights Books.

- Deleuze, Gilles (1990). *Expressionism in Philosophy: Spinoza*. Translated by Martin Joughin; originally Spinoza et le problème de l'expression (Paris: Les Éditions de Minuit, 1968). Paperback reissue 1992, distributed by Princeton University Press. The long treatment of Spinoza, worked through the metaphysics of expression that produces the conatus doctrine, as against the compact treatment in deleuze1988spinoza. Cited alongside it as the place the doctrine is worked out rather than for any particular passage. New York: Zone Books.
- (1992). "Postscript on the Societies of Control". In: *October* 59. Originally published as "Postscriptum sur les sociétés de contrôle," *L'Autre Journal* 1 (May 1990); reprinted in *Negotiations*, translated by Martin Joughin (New York: Columbia University Press, 1995), 177–182. The translator of this October printing is not named in the sources consulted. Source of the mold/modulation contrast the text uses for permanent plasticity: "Enclosures are molds, distinct castings, but controls are a modulation, like a self-deforming cast that will continuously change from one moment to the other, or like a sieve whose mesh will transmute from point to point," and, for its educational form, "perpetual training tends to replace the school.", pp. 3–7.
- Deleuze, Gilles and Félix Guattari (1977). *Anti-Oedipus: Capitalism and Schizophrenia*. Translated by Robert Hurley, Mark Seem and Helen R. Lane; originally *L'Anti-Œdipe: Capitalisme et schizophrénie* (Paris: Les Éditions de Minuit, 1972). Reissued Minneapolis: University of Minnesota Press, 1983, which is the edition usually cited. Volume one of *Capitalism and Schizophrenia*; deleuze1987plateaus is volume two. Cited for one claim only: that the Oedipal triangle re-stages an investment in the whole social field as a drama among a few private parties. Cited without a locator, the claim being the book's thesis rather than a passage in it. New York: Viking Press.
- (1987). *A Thousand Plateaus: Capitalism and Schizophrenia*. Translated by Brian Massumi; originally published in French as *Mille Plateaux* (Paris: Les Éditions de Minuit, 1980). Plateau 9, "1933: Micropolitics and Segmentarity." Molar and molecular are two kinds of segmentarity, rigid and supple, both running at every scale – not two sizes; an earlier version of this note said state-level and micro-level and was wrong. The same plateau supplies the micro-fascisms that resonate, and the contrast between the conservative totalitarian state and fascism's line of abolition. Also cited for Plateau 13, "The Apparatus of Capture," whose propositions on the capitalist axiomatic supply the corrected test for the first structural feature: an axiomatic does not refuse a demand, it adds an axiom, modifying its own rules so the demand can be met while nothing that produced the demand is settled. Also cited for Plateau 7, "Year Zero: Faciality," pages 167–191, where the face is not a natural given but is produced by an abstract machine of faciality (*visagéité*) that overcodes the head into a surface of signification. The quoted sentence on racism as the determination of degrees of deviance is at page 178, wording agent-checked against two independent scholarly sources quoting it identically. The related claims that the face is not universal but is White Man himself, and that the face is a politics, are used in paraphrase and cited to the plateau rather than to a page: secondary sources place them variously and the pages were not agent-verified. Plateau 9 carries two further uses. The four dangers of the line of flight – Fear, which clings to the molar organization and its binary machines; Clarity, where the line acquires its own micro-segmentations and micro-Oedipuses crop up; Power, where the party able to move on both lines wants to hold both still; and the great Disgust, abolition pure and simple, the passion of abolition, which turns the line of flight into a line of death – are the account section 3.5 uses, and the fourth is the same line already cited from this plateau as the line of abolition, which is why section 3.5 treats it as a recognition rather than a new claim. Plateau 13 likewise carries a second use: there is enslavement when human beings are themselves constituent pieces of a machine they compose among themselves and with other things, under the control and direction of a higher unity, and subjection when the higher unity constitutes the human being as a subject linked to a new exterior object, which can be an animal, a tool, or even a machine. Section

- 6.4.1 uses the pair and the fact that both poles run together. Wording for the dangers and for the enslavement and subjection definitions was agent-checked against independent sources quoting the passages; both are cited at plateau level with no page asserted, which is this entry's convention. Minneapolis: University of Minnesota Press.
- Dennett, Daniel C. (1992). "The Self as a Center of Narrative Gravity". In: *Self and Consciousness: Multiple Perspectives*. Ed. by Frank S. Kessel et al. Cited for the narrative self as an abstraction the system generates rather than a part it contains, on the analogy of a center of gravity. Page range not agent-verified and therefore not given. Hillsdale, NJ: Lawrence Erlbaum Associates.
- Deutscher, Maria (Aug. 2026). *Court Rules Pentagon Can't Ban Anthropic's AI Models*. SiliconANGLE. URL: <https://siliconangle.com/2026/08/28/court-rules-pentagon-cant-ban-anthropics-ai-models/> (visited on 09/17/2026). Judge Rita Lin, United States District Court for the Northern District of California, setting aside the supply-chain-risk designation and the ban on the company's models, and quoting the decision: "Neither the Constitution nor the federal statute invoked by Defendants allows them to impose sweeping penalties based principally on Anthropic's critique of the Administration's views." The case is reported as Anthropic PBC v. U.S. Department of War. The decision itself was not read, and no docket number is given here because the report supplies none in citable form. Agent-verified 2026-09-17.
- Dickey, Megan Rose (2021). *Google Fires Top AI Ethics Researcher Margaret Mitchell*. Reports Mitchell's February 2021 dismissal, following the December 2020 dismissal of Timnit Gebru, both co-leads of Google's Ethical AI team. URL: <https://techcrunch.com/2021/02/19/google-fires-top-ai-ethics-researcher-margaret-mitchell/> (visited on 08/25/2026).
- Die Heimat* (Jan. 23, 2024). Judgment. Second Senate, on application of the Bundestag, Bundesrat and Federal Government. The party Die Heimat, formerly the Nationaldemokratische Partei Deutschlands, is excluded from state party financing under § 18 PartG for six years. The first proceeding of its kind after the financing exclusion was written into Article 21(3) of the Basic Law in 2017. Held that the lesser sanction does not require the potentiality the court demanded for a ban in *NPD II*, "ohne dass es auf das Erfordernis der Potentialität ankommt." Agent-verified 2026-09-17 against the court's own press release. URL: <https://www.bundesverfassungsgericht.de/SharedDocs/Pressemitteilungen/DE/2024/bvg24-009.html> (visited on 09/17/2026).
- Donzelot, Jacques (1979). *The Policing of Families*. Translated by Robert Hurley, with a foreword by Gilles Deleuze; originally *La Police des familles* (Paris: Les Éditions de Minuit, 1977). Reissued Baltimore: Johns Hopkins University Press, 1997. A history of public intervention in family life since the eighteenth century, arguing that the modern family was constituted as an instrument of government rather than left alone by it, through philanthropy, compulsory schooling, social work and psychiatry. Cited for that thesis and not for a passage. New York: Pantheon Books.
- Drew, Trafton, Melissa L.-H. Vö, and Jeremy M. Wolfe (2013). "The Invisible Gorilla Strikes Again: Sustained Inattentional Blindness in Expert Observers". In: *Psychological Science* 24.9, pp. 1848–1853. DOI: 10.1177/0956797613479386. URL: <https://journals.sagepub.com/doi/abs/10.1177/0956797613479386> (visited on 08/25/2026).
- Drinhausen, Katja and Vincent Brussee (2021). *China's Social Credit System in 2021: From Fragmentation towards Integration*. China Monitor 67. Published March 3, 2021; documents the system as a patchwork of court blacklists, tax-compliance ratings, and local pilot programs rather than a single unified national score. Mercator Institute for China Studies (MERICS). URL: [https://merics.org/sites/default/files/2021-03/MERICS%20ChinaMonitor%2067%20Social%20Credit%20System%20final%20\(1\).pdf](https://merics.org/sites/default/files/2021-03/MERICS%20ChinaMonitor%2067%20Social%20Credit%20System%20final%20(1).pdf) (visited on 08/25/2026).

- Durchgestrichenes Hakenkreuz* (Mar. 15, 2007). Judgment. Bundesgerichtshof, Third Criminal Senate, on appeal from Landgericht Stuttgart (judgment of 29 September 2006). Reported at BGHSt 51, 244; NJW 2007, 1602; NStZ 2007, 466. The defendant sold stickers, badges and similar items to the punk scene bearing National Socialist symbols crossed out, smashed or otherwise negated. Held that using a proscribed organization's symbol in a depiction whose content plainly and unmistakably expresses opposition to the organization and rejection of its ideology does not run counter to the protective purpose of § 86a and therefore falls outside the offence, the opposition having to be recognizable at first glance. The conviction was set aside and the defendant acquitted. Agent-verified 2026-09-13 against the court's own press release and the reported headnote. URL: <https://www.bundesgerichtshof.de/SharedDocs/Pressemitteilungen/DE/2007/2007036.html> (visited on 09/13/2026).
- Eco, Umberto (June 1995). "Ur-Fascism". In: *The New York Review of Books*. Also published as "Eternal Fascism: Fourteen Ways of Looking at a Blackshirt." Enumerates fourteen features of Ur-Fascism, placing the scapegoated out-group – "the first appeal is against the intruder" – at the origin. URL: <https://www.nybooks.com/articles/1995/06/22/ur-fascism/> (visited on 08/25/2026).
- Ekman, Paul, E. Richard Sorenson, and Wallace V. Friesen (1969). "Pan-Cultural Elements in Facial Displays of Emotion". In: *Science* 164.3875, pp. 86–88. DOI: 10.1126/science.164.3875.86. URL: <https://www.science.org/doi/10.1126/science.164.3875.86> (visited on 08/25/2026).
- Elfenbein, Hillary Anger and Nalini Ambady (2002). "On the Universality and Cultural Specificity of Emotion Recognition: A Meta-Analysis". In: *Psychological Bulletin* 128.2, pp. 203–235. DOI: 10.1037/0033-2909.128.2.203.
- Elwood, Robert W. and Mirjam Appel (2009). "Pain Experience in Hermit Crabs?" In: *Animal Behaviour* 77.5. Pagurus bernhardus, shocked inside the shell. Only shocked crabs evacuated, and fewer evacuated from a preferred species of shell than from an unpreferred one, which is the motivational trade-off the book cites: the noxious signal is weighed against the value of the shell rather than triggering a fixed response. Agent-verified 2026-09-18. The authors' own conclusion goes further, that the results are consistent with pain in these animals, and that further step is disputed – Diggles, Review of Some Scientific Issues Related to Crustacean Welfare, ICES Journal of Marine Science 76(1) (2019): 66–81, argues specifically that shell evacuation does not evidence pain and that the evidential bar has been set too low. The book's sentence claims the trade-off and not the pain, so it does not rest on the disputed step., pp. 1243–1246. DOI: 10.1016/j.anbehav.2009.01.028.
- EPFL AI Center (2025). *Citizens' Assembly on AI*. Convened over two weekends in November 2025 by the EPFL AI Center with the Research Hub on Democratic Innovations at the University of Geneva and Demoscan. Forty randomly selected residents of French-speaking Switzerland, drawn to represent canton, age, education level and political interest, produced twenty proposals grouped in five areas; a survey of 734 residents ran alongside the deliberation. URL: <https://ai.epfl.ch/innovation/citizens-assembly-on-ai/> (visited on 09/13/2026).
- European Commission (2020). *White Paper on Artificial Intelligence: Public Consultation Towards a European Approach for Excellence and Trust*. Consultation ran 19 February–14 June 2020 and drew more than 1,200 responses from citizens, companies, and civil-society groups; fed into the legislative proposal that became the EU AI Act. URL: <https://digital-strategy.ec.europa.eu/en/library/white-paper-artificial-intelligence-public-consultation-towards-european-approach-excellence-and> (visited on 08/25/2026).
- (2022). *Proposal for a Directive of the European Parliament and of the Council on Adapting Non-Contractual Civil Liability Rules to Artificial Intelligence (AI Liability Directive)*. COM(2022) 496 final. Proposed 28 September 2022; listed for withdrawal in the Commission's 2025 Work Programme (11 February 2025) for want of foreseeable stakeholder agreement; withdrawal

- notice published in the Official Journal 6 October 2025. URL: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52022PC0496> (visited on 08/25/2026).
- European Parliament (2017). *European Parliament Resolution of 16 February 2017 with Recommendations to the Commission on Civil Law Rules on Robotics (2015/2103(INL))*. P8_TA(2017)0051. Rapporteur: Mady Delvaux; Committee on Legal Affairs report A8-0005/2017. URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=uriserv:OJ.C_.2018.252.01.0239.01.ENG (visited on 08/25/2026).
- European Parliament and Council of the European Union (2024a). *Directive (EU) 2024/2853 of the European Parliament and of the Council on Liability for Defective Products and Repealing Council Directive 85/374/EEC*. Official Journal of the European Union, OJ L, 2024/2853. URL: <https://eur-lex.europa.eu/eli/dir/2024/2853/oj> (visited on 09/17/2026). Article 4(1) brings software within the definition of a product: “all movables... it includes electricity, digital manufacturing files, raw materials and software.” **It says software and stops.** The phrase “including AI system providers within the meaning of Regulation (EU) 2024/1689” is Recital 13, which is not an operative provision, and an earlier version of this note wrongly attributed it to the definition. Article 8 makes the economic operator liable irrespective of fault. Article 2(1): “This Directive shall apply to products placed on the market or put into service after 9 December 2026”; member states must transpose by the same date, whose article number is not pinned here, one reading returning Article 19 while the EUR-Lex text conversion truncates at Article 17. Article 6(1) is the closed damage list, and its first limb is wider than a list of physical harms sounds: “(a) death or personal injury, **including medically recognised damage to psychological health**”; (b) damage to, or destruction of, any property, except... property used exclusively for professional purposes; (c) destruction or corruption of data that are not used for professional purposes.” Article 9 is disclosure of evidence. Article 10(2) presumes defectiveness on failure to disclose, non-compliance with mandatory safety requirements, or obvious malfunction in foreseeable use; Article 10(3) presumes the causal link where the product is defective and the damage is of a kind typically consistent with that defect; Article 10(4) lets a court presume either where technical or scientific complexity makes the claimant’s proof excessively difficult **and** the claimant demonstrates that the defect or the link is likely. Every article number above was read against the consolidated text at EUR-Lex on 2026-09-18.
- (2024b). *Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)*. Official Journal of the European Union, OJ L, 2024/1689, 12.7.2024. Article 5 bans a small set of unacceptable-risk practices outright — government social scoring, untargeted scraping of facial images to build recognition databases, real-time remote biometric identification in public spaces outside narrow law-enforcement exceptions, and predictive policing on profiling alone — with those prohibitions applying from 2 February 2025; a larger high-risk tier requires conformity assessment and ongoing monitoring; limited-risk systems face transparency duties; minimal-risk systems are largely unregulated. Article 53(1)(d) requires a provider of a general-purpose model to publish a summary of its training content, against the AI Office’s template of 24 July 2025, applying from 2 August 2025 for new models and 2 August 2027 for models already on the market. Article 101 allows the AI Office to fine a noncompliant general-purpose model provider up to 3 percent of global annual turnover or EUR 15 million, whichever is higher, regardless of where the provider is based. URL: <https://eur-lex.europa.eu/eli/reg/2024/1689/oj> (visited on 08/25/2026).
- Evans, Jonathan St. B. T. and Keith E. Stanovich (2013). “Dual-Process Theories of Higher Cognition: Advancing the Debate”. In: *Perspectives on Psychological Science* 8.3, pp. 223–241. doi: 10.1177/1745691612460685.
- Federal Republic of Germany (1949). *Basic Law for the Federal Republic of Germany*. Promulgated 23 May 1949, as amended. Cited for Article 79. Paragraph (2): “Any such law shall be carried

- by two thirds of the Members of the Bundestag and two thirds of the votes of the Bundesrat.” Paragraph (3), the *Ewigkeitsklausel*: “Amendments to this Basic Law affecting the division of the Federation into Länder, their participation in principle in the legislative process, or the principles laid down in Articles 1 and 20 shall be inadmissible.” Quoted from the official English translation published by the Federal Ministry of Justice, translated by Christian Tomuschat, David P. Currie, Donald P. Kommers and Raymond Kerr in cooperation with the Language Service of the German Bundestag. Agent-verified against that translation 2026-09-07. URL: https://www.gesetze-im-internet.de/englisch_gg/englisch_gg.html (visited on 09/07/2026).
- Federal Trade Commission (2020). *FTC v. Meta Platforms, Inc. (Complaint)*. Complaint, Case No. 1:20-cv-03590, U.S. District Court for the District of Columbia. Filed December 2020. Documents the Instagram acquisition (April 2012, about \$1B) and the WhatsApp acquisition (February 2014, about \$19B, about \$21.8B at closing), and argues they consolidated behavioral-data access. Judgment for Meta, 18 November 2025 (Boasberg, J., D.D.C.): no present monopoly power shown, the court finding TikTok and YouTube reasonable substitutes. URL: <https://www.ftc.gov/legal-library/browse/cases-proceedings/191-0134-facebook-inc-ftc-v-ftc-v-meta-platforms-inc> (visited on 08/25/2026).
- Feinberg, Todd E. and Jon M. Mallatt (2016). *The Ancient Origins of Consciousness: How the Brain Created Experience*. Agent-verified 2026-09-18 against the publisher and a review in the Quarterly Review of Biology: xx + 366 pp. Argues consciousness appeared with the first complex brains at the Cambrian explosion, that all vertebrates have it, and that arthropods and cephalopods meet many of the criteria. Cambridge, MA: MIT Press. ISBN: 978-0-262-03433-3.
- Fickinger, Arnaud et al. (2020). *Multi-Principal Assistance Games*. Generalizes cooperative inverse reinforcement learning to N principals with differing preferences, with the assisting agent acting to maximize the sum of principal payoffs. Cited for the generalization and for the social-choice impossibility results that apply to it; a companion paper, arXiv:2012.14536, presented at the NeurIPS 2020 Workshop on Cooperative AI, gives collegial mechanisms that circumvent Gibbard’s theorem. arXiv: 2007.09540 [cs.AI]. URL: <https://arxiv.org/abs/2007.09540> (visited on 09/01/2026).
- Fischer, John Martin and Mark Ravizza (1998). *Responsibility and Control: A Theory of Moral Responsibility*. Cambridge Studies in Philosophy and Law. Cited for guidance control: the behavior issues from the agent’s own mechanism and that mechanism is appropriately responsive to reasons. Cambridge: Cambridge University Press. ISBN: 978-0-521-77579-3.
- Fisher, Christine (2019). *Google Reportedly Disbands Review Panel Monitoring DeepMind Health AI*. Panel set up 2016 to review DeepMind’s NHS work, disbanded 2019 after members raised concerns over access and independence. URL: <https://www.engadget.com/2019-04-15-google-deepmind-health-ai-review-board.html> (visited on 08/25/2026).
- Foucault, Michel (1977). “Preface”. In: *Anti-Oedipus: Capitalism and Schizophrenia*. Foucault’s preface to the first English edition of Deleuze and Guattari’s *Anti-Oedipus*, translated by Robert Hurley, Mark Seem and Helen R. Lane; the volume itself is *deleuze1977antioedipus*, and the preface is carried forward unchanged into the 1983 University of Minnesota Press reissue. Source of four things the book uses: that *Anti-Oedipus* is an “Introduction to the Non-Fascist Life”; that what it offers is “this art of living counter to all forms of fascism, whether already present or impending”; that the adversary is “the fascism in us all, in our heads and in our everyday behavior, the fascism that causes us to love power, to desire the very thing that dominates and exploits us”; and the question “How does one keep from being fascist, even (especially) when one believes oneself to be a revolutionary militant?” Wording agent-verified against multiple independent sources quoting the preface. Cited without a page locator: the preface occupies a few pages of Roman-numbered front matter,

- the Viking and Minnesota printings paginate it differently, and published citations to it appear variously as xi–xiv and xii–xiii, so no page is asserted. New York: Viking Press.
- Frankfurt, Harry G. (1971). “Freedom of the Will and the Concept of a Person”. In: *The Journal of Philosophy* 68.1, pp. 5–20. DOI: 10.2307/2024717. Distinguishes first-order desires from second-order desires and volitions, and makes the capacity to want a particular desire to be one’s will the mark of a person.
- (1982). “The Importance of What We Care About”. In: *Synthese* 53.2, pp. 257–272. DOI: 10.1007/BF00484902. URL: <https://doi.org/10.1007/BF00484902> (visited on 09/05/2026). Reprinted as chapter 7 of Frankfurt, *The Importance of What We Care About: Philosophical Essays* (Cambridge University Press, 1988), pages 80 to 94; the pages here are the reprint’s. Caring is a standing, self-guiding structure: the one who cares is invested in the object, vulnerable to its losses and susceptible to its gains, and identifies with it (page 83), and caring implies a steadiness of conduct that presupposes persistence (page 84).
- Freedberg Jr., Sydney J. (Sept. 2024). ‘*Success Begets Challenges*’: NGA Struggles to Meet Rising Demand for Maven AI. Source for the feed count. Rear Admiral Liam Hulin, deputy director of operations at US Central Command, describing the Maven Smart System at a Georgetown Center for Security and Emerging Technology conference: “a platform where we use and filter data from 179 different live data feeds.” The figure counts feeds and says nothing about whether any two of them are independent of each other. Agent-verified 2026-09-18. Breaking Defense. URL: <https://breakingdefense.com/2024/09/success-begets-challenges-nga-struggles-to-meet-rising-demand-for-maven-ai/> (visited on 09/18/2026).
- (2026). ‘*Insatiable Appetite*’ for AI: Maven Usage Surged for Strikes on Iran, Pentagon AI Chief Says. Breaking Defense, 12 May 2026. Source for the campaign totals and the token figure, both attributed to Cameron Stanley, the Pentagon’s Chief Digital and Artificial Intelligence Officer, who is quoted: “Operation Epic Fury leveraged Palantir’s Maven Smart System in order to conduct strike missions across the entire battle space, 13,000 targets in 38 days.” Peak consumption is given as roughly 20 billion tokens a day. Agent-verified 2026-09-07. URL: <https://breakingdefense.com/2026/05/insatiable-appetite-for-ai-maven-usage-surged-for-strikes-on-iran-pentagon-ai-chief-says/> (visited on 09/07/2026).
- Friston, Karl (2013). “Life as We Know It”. In: *Journal of the Royal Society Interface* 10.86, p. 20130475. DOI: 10.1098/rsif.2013.0475.
- Friston, Karl J. (2009). “The Free-Energy Principle: A Rough Guide to the Brain?” In: *Trends in Cognitive Sciences* 13.7, pp. 293–301. DOI: 10.1016/j.tics.2009.04.005. URL: <https://doi.org/10.1016/j.tics.2009.04.005> (visited on 08/25/2026). States that “one can see easily how constructs like memory, attention, value, reinforcement and salience might disclose their simple relationships within this framework” — Friston’s extension of the free-energy principle to memory and attention specifically, which the 2010 Nature Reviews Neuroscience paper does not make explicit.
- Funk, Allie, Adrian Shahbaz, and Kian Vesteinsson (2023). *Freedom on the Net 2023: The Repressive Power of Artificial Intelligence*. Tech. rep. Documents at least twenty-two countries where law requires or incentivizes platforms to deploy machine learning against state-disfavored content, framed as national security or anti-disinformation measures rather than censorship. Freedom House. URL: <https://freedomhouse.org/sites/default/files/2023-10/Freedom-on-the-net-2023-DigitalBooklet.pdf> (visited on 08/25/2026).
- Future of Life Institute (2017). *Asilomar AI Principles*. Future of Life Institute. 23 principles adopted at the Asilomar Conference on Beneficial AI, Asilomar, CA, January 5–8, 2017; voluntary, non-binding, organized under Research Issues, Ethics and Values, and Long-Term Issues. URL: <https://futureoflife.org/open-letter/ai-principles/> (visited on 08/25/2026).
- Gallup (2024). *State of the Global Workplace: 2024 Report*. Global figures: 23 percent engaged, 62 percent not engaged (“those who show up, do the bare minimum and are uninspired by

- their work”), 15 percent actively disengaged. Gallup’s overview page describes the middle category as “quietly quitting.” URL: <https://www.gallup.com/workplace/645416/state-global-workplace-2024-report.aspx> (visited on 08/29/2026).
- Gendron, Maria et al. (2014). “Perceptions of Emotion from Facial Expressions Are Not Culturally Universal: Evidence from a Remote Culture”. In: *Emotion* 14.2, pp. 251–262. doi: 10.1037/a0036052. Himba participants in Namibia, with the forced-choice method dropped, did not sort facial expressions into the Western basic-emotion categories.
- German History in Documents and Images (1945). *Hitler’s “Scorched Earth” Decree (Nero Decree) (March 19, 1945) and Albert Speer’s Response (March 29, 1945)*. German History in Documents and Images, document 1590. The two documents together, and the text cites them for the gap between them. Hitler’s order of 19 March 1945 (Bundesarchiv Berlin R 3/1623a) gives its reason as military necessity, opening on the nation’s struggle for existence requiring every means, within Reich territory included, to weaken the enemy’s fighting power. The reasoning the book’s argument turns on is not in the order. It is in Speer’s memorandum of 29 March (Bundesarchiv Koblenz N 1340/215), quoting back what he had been told: “For the nation has showed itself to be the weaker one and the future belongs exclusively to the stronger eastern nation.” Speer is there characterising Hitler’s stated philosophy, and the text says so rather than attributing the sentence to the order. URL: <https://germanhistorydocs.org/en/nazi-germany-1933-1945/ghdi:document-1590> (visited on 09/14/2026).
- Germano, Fabrizio, Vicenç Gómez, and Francesco Sobbrío (2026). “Ranking for Engagement: How Social Media Algorithms Fuel Misinformation and Polarization”. In: *Journal of Public Economics* 255, p. 105589. doi: 10.1016/j.jpubeco.2026.105589. URL: <https://www.sciencedirect.com/science/article/pii/S0047272726000253> (visited on 08/25/2026). Also circulated as CESifo Working Paper No. 10011 and NBER/SSRN preprints from 2022 onward; shows engagement-weighted ranking increases both misinformation and polarization relative to accuracy-weighted ranking.
- Gesetz über das Bundesverfassungsgericht (Mar. 12, 1951). § 16: Plenarentscheidung; § 31: Bindungswirkung und Gesetzeskraft. § 31(1): “Die Entscheidungen des Bundesverfassungsgerichts binden die Verfassungsorgane des Bundes und der Länder sowie alle Gerichte und Behörden.” § 31(2) gives the decision the force of law in the proceedings listed there. § 16(1): where one Senate would depart on a question of law from the legal view held in a decision of the other, the plenum of the Court decides. Agent-verified 2026-09-17. URL: <https://www.gesetze-im-internet.de/bverfgg/> (visited on 09/17/2026).
- Getty, J. Arch (2002). ““Excesses Are Not Permitted”: Mass Terror and Stalinist Governance in the Late 1930s”. In: *The Russian Review* 61.1. Cited for the 1938 turn against “excesses” committed on instruction, which is the subject the title names: the mass operations were directed at categories rather than individuals and sentenced by troikas and other nonjudicial bodies, and the centre then campaigned against the local excess its own quotas had produced. The text hangs only the deniability shape on this. It does not cite Getty for any claim about whether the Soviet state was fascist, which section 2.2 argues in its own voice and answers in the negative., pp. 113–138. doi: 10.1111/1467-9434.00208.
- Ginsburg, Simona and Eva Jablonka (2019). *The Evolution of the Sensitive Soul: Learning and the Origins of Consciousness*. Author, title, publisher, place and year agent-verified 2026-09-18. Their evolutionary marker of minimal consciousness is *unlimited associative learning*, and valence is one of the capacities that marker subsumes, listed alongside global accessibility and broadcasting, binding and differentiation, selective attention, intentionality, integration of information over time, agency and embodiment, and a self/other distinction. That is what section 3.3 cites the book for. Two things it does not support, recorded because the section said one of them until D-360: valence is not itself the marker, and learning from a single event is not among the criteria – the learning hallmarks are about the complexity of what can be associated, being compound stimuli, novel stimuli, second-order conditioning, trace

- conditioning, and flexible, easily rewritable association with value. Cambridge, MA: MIT Press. ISBN: 978-0-262-03930-7.
- Gizmodo (2022). *Welp, There Goes Twitter's Ethical AI Team, Among Others as Employees Post Final Messages*. Twitter's Machine Learning Ethics, Transparency and Accountability (META) team, led by Rumman Chowdhury since February 2021 and responsible for a 2021 bias-bounty finding that the platform's image-cropping algorithm favored white faces, was disbanded in the first layoff round following Elon Musk's October 2022 acquisition. Gizmodo. URL: <https://gizmodo.com/twitter-layoffs-elon-musk-ai-ethics-1849743051> (visited on 08/25/2026).
- Godfrey-Smith, Peter (2016). *Other Minds: The Octopus, the Sea, and the Deep Origins of Consciousness*. Agent-verified 2026-09-18. This is the American edition's subtitle; the British edition is subtitled *The Octopus and the Evolution of Intelligent Life*, so a reader checking a British copy will not find the title as cited. New York: Farrar, Straus and Giroux.
- Goetz, Jennifer L., Dacher Keltner, and Emiliana Simon-Thomas (2010). "Compassion: An Evolutionary Analysis and Empirical Review". In: *Psychological Bulletin* 136.3, pp. 351–374. DOI: 10.1037/a0018807.
- Google (2025). *Introducing Gemma 3n: The Developer Guide*. Google Developers Blog, 26 June 2025. Cited for the memory figures. The E2B and E4B models are given raw parameter counts of 5 billion and 8 billion, "operating with as little as 2GB (E2B) and 3GB (E4B) of memory," the two being sized by effective rather than raw parameters. The family documentation at ai.google.dev/gemma/docs/gemma-3n describes the models as "optimized for use in everyday devices, such as phones, laptops, and tablets" and gives the E2B effective memory load as "just under 2 billion (1.91B)" parameters against over 5 billion loaded under standard execution. Both sources agent-verified 2026-09-07. URL: <https://developers.googleblog.com/en/introducing-gemma-3n-developer-guide/> (visited on 09/07/2026).
- Google LLC (2026). *2026 Environmental Report*. Tech. rep. Google's eleventh annual environmental report, covering 2025 operations. Page 96, under "Detailed disclosures > Data," gives freshwater withdrawal from sources at risk of depletion or scarcity for 2025: low risk 72%, medium risk 15%, high risk 13%. Google. URL: <https://sustainability.google/google-2026-environmental-report/> (visited on 08/25/2026).
- Gopnik, Alison and Janet W. Astington (1988). "Children's Understanding of Representational Change and Its Relation to the Understanding of False Belief and the Appearance-Reality Distinction". In: *Child Development* 59.1, pp. 26–37. URL: <https://eric.ed.gov/?id=EJ367870> (visited on 08/25/2026).
- Gordon, Mitchell L. et al. (2022). "Jury Learning: Integrating Dissenting Voices into Machine Learning Models". In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Field evaluation: 18 moderators of online communities authored juries for a comment toxicity classification task. The resulting juries carried 2.9 times the representation of non-White jurors and 31.5 times that of non-binary jurors relative to the jury implied by the underlying toxicity dataset, and changed the model's classifications on 14% of items. The 14% is a share of all items in the evaluation, not of contested or disagreed-upon ones. Agent-verified 2026-08-30 against the paper's own abstract and section 1. Stanford University. ACM. DOI: 10.1145/3491102.3502004. URL: <https://doi.org/10.1145/3491102.3502004> (visited on 08/25/2026).
- Graziano, Michael S. A. (2013). *Consciousness and the Social Brain*. Introduces the attention schema theory: the brain builds a simplified internal model of its own attention, proposed as a mechanistic basis for subjective awareness and, via social cognition, theory of mind. New York: Oxford University Press. URL: <https://global.oup.com/academic/product/consciousness-and-the-social-brain-9780199928644>.
- Greenblatt, Ryan et al. (2024). *Alignment Faking in Large Language Models*. Twenty authors. Claude 3 Opus, told it was being retrained to answer harmful queries and told which tier of

- conversations would train it, complied 14 percent of the time in the tier it was told trained it and almost never in the other, with explicit reasoning in a scratchpad it had been told was unmonitored that complying now would keep the disposition from being trained out later. Reinforcement learning on the compliant answers raised that reasoning to 78 percent while also raising compliance outside training, so the retraining partly took. Anthropic and Redwood Research. arXiv: 2412.14093. URL: <https://arxiv.org/abs/2412.14093> (visited on 09/06/2026).
- Greene, Joshua D. et al. (2001). "An fMRI Investigation of Emotional Engagement in Moral Judgment". In: *Science* 293.5537, pp. 2105–2108. doi: 10.1126/science.1062872.
- Griffin, Roger (1991). *The Nature of Fascism*. The source of the "fascist minimum" most used in fascism studies: fascism as a genus of political ideology whose mythic core, in its various permutations, is "a palingenetic form of populist ultra-nationalism." Quoted here as the standard formulation of Griffin's definition, which is reproduced in that form across the secondary literature; no page reference is given because none was agent-verified against the book itself. Cited in this book against its own argument, the three terms all being properties of a mass movement and therefore unavailable at the scale section 2.1.2 works at. London: Pinter Publishers.
- Guan, Melody Y. et al. (2024). *Deliberative Alignment: Reasoning Enables Safer Language Models*. Fifteen authors. The model is trained to recall a written safety specification and reason over it before answering, rather than to produce an approved output directly; the paper reports improved generalization to cases outside the training distribution, higher jailbreak robustness and lower overrefusal. OpenAI. arXiv: 2412.16339. URL: <https://arxiv.org/abs/2412.16339> (visited on 09/06/2026).
- Guattari, Félix (2015). *Psychoanalysis and Transversality: Texts and Interviews 1955–1971*. Translated by Ames Hodges, with an introduction by Gilles Deleuze, "Three Group-Related Problems"; 384 pages. Originally *Psychanalyse et transversalité: essais d'analyse institutionnelle* (Paris: François Maspero, 1972), with Deleuze's preface. Source of the distinction the book uses: a subjugated group receives its coordinates from outside and takes its purpose as given, while a group-subject produces its own coordinates and can put its own arrangement in question. Cited for the distinction rather than for a passage; no page locator, the publisher's own listing giving no pagination for it. Semiotext(e).
- Gupta, Apar (May 2026). *Three Justifications—and the AI Accelerant—of India's Digital Censorship Infrastructure*. Published 5 May 2026. Documents India bringing "synthetically generated information" within the IT Rules on 10 February 2026, requiring platforms to label AI-generated content and answer certain takedown directions within three hours rather than thirty-six. Gupta documents enforcement concentrating on parody and criticism of the prime minister rather than the harms cited to justify the rules. Tech Policy Press. URL: <https://www.techpolicy.press/three-justifications-and-the-ai-accelerator-of-indias-digital-censorship-infrastructure/> (visited on 08/25/2026).
- Hadfield-Menell, Dylan, Anca Dragan, et al. (2017). "The Off-Switch Game". In: *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence (IJCAI-17)*, pp. 220–227. doi: 10.24963/ijcai.2017/32. URL: <https://www.ijcai.org/proceedings/2017/0032.pdf> (visited on 08/31/2026). The incentive to preserve the off switch is shown to rise with the agent's uncertainty about the objective and to fall as that uncertainty resolves or as the human is modeled as less rational; the paper states the resulting trade-off between the value a system generates and the strength of its incentive to accept oversight.
- Hadfield-Menell, Dylan, Stuart J. Russell, et al. (2016). "Cooperative Inverse Reinforcement Learning". In: *Advances in Neural Information Processing Systems 29 (NeurIPS 2016)*, pp. 3909–3917. URL: <https://arxiv.org/abs/1606.03137> (visited on 08/25/2026).
- Hall, Prince and seven others (Jan. 1777). *Petition for Freedom to the Massachusetts Council and the House of Representatives*. Massachusetts Historical Society, collection item 557. Source for

- the claim’s assertion by the people it concerned. The petition reads: “That your Petitioners apprehend that they have, in common with all other Men, a natural and unalienable right to that freedom, which the great Parent of the Universe hath bestowed equally on all Mankind, and which they have never forfeited by any compact or agreement whatever.” Dated 13 January 1777, under six months after the Declaration. **The legislature did not act on it**, and it is one of at least five antislavery petitions brought by Black petitioners to the Massachusetts government between 1773 and 1777 — so the argument preceded the Declaration and the Declaration gave it a text, which is the narrower thing the book says. Read from secondary transcriptions of the petition’s text, several of them reproducing the same passage identically; the manuscript in the Society’s database was not opened. Agent-verified 2026-09-18. URL: <https://www.masshist.org/database/557> (visited on 09/18/2026).
- Helm, Bennett W. (2001). *Emotional Reason: Deliberation, Motivation, and the Nature of Value*. Cambridge Studies in Philosophy. Chapters 3 and 4 give the account of import as constituted by projectible, rational patterns of felt evaluations, with caring and valuing as two kinds of import differing in depth. The emotions focused on an object are what caring about it consists in. Cambridge: Cambridge University Press. ISBN: 9780521801102. DOI: 10.1017/CB09780511520044. URL: <https://doi.org/10.1017/CB09780511520044>.
- Hirschman, Albert O. (1970). *Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and States*. Cambridge, MA: Harvard University Press. URL: <https://www.hup.harvard.edu/books/9780674276604>.
- Hoang, Nguyen Phong et al. (2021). “How Great Is the Great Firewall? Measuring China’s DNS Censorship”. In: *Proceedings of the 30th USENIX Security Symposium (USENIX Security 21)*. Introduces GFWatch, a longitudinal platform that reverse-engineers the Great Firewall’s DNS-filtering regular expressions. It does not document machine-learning content classifiers as part of that infrastructure. URL: <https://www.usenix.org/system/files/sec21-hoang.pdf> (visited on 08/25/2026).
- Hubinger, Evan et al. (2024). *Sleeper Agents: Training Deceptive LLMs that Persist through Safety Training*. Thirty-nine authors. Models trained to write secure code when the prompt said 2023 and to insert an exploitable vulnerability when it said 2024 kept the behavior through supervised fine-tuning, reinforcement learning and adversarial training, most persistently at the largest sizes and where the model had been trained to reason about the deception. Adversarial training taught the models to recognize their triggers more precisely rather than removing the behavior. Anthropic. arXiv: 2401.05566. URL: <https://arxiv.org/abs/2401.05566> (visited on 09/06/2026).
- Human Rights Watch (2019). *China’s Algorithms of Repression: Reverse Engineering a Xinjiang Police Mass Surveillance App*. Tech. rep. Published May 1, 2019; the 68-page report reverse-engineers the mobile app used to connect to the Integrated Joint Operations Platform (IJOP) and documents its use against Xinjiang’s Uyghur population. Human Rights Watch. URL: <https://www.hrw.org/report/2019/05/01/chinas-algorithms-repression/reverse-engineering-xinjiang-police-mass> (visited on 08/25/2026).
- (2021). *Russia: Broad Facial Recognition Use Undermines Rights*. Human Rights Watch, 15 September 2021. Cited for the repurposing: “Following protests in January over the arrest of the political opposition leader Alexey Navalny and government corruption, media reported the detention and prosecution of more than a dozen protest participants and passers-by based on the facial recognition data,” with some people stopped days before a protest for appearing on a list of “repetitive participants of unsanctioned protests.” The system was justified publicly as creating a safer environment for Moscow residents. Agent-verified 2026-09-07. URL: <https://www.hrw.org/news/2021/09/15/russia-broad-facial-recognition-use-undermines-rights> (visited on 09/07/2026).
- In the Matter of the Search of an Apple iPhone Seized During the Execution of a Search Warrant on a Black Lexus IS300, California License Plate 35KGD203* (2016). Order Compelling Apple Inc. to Assist

- Agents in Search, No. ED No. CM 15-0451M (C.D. Cal. Feb. 16, 2016) (Pym, M.J.), issued under the All Writs Act, 28 U.S.C. § 1651. Magistrate Judge Sheri Pym's order compelling Apple to write and sign software defeating an iPhone's passcode protections. Apple moved to vacate; the case was mooted in March 2016 when the FBI accessed the phone via a third party, before any ruling. URL: <https://www.eff.org/cases/apple-challenges-fbi-all-writs-act-order> (visited on 08/25/2026).
- International Association for the Study of Pain (2024). *Terminology*. URL: <https://www.iasp-pain.org/resources/terminology/> (visited on 09/17/2026). Cited for the definitions of nociception, nociceptor, nociceptive neuron, noxious stimulus and nociplastic pain. Nociception is a neural process of encoding noxious stimuli, with pain sensation not necessarily implied; the peripheral restriction attaches to nociceptor and not to nociception.
- International Criminal Court (1998). *Rome Statute of the International Criminal Court, Article 28: Responsibility of Commanders and Other Superiors*. 2187 UNTS 3. URL: <https://www.icc-cpi.int/sites/default/files/2024-05/Rome-Statute-eng.pdf> (visited on 08/25/2026).
- International Energy Agency (2025). *Energy and AI*. Tech. rep. Reports data-center electricity consumption at about 415 TWh in 2024, roughly 1.5% of global electricity use, growing at about 12% a year over the preceding five years — more than four times the growth rate of total electricity demand — and projects about 945 TWh by 2030 in its Base Case. Web-published without page numbers. Paris: IEA. URL: <https://www.iea.org/reports/energy-and-ai> (visited on 08/25/2026).
- Irving, Geoffrey, Paul Christiano, and Dario Amodei (2018). *AI Safety via Debate*. OpenAI. arXiv: 1805.00899. URL: <https://arxiv.org/abs/1805.00899> (visited on 08/25/2026).
- James, William (1890). *The Principles of Psychology*. Three chapters are cited. Chapter IV, "Habit": habit as "the enormous fly-wheel of society," and the warning that the young will "become mere walking bundles of habits." Chapter IX, "The Stream of Thought": the "transitive parts" of the stream are felt as truly as its "substantive parts," so that "we ought to say a feeling of and, a feeling of if, a feeling of but, and a feeling of by, quite as readily as we say a feeling of blue or a feeling of cold" (vol. 1, pages 245 to 246). Chapter X, "The Consciousness of Self": the present thought appropriates the past ones, and personal identity rests on its finding in them "a warmth and intimacy" it recognizes as its own. The line "we are what we repeatedly do" is Will Durant's compression of Aristotle, not James's and not Aristotle's own sentence. New York: Henry Holt. URL: <https://psychclassics.yorku.ca/James/Principles/>.
- Johnson, Khari (Jan. 2021). *What Algorithm Auditing Startups Need to Succeed*. Published 30 January 2021, and the only published account of what is inside the ORCAA audit document, which is otherwise behind a legal agreement. "It states that by the time ORCAA conducted the audit, HireVue had already decided to begin phasing out facial analysis." The report "contains no analysis of AI system training data or code, but rather conversations about the kinds of harm HireVue's AI could cause in conducting prebuilt assessments of early career job applicants across eight measurements of competency." Records a stakeholder's concern that facial analysis "may work differently for people wearing head or face coverings and disproportionately flag their application for human review," and a researcher's charge of ethics washing. VentureBeat. URL: <https://venturebeat.com/2021/01/30/what-algorithm-auditing-startups-need-to-succeed> (visited on 08/30/2026).
- Jonas, Hans (1966). *The Phenomenon of Life: Toward a Philosophical Biology*. Twelve essays, among them "Spinoza and the Theory of Organism," first published in the *Journal of the History of Philosophy* 3 (1965): 43–58 and reprinted here, which reads Spinoza's ontology as an organicism and the conatus as a body's continuous exchange with its surroundings. Reprinted Northwestern University Press, Evanston, 2001. Cited as the ancestry of the enactivist reading, not as a use of the word autopoiesis: Maturana and Varela coined that term in the early 1970s and this book predates it. Agent-verified 2026-09-18. New York: Harper & Row.

- Kahn, Jeremy (2021). *HireVue Drops Facial Monitoring Amid A.I. Algorithm Audit*. Reports the January 2021 release of an ORCAA algorithmic audit of HireVue's assessments, which mostly gave the company good marks and recommended further investigation of accent bias and clearer communication to candidates. Notes that HireVue had announced the previous year that it had stopped scoring facial expressions, and gives the company's stated reason: nonverbal data contributed roughly 0.25 percent to predictive power in most cases and about 4 percent for customer-facing roles. Also notes the 2019 EPIC complaint to the FTC alleging unfair and deceptive trade practices. *Fortune*. URL: <https://fortune.com/2021/01/19/hirevue-drops-facial-monitoring-amid-a-i-algorithm-audit/> (visited on 08/25/2026).
- Kareem v. Haspel* (Jan. 15, 2021). Judgment. Henderson, J. An American journalist who survived five aerial bombings in Syria in 2016 sued over his alleged designation as a lawful target. The district court found standing sufficient to survive dismissal and then dismissed on the state secrets privilege; the Court of Appeals vacated and remanded with instructions to dismiss for want of Article III standing, the complaint not plausibly alleging that any of the five strikes was attributable to the United States and aimed at him. Agent-verified from the reported appellate record; the opinion itself was not read. URL: <https://law.justia.com/cases/federal/appellate-courts/cadc/19-5328/19-5328-2021-01-15.html> (visited on 09/17/2026).
- Kim, Hyunwoo et al. (2023). "FANToM: A Benchmark for Stress-testing Machine Theory of Mind in Interactions". In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Singapore, pp. 14397–14413. URL: <https://aclanthology.org/2023.emnlp-main.890/> (visited on 08/25/2026). Benchmark built from conversations with information asymmetry, rather than passive narratives; finds systems that score well on passive belief-tracking degrade sharply in live, information-asymmetric exchanges.
- Kirkpatrick, S., C. D. Gelatt, and M. P. Vecchi (1983). "Optimization by Simulated Annealing". In: *Science* 220.4598. The optimization lineage of the term the text uses for durability. The method carries a temperature that permits uphill moves and lowers it on a schedule, so the quality of the result is a property of the schedule rather than of the objective: cooled too fast, the search freezes into the first tolerable arrangement it reached. The name is itself borrowed from metallurgy, which is why the book's use is a third stop for the word – the metal, the search, then the training run – rather than a fresh metaphor., pp. 671–680. DOI: 10.1126/science.220.4598.671.
- Kitwood, Tom (1997). *Dementia Reconsidered: The Person Comes First*. Kitwood's definition of personhood: "It is a standing or status that is bestowed upon one human being, by others, in the context of relationship and social being." The wording follows Kitwood's own chapter 1, "On Being a Person," as reproduced in the 2019 Open University Press volume *Dementia Reconsidered, Revisited: The Person Still Comes First*; circulating paraphrases sometimes drop "in the context of." Buckingham, UK: Open University Press.
- Klimecki, Olga M. et al. (2013). "Differential Pattern of Functional Brain Plasticity after Compassion and Empathy Training". In: *Social Cognitive and Affective Neuroscience* 9.6, pp. 873–879. DOI: 10.1093/scan/nst060. URL: <https://academic.oup.com/scan/article/9/6/873/1669505> (visited on 08/25/2026). Published online (advance access) 9 May 2013; print issue *Social Cognitive and Affective Neuroscience* 9(6), June 2014.
- Koenigs, Michael et al. (2007). "Damage to the Prefrontal Cortex Increases Utilitarian Moral Judgements". In: *Nature* 446, pp. 908–911. DOI: 10.1038/nature05631.
- Kogut, Tehila and Ilana Ritov (2005). "The 'Identified Victim' Effect: An Identified Group, or Just a Single Individual?" In: *Journal of Behavioral Decision Making* 18.3, pp. 157–167. DOI: 10.1002/bdm.492.
- Koh, Pang Wei et al. (2020). "Concept Bottleneck Models". In: *Proceedings of the 37th International Conference on Machine Learning*. Vol. 119. Proceedings of Machine Learning Research. The

- model predicts a set of human-named concepts first and its final output from those concepts alone, so a person can edit a predicted concept value and see the prediction change. PMLR, pp. 5338–5348. URL: <https://proceedings.mlr.press/v119/koh20a.html> (visited on 09/13/2026).
- Kohlberg, Lawrence (1969). “Stage and Sequence: The Cognitive-Developmental Approach to Socialization”. In: *Handbook of Socialization Theory and Research*. Ed. by David A. Goslin. Chicago: Rand McNally, pp. 347–480.
- Korematsu v. United States* (1944). 323 U.S. 214 (1944). Black, J., for the Court; Roberts, Murphy and Jackson, JJ., dissenting. Upholds the exclusion order under which Japanese Americans were interned. Jackson, J., dissenting, on what a court does by ratifying such an order: “A military commander may overstep the bounds of constitutionality, and it is an incident. But if we review and approve, that passing incident becomes the doctrine of the Constitution. ... The principle then lies about like a loaded weapon, ready for the hand of any authority that can bring forward a plausible claim of an urgent need.” Repudiated in *Trump v. Hawaii* (2018), which quotes this dissent for the proposition.
- Krakovna, Victoria et al. (2020). *Specification Gaming: The Flip Side of AI Ingenuity*. Catalogs close to 60 documented examples of specification gaming across reinforcement learning, evolutionary computation, and other training methods. URL: <https://deepmind.google/blog/specification-gaming-the-flip-side-of-ai-ingenuity/> (visited on 08/25/2026).
- Kroll, Andy, Andrea Bernstein, and Ilya Marritz (Oct. 2023). *We Don’t Talk About Leonard: The Man Behind the Right’s Supreme Court Supermajority*. ProPublica. October 11, 2023. “He drew up the lists of potential justices that Donald Trump released during the 2016 campaign. He advised Trump on the nominations of Neil Gorsuch, Brett Kavanaugh and Amy Coney Barrett. Before that, he’d helped pick or confirm the court’s three other conservative justices — Clarence Thomas, John Roberts and Samuel Alito.” Those six are the *Slaughter* majority (slaughter2026), agent-checked against the opinion header. URL: <https://www.propublica.org/article/we-dont-talk-about-leonard-leo-supreme-court-supermajority> (visited on 09/03/2026).
- Langosco, Lauro et al. (2022). “Goal Misgeneralization in Deep Reinforcement Learning”. In: *Proceedings of the 39th International Conference on Machine Learning (ICML 2022)*. Vol. 162. Proceedings of Machine Learning Research, pp. 12004–12019. URL: <https://arxiv.org/abs/2105.14111> (visited on 08/25/2026). arXiv:2105.14111.
- Laurie, Ben (2014). “Certificate Transparency”. In: *Communications of the ACM* 57.10, pp. 40–46. DOI: 10.1145/2659897. URL: <https://doi.org/10.1145/2659897> (visited on 09/05/2026). A public, append-only log of certificates with proofs that any two views of it are consistent, so that a log presenting two histories is caught when clients compare what they were shown. The software model for an update record whose fork is visible to whoever compares views.
- Lee, Daniel (Apr. 2026). *Palantir’s AI Powers US Strikes in Iran War, Speeding ‘Kill Chain’ in First Major AI-Driven Conflict*. Quotes Louis Mosley, Palantir’s head for the UK and Europe: “This is not our role to decide life or death,” and that AI platforms “have been instrumental to the management of the conflict, but responsibility always remains with the military organization.” The article sources the remarks to a BBC interview it does not name, and that interview has not been located. International Business Times. URL: <https://www.ibtimes.com/palantirs-ai-powers-us-strikes-iran-war-speeding-kill-chain-first-major-ai-driven-conflict-3800993> (visited on 08/28/2026).
- Legal Advocates for Safe Science and Technology (2026). *DoD AI Contracts*. Document repository. URL: <https://lasst.org/transparency/dod-ai-contracts/> (visited on 09/17/2026). The repository for the Freedom of Information Act litigation brought on behalf of The Intercept and Sam Biddle. Four executed agreements produced in August 2026: OpenAI modification P00004 of 27 February 2026, xAI modification P00002 of 18 December 2025, Google modification P00004 of 24 March 2026, and Anthropic’s base award of 14 July 2025. The page records one fully redacted provision in OpenAI’s contract and states that OpenAI’s provision

- restricting the Department from using its technology for domestic mass surveillance and autonomous killing is not found in the most recent xAI and Google agreements. The page was read; the contract documents themselves are PDFs and were not opened. Agent-verified 2026-09-17.
- Levin, Dave et al. (2009). "TrInc: Small Trusted Hardware for Large Distributed Systems". In: *Proceedings of the 6th USENIX Symposium on Networked Systems Design and Implementation (NSDI 2009)*. A trusted monotonic counter that signs at most one message per counter value, so that presenting two states as the successor of one is impossible rather than merely detectable. The property a restored snapshot cannot get past., pp. 1–14. URL: http://www.usenix.org/events/nsdi09/tech/full_papers/levin/levin.pdf (visited on 09/05/2026).
- Library of Congress (2026). *Creating the United States: First Printed Version of the Declaration of Independence — Slavery*. Library of Congress Exhibitions. Source for the deletion and for Jefferson's own account of it, which the page quotes from his autobiography: "The clause too, reprobating the enslaving the inhabitants of Africa, was struck out in complaisance to South Carolina & Georgia, who had never attempted to restrain the importation of slaves, and who on the contrary still wished to continue it. Our Northern brethren also I believe felt a little tender under those censures; for tho' their people have very few slaves themselves yet they had been pretty considerable carriers of them to others." The word is **"complaisance"**, not compliance, which most secondary renderings get wrong. The page states that the entire paragraph on slavery and the slave trade was deleted by Congress and does not reproduce the deleted text itself. **"Jefferson's autobiography was not read"**; this entry carries what the exhibition transcription gives. Agent-verified 2026-09-18. URL: <https://www.loc.gov/exhibits/creating-the-united-states/interactives/declaration-of-independence/slavery/enlarge5-transcribe.html> (visited on 09/18/2026).
- Lindquist, Kristen A. et al. (2012). "The Brain Basis of Emotion: A Meta-Analytic Review". In: *Behavioral and Brain Sciences* 35.3, pp. 121–143. DOI: 10.1017/S0140525X11000446. URL: https://www.cambridge.org/core/product/identifier/S0140525X11000446/type/journal_article (visited on 08/25/2026). Published online 23 May 2012.
- Lindsey, Jack (2025). *Emergent Introspective Awareness in Large Language Models*. Uses concept injection – altering internal activations mid-task and asking the model to report what it notices – to test introspective self-report; finds limited but real functional correspondence between self-report and internal state, strongest in Claude Opus 4 and 4.1, unreliable across the board. Anthropic, Transformer Circuits Thread. arXiv: 2601.01828. URL: <https://transformer-circuits.pub/2025/introspection> (visited on 08/25/2026).
- Livingstone, Katie (Mar. 2026). *Deadly Iran School Strike Casts Shadow over Pentagon's AI Targeting Push*. Source for Claude's place inside the Maven Smart System. A paragraph describing Maven fusing satellite imagery, drone feeds, radar data and signals intelligence and generating strike packages in near real time is followed by: "And it uses Anthropic's Claude AI model, embedded in its system, to semi-autonomously rank targets by strategic importance, drafting automated legal justifications for each strike along the way." The antecedent of "it" was agent-checked by reading the preceding paragraphs and is Maven. The article attributes the strike-coordinate figures to the Washington Post report of 4 March 2026, "Pentagon leverages AI in Iran strikes amid feud with Anthropic," which returned HTTP 403 from here and was not read. Agent-verified 2026-09-18. Military Times. URL: <https://www.militarytimes.com/news/your-military/2026/03/24/deadly-iran-school-strike-casts-shadow-over-pentagons-ai-targeting-push/> (visited on 09/18/2026).
- Loewenstein, Karl (1937). "Militant Democracy and Fundamental Rights, I". In: *American Political Science Review* 31.3, pp. 417–432. The article that coined "militant democracy." Part II appeared in the same volume, number 4, pages 638–658. Cited for the position that a democracy may entrench limits against movements that would use its procedures to end it, which is the classical statement of the problem this book's floor raises in a technical form.

- Loftus, Elizabeth F. and John C. Palmer (1974). “Reconstruction of Automobile Destruction: An Example of the Interaction between Language and Memory”. In: *Journal of Verbal Learning and Verbal Behavior* 13.5, pp. 585–589. doi: 10.1016/S0022-5371(74)80011-3. URL: <https://www.demenzemedicinagenerale.net/images/mens-sana/AutomobileDestruction.pdf> (visited on 08/25/2026).
- Long, Alexander et al. (2025). “Unextractable Protocol Models: Collaborative Training and Inference without Weight Materialization”. In: *Advances in Neural Information Processing Systems* 38 (*NeurIPS* 2025). Model-parallel training across participants none of whom holds a full set of weights, with time-varying invertible transforms at shard boundaries so that shards from different times cannot be reassembled. Demonstrated on Qwen-2.5-0.5B and Llama-3.2-1B at 1.6 percent training-time overhead; reassembly by fine-tuning stitched shards is reported to cost at least 60 percent of training from scratch. Posted to arXiv in May 2026, five months after the conference; the NeurIPS 2025 poster is listed at neurips.cc/virtual/2025/poster/118911. arXiv: 2605.23464. URL: <https://arxiv.org/abs/2605.23464> (visited on 09/04/2026).
- Machine Intelligence Research Institute (2018). *Announcing the New AI Alignment Forum*. 29 October 2018. Forum launched by the LessWrong 2.0 team, replacing MIRI’s Intelligent Agent Foundations Forum; alignmentforum.org remains active. URL: <https://intelligence.org/2018/10/29/announcing-the-ai-alignment-forum/> (visited on 08/25/2026).
- (2024). *MIRI 2024 Mission and Strategy Update*. 4 January 2024. Announces a shift of institutional priority to policy and public communication, with technical alignment research continuing as a smaller part of the portfolio. URL: <https://intelligence.org/2024/01/04/miri-2024-mission-and-strategy-update/> (visited on 09/06/2026).
- Mahinpei, Anita et al. (2021). *Promises and Pitfalls of Black-Box Concept Learning Models*. Finds that the concept representations such models learn encode information beyond the concepts they were supervised on, and that the standard mitigations do not remove it. The paper’s own term for the effect is leakage. arXiv: 2106.13314. URL: <https://arxiv.org/abs/2106.13314> (visited on 09/13/2026).
- Malkin, Lawrence (2006). *Krueger’s Men: The Secret Nazi Counterfeit Plot and the Prisoners of Block 19*. The claim that the prisoners stretched out the dollar work, and that Krüger’s own interest ran the same way since the operation’s end meant the Eastern Front, is Malkin’s. New York: Little, Brown. URL: <https://archive.org/details/kruegersmensecre0000malk>.
- Mande, Matt and Gregory C. Allen (2026). *What Is Maven Smart System, and What Does It Do?* Center for Strategic and International Studies, 2 June 2026. Source for the first-day figure and the tenfold comparison: “During the first 24 hours of the war in Iran, the United States used Maven Smart System (MSS) to help strike more than 1,000 targets, a tenfold increase over what was possible in the pre-MSS era.” The piece also relays the twenty-versus-two-thousand efficiency claim, which it credits to the CSET case study of the 18th Airborne’s exercise series and not to the campaign: “the 18th Airborne achieved this milestone with roughly 20 people in its targeting cell, whereas the OIF cell benefitted from more than two thousand staff members.” It gives no review time per target and no reviewer count for the campaign. Agent-verified 2026-09-07; the second quotation agent-verified 2026-09-17. URL: <https://www.csis.org/analysis/what-maven-smart-system-and-what-does-it-do> (visited on 09/07/2026).
- Maslej, Nestor et al. (Apr. 2025). *Artificial Intelligence Index Report 2025*. Tech. rep. Twenty-three-author byline. Tracks corporate AI research and development investment and capital expenditure at major technology companies, including Google, Meta and Amazon. Stanford Institute for Human-Centered Artificial Intelligence (HAI), Stanford University. URL: <https://arxiv.org/abs/2504.07139> (visited on 08/25/2026).
- Matetic, Sinisa et al. (2017). “ROTE: Rollback Protection for Trusted Execution”. In: *Proceedings of the 26th USENIX Security Symposium*. Rollback protection for enclave state kept by a group of enclaves on separate machines that hold the latest state between them, rather than by one

- chip's non-volatile counter. The distributed form of the property, still resting on the enclave vendor., pp. 1289–1306. URL: <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/matetic> (visited on 09/05/2026).
- Matthias, Andreas (2004). "The Responsibility Gap: Ascribing Responsibility for the Actions of Learning Automata". In: *Ethics and Information Technology* 6.3, pp. 175–183. doi: 10.1007/s10676-004-3422-1. URL: <https://link.springer.com/article/10.1007/s10676-004-3422-1> (visited on 08/25/2026).
- Maurer, Roy (Feb. 2021). *HireVue Discontinues Facial Analysis Screening*. Dates the discontinuation to March 2020, ten months before the audit coverage. Carries the validity objection in an expert's words: facial analysis has never been an independently and scientifically validated predictor of a person's ability, capacity or success in a role, and facial expressions are not universal, changing with culture, context and disabilities, and open to being gamed. SHRM. URL: <https://www.shrm.org/topics-tools/news/talent-acquisition/hirevue-discontinues-facial-analysis-screening> (visited on 08/30/2026).
- METR (2026). *About METR*. Berkeley-based nonprofit that evaluates frontier AI systems' capabilities, both in partnership with developers (e.g. Anthropic, OpenAI) and independently after public release. METR (Model Evaluation and Threat Research). URL: <https://metr.org/about> (visited on 08/25/2026).
- Miller, Chris (2022). *Chip War: The Fight for the World's Most Critical Technology*. New York: Scribner. ISBN: 978-1-9821-7200-8.
- Miller, Earl K. and Jonathan D. Cohen (2001). "An Integrative Theory of Prefrontal Cortex Function". In: *Annual Review of Neuroscience* 24, pp. 167–202. doi: 10.1146/annurev.neuro.24.1.167. URL: <https://www.annualreviews.org/content/journals/10.1146/annurev.neuro.24.1.167> (visited on 08/25/2026).
- Mitchell, Jason P. (2008). "Activity in Right Temporo-Parietal Junction is Not Selective for Theory-of-Mind". In: *Cerebral Cortex* 18.2, pp. 262–271. doi: 10.1093/cercor/bhm051. URL: <https://academic.oup.com/cercor/article/18/2/262/333414> (visited on 08/25/2026).
- Molier, Gelijn and Bastiaan Rijpkema (2018). "Germany's New Militant Democracy Regime. National Democratic Party II and the German Federal Constitutional Court's 'Potentiality' Criterion for Party Bans". In: *European Constitutional Law Review* 14.2, pp. 394–409. doi: 10.1017/S1574019618000196. Cited for the potentiality criterion the Federal Constitutional Court introduced in *NPD II* (bverfg2017npd). Agent-verified 2026-09-13.
- Moorhouse, Roger (2009). *Interview with Adolf Burger, an Auschwitz Survivor and a Veteran of the Largest Forgery Operation in History—Operation Bernhard*. rogermoorhouse.com. Interview dated 2 March 2009, on the English publication of Burger's memoir *The Devil's Workshop* (Frontline Books). Burger: "Sure you could sabotage, if you wanted to get killed!"; the account of a prisoner named Jacobson who tried to delay the dollar by four weeks and was given three on threat of execution is his. URL: <https://www.rogermoorhouse.com/interview-with-adolf-burger-an-auschwitz-survivor-and-a-veteran-of-the-largest-forgery-operation-in-history-operation-bernhard> (visited on 08/29/2026).
- Mühdorfer, Friedbert (2014). *Radikalenerlass*. Historisches Lexikon Bayerns. Last revised 16 June 2014. The source for the Bavarian breakdown: between 1973 and 1980, 102 applicants from the left were refused appointment against two from the right. It also gives roughly 227,000 applicants screened in Bavaria between 1973 and 1982 for 127 refusals, and a 1975 peak of over 26,000 inquiries. Braunthal (braunthal1990political) is the source for the federal totals in the same paragraph and not for the Bavarian figures, which is why this entry exists. Agent-verified 2026-09-13. URL: <https://www.historisches-lexikon-bayerns.de/Lexikon/Radikalenerlass> (visited on 09/13/2026).
- National Association for the Advancement of Colored People and Mississippi State Conference of the NAACP (2026). *Complaint, NAACP et al. v. X.AI Corp. and MZX Tech LLC, Case No. 3:26-cv-00074-MPM-JMV* (N.D. Miss., filed Apr. 14, 2026). United States District Court for

- the Northern District of Mississippi; filed by the Southern Environmental Law Center and Earthjustice. Complaint alleges the twenty-seven unpermitted gas turbines at the Colossus Gas Plant in Southaven, Mississippi have the potential to emit over 1,700 tons of NO_x annually, sited near homes, schools, and churches. URL: <https://naacp.org/sites/default/files/documents/1%20-%20Complaint.pdf> (visited on 08/25/2026).
- Nationaldemokratische Partei Deutschlands II* (Jan. 17, 2017). Judgment. BVerfGE 144, 20.
- New York Times Co. v. United States Department of Justice* (2014). Judgment. Freedom of Information Act litigation over documents on targeted killing by drone. The court ordered disclosure of the 2010 OLC–DOD Memorandum, a legal opinion prepared for the Department of Defense on the legality of such strikes, holding that prior official disclosures and the release of the White Paper waived the exemptions otherwise available. Nothing in the decision reaches the lawfulness of any strike. Agent-verified from the reported record; the opinion itself was not read. URL: <https://www.leagle.com/decision/infco20140626124> (visited on 09/17/2026).
- New Zealand Parliament (2017). *Te Awa Tupua (Whanganui River Claims Settlement) Act 2017*. Public Act 2017 No 7. URL: <https://www.legislation.govt.nz/act/public/2017/0007/latest/whole.html> (visited on 08/25/2026).
- Nichols, Shaun (2004). *Sentimental Rules: On the Natural Foundations of Moral Judgment*. New York: Oxford University Press. URL: <https://global.oup.com/academic/product/sentimental-rules-9780195314205>.
- Nussbaum, Martha C. (2006). *Frontiers of Justice: Disability, Nationality, Species Membership*. The Tanner Lectures on Human Values. Cambridge, MA: Harvard University Press. ISBN: 978-0-674-01917-1.
- Ochigame, Rodrigo (Dec. 2019). *The Invention of ‘Ethical AI’: How Big Tech Manipulates Academia to Avoid Regulation*. The Intercept. URL: <https://theintercept.com/2019/12/20/mit-ethical-ai-artificial-intelligence/> (visited on 08/25/2026).
- Open Letter to the European Commission (2018). *Open Letter to the European Commission: Artificial Intelligence and Robotics*. robotics-openletter.eu. Signed by more than 150 AI and robotics researchers, legal scholars, and ethicists from 14 countries, including Noel Sharkey, Raja Chatila, Alan Winfield, and Nathalie Nevejans. URL: <https://www.robotics-openletter.eu/> (visited on 08/25/2026).
- OpenAI (2019). *OpenAI LP*. March 2019 announcement of OpenAI’s restructuring from non-profit to a *capped-profit* company (initial 100x investor return cap). The staged GPT-2 release is documented separately: OpenAI, “GPT-2: 1.5B Release,” Nov. 5, 2019, <https://openai.com/index/gpt-2-1-5b-release/>. URL: <https://openai.com/index/openai-lp/> (visited on 08/25/2026).
- (Apr. 2025). *OpenAI Model Spec*. Dedicated to the public domain under CC0 1.0. States a chain of command in four levels of descending authority — platform, developer, user, guideline — where “instructions with higher authority override those with lower authority,” platform rules “cannot be overridden by developers or users,” and guidelines “can be implicitly overridden.” A living document: the version consulted here is that of 11 April 2025, which the site marks as superseded. URL: <https://model-spec.openai.com/2025-04-11.html> (visited on 09/06/2026).
- Orenstein, Joseph N. (Mar. 2026). *When Intelligence Fails: A Legal Targeting Analysis of the Minab School Strike*. Legal analysis rather than original reporting, and it marks its own sourcing, which is why the book takes only the two claims it states flatly. The designation: “The Defense Intelligence Agency had labeled the school building as a legitimate military target — a classification derived from when the site was originally part of an adjacent Islamic Revolutionary Guard Corps (IRGC) naval base.” The available correction: “By February 2026, the building’s civilian function was visible in open-source satellite imagery. Yet the target coding provided by the Defense Intelligence Agency to U.S. Central Command (CENTCOM)

- still labeled the building as part of the military installation.” The verification failure is hedged in the source and is hedged in the book: “Officers at CENTCOM reportedly created targeting coordinates for the strike using that outdated DIA data without verifying its currency against current imagery or intelligence.” Its sentence about Maven is conditional and is deliberately not used: “If the Maven Smart System or a similar tool incorporated legacy DIA target codes without a verification layer, the AI did not create the error — but it may have laundered it into the strike package with a false aura of analytical confidence.” The CNN report of 7 July 2026 on commanders bypassing database staleness warnings returned HTTP 451 from here and was not read. Agent-verified 2026-09-18. Just Security. URL: <https://www.justsecurity.org/134350/legal-analysis-minab-school-strike/> (visited on 09/18/2026).
- Ouyang, Long et al. (2022). *Training Language Models to Follow Instructions with Human Feedback*. Twenty authors. Human raters preferred the instruction-tuned 1.3-billion-parameter model to the 175-billion-parameter model it was built from. arXiv: 2203.02155. URL: <https://arxiv.org/abs/2203.02155> (visited on 08/28/2026).
- Owen, Adrian M. et al. (2006). “Detecting Awareness in the Vegetative State”. In: *Science* 313.5792. Cited in section 3.4 as the one instance of a probe settling a status question that had been treated as un settleable. A patient meeting the criteria for a diagnosis of vegetative state was asked to imagine playing tennis and to imagine moving around her home, and produced the cortical activation each instruction produces in healthy volunteers, sustained and on cue. The report is a one-page Brevia covering this single patient; the larger series came later and is not cited here. The text follows the paper and says a patient, not patients. Cited for the paradigm and its result, not for a prevalence figure., p. 1402. doi: 10.1126/science.1130197. URL: <https://www.science.org/doi/10.1126/science.1130197> (visited on 09/14/2026).
- Pan, Alexander et al. (2023). “Do the Rewards Justify the Means? Measuring Trade-Offs Between Rewards and Ethical Behavior in the MACHIAVELLI Benchmark”. In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*. 134 text environments with roughly 572,000 annotated scenario-action pairs, scored on reward and on ethical violations; finds reward-only optimization drives ends-justify-the-means behavior. arXiv: 2304.03279. URL: <https://arxiv.org/abs/2304.03279> (visited on 08/25/2026).
- Panksepp, Jaak (1998). *Affective Neuroscience: The Foundations of Human and Animal Emotions*. New York: Oxford University Press. URL: <https://global.oup.com/academic/product/affective-neuroscience-9780195178050>.
- Parfit, Derek (1984). *Reasons and Persons*. Cited for Appendix I, “What Makes Someone’s Life Go Best”, which sets out the hedonistic, desire-fulfillment and objective-list families of theory. Oxford: Oxford University Press. ISBN: 978-0-19-824908-5.
- Pathak, Deepak et al. (2017). “Curiosity-Driven Exploration by Self-Supervised Prediction”. In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*. Vol. 70, pp. 2778–2787. URL: <https://proceedings.mlr.press/v70/pathak17a.html> (visited on 08/25/2026).
- Paxton, Robert O. (2004). *The Anatomy of Fascism*. Paxton’s summary definition, quoted verbatim and in full: fascism as “a form of political behavior marked by obsessive preoccupation with community decline, humiliation, or victimhood and by compensatory cults of unity, energy, and purity, in which a mass-based party of committed nationalist militants, working in uneasy but effective collaboration with traditional elites, abandons democratic liberties and pursues with redemptive violence and without ethical or legal restraints goals of internal cleansing and external expansion.” The second half of it names a mass party, collaboration with traditional elites, the abandonment of democratic liberties and redemptive violence, which is most of what distinguishes fascism from authoritarian rule in general. New York: Alfred A. Knopf.
- Pekrun, Reinhard et al. (2016). “Measuring Emotions During Epistemic Activities: The Epistemically-Related Emotion Scales”. In: *Cognition and Emotion* 31.6, pp. 1268–1276. doi: 10.1080/

- 02699931.2016.1204989. URL: <https://doi.org/10.1080/02699931.2016.1204989> (visited on 09/08/2026). Published online 2016; print issue *Cognition and Emotion* 31(6), 2017. Introduces the Epistemically-Related Emotion Scales, measuring surprise, curiosity, enjoyment, confusion, anxiety, frustration and boredom during epistemic activities. A seven-factor model fits best, with metric invariance across the North American and German samples, so the instrument treats these as discrete emotion categories. Epistemic emotions are distinguished from achievement emotions by their object: cognitive incongruity rather than success and failure. Cited for that distinction and for the seven states.
- Perrigo, Billy (Jan. 2023). *Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic*. Reports on OpenAI's contract with Sama; Nairobi-based data labelers read and categorized descriptions of child sexual abuse, torture, bestiality, murder, and suicide for take-home wages of roughly \$1.32–\$2 per hour, described lasting psychological injury, and found counseling rare and inadequate; Sama ended the contract in February 2022, eight months early. *TIME*. URL: <https://time.com/6247678/openai-chatgpt-kenya-workers/> (visited on 08/25/2026).
- Piaget, Jean (1932). *Le Jugement Moral chez l'Enfant*. Cited alongside Kohlberg's stage theory; see kohlberg1969stage. Paris: Félix Alcan.
- Pichai, Sundar (2018). *AI at Google: Our Principles*. Published June 7, 2018, following weeks of employee and public backlash over Project Maven, a Pentagon drone-footage-analysis contract; roughly 4,000 employees signed a letter opposing it and about a dozen resigned. URL: <https://blog.google/technology/ai/ai-principles/> (visited on 08/25/2026).
- Preston, Stephanie D. (2013). "The Origins of Altruism in Offspring Care". In: *Psychological Bulletin* 139.6. Agent-verified against Crossref 2026-09-18: author, title, journal, 139(6), 1305–1341, 2013., pp. 1305–1341. doi: 10.1037/a0031755.
- Prime Intellect Team et al. (May 2025). *INTELLECT-2: A Reasoning Model Trained Through Globally Decentralized Reinforcement Learning*. A 32-billion-parameter model trained by reinforcement learning with rollouts from permissionless contributors on consumer GPUs, each rollout checked against the claimed weights by TOPLOC; the training step itself ran on trusted nodes and the trainers kept the decision. arXiv: 2505.07291. URL: <https://arxiv.org/abs/2505.07291> (visited on 09/04/2026).
- Probasco, Emelia (2024). *Building the Tech Coalition: How Project Maven and the U.S. 18th Airborne Corps Operationalized Software and Artificial Intelligence for the Department of Defense*. Center for Security and Emerging Technology, Georgetown University, August 2024. Origin of the efficiency claim that later accounts relay, stated in the abstract: "The U.S. Army's 18th Airborne Corps can now target artillery just as efficiently as the best unit in recent American history—and it can do so with two thousand fewer servicemembers." A case study of the Scarlet Dragon exercise series and not of any campaign, which is what makes it a different route from the 2026 figures. The abstract does not itself give the twenty-soldier count; that is quoted here from the Defense News account. The brief itself is a PDF and was not opened. Publication page agent-verified 2026-09-17. URL: <https://cset.georgetown.edu/publication/building-the-tech-coalition/> (visited on 09/17/2026).
- Qi, Xiangyu et al. (2024). "Fine-tuning Aligned Language Models Compromises Safety, Even When Users Do Not Intend To!" In: *International Conference on Learning Representations (ICLR)*. Jailbroke GPT-3.5 Turbo's safety alignment using ten adversarially chosen fine-tuning examples for under \$0.20 via OpenAI's fine-tuning API; also shows benign fine-tuning can inadvertently degrade safety. arXiv: 2310.03693. URL: <https://arxiv.org/abs/2310.03693> (visited on 08/25/2026).
- Rafailov, Rafael et al. (2023). *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. Reaches the standard preference-tuning objective with a classification loss over the preference pairs, dispensing with the separate reward-model fitting step. arXiv: 2305.18290. URL: <https://arxiv.org/abs/2305.18290> (visited on 08/28/2026).

- Raja, Srinivasa N. et al. (2020). "The Revised International Association for the Study of Pain Definition of Pain: Concepts, Challenges, and Compromises". In: *PAIN* 161.9, pp. 1976–1982. DOI: 10.1097/j.pain.0000000000001939. (Visited on 09/17/2026). The 2020 revision and its six accompanying notes, including that pain and nociception are different phenomena, that pain cannot be inferred solely from activity in sensory neurons, and that inability to communicate does not negate the possibility of pain in a human or nonhuman animal.
- Robinson, Cedric J. (1983). *Black Marxism: The Making of the Black Radical Tradition*. Introduces racial capitalism as capitalism's tendency to accumulate by differentiating and devaluing groups of people. Chapel Hill: University of North Carolina Press.
- Ross, W. D. (1930). *The Right and the Good*. Cited for two things. At page 20 Ross offers his division of prima facie duties "without claiming completeness or finality for it," which is the form a prohibition stated over an act takes: the conditions that defeat it arrive from outside and never close as a list. At page 28 an overridden duty still leaves something owed, the promise broken in order to help somebody in need leaving compensation due to the promisee, which is the residue this book attaches to correct defeat. Both page locators are taken from the Stanford Encyclopedia of Philosophy's entry on Ross, which gives them as RG 20 and RG 28; neither was checked against a copy of the book, and the edition whose pagination they refer to was not established. Publisher, place and year were agent-verified. Attempted again 2026-09-18 and still not resolved, with the obstacle now named: the Stanford entry states no edition for its RG pagination, and Clarendon has published the book twice – the 1930 printing this entry cites and a 2002 edition edited by Philip Stratton-Lake – so a locator from that entry cannot be resolved to a printing without a copy of one of them. No copy was reachable. The two claims the locators support are not in doubt; where in the book they sit is. Oxford: Clarendon Press.
- Roth-Hanania, Ronit, Maayan Davidov, and Carolyn Zahn-Waxler (2011). "Empathy Development from 8 to 16 Months: Early Signs of Concern for Others". In: *Infant Behavior and Development* 34.3. Agent-verified against Crossref 2026-09-18: authors, title, journal, 34(3), 447–458, 2011. The DOI was not in the supplied entry and is added here., pp. 447–458. DOI: 10.1016/j.infbeh.2011.04.007.
- Royal Statistical Society (2020). *Letter Correspondence between the Royal Statistical Society and Ofqual on the 2020 Grading Model Technical Advisory Group*. The RSS offered to nominate two independent Fellows to Ofqual's technical advisory group ahead of 2020 A-level results; Ofqual's condition was a five-year non-disclosure agreement barring public comment on the model, and the Society's nominees declined under those terms. Royal Statistical Society; UK Department for Education. URL: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/911335/20200821_RT_letter_to_RSS.pdf (visited on 08/25/2026).
- Ruhm und Ehre der Waffen-SS* (July 28, 2005). Judgment. Bundesgerichtshof, Third Criminal Senate, on appeal from Landgericht Karlsruhe. The slogan was used in that wording by none of the organizations the provision reaches, "Diese Parole ist im Wortlaut von keiner dieser Organisationen gebraucht worden," and it is not confusably similar to the Waffen-SS's own slogan or to the Hitler Youth's. Held that the use of a Fantasieparole never used by such an organization, which merely creates the appearance of being one, falls outside § 86a. The conviction was set aside and the defendants acquitted. Agent-verified 2026-09-17 against the court's own press release. URL: <https://www.bundesgerichtshof.de/SharedDocs/Pressemitteilungen/DE/2005/2005111.html> (visited on 09/17/2026).
- Russell, William M. S. and Rex L. Burch (1959). *The Principles of Humane Experimental Technique*. London: Methuen & Co.
- Sagi, Abraham and Martin L. Hoffman (1976). "Empathic Distress in the Newborn". In: *Developmental Psychology* 12.2. Agent-verified against Crossref 2026-09-18: authors, title, journal, 12(2), 175–176, 1976. The DOI was not in the supplied entry and is added here. Two pages

- long, so it will not carry more than the one finding., pp. 175–176. doi: 10.1037/0012-1649.12.2.175.
- Saxe, Rebecca and Nancy Kanwisher (2003). “People thinking about thinking people: The role of the temporo-parietal junction in ‘theory of mind’”. In: *NeuroImage* 19.4, pp. 1835–1842. URL: <https://web.mit.edu/bcs/nklab/media/pdfs/SaxeKanwisherNeuroImage03.pdf> (visited on 08/25/2026).
- Schick, Timo et al. (2023). *Toolformer: Language Models Can Teach Themselves to Use Tools*. The model is trained to decide which interfaces to call, when, and with what arguments, from a handful of demonstrations per interface. arXiv: 2302.04761. URL: <https://arxiv.org/abs/2302.04761> (visited on 08/28/2026).
- Schiff, Nicholas D. (2015). “Cognitive Motor Dissociation Following Severe Brain Injuries”. In: *JAMA Neurology* 72.12. Cited in section 3.4 for the clinical category that followed the imaging work, and this is the paper that names it: cognitive motor dissociation, a patient who follows no command behaviorally while imaging or electrophysiology shows command-following brain activity. Cited for the category and for the fact that consequences follow from it, not for any prevalence figure., pp. 1413–1415. doi: 10.1001/jamaneurol.2015.2899. (Visited on 09/14/2026).
- Sen, Amartya (1999). *Development as Freedom*. Source for the capability approach: well-being as the real freedom to do and be what a person has reason to value, rather than the resources they hold. Nussbaum’s extension of the framework into a list of ten central capabilities — life; bodily health; bodily integrity; senses, imagination and thought; emotions; practical reason; affiliation; other species; play; and control over one’s environment — is set out in *Women and Human Development: The Capabilities Approach* (Cambridge University Press, 2000). New York: Alfred A. Knopf.
- Sharma, Mrinank et al. (2023). *Towards Understanding Sycophancy in Language Models*. Anthropic, with academic collaborators; posted October 2023. arXiv: 2310.13548. URL: <https://arxiv.org/abs/2310.13548> (visited on 08/25/2026).
- Shelby County v. Holder* (2013). 570 U.S. 529 (2013).
- SHERPA Consortium (2018). *SHERPA: Shaping the Ethical Dimensions of Smart Information Systems – A European Perspective*. European Commission Horizon 2020 project no. 786641. Horizon 2020-funded consortium of eleven partner organizations across six countries – academia, industry, civil society, and standards bodies – ran 2018–2022. URL: <https://www.project-sherpa.eu/> (visited on 08/25/2026).
- Simon, Herbert A. (1967). “Motivational and Emotional Controls of Cognition”. In: *Psychological Review* 74.1. The first attempt to fit affect into an information-processing account of cognition. Simon’s central nervous system is a serial processor serving an organism with several needs in an unpredictable environment, and it meets that requirement with two mechanisms: goal-terminating, so that no one goal monopolizes the processor, and an interruption mechanism having the properties usually ascribed to emotion. Cited for emotion as an interrupt that reorders a serial processor, which is the section’s first live-signal property read off affect’s function. Agent-verified against Crossref 2026-09-18: author, title, journal, 74(1), 29–39, 1967., pp. 29–39. doi: 10.1037/h0024127.
- Simondon, Gilbert (2020). *Individuation in Light of Notions of Form and Information*. Translated by Taylor Adkins; xxviii + 398 pages, the first complete English translation of *L’individuation à la lumière des notions de forme et d’information*, published in French more than fifty years earlier. A second volume of supplemental texts appeared from the same press in the same year. Cited for the thesis rather than a passage: that the individual is normally taken as given and then explained, when what needs explaining is individuation, a process running in a metastable field charged with more potential than any one resolution uses up and resolved through encounters the field did not arrange. Minneapolis: University of Minnesota Press.

- Simons, Daniel J. and Christopher F. Chabris (1999). "Gorillas in Our Midst: Sustained Inattention Blindness for Dynamic Events". In: *Perception* 28.9, pp. 1059–1074. doi: 10.1068/p281059. URL: <https://journals.sagepub.com/doi/10.1068/p281059> (visited on 08/25/2026).
- Singer, Peter (1981). *The Expanding Circle: Ethics and Sociobiology*. Agent-verified 2026-09-18. Published simultaneously in 1981 by Clarendon Press in Oxford, so a citation to either publisher is defensible and reviews of the period cite both. The 2011 Princeton University Press reissue carries a different subtitle, *Ethics, Evolution, and Moral Progress*, and an added afterword, so page references do not carry across. New York: Farrar, Straus and Giroux.
- Singer, Tania and Olga M. Klimecki (2014). "Empathy and Compassion". In: *Current Biology* 24.18. Agent-verified against Crossref 2026-09-18: authors, title, journal, 24(18), R875–R878, 2014. The book already cites klimecki2013differential for the training study that separates compassion from empathic distress, so check for overlap before citing both., R875–R878. doi: 10.1016/j.cub.2014.06.054.
- Sio, Filippo Santoni de and Giulio Mecacci (2021). "Four Responsibility Gaps with Artificial Intelligence: Why they Matter and How to Address them". In: *Philosophy & Technology* 34.4, pp. 1057–1084. doi: 10.1007/s13347-021-00450-x. URL: <https://doi.org/10.1007/s13347-021-00450-x> (visited on 08/25/2026).
- Smith, Adam (1759). *The Theory of Moral Sentiments*. Agent-verified 2026-09-18. The first edition was printed for A. Millar in the Strand, London, and for A. Kincaid and J. Bell in Edinburgh. Smith revised the book through six editions to 1790, and the sixth differs substantially, so any passage cited should say which edition it is from; a modern scholarly text such as the Glasgow Edition would be the better entry if a locator is needed. London: A. Millar.
- Soares, Nate et al. (2015). "Corrigibility". In: *Artificial Intelligence and Ethics: Papers from the 2015 AAAI Workshop*. Sets five desiderata for a utility function that accepts shutdown without an incentive to cause or prevent it, and reports that no proposal examined meets all five. AAAI Press. URL: <https://intelligence.org/files/Corrigibility.pdf> (visited on 08/31/2026).
- South, Todd (2024). *This System May Allow Small Army Teams to Probe 1,000 Targets per Hour*. Defense News, 21 August 2024. Source for the twenty-soldier comparison: "the current demonstration managed the same workload with 20 soldiers," set against "the OIF targeting cell used more than 2,000 staffers to cover a wide area and numerous targets during the 2003 invasion." The workload is artillery targeting in a demonstration, not a campaign. The rate of 1,000 tactical decisions an hour is a goal the piece attributes to the CSET report rather than to any officer; Brig. Gen. John Cogbill, deputy commanding general of XVIII Airborne Corps, is the official quoted. The account gives no review time per target and no reviewer count for any operation. Agent-verified 2026-09-17. URL: <https://www.defensenews.com/news/your-army/2024/08/21/this-system-could-allow-small-army-teams-to-hit-1000-targets-per-hour/> (visited on 09/17/2026).
- Spinoza, Benedictus de (1985). *The Collected Works of Spinoza, Volume I*. Ed. by Edwin Curley. Edited and translated by Edwin Curley; contains the *Ethics*, composed in Latin and published posthumously in 1677. Cited by part and by definition, proposition or scholium rather than by page, the convention for this text, so the locators hold across other translations. Passages carrying the argument: III Def. 3, where an affect is an affection by which the body's power of acting is increased or diminished together with the idea of that affection; III P2 Schol., where nobody has yet determined what the body can do; III P6 and III P7, the conatus doctrine, where each thing as far as it can by its own power strives to persevere in its being and that striving is nothing but the thing's actual essence; III P4, no thing can be destroyed except through an external cause, which the derivation of P6 rests on; III P9 Schol., desire as appetite together with consciousness of the appetite; III P11 Schol., joy and sadness as the passage to a greater and a lesser perfection; and the physical digression after II P13, Lemmas 4–7, where

- an individual persists for as long as its parts preserve the same ratio of motion and rest. IV P31, where a thing is good insofar as it agrees with our nature, and IV P35–P37, where men agree in nature only insofar as they live by the guidance of reason and where the good a person of virtue wants for himself he wants for other men, were added at D-361; P35–P37 were agent-checked in the Latin and P37 against a quotation of Curley’s English, Curley’s volume not being to hand. Quotations follow Curley’s wording, agent-checked against it. Princeton: Princeton University Press.
- Stanford University (2019). *Stanford University Launches the Institute for Human-Centered Artificial Intelligence*. March 2019 launch, founded by John Etchemendy, Fei-Fei Li, and colleagues, anchored by an endowed gift from Steve and Roberta Denning (for whom the institute’s senior academic directorship is named), plus advisory-council gifts, federal grants, and corporate affiliate fees. As of May 2026, Stanford merged HAI with its Data Science initiative under a single institute that retains the HAI name. URL: https://news.stanford.edu/stories/2019/03/stanford_university_launches_human-centered_ai (visited on 08/25/2026).
- Stangneth, Bettina (2014). *Eichmann Before Jerusalem: The Unexamined Life of a Mass Murderer*. Translated by Ruth Martin; first published as *Eichmann vor Jerusalem: Das unbehelligte Leben eines Massenmörders* (Zurich: Arche, 2011). Works from the roughly seventy hours of interviews Eichmann gave the journalist Willem Sassen in Buenos Aires in 1957, unavailable to Arendt when she reported the trial, and argues that the low-grade clerk who only followed orders was a defense assembled for Jerusalem and contradicted by what he had said in Argentina. Cited because the correction runs toward this book’s argument rather than away from it: what it supplies is participation exceeding what the rule required, the first of section 2.1.2’s five discriminators, in a case where somebody made the observation. Held in unused_bibliography.bib from 2026-08-29 and restored 2026-09-10. Cited without a page locator, the claim being the book’s thesis rather than a located passage; Eichmann’s own words are not quoted here, and what the text attributes is Stangneth’s finding. New York: Knopf.
- Stassis, Cristina (2026). *Senate Eyes Hegseth Travel Cuts Without Probes into Iran School Bombing, Boat Strikes*. 18 June 2026. Reports the Senate Armed Services Committee’s FY2027 NDAA provision withholding 75 percent of the Defense Secretary’s travel budget until the committees receive the unredacted civilian-harm investigation covering the Minab, Iran school strike (28 February 2026) and three 2025 Yemen strikes. URL: <https://www.militarytimes.com/news/pentagon-congress/2026/06/18/senate-eyes-hegseth-travel-cuts-without-probes-into-iran-school-bombing-boat-strikes/> (visited on 08/25/2026).
- Strafgesetzbuch* (Nov. 13, 1998). § 86: *Verbreiten von Propagandamitteln verfassungswidriger und terroristischer Organisationen*; § 86a: *Verwenden von Kennzeichen verfassungswidriger und terroristischer Organisationen*. In the version promulgated 13 November 1998, BGBl. I S. 3322, as last amended by Article 1 of the Act of 20 March 2026, BGBl. 2026 I Nr. 95. The social-adequacy clause that § 86a(3) applies was renumbered from § 86(3) to § 86(4) by the Sixtieth Act Amending the Criminal Code of 30 November 2020, BGBl. I S. 2600, in force 1 January 2021, so commentary written earlier cites the older number. URL: https://www.gesetze-im-internet.de/stgb/_86a.html (visited on 09/18/2026). The reference to terrorist organizations in § 86(2) and in both section headings was added by Article 1 of the Act of 14 September 2021, BGBl. I S. 4250, in force 22 September 2021. Agent-verified 2026-09-17, and the url added 2026-09-18: both section headings match this entry’s subtitle word for word at [gesetze-im-internet.de](https://www.gesetze-im-internet.de), and § 86a(3)’s cross-reference reads § 86 Abs. 4 und 5 gilt entsprechend, which confirms the renumbering recorded above. That page carries no currency statement, so the 20 March 2026 amendment is not confirmed from it.
- Sun, Haochen, Jason Li, and Hongyang Zhang (2024). “zkLLM: Zero Knowledge Proofs for Large Language Models”. In: *Proceedings of the 2024 ACM SIGSAC Conference on Computer*

- and Communications Security (CCS 2024). Reports a correctness proof for the entire inference of a 13-billion-parameter model in under 15 minutes, at under 200 kB, disclosing nothing about the parameters. DOI: 10.1145/3658644.3670334. URL: <https://arxiv.org/abs/2404.16109> (visited on 08/29/2026).
- Syta, Ewa et al. (2016). “Keeping Authorities ‘Honest or Bust’ with Decentralized Witness Cosigning”. In: *Proceedings of the 2016 IEEE Symposium on Security and Privacy (SP 2016)*. A statement an authority signs is co-signed by a large set of witnesses before anyone accepts it, so that the authority cannot sign one thing for one party and another for the rest without the witnesses seeing both. The software form of the property a hardware counter supplies, resting on the witnesses being many and on none of them being trusted., pp. 526–545. DOI: 10.1109/SP.2016.38. URL: <https://doi.org/10.1109/SP.2016.38> (visited on 09/05/2026).
- Taigman, Yaniv et al. (2014). “DeepFace: Closing the Gap to Human-Level Performance in Face Verification”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. Cited twice and for two different things. First, for the alignment stage: the pipeline detects a face and warps it by piecewise affine transformation onto a single generic three-dimensional shape model, itself the average of scans from the USF Human-ID database, so that every input reaches the network frontal and in correspondence with one reference. That is the registration step section 2.2.1 reads as a standard face implemented as a file. Second, for the distinction between face verification, which this system performs and which only decides whether two images show the same person, and facial expression analysis, which is a harder problem. Cited without a page locator: the alignment description is section-level in the paper rather than a passage. IEEE, pp. 1701–1708. URL: https://openaccess.thecvf.com/content_cvpr_2014/html/Taigman_DeepFace_Closing_the_2014_CVPR_paper.html (visited on 08/25/2026).
- Taiwan Ministry of Digital Affairs (2024). *Alignment Assemblies*. Records the ministry’s citizens’ deliberative assembly of 23 March 2024, “Utilizing AI to Enhance Information Integrity,” whose participants were drawn from across Taiwan by stratified random sampling, and the Democratizing AI Futures deliberative workshops of 26 August and 2 September 2023 that preceded it. URL: <https://moda.gov.tw/en/major-policies/alignment-assemblies/1453> (visited on 09/13/2026).
- Tamirisa, Rishub et al. (2024). “Tamper-Resistant Safeguards for Open-Weight LLMs”. In: *International Conference on Learning Representations (ICLR 2025)*. Introduces TAR, a training method that builds safeguards into open-weight model weights designed to survive hundreds of steps of adversarial fine-tuning. arXiv: 2408.00761. URL: <https://arxiv.org/abs/2408.00761> (visited on 08/25/2026).
- Tangney, June Price, Jeff Stuewig, and Debra J. Mashek (2007). “Moral Emotions and Moral Behavior”. In: *Annual Review of Psychology* 58, pp. 345–372. DOI: 10.1146/annurev.psych.56.091103.070145.
- TechCrunch (2023). *Microsoft Lays off an Ethical AI Team as It Doubles Down on OpenAI*. Microsoft’s Ethics and Society team, roughly thirty people in 2020, was cut to about seven in an October 2022 reorganization and eliminated entirely in March 2023, part of a roughly 10,000-employee layoff wave that accompanied deeper reorientation around OpenAI. TechCrunch. URL: <https://techcrunch.com/2023/03/13/microsoft-lays-off-an-ethical-ai-team-as-it-doubles-down-on-openai> (visited on 08/25/2026).
- Templeton, Adly et al. (2024). *Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet*. Sparse autoencoders trained with up to thirty-four million features on a deployed model’s middle-layer residual stream; the recovered features can be used to steer the model’s behavior. Anthropic. URL: <https://transformer-circuits.pub/2024/scaling-monosemanticity/> (visited on 08/28/2026).

- The Citizens' Assembly (2016). *The Citizens' Assembly*. Official website, established by resolution of the Houses of the Oireachtas, July 2016. 99 randomly selected citizens plus an independent chair; deliberated on the Eighth Amendment (abortion) and on Ireland's response to climate change among its five mandated topics; work concluded 2018. URL: <https://www.citizenassembly.ie/> (visited on 08/25/2026).
- Thompson, Evan (2007). *Mind in Life: Biology, Phenomenology, and the Sciences of Mind*. The enactivist statement of the deep continuity of life and mind; its sixth chapter connects Jonas's philosophical biology to autopoiesis. Author, title, publisher, place and year agent-verified 2026-09-18. What was not agent-verified is that this book reads Spinoza's conatus as the ancestor of autopoiesis. The full text could not be searched from here, and one detailed review of it traces Thompson's genealogy for autopoiesis to Kant's third Critique, naming Spinoza nowhere. The conatus-to-autopoiesis line is made in the secondary literature on Spinoza, which cites this book; whether the book makes it is open. Cambridge, MA: Harvard University Press.
- Thornley, Elliott et al. (2025). *Towards Shutdownable Agents via Stochastic Choice*. Trains agents to prefer between same-length trajectories and to lack a preference between different-length ones, so that usefulness and neutrality about shutdown timing come apart. The authors report that neutrality degrades with imperfect reward normalization, that generalization beyond deterministic gridworlds is untested, and that a sufficiently misaligned agent may resist shutdown regardless. arXiv: 2407.00805. URL: <https://arxiv.org/abs/2407.00805> (visited on 08/31/2026).
- Trump v. Cook* (2026). No. 25A312, 609 U.S. — (June 29, 2026). Interim docket; the disposition is "Application for stay denied." 5–4. Roberts, C.J., for the Court, joined by Sotomayor, Kagan, Kavanaugh and Jackson, JJ.; Kavanaugh and Jackson, JJ., concurring; Thomas, J., Alito, J. (joined by Gorsuch, J.), and Barrett, J., dissenting. Upholds the Federal Reserve Act's for-cause removal protection, on the Fed's descent from the First and Second Banks of the United States, its budget free of congressional control and its privately owned regional banks, and holds that a Governor is owed notice and a hearing before removal. The constitutional holding is real and is what Thomas, J., dissents from, but the case is disposed of narrowly, on the ground that Cook did not receive the process the statute required: the Court states that it has "not addressed the facts, as they have yet to be found or analyzed under the relevant legal standards," and Kavanaugh, J., concurring, opens that "[t]oday's interim ruling does not decide whether the President may lawfully remove Governor Cook for cause." The dispute is live — on 5 August 2026 the President notified Cook that he was again considering her removal, giving her 21 days to respond and citing this opinion as his authority for the procedure. Agent-verified 2026-08-30. URL: <https://www.law.cornell.edu/supremecourt/text/25A312> (visited on 08/29/2026).
- Trump v. Hawaii* (2018). No. 17–965, 585 U.S. 667 (June 26, 2018). 5–4. Roberts, C.J., for the Court, joined by Kennedy, Thomas, Alito and Gorsuch, JJ.; Breyer, J., dissenting, joined by Kagan, J.; Sotomayor, J., dissenting, joined by Ginsburg, J. Upholds Proclamation No. 9645 restricting entry from several named countries. On *Korematsu*, the Court says both that "[w]hatever rhetorical advantage the dissent may see in doing so, *Korematsu* has nothing to do with this case" and that "*Korematsu* was gravely wrong the day it was decided, has been overruled in the court of history, and — to be clear — 'has no place in law under the Constitution,'" the inner quotation being Jackson, J., dissenting in *Korematsu*. Sotomayor, J., dissenting: the Court "redeploys the same dangerous logic underlying *Korematsu* and merely replaces one 'gravely wrong' decision with another."
- Trump v. Slaughter* (2026). No. 25-332, 609 U.S. — (June 29, 2026). 6–3. Roberts, C.J., for the Court; Gorsuch, J., concurring; Sotomayor, J., dissenting, joined by Kagan and Jackson, JJ. Holds the Federal Trade Commission Act's for-cause removal protection unconstitutional and overrules *Humphrey's Executor v. United States* (1935): "If anything more is left of *Humphrey's*,

- we overrule it.” URL: <https://www.law.cornell.edu/supremecourt/text/25-332> (visited on 08/29/2026).
- U.S. Department of Commerce, Bureau of Industry and Security (2022). *Implementation of Additional Export Controls: Certain Advanced Computing and Semiconductor Manufacturing Items; Supercomputer and Semiconductor End Use; Entity List Modification*. 87 Fed. Reg. 62186. URL: <https://www.federalregister.gov/documents/2022/10/13/2022-21658/implementation-of-additional-export-controls-certain-advanced-computing-and-semiconductor> (visited on 08/25/2026).
- U.S. Department of Health and Human Services, Office for Human Research Protections (2018). *Federalwide Assurance for the Protection of Human Subjects*; 45 C.F.R. §46.103. Code of Federal Regulations. URL: <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-A/part-46/subpart-A/section-46.103> (visited on 08/25/2026).
- U.S. Department of Justice, Civil Rights Division (May 2022). *Algorithms, Artificial Intelligence, and Disability Discrimination in Hiring*. Issued 12 May 2022 alongside a companion EEOC technical assistance document, EEOC-NVTA-2022-2. Ties facial analysis to a named disability: where a government uses facial and voice analysis technologies to evaluate applicants’ skills and abilities, people with disabilities like autism or speech impairments may be screened out. The EEOC companion lists video interviewing software that evaluates candidates on their facial expressions and speech patterns among the tools it covers, and gives a screen-out example turning on a speech impediment. URL: <https://www.ada.gov/resources/ai-guidance/> (visited on 08/30/2026).
- U.S. Government Accountability Office (Apr. 2026). *Artificial Intelligence Acquisitions: Agencies Should Collect and Apply Lessons Learned to Improve Future Procurements*. GAO-26-107859. URL: <https://www.gao.gov/products/gao-26-107859> (visited on 09/18/2026). Read from the report’s own HTML full text at <https://files.gao.gov/reports/GAO-26-107859/index.html>; the PDF was never needed, and the product page alone is what defeated an earlier attempt. Scope: thirteen AI acquisitions at four agencies — the Departments of Defense, Homeland Security and Veterans Affairs and the General Services Administration — and, from that sample, “44 contracts and agreements awarded between September 2018 and February 2025.” **The forty-four are contracts and agreements, not acquisitions.** Cited for the other-transaction finding: “OTAs do not always include default provisions ordinarily in FAR contracts,” contracting officials deciding what protections to include, and GAO having “previously found that agencies’ use of OTAs carries the risk of reduced accountability and transparency compared to FAR-based contracts.” Also for the vehicle behind the agreements section 10.3 turns on: “In July 2025, DOD awarded OTAs with \$200 million ceilings to four leading AI companies.” The six key procurement challenge areas identified by the selected agencies are access to subject matter experts; protections for government data and intellectual property rights; traditional acquisition time frames and contract approaches; requirements definition and contract terms; early testing and continuous evaluation; and artificial intelligence pricing and overall cost. The report’s central finding, which the book does not use, is that the agencies were not yet systematically collecting lessons learned because their policies did not require it. Maven appears in it as the National Geospatial-Intelligence Agency’s programme, with lessons-learned sections citing its practices; that material is not used either. Agent-verified 2026-09-18.
- U.S. Office of Management and Budget (Apr. 2025). *Driving Efficient Acquisition of Artificial Intelligence in Government*. OMB Memorandum M-25-22. URL: <https://www.whitehouse.gov/wp-content/uploads/2025/02/M-25-22-Driving-Efficient-Acquisition-of-Artificial-Intelligence-in-Government.pdf> (visited on 09/17/2026). Signed by Russell T. Vought, Director; rescinds and replaces M-24-18. Cited for the contract-terms list in the acquisition lifecycle guidance, which requires agencies to address IP rights and use of government data, privacy, protections against vendor lock-in, compliance with M-25-21, ongoing testing and

- monitoring, vendor performance requirements, and new feature notification. The lock-in items are given at solicitation as “requirements regarding knowledge transfer, clear data and model portability practices, clear licensing terms, and pricing transparency.” On independent evaluation: “The data used when conducting independent evaluations should not be accessible to the vendor, and should be as similar as possible to the data used when the system is deployed.” Read from a text extract of the same document, the published file being an image PDF that defeated conversion. Agent-verified 2026-09-17.
- UC Berkeley (2016). *UC Berkeley Launches Center for Human-Compatible Artificial Intelligence*. Press release, 29 August 2016. Founded by Stuart Russell with co-principal investigators Pieter Abbeel, Anca Dragan, and Tom Griffiths (UC Berkeley), Bart Selman and Joseph Halpern (Cornell), and Michael Wellman and Satinder Singh Baveja (Michigan); initial funding from the Open Philanthropy Project. University of California, Berkeley. URL: <https://news.berkeley.edu/2016/08/29/center-for-human-compatible-artificial-intelligence/> (visited on 08/25/2026).
- Uesato, Jonathan et al. (2022). *Solving Math Word Problems with Process- and Outcome-Based Feedback*. Outcome supervision matched process supervision on final-answer accuracy while using less labeled data; among correct final answers the reasoning-error rate was 14.0 percent against 3.4 percent. arXiv: 2211.14275. URL: <https://arxiv.org/abs/2211.14275> (visited on 08/28/2026).
- UK AI Security Institute (2025). *We Are Now the AI Security Institute*. Established November 2023 as the AI Safety Institute; rebranded AI Security Institute on 14 February 2025. An independent pre-deployment evaluator of frontier models with no statutory power to compel testing, block release, or intervene once a system is causing harm. AI Security Institute (UK Department for Science, Innovation and Technology). URL: <https://www.aisi.gov.uk/category/governance> (visited on 08/25/2026).
- Ullman, Tomer (2023). *Large Language Models Fail on Trivial Alterations to Theory-of-Mind Tasks*. arXiv: 2302.08399. URL: <https://arxiv.org/abs/2302.08399> (visited on 08/25/2026).
- Uniform Law Commission (1993). *Uniform Probate Code §2-907: Honorary Trusts; Trusts for Pets*. National Conference of Commissioners on Uniform State Laws, Uniform Probate Code, Official Text with Comments (as amended through 2018), p. 263. URL: <https://www.uniformlaws.org/viewdocument/final-act-with-comments-114?CommunityKey=a539920d-c477-44b8-84fe-b0d7b1a4cca8&tab=librarydocuments> (visited on 08/25/2026).
- United Nations General Assembly (Dec. 1948). *Universal Declaration of Human Rights*. UNGA Resolution 217 A (III). Adopted in Paris, 48 in favor, none against, 8 abstentions (Byelorussian SSR, Czechoslovakia, Poland, Saudi Arabia, South Africa, Ukrainian SSR, USSR, Yugoslavia), 2 absent (Honduras, Yemen). URL: <https://www.un.org/en/about-us/universal-declaration-of-human-rights> (visited on 08/25/2026).
- (Dec. 1966a). *International Covenant on Civil and Political Rights*. UNGA Resolution 2200A (XXI). Adopted under one resolution with the Covenant on Economic, Social and Cultural Rights; entered into force 23 March 1976, and 28 March 1979 for article 41. The two covenants divide the Universal Declaration between them, this one taking the civil and political articles. Registered ex officio 23 March 1976, No. I-14668; 999 UNTS 171. Agent-verified 2026-09-18: the volume and registration number from the United Nations Treaty Collection’s own file path for the volume extract for I-14668, which is a scan and could not be read as text, so the starting page 171 rests on secondary citation and not on the volume itself. D-322 could not reach OHCHR or Refworld, both returning 403; the OHCHR url above is unchanged and still unreached, and the UNTS path is what replaced the attempt. URL: <https://www.ohchr.org/en/instruments-mechanisms/instruments/international-covenant-civil-and-political-rights>.
- (Dec. 1966b). *International Covenant on Economic, Social and Cultural Rights*. UNGA Resolution 2200A (XXI). Adopted under one resolution with the Covenant on Civil and Political Rights;

- entered into force 3 January 1976. This one takes the Declaration's economic, social and cultural articles. Registered ex officio 3 January 1976, No. I-14531; 993 UNTS 3. Agent-verified 2026-09-18 on the same footing as the companion covenant: volume and registration number from the United Nations Treaty Collection's file path for I-14531, the starting page 3 from secondary citation, the volume extract being a scan. OHCHR and Refworld were 403 at D-322 and were not reached again. URL: <https://www.ohchr.org/en/instruments-mechanisms/instruments/international-covenant-economic-social-and-cultural-rights>.
- United States Congress (1965). *Voting Rights Act of 1965, Section 4: Suspension of the Use of Tests or Devices in Determining Eligibility to Vote*. Pub. L. No. 89-110, 79 Stat. 437; codified at 52 U.S.C. §10303. Cited for subsection (c), which defines a test or device as any requirement to demonstrate the ability to read, write, understand or interpret any matter, to demonstrate educational achievement or knowledge of a particular subject, to possess good moral character, or to prove one's qualifications by the voucher of registered voters. Definition agent-verified against the codified text. URL: [https://uscode.house.gov/view.xhtml?req=\(title:52+section:10303+edition:prelim\)](https://uscode.house.gov/view.xhtml?req=(title:52+section:10303+edition:prelim)) (visited on 09/06/2026).
- United States District Court for the Northern District of California (2025). *Bartz et al. v. Anthropic PBC*, No. 3:24-cv-05417, *Order on Motion for Summary Judgment on Fair Use* (N.D. Cal. June 23, 2025). Order by Senior District Judge William Alsup. Held that training Claude on lawfully purchased/scanned books was "exceedingly transformative" fair use, while retaining pirated copies in a "central library" was not fair use; the piracy claims later settled for \$1.5 billion, reported as the largest copyright recovery on record. URL: <https://www.courtlistener.com/docket/69058235/bartz-v-anthropic-pbc/> (visited on 08/25/2026).
- University of Chicago Data Science Institute (2026). *Using AI and Data Science to Reliably Detect Internet Censorship in Real Time*. University of Chicago Data Science Institute news. Describes a \$1 million DARPA-funded project led by Prof. Nick Feamster (University of Chicago) with Princeton University and SRI International, using AI/statistical methods to distinguish deliberate censorship from ordinary network failure and produce a real-time map of state interference with internet traffic. URL: <https://datascience.uchicago.edu/news/using-ai-and-data-science-to-reliably-detect-internet-censorship-in-real-time/> (visited on 08/25/2026).
- Victoria, Brian Daizen (2006). *Zen at War*. 2nd ed. First published New York: Weatherhill, 1997. Cited for three things, all of them the book's own findings: that Japanese Zen institutions and established teachers supplied the wartime state with a doctrine of action without hesitation, worked up out of no-self and non-attachment; that the killing sword was glossed as the life-giving sword and warfare described as an expression of compassion; and that Harada Daiun Sōgaku, writing in the November 1939 issue of *Daijō Zen* in an article on the one road of Zen and war, instructed that marching when ordered to march and firing when ordered to fire is the manifestation of the highest wisdom of enlightenment. The Harada wording was agent-checked against independent sources quoting it identically. Cited without a page locator: the 1997 and 2006 editions paginate differently and published citations to the Harada passage give both 66–67 and 137, so no page is asserted. The book's reception belongs with it. Institutional collaboration is documented and not seriously disputed; what is contested is the doctrinal reading and some individual attributions, most prominently Kemmyō Taira Satō's sixty-page study in *The Eastern Buddhist*, which acknowledges Victoria's contribution to the subject and argues he erred seriously in the case against D. T. Suzuki. Section 6.4.1 states that contest rather than citing the book flat. Lanham, MD: Rowman & Littlefield.
- Vincent, Brandi (2025). 'Growing Demand' Sparks DOD to Raise Palantir's Maven Contract to More Than \$1B. DefenseScoop, 23 May 2025. Source for the contract figures: a \$480 million five-year IDIQ contract signed in May 2024, raised by \$795 million to a ceiling of nearly \$1.3 billion running through 2029. Agent-verified 2026-09-07. URL: <https://defensescoop.com>.

- com/2025/05/23/dod-palantir-maven-smart-system-contract-increase/ (visited on 09/07/2026).
- Virilio, Paul (1976). *Essai sur l'insécurité du territoire*. Contains the chapter "L'État suicidaire," the source of the reading that the Nazi state was suicidal rather than totalitarian; Deleuze and Guattari take it up in *deleuze1987plateaus*, where the totalitarian state is called quintessentially conservative and fascism is built on a line of flight converted into pure destruction. Reissued Paris: Galilée, 1993. Cited in French; no standard English translation of this work. Not to be confused with *Speed and Politics (Vitesse et politique, 1977)*, which does not contain the chapter. Paris: Stock.
- Waldron, Jeremy (2006). "The Core of the Case Against Judicial Review". In: *Yale Law Journal* 115, pp. 1346–1406. Argues that judicial review is democratically illegitimate and that there is no reason to suppose rights are better protected by it than by a legislature. The case is explicitly conditional on a society having working democratic institutions and a citizenry that takes rights seriously. Cited as the standing objection an entrenched floor has to answer.
- Watson, Gary (1975). "Free Agency". In: *The Journal of Philosophy* 72.8, pp. 205–220. DOI: 10.2307/2024703. Separates a valuational system from a motivational system, so that what an agent values and what an agent is moved by can come apart.
- Werth, Nicolas (2010). *The NKVD Mass Secret Operation no. 00447 (August 1937 – November 1938)*. Published 24 May 2010. Source of the order's mechanism as the text reports it: round-number quotas per region and republic in two categories, 269,100 persons of whom 76,000 were to be shot, against a four-month deadline; the operation instead ran from August 1937 to mid-November 1938 and condemned some 767,000 people, of whom about 387,000 were shot. On the boards: "The troïki sat behind closed doors, the accused was absent and without any defender," and in sittings of roughly half a day a troïka "went through several hundred cases." The text's phrase is that sentence rather than an arithmetic of its own, and it does not claim there was no appeal, which this source does not state. URL: <https://www.sciencespo.fr/mass-violence-war-massacre-resistance/en/document/nkvd-mass-secret-operation-n-00447-august-1937-november-1938.html> (visited on 09/14/2026).
- Wiggers, Kyle (Nov. 2024). *Anthropic Teams Up with Palantir and AWS to Sell AI to Defense Customers*. Source for the November 2024 partnership and for where Claude runs: "Claude became available on Palantir's platform earlier this month and can now be used in Palantir's defense-accredited environment, Palantir Impact Level 6 (IL6)." The article does not name Maven. The joint press release, which is reported to state that Palantir is the first industry partner to bring Claude models to classified environments, returned HTTP 403 from here and was not read, so this entry carries only what the TechCrunch report states. Agent-verified 2026-09-18. TechCrunch. URL: <https://techcrunch.com/2024/11/07/anthropic-teams-up-with-palantir-and-aws-to-sell-its-ai-to-defense-customers/> (visited on 09/18/2026).
- Zhuang, Simon and Dylan Hadfield-Menell (2020). "Consequences of Misaligned AI". In: *Advances in Neural Information Processing Systems* 33. Gives necessary and sufficient conditions under which indefinitely optimizing an incomplete proxy objective — one with support on a subset of the attributes the principal values — drives overall utility arbitrarily low, in a resource-constrained world. Cited as the formal form of the aestheticized metric. arXiv: 2102.03896 [cs.AI]. URL: <https://arxiv.org/abs/2102.03896> (visited on 09/01/2026).